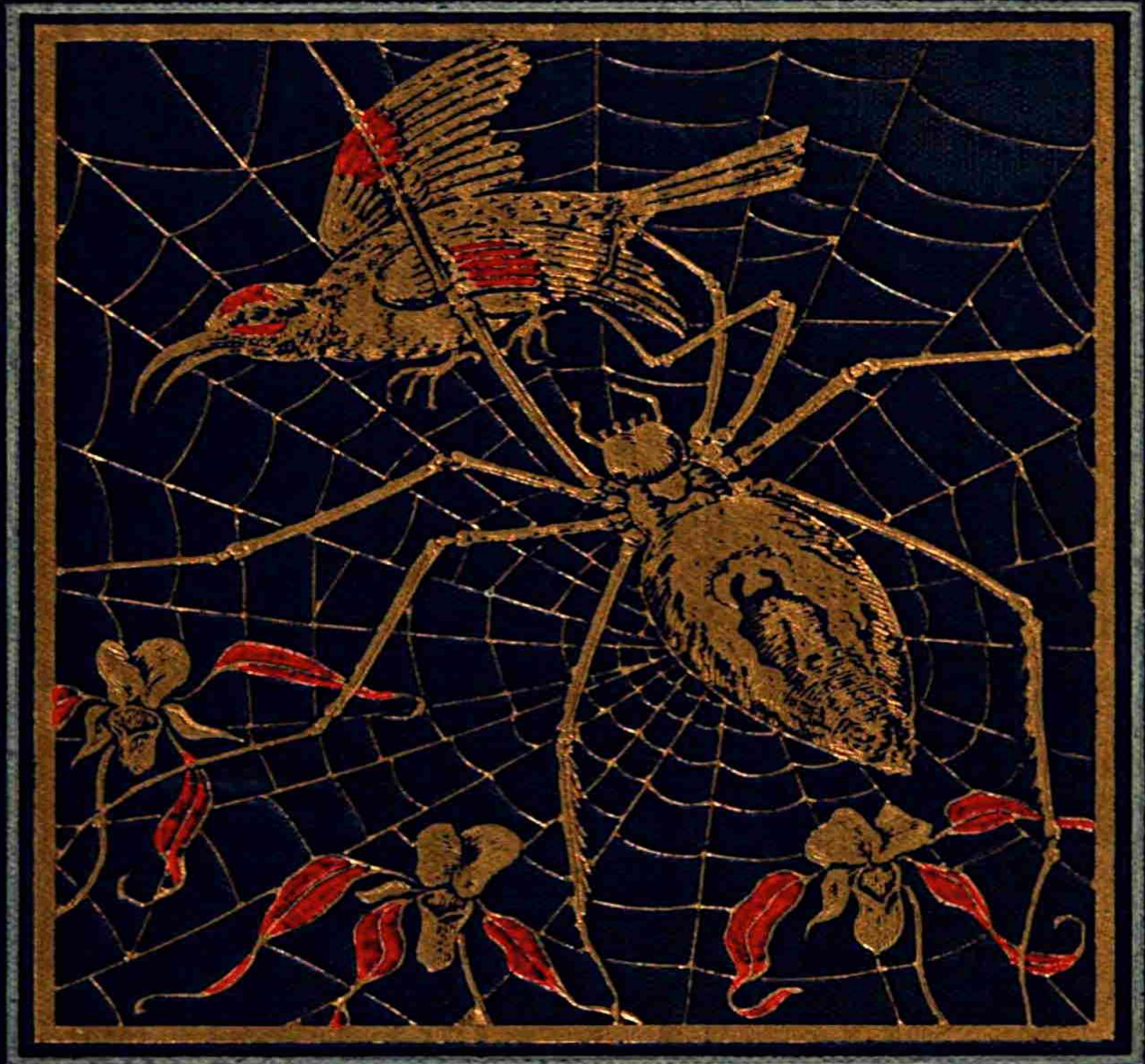
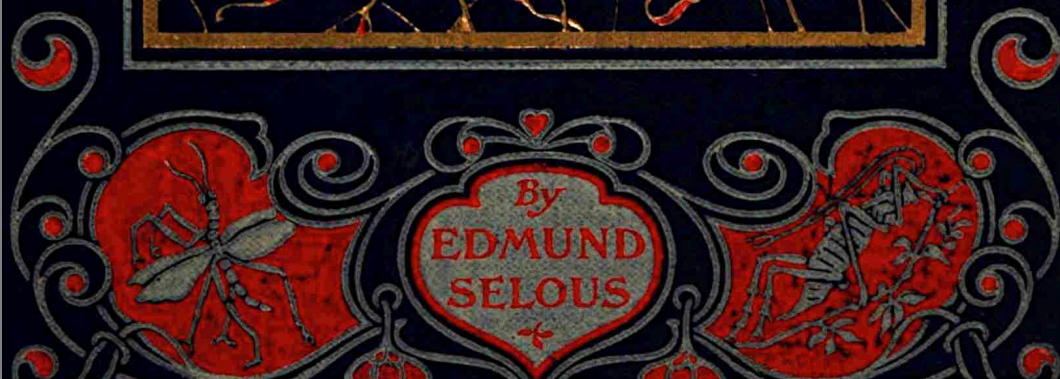


THE ROMANCE OF INSECT LIFE



By
EDMUND
SELOUS

THE ROMANCE OF INSECT LIFE



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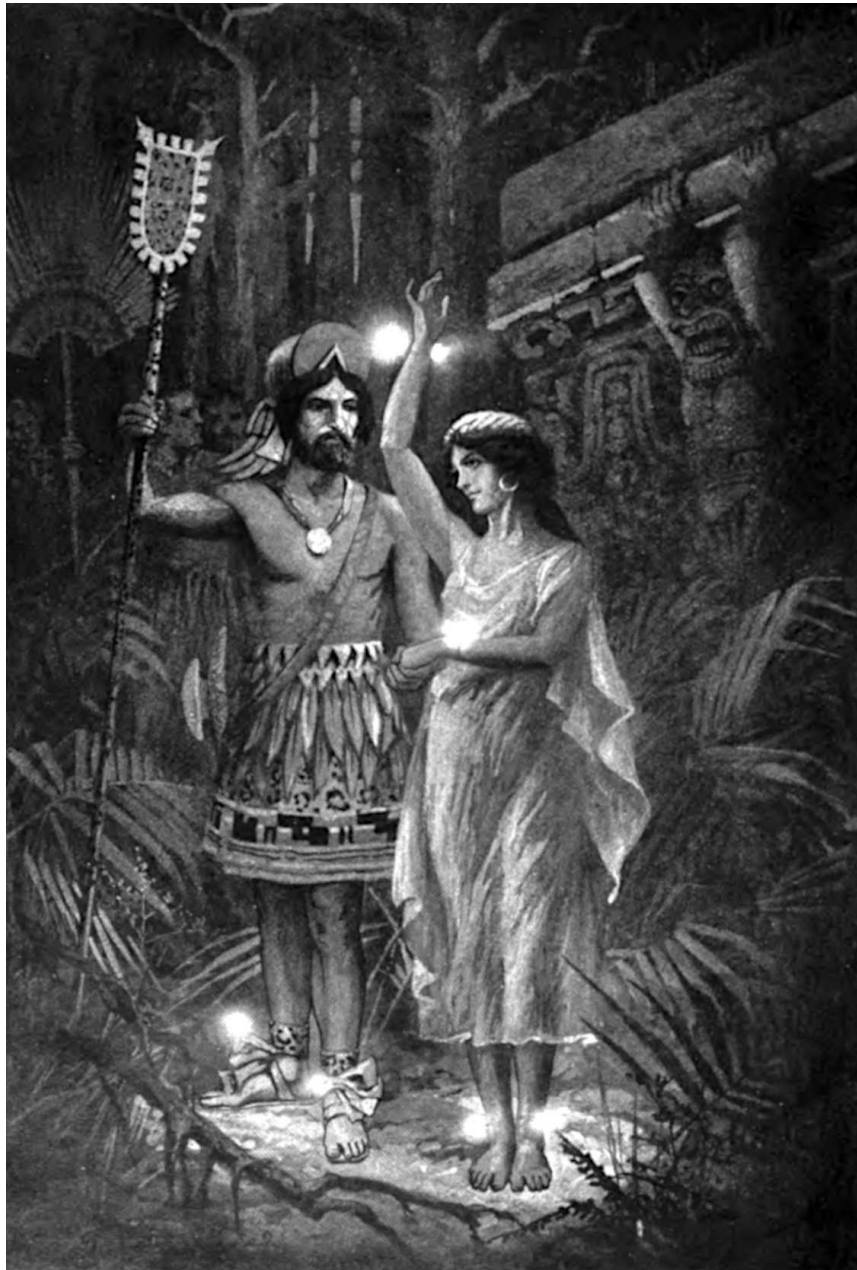
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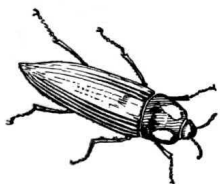
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OF INSECT LIFE ***

THE ROMANCE OF INSECT LIFE



FIRE BEETLES AS LANTERNS.



The Aztecs of Mexico were accustomed to use these insects to light them through the forests by night. Fastening them to their hands and feet, they passed flaming along. It is said that the Mexicans still use them for this purpose. The fire beetle is shown to the left of this inscription.

THE ROMANCE OF INSECT LIFE

INTERESTING DESCRIPTIONS OF
THE STRANGE AND CURIOUS IN
THE INSECT WORLD

BY

EDMUND SELOUS

AUTHOR OF

“THE ROMANCE OF THE ANIMAL WORLD,” “BIRD LIFE GLIMPSES,”
“BIRD WATCHING,” “TOMMY SMITH’S ANIMALS,” &c.

WITH TWENTY-ONE ILLUSTRATIONS BY

LANCELOT SPEED

&

CARTON MOORE PARK

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1914

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By Lancelot Speed & Carton Moore Park

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THE ROMANCE OF INSECT LIFE

CHAPTER I

“The natural system”—A middle course—Neuropterous insects—White ants and their ways—Kings and queens—A royal diet—Secondary majesties—Soldiers and workers—Ant invaders—Methods of warfare.

IF there is any plan in this little book it will, no doubt, appear in time to its readers, but I myself am only quite clear as to this, that, not being of a scientific nature, it will not include a definition of an insect. Why should it? Everybody knows what he thinks an insect is, and those who may be willing to have their ideas on such a fundamental subject disturbed will rightly consult some work of greater authority than this can pretend to. So instead of worrying myself, and others, about what insects really are, or what are not really insects, as, for instance, spiders, centipedes, scorpions, and the like, all which I propose to include in my tale—should they happen to present themselves—I shall confine myself to saying something about what some insects do, and I shall let one suggest and lead to the discussion of another, quite at haphazard, and without any attempt at system or classification whatever. This, in fact, is my own idea as to what is “the natural system,” and the only trouble about it is knowing where to begin, because, as there are some 300,000 known insects,^[1] and any one of them will do as well to start from as any other, there is a great *embarras de richesses*. In such cases the usual thing to do is to take either the head or the tail of the series—to commence with the *Hymenoptera*, which include the highest and most intelligent forms, such as the ants and bees, or else with the *Collembola* or *Thysanura*, which are understood to contain the lowest. I shall not adopt either of these methods. The *Neuroptera*, as far as I can make out (and if they don’t it doesn’t matter), stand somewhere about the middle, and with them accordingly—as being between the two extremes—I decide to break ground. Having done so, as I said before, I may go anywhere—absolute freedom will be mine. Like Plato, I can follow the argument whithersoever it leads; inspired with which reflection I hasten to begin it.

Though the order of neuropterous—which, by the way, means nerve-winged—insects does not contain any ants, yet the so-called white ants or termites—which are very like ants in their ways, and almost, or quite, as interesting to talk about—are included in it. They are commonest in tropical or, at any rate, very hot countries, such as Africa, Australia, and South America, and here the conical, or dome-shaped structures, made of red earth, which they erect above the surface of the ground, and which contain the greater part of the nest, are of such dimensions as to take a very prominent part in the features of the landscape. Often they are covered with vegetation, including bushes, or even small trees, on which, in Africa, antelopes are accustomed to browse. In Australia there is no reason, that I can see, why kangaroos should not, at least upon the grass which must often clothe them, and which is their staple of food.



WHITE ANT HEAPS

These great mounds are made by the white ants, and contain their nests; but large and strong as they are, the ant-eater breaks them down and devours the ants. A queen white ant is shown at the right-hand corner with the extraordinary development in which the eggs are carried.



These great mounds are pierced in every direction with innumerable galleries, leading to and from the various cells and chambers in which the

domestic economy of the white ants is principally performed, one of which, known as the royal cell, contains the king and queen, and is situated beneath all the others. Not all white ants, however—for there are several species—are governed or presided over in this way. Grassi, who studied them in Sicily,^[2] declares that the whole of the Termitidæ, whether belonging to Southern Europe or the still hotter countries from which they have, no doubt, been unknowingly imported, fall into two primary types. In the first of these the colony is presided over by a king and queen, representing the fully developed male and female forms, which have once, unlike the workers and soldiers—for, like ants, these insects are divided into castes—possessed fully developed wings, which they have subsequently got rid of in the same way that the queen ant does hers. In the second type the colony possesses several kings and queens, but these, though they marry and produce offspring, are not perfect males and females, and never possess wings. They are, in fact, produced artificially by the working termites, just as the hive-bees are able to make themselves a new queen—should they require one—by feeding an ordinary worker with royal jelly, and by a method somewhat similar though not precisely the same, the royal substitutes being fed, not on any extraneous substance, but on a salivary fluid secreted by the workers themselves—saliva, in fact. The colony, however, is, in this case, not founded by the royalties thus bred up, but by a portion of a pre-existent colony which, migrating from the parent nest, takes this method of augmenting its numbers.^[3]

In the termite nest, as amongst ants, all members work for the good of all. The soldiers, which are furnished with large heads and long scythe-like jaws, take upon themselves the duties of attack and defence, though in some species they only do so when the enemy is of a formidable nature, leaving unimportant foes to their less specialised companions. These are equal to such inglorious tasks, but when the colony is invaded by hostile members of their own race, or by some fierce ant enemy, they retreat into the inner recesses, leaving the danger and honour to others. Such an enemy is *Cremastogaster scutellaris*—or call him Cremas—who, though never invaded by the white ants, enters their nest—or termitary, to use the learned word—intent upon massacre. Under such circumstances “the soldiers place themselves, with gaping mandibles, waiting for any ant that may come within reach. They then snap their jaws rapidly, shearing off antennæ and

legs, tearing the abdomen, or even cutting the ants in two. The soldiers' mandibles are seen to act like extremely sharp shears."^[3] This should be somewhat discouraging for the ants, and, indeed, they seem rather shy of the soldiers, avoiding their heads, and "only daring occasionally to attempt to lop off their mandibles." Their more considered method, which they adopt whenever practicable, is to approach them from behind, and bite their abdomens, the soldiers, on their part, endeavouring to protect this vulnerable portion—and it is a fairly large one—of their anatomy by creeping backwards under pieces of wood or stones, from which the head, with its murderous jaws, is alone allowed to project.

In these encounters the advantage does not seem to lie so decidedly with the ants as to explain their conduct in making the invasion, since peace, according to Professor Grassi's observations, is usually concluded "after about an hour's conflict, with a certain number of killed and wounded on both sides."^[3] As a result, however, it would appear that the ants often remain in possession of a portion of the nest, whilst the original occupants have to be contented with what remains. If this, therefore, is their object, the invaders have carried the day, but if, as seems likely under natural conditions, they should prefer to return to their own home, they can hardly be said to have done so. Information seems wanting on these points.

As with ants, war is also waged between the various species of Termitidæ. *Termes lucifuga*, for instance—for where there is no English name there is nothing for it but to speak Latin—is, though much smaller, a terrible enemy of *Calotermes*. The soldiers of the latter can, indeed, without much difficulty, cut their own in two, but their greater activity is often more than a match for the superior strength of their opponents. The workers are more easily disposed of, but with these the soldiers of *Calotermes* do not often concern themselves. They are left to the nymphs^[4] and larvæ, the equivalents, with the latter species, of a true worker caste which has not yet been developed amongst them, as it has with others of the family. When Professor Grassi placed a worker of *Termes* in one of his *Calotermite* tube-nests^[5] it was at once placed *hors de combat* by a nymph (somewhat a shrewish one) of the latter, which, rushing upon it, cut off a portion of its mouth. Other nymphs, as well as several large larvæ, then hurried up and proceeded to further the good work by severing the unfortunate creature's legs, and tearing open its abdomen. In all this the soldiers took no part until

one, towards the end of the struggle, advanced and added his single bite to those which had been so plentifully bestowed. Similar observations were made upon various other occasions, from which it appears plain that, as before remarked, the soldiers of this—very probably of all the termites—are accustomed, purposely, to reserve their strength for foemen worthy of their steel.

It will be seen from the above account that termites differ from true ants in one very important particular, namely, that they are as active and free-moving in the larval and pupal states as in the mature, or imago, one. “The termite society,” indeed, “consists, for the most part, of wingless sexually immature individuals, children potentially of both sexes, which do not grow up.”^[6] Out of the majority of these the worker caste, when it exists, is formed, whilst a much lesser number develop into the large-headed, long-jawed soldiers. Both of these castes, apparently, are produced independently of sex, that is to say, they are potentially either males or females, and not composed exclusively, as is the case with ants and bees, of undeveloped females. Only the genuine king and queen of the termitary would seem to have attained the true imago state; such substitute royal forms as the workers, by feeding the larvæ with saliva, are able to produce, retaining larval characteristics, though sexually mature—a phenomenon scientifically known as *neoteinia*. As with the bees, these potential future royalties are bred up by the working termites to meet possible future emergencies. They are never allowed to leave the nest, and, should any accident befall the reigning king and queen, a pair of them are chosen to rule and produce offspring.

CHAPTER II

Ant language—Stridulatory organs—How white ants communicate—Conversation through convulsions—Nests in tubes—Detection of a “crepitus”—Mutual recognition—Cannibalistic propensities—Royal jealousy—Loyal assassins—A kingly feast—Methods of feeding—Foundation of colonies—Swarming habits.

IT used to be supposed that such communication as ants are capable of holding with one another took place entirely, or almost entirely, through the mutual stroking of the antennæ, and Sir John Lubbock (now Lord Avebury) was unable to satisfy himself, after numerous experiments, that they could either hear or utter any sound. It is now known, however, that not only can some ants emit various sounds at their pleasure—as, indeed, is sufficiently obvious in the case of one or two species—but also that they possess special structures enabling them to do so, and the existence of which is inconceivable, except on the supposition that they both hear and attach a meaning to the notes thus evolved. Thus at a meeting of the Entomological Society held in the year 1893, Dr. David Sharp (author of the “Insects” portion of *The Cambridge Natural History*) declared that “examination revealed the existence in ants of the most perfect stridulating or sound-producing organs yet discovered in insects, these being situated on the second and third segments of the abdomen in certain species. The sounds produced were of the greatest delicacy, and it appeared doubtful whether the microphone would be able to assist the human ear in their detection”—which, indeed, it has not yet done.^[7] Later, in the work above mentioned, Dr. Sharp remarks, “In many ants these parts”—that is to say the abdominal segments—“bear highly developed stridulating organs, and the delicacy and perfection of the articulations allow the parts to be moved, either with or without producing stridulation.”^[8]

As these ant utterances are not sufficiently loud to be audible to our human ears, they must, I suppose, be inferred from the existence of the organs above-mentioned, and the way in which they work; but this is surely

sufficient data to go upon, since it is hardly possible for one hard substance to grate upon another silently. Forel, accordingly, as well as Janet and other observers, now believe sound to be one of the principal means by which ants hold converse with each other, and it is interesting to find that Grassi and Sandias have arrived at the same conclusion in regard to white ants, or termites. Their opinion, together with the facts upon which it has been founded, is thus expressed:—

“Several writers have mentioned the convulsive movements characteristic of Termites. These movements, or quiverings, are easily observed in Calotermes, and may be repeated periodically at very short intervals, almost at the frequency of the pulse-rate. In the act of quivering, the legs are held motionless, whilst the body is shaken forwards and backwards. Sometimes a white ant may stop, whilst running, in order to quiver one or more times. Occasionally these convulsive movements are repeated a few times only, and then stop altogether; but at other times they recur after a few seconds’ or, at most, a few minutes’ rest, and may thus be continued, sometimes, for hours, at regular or irregular intervals. In the intervals between successive convulsions the insect remains still, or progresses for a short distance only. These movements are executed by all members of the colony except the newly hatched ones. I have satisfied myself,” continues Professor Grassi, “by careful observation of the phenomena exhibited in tube-nests, that these convulsions serve as a cry to summon help or give alarm, or as a lament: in short as a mode of intercommunication.”^[9]

The same observers then go on to tell us that if white ants are disturbed in any sudden way, as by the too rough shaking of their nest, or by a light being suddenly flung upon it, or if otherwise annoyed, “all the members of the colony begin to quiver, except those that are running briskly about in search of a better situation.”^[9] When dying, too, they will sometimes quiver in this way, at intervals of a few minutes, for as much as an hour or two, or even longer. Should an enemy—such as those we have been speaking of—be introduced of a sudden into the nest, the less valiant members of it prefer to run away, but in the midst of their retreat they may often be seen to stop and quiver with unusual energy. Their object in these cases seems to be to raise a general alarm, nor is it long before they are successful. Again, if whilst one insect is burrowing into wood another outside should quiver in this way, the burrower quickly comes out, as though in response to some

signal of alarm. From all this it seems evident that these curious movements must be accompanied by some sound, or sounds, inaudible to our human ears, and perhaps having a varied range, and with considerable power of modulation. To produce them, however, some stridulating or other organs would seem to be necessary, and of these, though they must, if there, be visible under the microscope, Professor Grassi says nothing. Possibly, however, sounds may be produced by the rubbing together of various parts of the body without any special apparatus having been developed, in which case the language, if we may call it so, cannot be so rich or copious.

The above remarks apply more especially to the larger of the two white ants of Southern Europe. In regard to the smaller one, Professor Grassi makes the following interesting remarks: “*Termes* makes the same convulsive movements as does *Calotermes*, but the soldier of this species is able to produce a special creaking sound, which arises, whenever the head is held horizontally, during the act of quivering, by friction between the back of the head and the front part of the thorax. But whenever the head, during this act, is held in the usual position, which is not quite horizontal, no perceptible sound is produced, owing to the absence of such friction. The soldiers of *Termes*, therefore, possess two distinct modes of communication, whilst those of *Calotermes* have only one, in which no perceptible sound is produced. This characteristic *crepitus*,” continues the Professor, “may be heard, at frequent intervals, by applying the ear to a tube containing a nest of Termites. This proves that the quivering motions are a constant feature in undisturbed nests, so that they cannot be employed only as signals of alarm or distress. I conclude, therefore, that besides such special significations these convulsive movements must also have the value of ordinary speech; that they constitute, in short, a means of intercommunication. The same conclusion holds good for *Calotermes* (the one we have hitherto been talking about), and I imagine that the quivering of both species produces a sound which is perceptible to the insects themselves, but inaudible to the human ear.”^[9]

Members of the same ant community are known to recognise each other, and this is no less the case with the white ants, or termites. Thus when a few of the latter were removed from the termitary and returned to it after five or six hours, the population showed no signs of alarm—not scurrying wildly about as they would have done had strangers been introduced—but

remained quiet and orderly. It was objected, however, though I cannot see the force of such an objection, that the exiles, on their return, would have instantly recognised their old nest, and thus, knowing exactly where to go and what to do, they would have created no disorder, and consequently roused no suspicion, amongst the other members of the colony. To meet this theory Professor Grassi provided one of his colonies with a new nest from which he excluded a certain number of individuals, so that when these were introduced into it, an hour or two after their companions had settled down in their fresh abode, it was, of course, quite unfamiliar to them. In spite of this, however, they caused no disturbance, but were clearly recognised as friends. When, however, a few strangers of the same species were introduced, they created great alarm amongst the rightful proprietors, who scattered in all directions. In a little while, however, all was again quiet, and as no fighting was observed, it would appear that, amongst the termites, strangers from different nests soon become friendly with one another. This, however, applies to the commoners only, it is not the same where royalty is concerned. Thus when a second king and queen were introduced into a termitary already provided with a pair, they were at once attacked by the subjects of the latter, who loyally bit off their legs. Two days afterwards the reigning queen was herself seen to attack the male pretender, or rather unfortunate victim of scientific curiosity. He, however, though without legs to assist him, managed to drag himself away, but was afterwards found dead, with the outraged queen nibbling vindictively at his mutilated stumps. Next day the stranger queen was also found dead, and the same thing always happened whenever the experiment was repeated. Sometimes, indeed, the supernumerary royal pair, or pairs, had disappeared altogether, from which it seems clear that they must have been not only killed, but eaten.

Cannibalism, indeed, is rather an institution than a vice in the termitary. To begin with, the cast skin of every member is eaten by the others as a matter of course. With this view, any individual who is ready to moult receives the skilled aid of two or more assistants, who either eat the outer portion of their friend, bit by bit, as they shred it off, or else carry it away whole and devour it at their leisure. Sometimes, moreover, one, after licking another affectionately, in the way that ants do, may be seen to give it a covert bite, as though desirous of something more filling, whilst any sick member is eaten by its companions before it is dead. Royalty is not exempt from this

treatment, and, on one occasion, nine individuals, including one soldier, were observed by Professor Grassi in the act of enjoying a meal on the body of a substitute king who was in process of moulting. The wretched animal was still alive, and writhed all over its body, to free itself from the torture. The nine assassins were probably annoyed at the light, for they at once stopped eating, and jointly carried off the victim to a darker part of the nest. Meanwhile many others crowded up to partake in this feast of royal flesh.^[9]

A soldier, too, has been observed to kill and partially eat one of its worker companions, nor is it altogether uncommon for an individual of any class, after licking, for some little while, the leg of another, suddenly to snap it off. The bond of union, therefore, though sufficiently developed to allow of an elaborate social organisation, is not so strong between members of the same termitary as it is in the case of ants, amongst which latter such unseemly conduct is never known to occur. So, too, unless a particular chemical substance, which seems to have a maddening effect, be flung amongst them, ants of one community never attack each other. Amongst white ants, however, warfare will occasionally break out within the nest, more especially if this be disturbed, in which case the soldiers are apt to turn savagely on those nearest to them, perhaps considering them as the cause of the calamity. Still, upon the whole, order, and, if not friendship, at least co-operation, is conspicuously displayed, and the majority often interfere to put a stop to such individual or partial combats as may from time to time break out.

There is more excuse for the soldier termites in their cannibalistic propensities, since owing to the special development of their jaws, which are long and slender, they are unable to triturate wood, which is the basis of diet of these insects. They might die, therefore, but for such occasional lapses, were it not the common practice for all members of the community to feed one another, though the soldiers, for the above reason, are much more dependent on such aid. The food thus administered has just been swallowed by the individual who parts with it. Such transfer is performed in two ways, the first of which is familiar enough—that process, namely, known as regurgitation—but the second and more staple one is too peculiar to be dealt with in a non-scientific work. When a termite regurgitates, an exceedingly small round pellet of reddish-brown colour may be seen, by attentive observation, to form about the mouth, and gradually to increase in

size till it becomes plainly apparent, and is seen to consist of food—that is to say, wood—which has previously been swallowed, in a moistened and softened condition. Sometimes this pellet is used for building purposes, but often another termite comes forward, receives, and swallows it.

Another article of diet which has a peculiar efficacy, and is used for a certain purpose, has been already alluded to—viz. saliva. This, we are told, “issues,” when required to do so, “as a colourless and distinctly alkaline liquid. It collects on the labium (the insect equivalent of lips) as a small drop, which may be employed either as a cement in building or as food for others. These may either possess themselves of the drop and then retreat a little way in order to swallow it gradually, or they may receive it from the one which secretes it and clearly provides it for them as an article of diet. The assimilation of a drop requires a certain number of acts of deglutition, which may be counted, and are usually four or five.”^[9] Very young larvæ (the whole community, it must be remembered, are either in this or the pupal state) are fed after this fashion, until sufficiently advanced to be able to swallow wood-meal. Under this course of diet the abdomen becomes remarkably transparent, and this, in older individuals, is an indication that they are being bred up by the workers to become royal substitutes. The development, therefore, of termites from the larval to the perfect, or, at least, the sexually perfect form, seems to be wholly dependent on their being fed with this substance.

As is well known, the body of the queen termite, in the African and other tropical species, swells, when about to lay, to an enormous size, but this is not nearly so noticeably the case with her European representatives. Neither is a cell, in this case, constructed for her accommodation, but the royal pair, whether they are true king and queen, or only substitutes, “remain, in close proximity, in the heart of the nest, where the inmates are always most crowded.” They are not imprisoned, therefore; but can go from one place to another, should they, as sometimes happens, wish to change their situation. In this they would seem to be happier than their more specially accommodated royal cousins, but no doubt, with the latter, or at any rate with the queen, the instinct of locomotion ceases with the capacity to indulge in it. The purpose of the specially made cell is probably rather to guard than to restrain the queen.

In regard to the swarming of white ants—another habit in which we are reminded both of ants and bees—with the subsequent founding of a new colony, Professor Grassi has the following remarks to make. They apply more especially to the larger of the two European species, viz. *Calotermes*. “Before swarming,” he tells us, “they collect near one of the exit-holes of the termitary, and when the proper time comes, issue from it in ones or twos, so that the twenty or thirty members who are ready to take flight emerge in perhaps a quarter of an hour. Once outside, they run upwards, if the locality admits of it, for a few metres, and then only do they take wing. In a room they fly towards the light, and if a wind is blowing they follow its direction. Some, becoming tired, settle soon upon trunks of trees, and all may do this eventually. Here they group themselves into pairs, the males and females of which must frequently be derived from separate nests, since the sexes swarm separately; this acts as a safeguard by which *Calotermes* habitually avoids in and in breeding. Matrimonial alliances having been thus formed, the work of excavation commences, each pair seeking for some decayed spot in which to bury themselves and become, in time, the parents of a fresh community. The wings, by this time, have been got rid of. They may be shed by coming into contact with an obstacle, or by getting damp and adhering to some spot, while the insect continues to move. But, if not favoured by chance, the *Calotermite* purposely rids itself of these now useless encumbrances. Thus four perfect insects were captured after flying about the room, and put under a piece of rotten wood. Hardly had they settled when they stripped off their wings by resting the tips against some projecting corner of the wood, and then moving backwards a little, so that the wings bent near the base, broke, and dropped off. When rid of them they began to gnaw the wood, at first along and then across the grain. When they encountered each other by chance they first threatened to bite one another, and then ran off in opposite directions. This was because they were of the same sex. Had they been of opposite ones, an attachment, under such circumstances, would no doubt have been formed between them.”^[9]

This is all the space which I can afford to these interesting insects. There are many other points in connection with them which I might have touched upon, but I thought it better to say less about what may be read by anyone in a score or so of works, and select as my text-book a series of the closest and most interesting observations, which lie buried in the pages of a scientific journal not at all likely to meet the public eye. Where possible, I

shall be guided by the same or a similar principle throughout this small work.

CHAPTER III

Ants and white ants—Guest insects—Ants’-nest beetles—Doubtful relations—A strange forbearance—Yellow ants and white wood-lice—Beetles fed by ants.

FROM what has been said about the Termites in the last chapter, it is clear that they very much resemble ants in their habits, so that it is no wonder that they have long passed for ants in popular estimation. Such a similarity is quite enough to justify one part of the name, as names go; and as for the word white, which entomologists are always complaining about, that is quite near enough too, for though their bodies are not white, but yellow, yet the greater part of them—the soft fat abdomen, which particularly catches the eye—is of such a light yellow that it suggests white in contrast to the darker colouring of most ants. Scientific men—unless their particular science is philology—are dreadful pedants in regard to names, and always want to substitute their own manufactured ones, which have no real life in them, for what has sprung up naturally on the lips of the people. Thus, instead of hedge-sparrow—a name that explains itself to anyone who has seen the bird and knows something of its ways—ornithologists would have us say “hedge-accentor”—a preposterous concoction—and stormy petrel should, according to them, be “storm-petrel,” because the bird itself cannot be stormy, whatever the sea may be. No imagination behind the common use of language, then. No poetic transference of attributes. All is to be as prosy as professors can make it, and “we must speak by the card, or equivocation will undo us.” But names, which are a part of language, come into being as language itself does—spontaneously, that is to say, and by a natural growth. They are right because they exist; and the very errors contained in them—telling, as they do, of popular beliefs and superstitions—are of greater and wider interest than the rectitudes of a few pedants. Could they play with substance as they can with breath, these wise simpletons would first draw up a theory of anatomy, and then annul all bodies that did not conform to it. Such and such a word or name is wrong, in their eyes, though it exists quite as naturally as any nerve or muscle, and

is quite as tough though only made of air. This last they will find if they live long enough, and “hedge-sparrow” and “stormy petrel” will survive all their lifeless substitutions, though embalmed in many dull paragraphs of many dull books.

But let us come back from words to things. Much as the white ants resemble real ones in many of their habits, the more remarkable ones that distinguish the latter are not practised by them. They make no slaves and keep no domestic animals—at least I have never heard of their doing so, though in natural history one must always be prepared for new discoveries. Many insects do, in fact, live with them in the termitary, just as others live in the formicaries of ants, and it is quite possible that, when these have been better studied, some of them will be found to have special relations—involving mutual intelligent action—with their landlords.

At present, however, we seem to have little or no information on this head. With ants it is different, and perhaps one of the most interesting chapters in their history is that which has to do with these *myrmecophilous*, or guest insects, as they are called.^[10] Take, for instance, the ants’-nest beetles, and especially one family—the *Paussidæ*—which numbers some 200 species, every one of which passes the whole of its life, when not flying by night, within the nest of some species of ant. These beetles are small, as might be expected, the largest being not more than half an inch in length, but present an extraordinary appearance owing to the antennæ ending in two broad palmated surfaces, like the horns of a moose deer, which project outwards, one on each side, at right angles with two short stalks, forming the only serviceable joints of these strangely modified feelers. All the other ones (in some species, at any rate) have been fused and welded together to form these flattened club-like structures, the use of which is not at first-sight apparent, and may not be fully understood. If, however, a *paussus* is laid on its back upon a flat surface, a predicament which would be as embarrassing to many beetles as it is to a turtle, one of their special functions is at once seen. Turning back the two clubs till they rest on the ground, and making the joint rigid, the insect uses the one most conveniently placed as a lever, and soon gets on to its legs again. Could we imagine that such an expedient would often need to be resorted to, the curious modification of the antennæ is at once explained; but it probably rarely happens that any small beetle finds itself on its back in a place where there are no irregularities to aid it in

righting itself. Possibly, however, the smooth galleries or chambers of some of the larger ants might expose these *Myrmicophilæ* to such a catastrophe, though, for my part, I suppose that the antennæ are used in some other special way which is of far more importance to their owners.

The relations existing between the ants and these curious beetles has not yet been fully made out. It is true that the various species of *Paussidæ* have upon some part of their bodies a smooth downy substance—a *pubescence*, to use the word dear to entomologists—which in other ants'-nest beetles is known to exude a sweet honey-like dew which the ants, not unnaturally, are very fond of, and for which they assiduously lick them. As they have also been seen to lick their *Paussi*, we seem, here, to have at least the root of the matter, nor does the fact that, at other times, when perhaps these have ceased to supply the attraction, they pay little attention to them, seem of much importance, since we are all neglected when we have given what we have to give. But there are other circumstances not of so straightforward a nature. It has been lately discovered by a French observer—M. Péringuey—that these *Paussidæ*, welcome guests as they generally are, will yet, sometimes, eat the larvæ of the ants with whom they live, when any small worker is engaged in carrying them from one place to another. The ants resent this, and occasionally a large one, who feels himself equal to the undertaking, will attack and even kill a *Paussus* that he sees behaving in such a manner. Yet, with all this, so valued are these beetles by the ants that they often drag them back into their nests, when they have approached, or emerge from, the entrance. On such occasions, and also when the ants attack and even dismember them, the *Paussidæ* make no sort of resistance. Yet they are extremely well able to do so, being armed with a weapon of tremendous efficiency, by which in a moment they could kill or stun a whole crowd of ants round about them. For they are bombardier beetles, having the power at any moment of discharging a fluid of a highly acrid nature, and so volatile that, on coming in contact with the air, it explodes with a puff of blue smoke, exhaling at the same time a very pungent and unpleasant odour. When they are tickled with a straw, even, this bombardment at once takes place, and ants all round are seen to stagger or drop to the earth. Small workers are killed, large ones retreat in confusion; yet the owner of this deadly battery, which can only have been developed for the express purpose of overwhelming an enemy, will not, even to save

life or limb, discharge it against an ant—not one, at least, to whom it stands in these somewhat doubtful relations.

How have these relations—whatever in their entirety they may be—come about? My own idea is that these beetles, like some other creatures—amongst them the little white wood-louse that lives with our own *Formica flava*—found ants' nests very comfortable places of retirement, since, by reason of their peculiar weapon of defence, they could defy any attempt to interfere with them, on the part of the ants. The ants, on their side, would soon have given up molesting them, so that, never requiring to defend themselves against the creatures by whom they were surrounded, the intruders got to associate them with quite other ideas, and, having first lost the habit, at length lost the power of turning their artillery in this direction. Meanwhile *Paussus*, owing to its sweet secretion, which, after relations had once become amicable, the ants would soon have discovered, had got to be a very welcome guest, so much so that, even when it took to eating their larvæ, they retained their love for it, as a species, though resenting such conduct upon the individual. And now the once redoubtable invader could be punished with impunity, for the habit of never discharging against an ant had become a fixed, inherited instinct, not to be got rid of even though life were at stake. Thus, as it appears to me, it may have come about that, though armed with dynamite, and carrying bombs, no living *Paussus* has ever defended itself against an ant, and no living ant, perhaps, ever seen a *Paussus* discharge its artillery. Of course these are only conjectures, and the last, especially, may be opposed to fact, since it has been suggested that one way in which *Paussus* may make itself useful within the nest of its hosts, may be by bombarding certain obnoxious parasites, or other would-be invaders. This does not, however, appear to me to be likely, for how could these explosion take place, under such circumstances, without doing damage to the ants themselves? In one's own house one would hardly wish a bomb to be thrown, even against one's greatest enemy—at any rate not in the drawing-room. That the ants should, by this means, be able, or, if able, willing to rid themselves of the mites which infest them, as has been conjectured, seems especially unlikely—indeed, hardly possible. On the whole, it seems to me that the relations at present existing between the two insects could only have grown up through *Paussus* having ceased to discharge, not only at an ant, but even—owing, probably, to there never

being any occasion for it—in an ants' nest. The experimental tickling with a straw was, of course, an artificial stimulus. In spite of its sweet secretion, I cannot see how a beetle with such a power at its command as *Paussus* has, can have been originally selected by the ants for domestication, but, on the other hand, an armed invader might easily, by coincidence, possess some property which would make it, in time, of use and value to the population on which it forced itself.

An example of an invader having no such merit, but harmless, and that has become tolerated through necessity, is, in my opinion, the little white wood-lice before mentioned. It apparently has now lost the power of rolling itself into a ball, but when it first began to penetrate into the galleries of *Formica flava*—our little yellow ant—it may very well have had it, and this would have rendered it impervious to attack, whilst its weight and round scaly surface would have made the task of removing it almost an impossible one. Thus, perforce, it stayed where it wished to stay, penetrating, perhaps, deeper and deeper into the labyrinth of galleries, as successive generations of cave animals have retreated farther and farther from the light of day, until at length, finding the wherewithal to live, it became wholly subterranean in its habits, lost the power of doing what it never required to do—namely, of rolling itself into a ball—and, through the absence of all sunlight, lost, too, its colouring matter, and became of its present dead, bleached white. Whether its eyesight, if it ever had any, is also gone, I do not know; but it can hardly, under present conditions, have any use for it, whereas its antennæ are constantly moving, and seem to be of extreme delicacy. I could never observe—for I have kept nests of *Formica flava*—the smallest sign of any kind of relations between these wood-lice and their hosts; and if any scavenger work is done by the former, from which the latter derive benefit, I believe that this is merely incidental, and that the ants know nothing about it. But they have got accustomed to the wood-lice being there, and put up with them because they cannot help it.

It must be remembered, in regard to *Paussidæ*, that the family is represented by some two hundred species, all of which pass the greater part of their lives with ants. In regard to this Mr. Kirby remarks: “The observations made upon the family are so contradictory that the discrepancies can only be accounted for by supposing that different species have very different habits. Possibly some species may perform various useful services to the

ants, while others are hostile; or they may be so useful that the ants are willing to pay toll of a certain number of their offspring, in return.”^[11]

This last, however, does not seem very well to accord with recorded observations as to ants attacking any individual *Paussus* whom they may chance to see devouring their larvæ, nor with the latter refusing to bombard, under these circumstances, even when in danger of their lives. It is impossible to imagine a hostile *Paussus* not bombarding, in such a case, unless, indeed, we suppose it to have first lost its hostility, and then again to have become hostile, without, however, regaining the power of using its natural weapon. But this is a state of affairs hardly to be conceived.

We probably do not know the whole round of occupations which make up the life by day, of the *Paussidæ*; of their life by night, we know nothing at all. The nefarious raids upon the larvæ or eggs (for both are appreciated) of the ants can hardly be of frequent occurrence, or the partnership, one would think, must come to an end. Other ant-guests, however, including sometimes smaller members of their own family, are likewise preyed upon by these curious beetles, but very frequently, when observed, they seem to be asleep, nor do they appear then to be taken much notice of by the ants. Where or under what conditions their eggs are hatched, or what is the larval and pupal history of each species, we do not know, but only the perfect insect has as yet been found in any ants' nest.

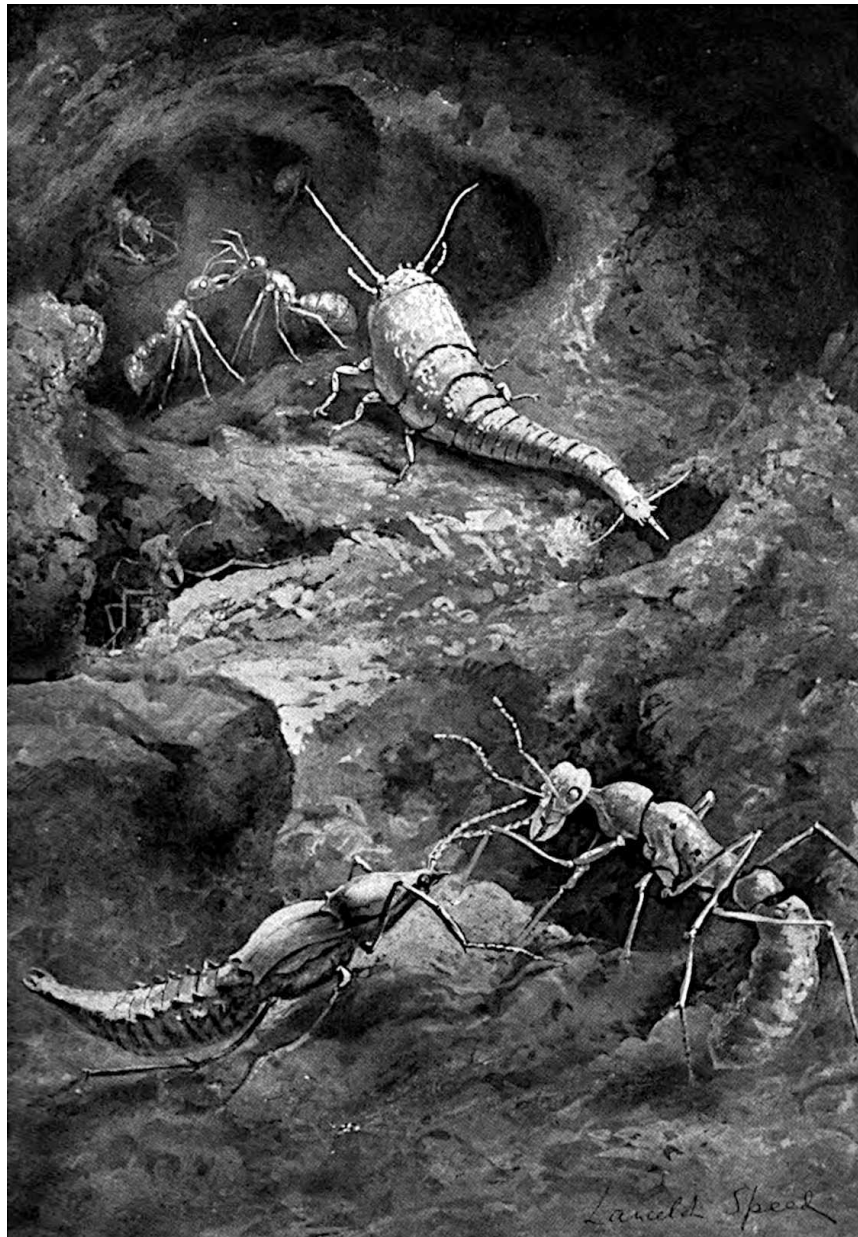
Other beetles that live with ants are either indifferent or hostile to them, but others, again, are kept and tended in the same manner as are the aphides, and for a similar purpose. All or most of these secrete some sweet substance, which their hosts lick up, and, in return, offer them an asylum from all enemies, and are ready to give them their personal protection, should this be necessary. They go even further than this, and actually feed them as they do their own larvæ, with honey, or something of a similar nature, which they regurgitate from their crops. One little beetle—*Atemeles* by name—is extremely fond of such a meal, and solicits it from the ants by stopping in front of them and assuming a certain attitude, accompanied with insinuating motions of the antennæ. Whether *Atemeles* is able to feed itself, or must live wholly upon these ministrations, I am not quite sure; but another beetle—poor *Clavigertestaceus*—is, according to Janet, so entirely

dependent upon the ants for subsistence that, if separated from them, he has nothing to do but to die.

CHAPTER IV

Ant parasites—Fleet-footed brigands—Honey-stealing mites—A strange table companion—Privileged cockroaches—Ants and their riders—A fly-ride on beetle-back.

LEAVING the beetles—though as there are probably some thousands that live habitually in ants' nests, we have said very little about them—we may glance at an extraordinary little creature, in appearance something like a wood-louse with a fish's tail, that resides with certain ants on the footing of a freebooter, constantly stealing from them, and eluding their resentment by extreme activity, living, as it were, in a state of perpetual motion. The legs of these persistent yet withal timid brigands are many and long, which, together with their shape and general lightness of build, enables them to run with great speed, so that they easily outdistance the ants, and, escaping to some less frequented part of the nest, with which they are always well acquainted, remain there quiet for a time. Should a single ant approach them, however, they immediately run away, or, if forced by circumstances to be near one or more—which, in an ants' nest, must be often difficult to avoid—make a point, apparently, of never keeping still, as though to confuse them, or, perhaps, to be the better able to dash off at any instant.



AN INSECT FREEBOOTER, AND AN INSECT BEGGAR.

The extraordinary looking insect shown towards the top is the lepismid, or fleet-foot, who lives by stealing food from ants when they are in the act of passing it from one to the other. The atemeles beetle shown below is begging food, which will not be refused, from the ant in front of him.

The way in which these fleet-foots secure their food is highly remarkable, each little theft—which has about it more of the parasite than the brigand—

occasioning a group of three. The ants upon which they live are of the species known as *Lasius umbratus*, and, like many other kinds, often feed one another, the hungry asking of the full, by whom he is rarely, if ever, denied. In the process of regurgitation—with which we are now familiar—the two stand fronting each other, with mandibles interlocked, and a drop of honey passes from mouth to mouth. For an instant it trembles between the two, resting on both, and that instant is the opportunity of the *Lepismid*. Darting forward, he interposes his own, and having absorbed some portion of the drop *in transitu*, speeds swiftly away to make a third elsewhere. Such a life, however great may be the thief's agility, is full of danger, and, from time to time, an individual is captured and killed. In nests under observation such executions may be witnessed, and *Lepismid* corpses—or, as various professors prefer calling them, *cadavers*—are sometimes noted. Under artificial conditions, however, opportunities of escape are much more limited, unless, indeed, some special provision is made. Thus, when Professor Wheeler first introduced a colony of *Lasius umbratus* into one of his *formicariums*, he found, after a couple of *dies*,^[12] five *Lepismid* *cadavers*. But having, by the addition to the said *formicarium* of a *refugium*, or asylum, made it more as *in natura*, this mortality ceased, and the remaining *Lepismids* continued henceforth *existentes*.

A similar mode of feeding, but under circumstances of much greater security, is indulged in by *Antennophorus*, another ant-guest, whose relations with its host are of a still closer description. *Antennophorus* is a mite which, according to M. Janet, fixes itself on to the head, or the sides of the abdomen, of the ant which it affects, and clings there as long as it sees fit. This it is enabled to do owing to a special adaptation of the feet, which end in little horny cups (*cornicula* is the word here) furnished with some substance of so adhesive a quality that it might well be called “stickphast,” if no Latin word were at hand. Not all the feet, however, are of this description, for the anterior ones are transformed into a pair of long waving antennæ, which contain olfactory organs of the greatest sensibility. With these their owner makes up for the want of eyes, and, smelling and feeling its way, walks, when it wishes to, along the bodies of its hosts, passing from one to another. Sometimes, either by accident or otherwise, it becomes detached, and is then helpless as far as locomotion is concerned, but by no means so in other respects. Its object, now, is to reaffix itself: nor is it long

before it succeeds in doing so. As it “lies upon the soil in one of the galleries of the nest it raises and stretches forward its first pair of ambulatory feet, and, at the same time, explores the space around it with its long antenniform ones. These appendages are much more agitated when an ant passes close by. Should it pass near enough, the *Acarid* (which has a finer sound than ‘mite’) glues itself on to its body by means of the cup of sticky material at the end of one of its ambulatory feet, which it holds up ready for this operation, and it can, in this way, soon climb up and fix itself in a good position on its host. The latter is surprised, and seeks to rid itself of its strange companion, but failing in this, it becomes resigned very quickly (as we do to increased taxation) as soon as the *Acarid* has taken up one of its normal positions.”^[13] It will carry two indeed, or even three, without complaining. An ant with one of these burdens fixed, like the income-tax, to the under side of its head, and two others, which may stand for a rise in tea and sugar, is a very common sight.

The feeding of *Antennophorus* has been closely observed by M. Janet in his artificial nests, and is thus described by him: “The ants had acquired a habit of placing themselves, crowded one against another, in one corner of the nest, and thither came such as had their crops well filled, after a meal of honey, and disgorged it before the mouths of their comrades who had none. While the fasting ant was eating the honey thus disgorged, *Antennophorus*, riding on its head, took its share. To do this, it pushed itself forward, and thrust its rostrum into the droplet, and generally, whilst holding itself in position by means of the two hinder pairs of legs, it attached itself by means of the forward pair (which in this case, however, would represent antennæ) to the head of the disgorging ant.”^[13] Perhaps there is some little mistake here—possibly I have not copied the passage correctly. There has been no hint before as to the modified antenniform legs of the parasite performing any other office than that of feeling and smelling, whilst the word “attach” or “affix” is that always used to describe the working of the sticky, cup-footed ones. In the position described the antennæ might very well act as supports, but hardly, one would think, in such a way as that their owner could be described as attaching himself through their means. Possibly it is the first pair of true legs that act in this way, but the matter is of no great consequence—not more than a war, say, or the fall of a ministry, in the general run of things. Suffice it that we have our picture, the little parasite

stretched, like a bridge, between the heads of the feeding and disgorging ants, and taking its share with the latter.

Lasius something is the name of the ant which *Antennophorus* utilises in this way, and it is, I think, a European species. Another one—*Pachycondyla harpax*—the large, black ant of America—wears its parasite round its neck, like an Elizabethan ruff. In this case both host and guest are in the larval state, and the involuntary partnership between them—involuntary probably on the part of either—is not dissolved until both have attained full maturity. The position of affairs is this: the ant larva apparently lies on its back upon earth a little hollowed, to receive it, by the workers of the nest. The parasitic larva—that of an unknown species of fly—has a long, slender neck, as we may call the anterior part of the body, and whilst this is wound about the corresponding portion of its host, the body, which broadens out after the manner of an oil-flask, is affixed by a disc at its end to some part of the back of the latter. When the ants feed the larvæ, they place the food—which consists either of grain that has been stored, or of insects captured and torn up by them—on the broad surface of the abdomen, which forms a sort of trough for its reception. Immediately upon feeling the welcome load, the hungry larva stretches down its head to the banquet, but that of the parasite moves with it, and its small, sharp jaws take eager toll of each dish. Thus the two feed together, cheek by jowl, and should what has been provided prove insufficient for this double onslaught, the unbidden guest will stretch its snake-like neck, and move it ceaselessly until the ever-ready jaws come into contact with a second feast, upon the table next it. Should none, however, be within reach, the guest will give vent to its irritated feelings by biting the bodies of such unbounteous Timons, or even that of its own host. They wriggle with pain, and this may possibly induce the ants to bring them fresh supplies, under the impression that they are hungry, as indeed they may be, with meals shared in this way. If so, we can hardly suppose a parasitic larva to act with such a motive, but as the best biters would in this case get the most food, natural selection may possibly have helped to develop the habit, which would have a compensating advantage for the wrigglers too. As the French say, “*Il y a compensation en tout.*”

The parasite, whilst stretching out as far as it can from the body of its host, in quest of food, remains, all the while, attached to the latter by the disc in which its body ends. It can, however, leave one ant larva for another, though

Professor Wheeler, to whom we owe this interesting discovery, believes that it does this “with great reluctance, and only under urgent circumstances, such as extreme hunger, the death of the larva to which it is attached, and perhaps, when fully mature, and about to pupate.”^[14] So long, indeed, as its original host, on whose body, when quite young, it was probably hatched from the egg, continues well and is well fed, it has no reason to seek farther, since all its wants are provided for. It is not only fed by the worker ants, but shares in any other of the benefits which these may bestow upon the rising generation of the nest. Thus, if they move larvæ, as is customary, to give them change of temperature, and produce the requisite hygienic conditions, the parasite is moved along with them, and it is cleaned also—a still more important advantage possibly—at the same time as they are. At such times the ants never seem to notice the uncouth incubus upon the bodies of their infant sisters, though one would suppose the difficulty would be not to do so. They are, it is true, blind, or nearly so, but it seems strange that their sense of touch, which is no doubt delicate, should not be able to inform them, since the parasite, though small enough, absolutely, is of great size regarded as an excrescence on its host’s body. This probably is the way in which the matter presents itself to the ants, if they think about it at all, for since the two lives are passed constantly together, and are subjected to the same conditions, it is likely that they share one smell between them.

But this curious parasitic relation between ant and fly is not confined to the larval stage of each. Continued observation led to a further discovery which I give in Professor Wheeler’s own words: “As the days passed, the mature ant-larvæ spun their brown cocoons one by one, and one by one the mature commensals (the larval parasites, that is to say) disappeared. No traces of them could be discovered. The only remaining resource was to open the cocoons. Five were opened, and in two of recent formation commensals were found! Having shared the table of their host, they had come to share its bed as well. The dipteran (the parasite, as I have said, is a fly) had pupated after the manner of its kind, forming a puparium, that is, instead of spinning a cocoon like the ant larva: the dead larval skin, somewhat shrivelled and contracted, was used as an envelope, and within this the pupa proper was found. In all cases the puparium was located in the caudal pole (at the bottom) of the ant cocoon, and was immovably stuck to the wall of the cocoon, its anterior end directed towards the cephalic pole”^[14] (the top). But what, asks Professor Wheeler, does the commensal larva do “while the

ant-larva is weaving its cocoon? Does it move about to avoid the swaying jaws of the spinning larva? or does it take up its position, from the first, at the posterior end of the larval ant, and there remain motionless while the posterior pole of the cocoon is being completed? It is very difficult to answer these questions.”^[14]

One might think that young ants thus deprived, day by day, of a portion of every meal, would be stunted in their growth, and not make such large and healthy workers as those who had never been encumbered with a parasite. This, however, does not seem to be the case, and no difference can be detected between the one and the other. Perhaps, therefore, ants habitually eat, if not more than is good for them, at least more than they require. This is the case almost universally amongst civilised men, at least in Northern Europe, and with savages to a still greater extent whenever the wherewithal is at hand. In the above case we have, as Professor Wheeler remarks, a very perfect example of what is termed commensalism, in the original meaning of the word—that is to say, of two or more individuals dining together at the same table. As applied to natural history, the individuals in question must be of different species, but it is not often that the definition otherwise is so rigorously adhered to.

This curious parasite inhabits the nests, or, more strictly speaking, the bodies, of an ant, native to Texas, that has long been famous as a storer of grain, but whose supposed still further achievements in an agricultural direction would now seem open to doubt. In the nest of another American ant, which most certainly does grow mushrooms, the same observer found another “*myrmecophile*,” or ants’ nest insect, viz. a minute species of cockroach that lives its life amongst the caves and galleries of the great vegetable mass which forms, and is designed to form, the mushroom bed, upon the product of which it feeds. Here again the ants have become thoroughly reconciled to the presence amongst them of a guest from which, as far as can be seen, they derive no benefit, whilst having to submit to a loss, through its agency, of some part of the fruits of their labours. These little cockroaches are fairly numerous, and have become so adapted to living in darkness that their eyes have almost disappeared. Another loss, or partial loss, is of a more curious nature, and, one might think, would be a great privation to them. Their antennæ, namely, are always incomplete, but this does not seem to have come about by a gradual process of atrophy, but

rather to have been caused by mutilation during their owners' lifetime. But how has this happened, and what has been the mutilating agency? Professor Wheeler's explanation, which he believes to be the only one, is that their antennæ have been unconsciously sheared off by the ants, whilst engaged either in clipping their mushrooms or in cutting up the pieces of leaves which they are continually bringing into the nest, to add to the bed on which they grow. "It is easy," he says, "to understand how an insect like a cockroach, living in the midst of thousands of ants which are continually opening and closing their scissor-like mandibles, should be certain, sooner or later, to have its long antennæ cropped. One wonders how the tarsi (the legs, that is to say) of the cockroach escape the same treatment."^[15] This wonder, however, if there is really any reason for it, suggests a doubt as to the sufficiency of the explanation here offered. The antennæ, one would think, might be held high, in which case, if sheared at all, it could only be at the base, but if here (as would not, however, seem to be the case) why should not the legs be sheared too? Again, it seems possible that the insects themselves may be in the habit of gnawing one another's antennæ. As the cockroaches live and flourish it would seem that this mutilation of their antennæ, if that, indeed, be the explanation, can do them no great injury. Yet these organs are supposed to be of great importance to insects, and, judging by their length and delicacy, one would think that they were especially so to the members of the cockroach family. In this case they would probably be extremely careful of them, and the fact that these ants'-nest cockroaches do not seem to be so, may show that subterranean conditions, contrary to what one might have expected, have affected their efficacy.



A RIDE ON BEETLE-BACK, AND A LIVING SWEET-SHOP.

Enjoyment seems to be the only motive the fly has for riding on the back of the African beetle shown in the upper part of this illustration. Beneath is shown the well named honey-pot ant with its distended body full of honey, which it gives away to any hungry working ant.

A diet of mushrooms, or fungus, is not the only thing for which these little blind, light-shunning cockroaches are indebted to their landlords, the ants, for often one of them may be seen to mount upon one of the latter, and take

a ride on its back. They seem especially fond of the soldiers, as horses, and will sit perched on their enormous heads, as they walk up and down in a stately sort of way, sometimes for quite a long time. Enjoyment seems here to be the only motive, and perhaps it is a natural one, since there is a fly in Africa which seems to have quite a passion for riding on the back of a beetle. “Across the mouth of the Seyhouse,” says the Rev. Mr. Eaton, “on sandy pasture-land bordering the seashore, big *coprophagous* beetles—it sounds abusive, but no harm is meant—are common, sheltering in large holes in the soil, when at rest, and running about on business. A small species of *Borborinæ* (that is the fly) may often be seen riding on their backs, chiefly on the pronotum and about the bases of the elytra, sometimes half a dozen females on one beetle. The beetles occasionally throw themselves on their backs, and try to get rid of them by rolling; but the flies elude all their efforts to dislodge them, dodging out of harm’s way into the jointures of the thorax, and darting from back to breast, and back again, in a way that drives the beetle nearly mad. In vain she scrapes over them with her legs, in vain does she roll over, or delve down amongst the roots of the herbage: the flies are as active as monkeys (not perhaps a very striking simile here), and there is no shaking them off. It is difficult (such is their strange predilection) to get them off into the killing-bottle. Nothing (not even the killing-bottle) persuades them to fly, and they would very much rather stick to the beetle than——” what? Not go to heaven, but “be driven off it down the tube.”^[16] The tube must be the neck of that same bottle. This, surely, is a case of infatuation if ever there was one. Eccentric fly! And what must be the charms of a beetle that can prevail over those of cyanide of potassium! But the beetle, it must be remembered, is a *coprophagous* one. There may be a world of explanation in a word like that.

CHAPTER V

From biped to quadruped—Flies that borrow wings—Sit-o'-my-head—A novel cradle
—Flies that kill bees—Nature's sadness—Consolations of the future—The Tachina
fly and the locust.

ALTHOUGH from the way in which the story is told, one might imagine that the fly here was merely enjoying a ride upon beetle-back, yet, from the efforts made by the latter to shake off its persecutors, and, still more, because these were of the female sex, the probability is that we have here to do with a case of parasitism. The fly, we may almost feel certain, was endeavouring to lay its eggs, and the reason why she took so long about it was that she required a certain spot upon the beetle in order to do so, and that the beetle's efforts, though appearing futile, were more or less successful in guarding this spot. At any rate, if this was not the case here, it is so in many other instances, various flies being parasitic on various other insects. Not all of these are fatal to the object of their choice, which, if it affords them board as well as lodgings, may only do so to the extent of its blood. Such are the curious family of *Hippoboscidae*, or Bird Ticks, who begin life with wings, but are so little appreciative of the powers which these confer that, having found the creature upon whom they elect to live, they bite them off, or otherwise wilfully rid themselves of them, after the manner of ants and termites, thus offering yet another example in the insect world of

“one whose hand,
Like the poor Indian, flung a pearl away
Richer than all his tribe.”

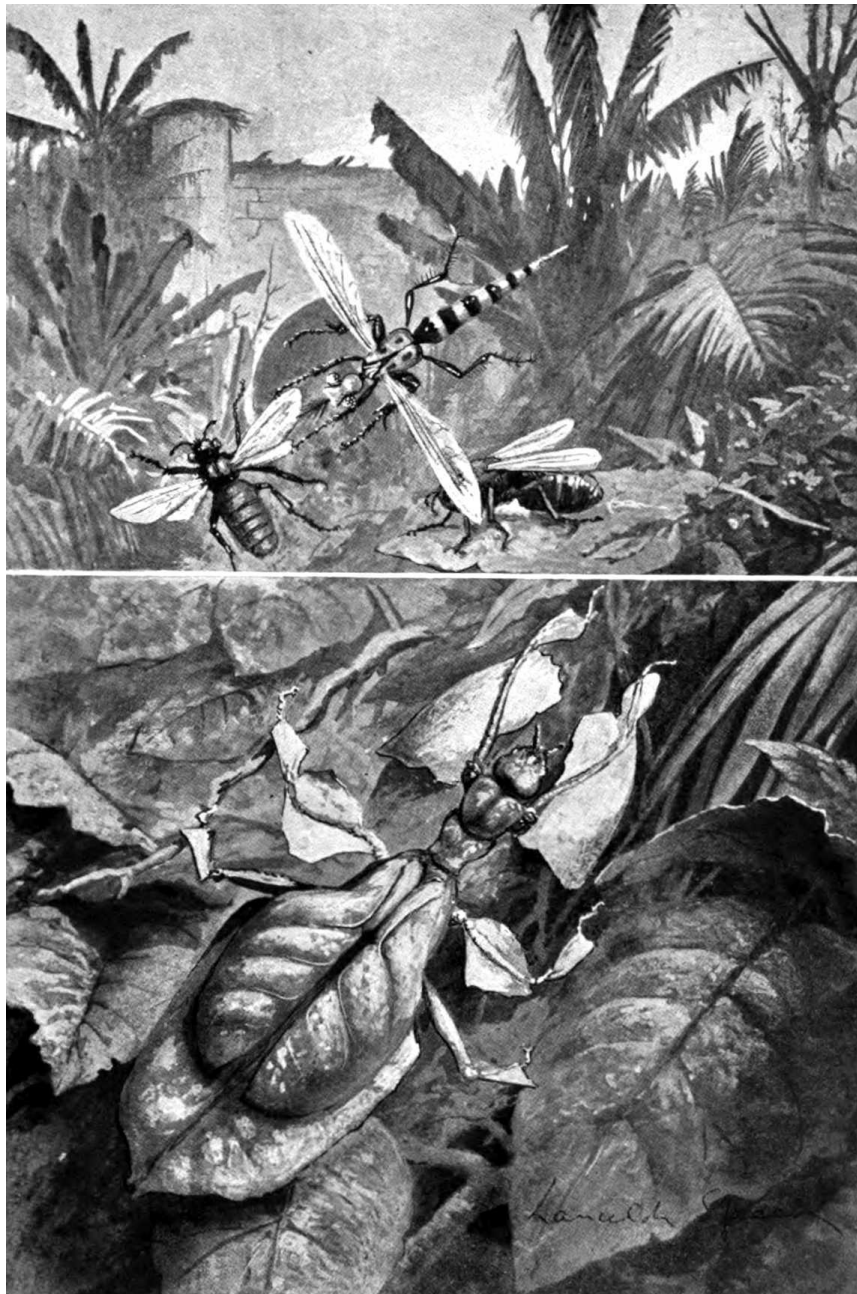
For what can be imagined more glorious to possess, speaking of physical attributes, than the power of flight?

The course of life of these flies—if all be truth that is spoken of them—is, indeed, very extraordinary, for during the first or winged stage of their adult life they live on birds, but migrate from them to some quadruped—as, say, a

deer—as soon as they find themselves within easy reach of it, and then, as having reached their final place of abode, do away with their wings. Thus, being too lazy or lethargic to fly themselves, they choose rather to stand indebted to another being for a power which they no doubt once possessed in perfection, and which they are still quite capable of exercising. What the larval stage of these flies is, whether they lay their eggs upon their first or last habitation—or on both, and if not, where or in what manner the larva passes its life, I do not know, and as my authority, who should be up to date, holds his peace upon the matter, I conclude that it is not yet made out. Possibly the grub is a vegetable feeder, or possibly, again, it is as fatal to some other insect as is that of a little fly with a big name—*Apocephalus pergandei* to wit—to ants. The victim here chosen—if there be not others also—is a black tree-climbing ant, common in Pennsylvania. As it runs over the ground or up and down the trunks of trees, the fly darts after it on tiny wings, intent on laying her egg upon its neck. The ant tries to elude her endeavours, but *Apocephalus*—or Sit-o'-my-head—has a mission to fulfil, and will take no denial. The egg is laid, it cannot be detached, and, when hatched, the issuing grub bores, with enthusiasm, into the head of the ant. Coming to the brain he has nothing to do but to eat it, and he does so until the whole cavity of the skull has become an empty chamber, except for his own presence there. The movements of the ant during this process—of its feelings we have no record—have become more and more erratic, and it feels itself less and less capable of performing its duties as a member of an active and industrious community. At length it falls down, and not long afterwards its head falls off, giving to the maggot inside it its first opportunity of looking out into the world through the window of the neck-hole. Hitherto its life, however easy and pleasant, has been of a sedentary nature, but now it can enjoy the pleasures of a walk, and moves about something after the manner of a snail, dragging its cephalic shell behind it. But this active state is not of long duration. The time of change is at hand, and snug within the ant's head and its own last larval skin, which, as is the way with fly caterpillars, serves it in lieu of a cocoon, the fortunate little creature turns into a chrysalis, and dreams away its time till, on some sunny day, it issues from its cradle a happy, active fly, feeling strangely attracted by ants.

Another little fly belonging to this same family group—the hump-backed flies or *Pharidæ*—has its fate linked with that of bees, in whose hive it is hatched and on whose eggs and larvæ it feeds; nor does the grown bee itself, though armed with its sting, escape from the more rapacious members of the order. These are known by the name of Robber Flies, though as the robbery involves the death of the victim, and consists of the juices of its body, murder would seem to be the better word. These flies, though of somewhat slender build, which the better fits them for their swift and darting flight, are excessively strong, as might be expected from their long muscular-looking legs and rough hairy bodies. All sorts of insects are their prey, for the despatch of which they are furnished with a hard tubular beak, enclosing, as in a sheath, a lancet-like instrument, which, being protruded at will, severely lacerates the body of the captive. The beak, or sheath, is also struck some way into the wound, and being tipped with bristles, these serve as so many barbs to keep it in position, whilst the blade continues to probe and hack the victim, on whose back the fly has descended, embracing it with its powerful legs. “These flies,” says Dr. Fitch—who seems strangely unalive to the moral beauty underlying the mere mechanical expression of it—“are inhuman murderers, they are savages of the insect world, putting their captives to death with merciless cruelty. Their large eyes, divided into such a multitude of facets, probably give them the most acute and accurate vision for espying and seizing their prey; and their long, stout legs, their bearded and bristly head, their whole aspect indicates them to be of a predatory and ferocious character. Like the hawk, they swoop upon their prey, and grasping it securely between their fore feet, they violently bear it away.”^[17] Bees, beetles, butterflies, moths, even grasshoppers are thus treated, and sometimes, by a beautiful retributive arrangement—enough to throw one into ecstasies—they turn cannibals, and prey upon each other. Nay, there is even more than this to arouse our admiration, for so stern and unbending is the law of eternal justice, that even the softest feelings of nature are not allowed to interfere with it, and the female, wooed by the male, is frequently compelled to eat him. Thus the noble maxim of *fiat justitia ruat cælum*, though, for a time, it may seem to be in abeyance, finds, at last, unconscious expression, if not in the breast, at least in the appetite of a cruel and murderous insect; and thus in the animal world, not less than in our human one, “the whirligig of time brings in his revenges.”

To bee-keepers—and to bees perhaps still more so—these terrible buccaneer flies are especially obnoxious. Poised in air, in the neighbourhood of some hive, they watch the issuing and returning stream, and, making swift choice of a victim, sweep, like the wingéd furies that they are, upon him. There is a sharp, shrilly sound, as the bee's wings vibrate, for a moment, more rapidly, then the fatal legs wrap her round, and, pressed tightly to the oppressor's body, she is borne to some shrub or flower, in the shade or pleasant fragrance of which the juices of her body are sucked out, through a hole specially made to allow of their passage. When nothing remains but the empty shell, the fly drops this, and returns to the scene of its labours. Through all the hot sunny hours these raids are continued, till hundreds of empty bee-shells strew the ground. As the sun declines the sport flags and gradually ceases, but it begins again the following morning as merrily as ever. America seems to be the home *par excellence* of these flies, but they are represented, under various forms, in many parts of the world. The United States has been accorded its fair share of them, and according to their numbers, each season, the labours of the bee-farmer are rewarded or otherwise. So much is this the case that the fact that “during certain seasons, in a bee-raising district of New York, not a single hive threw off a swarm”^[17] has been attributed to this cause alone.



A BUCCANEER FLY, AND A LEAF-RESEMBLING INSECT.

Poised in the air, the buccaneer fly selects its victim from the bees issuing from a hive, pounces on it like a winged fury, and kills its hapless prey. The insect depicted beneath is protected from its enemies by its strange resemblance to a dead leaf.

It would appear from these facts either that no bee ever succeeds in stinging its assailant, or else that the latter is proof against the injection of poison.

The former seems to me the most probable, since the system of the bee itself has no such immunity. It seems strange that so deadly a weapon should fail thus constantly, at a pinch, and it would be interesting to know if these redoubtable adversaries attack wasps as well as bees. As it is not stated that they do so—as wasps are pointedly omitted from the list made out of their victims—the contrary may, I think, be assumed, and also, as a corollary, that if wasps were attacked they would be able to use their sting, probably with fatal effect. This superior capability is, no doubt, owing to the superior flexibility of a wasp's abdomen over that of a bee; and if we ask ourselves what is the cause of this—how and for what reason the superiority has been acquired—the answer seems “as ready as a borrower's cap,” viz. “as a means of self-defence through a process of natural selection.” Nothing could be better adapted to bring this process into play than the very ordeal through which the bee is passing; for if some could only succeed, through superior flexibility, in stinging the flies, they ought to increase at the expense of those unable to do so. As far as it goes, this seems to point to the wasp having gone through a longer course of development than the bee—to its ancestry dating farther back in time; but when we think of the latter's more elaborate social organisation and the greater perfection of its cells, one feels inclined to reverse this opinion. As no bees possess such powers of twisting about and doubling round their abdomens as do wasps, though some can do so in a very fair degree, it seems probable that the common ancestor of all the species was more thickly built than that of the wasps, or at least that the potential capacity handed down by it of development in this direction was less. But precisely the same argument may be used in regard to the brain of the ancestral wasp, and thus we see that unless we have geological evidence on the subject it is very difficult to say which of two species has the more ancient descent.

The Robber Flies—whose scientific name I have forgotten—however disagreeable they may be, are at least not parasites. They attack their prey and kill it quickly, instead of handing it over to prolonged torture at the hands of the next generation. This last is what the Tachina flies—to say nothing of other kinds—do, who, as they principally attack caterpillars, may be considered beneficial to man. In the United States of America there is no greater destroyer of all sorts of trees than the so-called army-worm caterpillar, or rather grub—for it represents a fly merely—which gathers together in enormous numbers when about to enter the pupal state. “I have

seen,” says Mr. Leland Howard, “vast armies of the army-worm, comprising, unquestionably, millions of individuals, and have been unable to find a single specimen which did not bear the characteristic eggs of a Tachina fly. These flies were present in such numbers that their buzzing as they flew over the army of caterpillars could be heard at some distance, and the farmers were unnecessarily alarmed, since they conceived the idea that the flies were the parents of the caterpillars, and were flying everywhere and laying their eggs in the grass and wheat. As a matter of fact, one great outbreak of the army-worm in northern Alabama in the early summer of 1881 was completely frustrated by the Tachina flies, aided by a few other parasites and predatory insects. They also attack grasshoppers, bugs, beetles, saw-flies and saw-fly larvæ, humble bees, and wasps. (How they avoid the sting of the latter we are not told; perhaps their insignificant size is a protection.) The eggs are stuck by some sort of gummy substance to the surface of the preyed-on insect; and the small white eggs are frequently seen sticking to the back of some unfortunate caterpillar. From the under side of each egg there hatches a little maggot, which bores its way through the skin of the host, and penetrates into its body, where it lives, nourishing itself upon the fatty matter and lymph until it reaches full growth, usually, if not always, destroying before it emerges some vital organ, so as to cause the death of the host insect. It almost invariably issues, when full grown, from the body of the insect attacked, and pupates, at or near the surface of the ground, within the last larval skin, which hardens into a brown oval puparium.”^[17] There are some points of special interest about the parasitism of these Tachina flies, which seem to be directed by a less perfect instinct than that which guides other insects of similar habits; for instance, the Ichneumon flies, which, however, are such merely in name, being members of the order *Hymenoptera*, which includes the bees and ants.

These latter, by merely touching an insect with their antennæ, can tell if it is already occupied—in which case they withdraw—nor do they ever lay eggs in excess of the number of issuing larvæ that can be supported by the little world of provender into which they will be born. Neither do they choose a caterpillar to lay on, which is just about to cast its skin, by which manœuvre the host would escape, and the guests be left to perish. All these mistakes, however, are frequently made by the Tachina fly, the consequence being that many poor children die of starvation; whilst others, from wanting their necessary complement of food, have their growth checked and become poor

pitiable objects, less than half the size that, with a more generous diet, they would certainly have attained to. It is painful to know that such privation exists and to have no means of relieving it; but nature is full of sadness, and it is best to look truth in the face. Some comfort may perhaps be derived by looking forward to a distant future, when the instinct which is now liable to these errors shall have been perfected. Such comfort, at any rate, lives in Mr. Leland Howard's views that "the parasitic mode of life in the Tachina fly is one of comparatively recent acquirement, and that sufficient time has not elapsed since they began to take on this habit"^[17] to allow of its having reached the final goal towards which it is always advancing. It is difficult, however, to console oneself for the imperfections of a work-a-day world in a far distant prospect of Elysium.

In the somewhat numerous list of insects distinguished by the attentions of the Tachina fly, grasshoppers have been mentioned. In Africa they, or, at any rate, one species of the family, attack the terrible plague locust, that has from time to time committed, and still apparently commits, such terrible devastations. The latter seems quite aware of the fate in store for it, and makes vigorous efforts to evade its destiny. Buzzing in the air, above the ravenous horde, the fly waits for one to hop or rise on the wing, and then darts swiftly upon it. To avoid her, the locust rises or sinks, tacks suddenly to right or left, scudding this way and that like a ship to meet a varying breeze. The Tachina, in the meanwhile, circles about her quarry, awaiting a favourable opportunity, which generally arises just as the locust alights, or is on the point of alighting, when, descending upon it before the lost impetus can be renewed, she clings lovingly, and deposits her eggs, either on the neck or under one of the wings,

"—and with a little pin
Bores through his castle wall, and farewell king."

It is not, however, as a rule, till after the grub or grubs have made their exit from the body that the locust dies, though it has drooped and become languid for some time. Of the vast swarms that darken the sky and descend upon the country, like a mantle, a very small proportion would seem to perish in this way, since everywhere the females may be seen drilling with their abdomens into the ground, preparatory to laying their eggs. The check upon their numbers, whatever it may be—and on the whole it must be very

effective—supervenes, for the most part, at this early stage, before the egg is hatched, that is to say.

CHAPTER VI

The burden of the locusts—Classical nonsense—Address to Mahomet—Locusts in Europe—Succumb to the English climate—Described by Darwin—Locusts in Africa—The wingless host do greatest damage—Hoppers and jumpers—“An army on the march.”

LOCUSTS are insects famous in story, and when one reads about them in various entomological or other writings, one might imagine that the whole world had been doing little else, ever since it began, than play a losing match with these creatures. It is only after one has gone a little about the world, and lived for some time in regions noted as their head-quarters without seeing anything whatever of them, that one begins to doubt this view, and lean towards another one, viz. that they are fabulous animals; but truth, as in other cases where two extreme views are held, lies somewhere betwixt and between. The whole matter is this, that when one reads one narrative after another, with its burden of a darkened sun, devastated territories, strong winds, drownings in the sea, and pestilences engendered by innumerable carcasses cast up along hundreds of miles of beach, the intervals, as well as the countries, between each one of these occurrences, are annihilated in the imagination, and the dates, if seen, are forgotten. Thus, to use the Kaffir expression—which has not yet lost its meaning for a civilised European—one sees everything red; locusts are very convincing—“you may almost hear the beating of their wings.”

However, there is no doubt that these insects, in relation to man, have played what the Germans call “*eine bedeutende Rolle*” in the world, and are worth saying something about, if only one has something not too desperately antique to say, and this, by virtue of a work which I, at any rate, have never seen quoted, and a paper in a certain Antipodean organ, which for the majority of people here might as well be in the *Faerie Queene* or *Paradise Lost*, I think I may have. But first let us turn to what, though it be antique, is also classical, and—though this would not be a corollary for everyone—very delightful: “To look,” say the authors of the famous

Introduction, “at a locust in a cabinet of insects, you would not, at first sight, deem it capable of being the source of so much evil to mankind as stands on record against it. ‘This is but a small creature,’ you would say, ‘and the mischief which it causes cannot be far beyond the proportion of its bulk. The locusts so celebrated in history must surely be of the Indian kind mentioned by Pliny, which were three feet in length, with legs so strong that the women used them as saws. I see, indeed, some resemblance to the horse’s head, but where are the eyes of the elephant, the neck of the bull, the horns of the stag, the chest of the lion, the belly of the scorpion, the wings of the eagle, the thighs of the camel, the legs of the ostrich, and the tail of the serpent, all of which the Arabians mention as attributes of this widely dreaded insect destroyer, but of which, in the insect before me, I discern little or no likeness?’” Personally, I do not for a moment imagine that even in 1815, the date of the first edition of the work in question, any reasonably educated person would have spoken or thought in this way, without any conception, apparently, of what numbers can effect, but it is interesting to know what the Arabs think, or say, about the locust, and especially that they represent it—as we are told a few lines on—as thus addressing Mahomet: “We are the army of the Great God; we produce ninety-nine eggs; if the hundred were completed, we should consume the whole earth and all that is in it.”

The authors then proceed to give a short résumé of the various locust plagues under which the earth, over a large part of its surface, has at different times groaned. The first and best authenticated goes back to a very early period—about 4000 B.C.—after which the evidence does not conform quite so strictly to the test demanded of it by the modern scientific spirit. Pliny, however, we are told, “mentions a law in Cyrenaica by which the inhabitants were enjoined to destroy the locusts in three different states, three times in the year—first their eggs, then their young, and lastly the perfect insect. And not without reason was such a law enacted; for Orosius tells us that in the year of the world 3800 Africa was infested by such infinite myriads of these animals that having devoured every green thing, after flying off to sea they were drowned, and being cast upon the shore, they emitted a stench greater than could have been produced by the carcasses of 100,000 men (a very confident statement, surely). St. Augustine also mentions a plague as having arisen in that country from the same cause, which destroyed no less than 800,000 persons (*octingenta hominum millia*)

in the kingdom of Masanissa alone, and many more in the territories bordering upon the sea.” After this we make a jump to A.D. 591, and find the locusts in Europe. In that year “an infinite army of them, of a size unusually large, grievously ravaged part of Italy; and, being at last cast into the sea, from their stench arose a pestilence which carried off near a million of men and beasts. In the Venetian territory also, in 1478, more than 30,000 persons” (but this seems pitiful) “are said to have perished in a famine occasioned by these terrific scourges.”

Many other instances of their devastations in Europe, in France, Spain, Italy, Germany, etc., are recorded by the same authors. “In 1650 a cloud of them was seen to enter Russia in three different places, which from thence passed over into Poland and Lithuania, where the air was darkened by their numbers. In some places they were seen lying dead, heaped one upon another to the depth of four feet; in others they covered the surface, like a black cloth, the trees bent with their weight, and the damage they did exceeded all computation.” Nay, “even this happy island (*lucus a non lucendo*), so remarkably distinguished by its exemption from most of those scourges to which other nations are exposed (as fog, sunshine, etc.), was once alarmed by the appearance of locusts. In 1748 they were observed here in considerable numbers, but providentially they soon perished without propagating”—the “happy island” apparently having been too much for them. These unfortunates would appear to have been stragglers from far vaster numbers which a year before had devastated Eastern Europe, one swarm of which, “entering Transylvania in August, was several hundred fathoms in width. At Vienna the breadth of one of them was three miles, and extended to so great a length as to be four hours in passing over the Red Tower: and such was its density that it totally intercepted the solar light, so that when they flew low one person could not see another at the distance of twenty paces.” Another host that appeared in India is said to have formed a column five hundred miles long, and “so compact was it when on the wing that, like an eclipse, it completely hid the sun, so that no shadow was cast by any object, and some lofty tombs not more than 200 yards off were rendered quite invisible.”

Dr. Clarke in his *Travels* speaks of locusts covering “his carriage and horses, and says the Tartars assert that people are sometimes suffocated by them.” He mentions two species, “the first of which is almost twice the size

of the second, and, because it precedes it, is called by the Tartars the herald or messenger.” From 1778 to 1780 a dreadful curse of locusts, alluded to by Southey in his “Thalaba”—or, perhaps, forming the subject of that poem—I really don’t know—fell upon the Empire of Morocco. “Everything green was eaten up, not even the bitter bark of the orange and pomegranate escaping. A most dreadful famine ensued. The poor were seen to wander over the country deriving a miserable subsistence from the roots of plants; and women and children followed the camels, from whose dung they picked the undigested grains of barley, which they devoured with avidity; in consequence of which numbers perished, and the roads and streets exhibited the unburied carcasses of the dead.” Again, “From Mogador to Tangier, before the plague of 1799, the face of the earth was covered by them. At that time a singular incident occurred at El Araiche. The whole region from the confines of the Sahara was ravaged by them; but on the other side of the river, El Kos, not one was to be seen, though there was nothing to prevent their flying over it. Till then they had proceeded northwards; but upon arriving at its banks they turned to the east, so that all the country north of El Araiche was full of pulse, fruits, and grain—exhibiting a most striking contrast to the desolation of the adjoining district.” Lastly—that is to say, to make a last quotation from the classics—“The Arabs of the Desert, whose hands are against every man, and who rejoice in the evil that befalls other nations, when they behold the clouds of locusts proceeding from the north, are filled with gladness, anticipating a general mortality, which they call *El Khere* (the benediction), for when a country is thus laid waste they emerge from their arid deserts and pitch their tents in the desolated plains.”

Darwin, in his *Journal of Researches*, gives the following account of a flight of locusts: “Shortly before we arrived at the village of Luxan we observed to the south a ragged cloud of a dark reddish-brown colour. At first we thought that it was smoke from some great fire on the plains; but we soon found that it was a swarm of locusts. They were flying northward, and with the aid of a slight breeze they overtook us at a rate of ten or fifteen miles an hour. The main body filled the air from a height of twenty feet to that, as it appeared, of two or three thousand above the ground; ‘and the sound of their wings was as the sound of chariots of many horses running to battle,’ or rather, I should say, like a strong breeze passing through the rigging of a ship. The sky, seen through the advanced guard, appeared like a

mezzotinto engraving, but the main body was impervious to sight; they were not, however, so thick together but that they could escape a stick waved backwards and forwards. When they alighted, they were more numerous than the leaves in the field, and the surface became reddish instead of being green: the swarm having once alighted, the individuals flew from side to side in all directions.” At that time—the year was 1835—locusts were “not an uncommon pest in this country: already during this season several smaller swarms had come up from the south, where, as, apparently, in all other parts of the world, they are bred in the deserts. The poor cottagers in vain attempted by lighting fires, by shouts, and by waving branches to avert the attack.” This locust, Darwin tells us, closely resembled the famous *Gryllus migratorius* of the East—the one that spoke to the Prophet—if, indeed, it was not identical with it.

Though these accounts are all interesting in their way, they none of them tell us much—or, indeed, anything—about the locusts themselves, for which reason I will supplement them with some which have that advantage, and are also, in some sort, a check or commentary upon the others. It is to be noted that, in all these, we hear only of flying locusts, and anyone would imagine by reading of them that it was by such, and no others, that all the damage was done. In Africa, however, and also in Cyprus—from which we may assume that it is the same elsewhere—the case is widely different. Writing evidently as a locust expert of the former country, Dr. Æneas Munro tells us that it is in their early wingless state—answering to the caterpillar one, though far less differentiated from the perfect form—that the most terrible, the overwhelming, injury is inflicted by these insects. Of the full-grown flying locusts he says, “To a certain extent, they do injure here and there, where they select to settle and feed; but they do not devour everything clean before them, like the army of the larval stage or jumpers.”^[18] Of the latter and its doings *ab ovo* we have the following interesting account: “When the tiny creatures issue from their nest they are of a greenish white or creamy colour, about an eighth of an inch in length,” and on the day that they do so “the very dust of the ground, which was so still before, now seems to waken into life. They begin to move by a process of twisting or rolling over one another, so that, for the first few days, they receive the name of twisters. Within eight or ten days, however, they can jump four or six inches, and at the age of three or four weeks a new characteristic makes its appearance. A desire to explore manifests itself, and

in a surprising manner. The whole company moves in a body in one general direction, and more or less in a straight line, as if by one common instinct, without apparently having any recognised leader or commander,”^[18] which is just the way, in my opinion, that rooks and starlings move.^[19]

Marching in this way they spread themselves out over the country, “eating everything that comes in their way—wheat (if sufficiently young and tender), maize (even if strong and old), corn, sugar-cane, linseed, alfalfa (lucerne), pasture of all kinds, vegetables of all kinds (tomatoes and celery), and all garden produce, potatoes (ordinary and sweet), the leaves and sometimes even the bark of the trees, causing their ruin. The fruit, of course, is lost for the season. Orange, willow, poplar, palm, banana, peach, pear, plum, vine, acacia, roses, etc., are stripped,” but not “the gum and paradise trees, which seem to be poisonous to them. They make everything ‘clean bare’; sometimes they will enter houses, and eat the very clothes and curtains at the windows.”^[20] They will even eat the wool off the backs of the sheep, and “last stage of all that ends this strange, eventful history,” on a pressing occasion they will eat one another. Continuing his interesting account—the graphic and convincing one of an eye-witness— Dr. Munro tells us that “when these hoppers and jumpers (*voetgangers*, as the Boers call them) are on the march, they sometimes appear so determined and bent on the fearful execution of their work, that they resemble and have got the name of ‘an army on the march.’ They move in open file, and carry themselves in a proud, haughty way, with heads high up and fixed. It is beautiful and interesting to see them on the march, if we only divest ourselves for the moment of the idea of their devastating object.”^[20] And again, “The whole of the company begin to walk at the same time, as if by order; the head is kept erect, and the neck is as if stiffened. They go straight on, irrespective of danger,”^[20] and are deterred, as is well known, by hardly any obstacle. “The sight of this army is a very impressive one, and once seen will never be forgotten. In some respects it is an awful sight; the spectacle strikes you with pity and sorrow to see at once before you that the toil and the labour for the season, or, indeed, the year, is lost.”^[20]

“It is in this marching stage,” continues Dr. Munro, “that the *voetgangers* do enormous damage and eat every edible thing in their path, and completely destroy the work of the husbandman. They are not content with levying toll merely, but they will have all, and will leave nothing behind but

desolation. They are therefore unlike the flying company of locusts, which only levy toll here and there, but these, when they pass, leave nothing.”

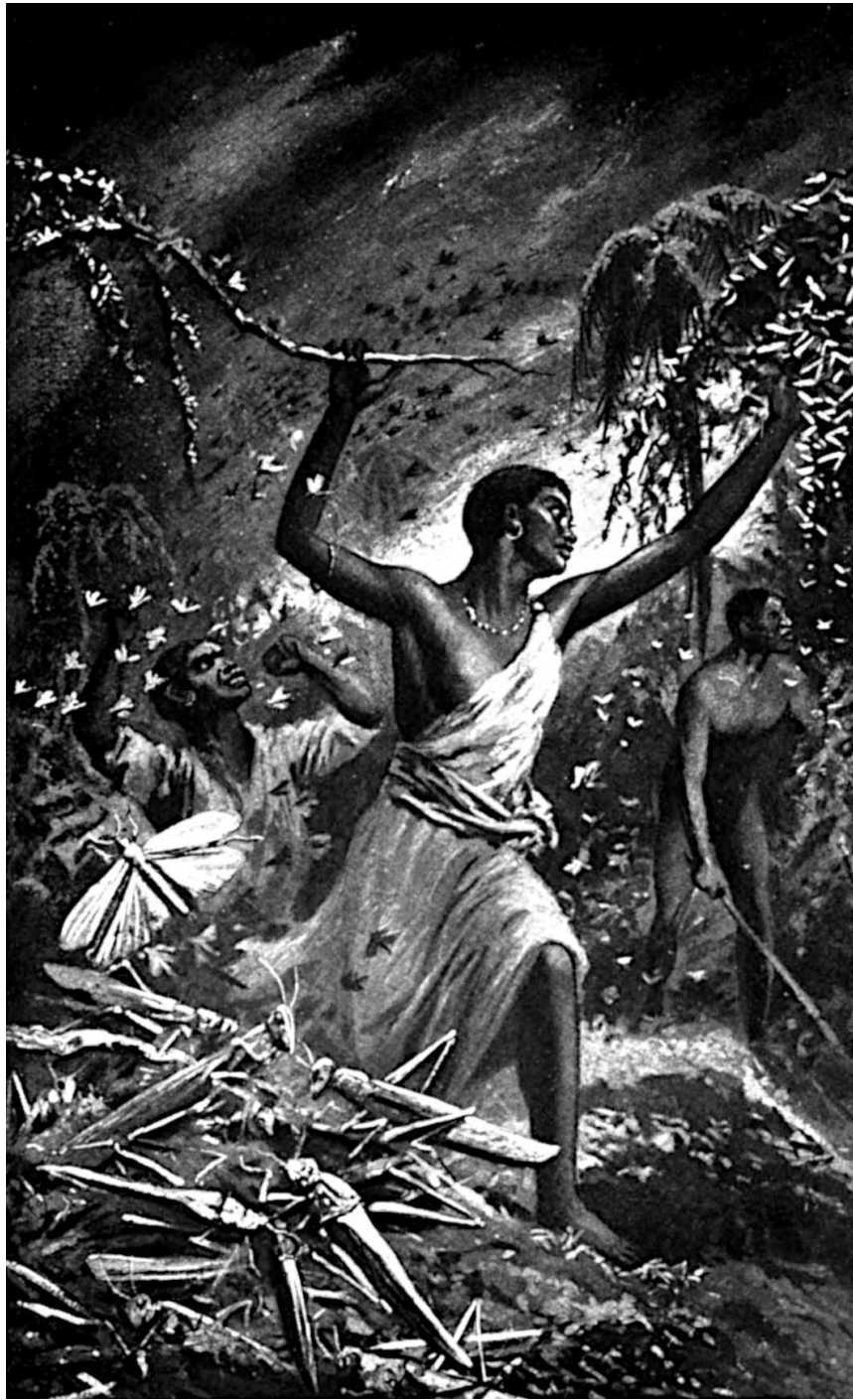
Some curious facts are then given in regard to the uniform direction—varying according to the country—in which these wingless locusts march. The account will be remembered of how a flying host came to the banks of a river which they refrained from crossing. One might almost think that a mistake had here been made, and that the locusts were really *voetgangers*, but had they been so, the river, unless a large one, need not have deterred them—at least, they will pass streams, though doubtless great numbers are sacrificed in doing so. If, however, the stream had run parallel to the direction in which the swarm was advancing, we can understand, in the light of what seems to be now established, their not crossing it.

CHAPTER VII

The sense of direction—How locusts look flying—Follow no leader—Unanimity of movement—Flight by moonlight—Roosting at night—Extirpated in Cyprus—The “Chinese Wall” system—Not adapted to Australia—Deference to aboriginal feeling—Locusts in Australia—Strange ceremony of egg-laying—Inadequate explanation.

IN regard to the faculty of direction with which locusts seem endowed, Dr. Munro says: “The flying locusts in the Argentine come from a northerly direction, and the hoppers or creepers march towards the south, although it might be, so far as abundance of suitable food goes, to their manifest advantage to go in an opposite direction. In certain countries the direction may be known. In this country (South Africa) it will be found that they march towards the south, and not towards the north, east, or west, though either of these directions might have been better for them. The direction may not be true south; it may incline at one time to the south-east, at another to the south-west; but, taken as a whole, it will be southwards.” And he adds: “If proof be needed that the ‘*saltonas*,’ another name—perhaps the Portuguese one—for these wingless armies, march in one direction, it is abundantly found in the experience of the screen and trap, or Cypriote system of destroying locusts, which is based on this fact, and on this alone. This is conclusive demonstration.”^[21]

The distance that these footgangers—to translate, almost without changing, the Dutch word—go in a day depends upon the amount of food they find upon the road, but fluctuates, as a rule, between one mile and two. They start about eight o’clock, when the sun begins to get hot; and halt for the night a little before the sun sets. Dr. Munro describes the way in which the female locust, before laying her eggs, drills a hole in the hard ground with the disc-like extremity of her abdomen, but he mentions nothing very peculiar in connection with the laying of the eggs such as characterises the performance of that ceremony by the Australian plague locust, as will be mentioned shortly.



A PLAGUE OF LOCUSTS

The dark cloud is entirely composed of locusts, which sometimes fill the air from twenty feet to two or three thousand feet above the ground. The poor people attempt in vain, by shouts, by lighting fires, and waving branches, to avert the attack.

The first appearance of these locusts is in enormous hosts, which may sometimes be seen at a distance of from seven to ten miles, and then appear as a black cloud in the clear and rarefied air of South Africa. “It is impossible,” says Dr. Munro, “to estimate the number of locusts in these clouds, but some idea may be formed from the fact that when they are driven, as sometimes is the case in a storm, into the sea, so many are washed ashore that they lie on the beach as a bank from three to four feet thick and from fifty to one hundred miles in length, and the stench from the corruption of their bodies, it is affirmed, is sensibly perceived for a hundred and fifty miles inland.”^[22] The aerial movements of the locusts, when they fairly surround one, are described as “curious, interesting, and pretty.” Distant vision (more especially overhead) is impeded on account of their numbers. The effect when you look on them in the sun’s rays resembles “snow falling thickly and gently,” and the sun is only seen as though it were in eclipse. “Its light is darkened and shadows cannot then be cast from it.”^[23] The height at which the swarm flies may be anything between forty feet and two miles from the ground, but as a rule it is not greater than 400 feet, though from 500 to 800 is not uncommon. Sometimes they fly by moonlight, but this is not their usual practice. As in their earlier wingless state, they seem to act by one common impulse, which prevents confusion. It is obvious, indeed, that with such myriads filling the whole air, a leader could neither be perceived nor followed, and from my own observations I am convinced that the same difficulty applies to this way of explaining the movements of flocks of birds. I have never, myself, seen any evidence of birds being led by one or more of their number, but much to convince me that when banded together, in numbers, their movements are governed by a totally different principle, viz. that of thought transference or thought-unity—collective thinking, as I have elsewhere called it—for that is what it most suggests. If this is not the case with locusts, what, I would ask, is the alternative explanation? If great hosts of men be neither led nor of one mind where to go, they must fall into confusion, impeding one another’s movements, and this is a law which has to do with numbers merely, without respect to the species of which they are composed. It has often been noticed, however, that large crowds seem liable to be swayed suddenly by some common impulse.

Locusts may fly about a district all day doing but little harm, “and at sundown,” says Dr. Munro, “the sight becomes interesting beyond description, for the whole company then appear to vie with one another in order to roost quickly.”^[23] When all have found a resting-place, “every twig, branch, bush, or separate stalk of the corn or wheat or flax are completely covered, and sometimes they stick to each other”^[23]—three or four deep even. “As far as the eye can see, the surface assumes a brownish-red hue. Pillars, posts, or the walls of houses are all alike to them at the time of roosting for the night.”^[23]

Such, then, is the Plague Locust of South Africa, which is, when at maturity, about three inches long. Some years ago, however—the exact date is not given—a larger and handsomer species made its appearance, and is thus referred to in a letter which was sent by “A Disgusted Farmer” of Grahamstown to one of the South African papers: “The new red locust, which, during the last month, has spread from the Orange River to the sea, coming apparently from the north as well as from Natal, is doing terrible damage. Everywhere fruit-trees are being destroyed—quince, apricot, fig, orange, lemon, naartje trees. Not only are the leaves eaten, but young branches are all barked, so that they are probably killed. A splendid crop of mealies, covering the whole of Peddie, Lower Albany, Alexandria, and other districts, has been entirely destroyed. Pumpkin plants are being eaten too. Vegetables of all kinds—lucerne, cattle-cabbage, and kale—are also swept away. The locusts are laying everywhere, and, no doubt, the plague will continue some years. What is the agricultural farmer to do?”^[23] I do not know, but here, probably, were there locusts, he would pick out all such birds as fed on them and try to get them taken off the list of protected species, shooting them illegally all the while he was petitioning.

Dr. Munro’s work was published in 1900, and its principal object was to induce the South African Government to adopt the system of dealing with the locust plague which had been practised with such entire success in the isle of Cyprus. Whether this has since been done, and with what results, I am unable to say. In Cyprus, however, the locusts, which from the year 1600, especially, have changed the country from a garden into a wilderness, were in one season almost entirely swept away. The method by which so great a result was effected is the invention of an Italian gentleman—Signor Matthei by name—resident in the island, and is based upon the inability of

the immature locusts, the footgangers—probably the grown ones too, but this is immaterial—to crawl up a smooth perpendicular, still more an overhanging surface. Such a surface was supplied by a long band of leather, glazed and polished, surmounting a strip of calico, which was made about four feet high, but need not, as it was afterwards found, have been more than two. This insurmountable obstacle, supported at intervals by sticks set in the ground—not upright, but slanting a little towards the path of the locusts—was set up over a large area of country like a miniature Chinese Wall, and proved even more insurmountable. At intervals along the inner side of the barrier deep pits were dug, whilst at wider ones stood men provided with brooms, spades, brushwood, and all else requisite. When the locusts arrived at the Chinese Wall they climbed up the canvas part of it, but being unable to pass the smooth band of leather they fell down in heaps, and their ever-increasing multitudes soon filled the pits, in which they were buried, burnt, stamped down, or otherwise provided for. Afterwards their carcasses were dug out and heaped on carts, and the pits, being empty again, were ready for more. In this way two hundred million quadrillion billions—or something of that sort—of locusts were destroyed, and next year when everything was again ready for them hardly any appeared. By this invention, as simple as it is ingenious and inexpensive, the locust plague in Cyprus has become a thing of the past, and if the conferrer of so great a benefit was ever not a man of large fortune, let us hope that that has become a thing of the past, too, for he must have saved several to the British Government. If the locusts, after coming to the Chinese Wall and finding themselves unable to climb it, had turned round and walked in another direction, this would have made a capital instance of intelligence shown by insects—but they did not do so.

With native labour, the above system, which has been so entirely successful in Cyprus, could, Dr. Munro makes no doubt, be put in operation in Africa; but Mr. W. W. Froggatt, the Government entomologist of Australia, does not think it adapted for that country. Writing in *The Agricultural Gazette* of New South Wales (March, 1901), Mr. Froggatt says, “Though they have been successfully dealt with in Cyprus, Egypt, Algeria, and India by means of trenches, traps, and burning in the hopper state, and digging up and destroying the eggs in the earlier stage, in nearly all cases the areas infested were comparatively small; the labour employed was so cheap that small armies of natives could be employed at a small cost to destroy them, while

in several instances an autocratic government made the natives, whether they were inclined or not, work at their plan of destruction.” In Australia, where, “whether they were inclined or not,” the natives have been got rid of, very much as though they were locusts—or some less stubborn insects—themselves, this would not do.

It was in the summer of 1899 that Mr. Froggatt, in consequence of reports received of the advent of locusts in various parts of the country, left Sydney for Condobolin. On the way there many “mobs”—to use the Australian word—were encountered, and numbers of locusts flew in at the railway carriage windows. Upon alighting, Mr. Froggatt became the witness of a very interesting spectacle—a ceremony, as it may well be called, in which vast numbers of the insects were engaged—of which he gives the following description:—

“In the open red soil we found them laying their eggs in thousands, and the operation was very remarkable. The female set to work by pressing the tip of her abdomen into the soil, and working the plates at the apex, so that she gradually bored a regular circular shaft, slightly over an inch in depth and under a quarter of an inch in diameter, the segments of the abdomen extending and stretching as the work progressed. But the most extraordinary part of the operation was that each female, while boring the chamber to deposit the eggs, was attended by two males, each of which rested his head against hers, with his antennas resting over her head, and the inner foreleg clasped over the prothorax behind the base of the head. Resting like this, with the tails of the two attendant males pointing outwards, the three formed a three-rayed star. Wherever the business of egg-laying was going on, each female and her attendants were surrounded by a cluster of admiring males, averaging from thirty to fifty in number, generally in bunches of fours or fives, forming an irregular ring round them, but separated from her by a clear space of three or four inches. In no instance were there ever more than two males touching the female, though we examined thousands of them at work.”^[24]

What is the meaning of this odd performance?—this ceremony, as it appears to me, though Mr. Froggatt takes a utilitarian view of it. “The probable and only reason,” he remarks, “that I can see for the attendance of the two males upon the egg-laying female is that it enables her to get a firmer grip of the ground, and, in fact, holds her in position till she completes her task.”^[24] But

why, then, should the females of no other species of locust, as far as we are aware, require this aid, and should not the soil of Africa be as hard as that of Australia? “I can find,” says Mr. Froggatt, “no record of this habit in any of our described species, which have the same habits.”^[24] Again, besides the two chief actors, we have the admiring ring of from thirty to fifty males, who can be of no possible service, but whose conduct shows that they take a strong interest in what the female is doing. What is it, too, that regulates the number, or, at any rate, the personality of the assistant males? If it is a matter of rendering assistance only, and the two males who do so are bound to the female by no more special tie than the crowd of interested spectators, why do not these, or some of them, push forward? Why is there never any contention between them? These considerations make me think that there is something of a formal and ceremonious character about these queer proceedings, and that they are governed by the same general law as are certain antics or set figures amongst birds, wherein three individuals take a part. What one requires to know is the courting and marital relations of the male and female locust before the egg-laying takes place.

These little locusts—*Epacromia terminalis* is the specific name—are only about an inch in length, and the male, from the description, seems a little brighter than the female, which may be due to sexual selection. The female appears to lay nineteen eggs only, neither more nor less, which is not so many as one would have expected from the Arabian legend. With some other species, however, the number more conforms to the statement said to have been made to Mahomet.

There is no vanity at all in my thinking that this has been an interesting account of locusts, since I myself have had nothing to do with it. In giving a general description, from general reading, of things generally known, and that have been described scores of times before, one is entitled to use one’s own language, and to think, perhaps, that one stands at no particular disadvantage in doing so. But when, in regard to something specially curious or interesting, the graphic words of an eye-witness are before one, the best thing one can do, in my opinion, is to copy them out. If it be suggested that this is but a lazy way of writing a book, my reply is that a compiler best shows his industry in the searching out of material. The late Professor Romanes was alive to this fact, and has left us in consequence his *Animal Intelligence*—one of the most interesting books that exist, in my

opinion—about one-eighth of which, or perhaps a little more, is written by himself.

CHAPTER VIII

Locusts and *locustidæ*—The most musical grasshoppers—Katydid concerts—A much-resembling note—Cricket thermometers—Cicadas and sounding-boards—Admired musicians—An appreciative audience.

LOCUSTS, as everybody knows, belong to the grasshopper family, but it may surprise some who have read the grumblings of the learned over popular names—white ants, hedge-sparrows, etc.—to find that entomologists have so managed matters that they do not belong to the *locustidæ*—which is one of the two groups into which all grasshoppers are divided—but to the other group. There are long-horned grasshoppers and short-horned grasshoppers. The long-horned ones, which are not locusts, are all of them *locustidæ*, but none of the *locustidæ* are locusts, because locusts have short horns. Entomologists think it would be absurd to alter this, after it has gone on so long, a view in which ornithologists, with their *storm-petrels* and hedge-*accentors*, no doubt agree with them. A mere popular name, with its roots in the Saxon or Celtic, can be changed, and there an end, but scientific nonsense, in Latin, and begun by Linnæus, as is generally the case, let no man presume to meddle with.

It is amongst the *locustidæ* that we find the most musical of the grasshoppers, the Katydids—so well known and highly appreciated in the United States—standing on a far higher level in this respect than the comparatively unmusical locusts. Not that the *locustidæ*—however musical—use their long horns for blowing purposes. Properly speaking, these are only antennæ, and function as such, the musical apparatus being situated elsewhere. The Katydids, for instance, rasp their fore wings against each other, according to the general idea, three times in succession, producing the three syllables, Ka—ty—did, which have given the insect its name, but according to Mr. Scudder^[25] only twice, which makes either “Katy,” or “She did”; that is to say, as a general rule, for he admits the three on occasions. The notes are uttered with great emphasis, and at the rate of some two

hundred in the minute, the performance continuing, at least in the case of some species, all day and all night long.

A number of grasshoppers go by the name of Katydids in America, but the general type of the insect is a graceful, green, fragile-looking creature, with very long, slender antennæ, and, in the female, a long ovipositor at the other end, as if to balance matters. There are many species, and all, or most of them, sing both by night and day, and what is very remarkable, or, at least, very interesting, they have a different note for either. Speaking of one—or, rather, of a long-horned grasshopper nearly related to the Katydids, but not actually a member of the sisterhood—which he had been watching in the sunshine, Mr. Scudder says: “As a cloud passed over the sun he suddenly changed his note to one with which I was already familiar, but without knowing to what insect it belonged. At the same time, all the individuals around, whose similar day-song I had heard, began to respond with the night-cry. The cloud passed away, and the original note was resumed on all sides.”^[26] *Scudderia angustifolia* is the name of this little musician, so called, perhaps, because so sensitive to scudding clouds. But the Katydids do more than merely play an individual tune, each on his own instrument. They hold concerts, at which many join together to make an elaborate musical display, a certain number commencing on one note, and others joining in harmoniously on another. There are leaders, whose business it is to hold the time-measure, and, by a steady insistence on the right note, to draw back any who may happen for a moment to get out of tune. The orchestra is divided into so many companies, who support and assist one another, so that the whole makes a concerted harmony, in which there are many different movements. As a rule the performance is most creditable, though occasionally the effect is marred by a careless player. Before commencing, the company always tunes up.

Possibly it may be thought that there is some mistake here—that things cannot be quite like this. Personally I have no knowledge on the subject—never having been to America—but here is what Dr. George M. Gould says, writing in *Science* for October or November, probably 1895, since the number is referred to as “recent” in *Nature* for December 5th of that year. “As soon as the sun has set and twilight is advancing, the Katydids in the trees begin to ‘tune up.’ The first notes are scattered, awkward and without rhythm, but if no wind is blowing thousands soon join in, and from time to

time, until daylight breaks, there is no intermission.... In order to make my description clearer, let us suppose a thousand Katydids, scattered through the trees, to utter their several notes all at once, and call them Company A. Another thousand—Company B—at once answers them, and this swing-swing is kept up, as I say, all night. Company A's note is the emphatic or accented note, and is more definitely and accurately a precise musical note, whilst the note of Company B varies from one to five half-tones below, the most conspicuous note being five. In the old-fashioned musical terms I learned as a boy, Company A is, *e.g.*, clearly and definitely *do*, while the note of Company B is either *la*, or more certainly *sol*. Not only is Company A's note more unisonal and definite, but it is firmer, more accented, and it seems to me that more insects join in this note than in the second. Careful observation has convinced me that no insect of Company A or Company B ever joins in the other company's note. The rhythm is usually perfect, unless there is a disturbance by a breeze. A sharp gust upsets the whole orchestra, and confusion results, but the measured beat is soon refound. In the instants of confusion one can detect the steady see-saw of certain ones, as it were, 'leaders,' or first violinists, who hold the time-measure, despite the wind, and who soon draw the lost notes of the others once more into the regular measure or beat. I do not mean to say that by diligent attention one may not at times detect individuals sawing out of tune, stray fellows that are indifferent or careless, but the vast majority, usually even without a single exception, if there is no wind or rain, thus swing along, hour after hour, in perfect time. I have counted the beats several times, and find the number is always identical: thirty-four double beats, or sixty-eight single ones, in sixty seconds. The effect of the rhythm upon the mind is not unlike that of the woodman's cross-cut saw, handled by two steady, tireless pairs of hands, although the Katydids give a larger volume of sound, and the *timbre* is harsher." Such is the account, and upon it Dr. Gould asks two questions: "What function does the orchestration subserve?" and "Is there anything comparable to it among other animals?"

In view of these performances of the Katydids one may perhaps question the statement, often made, that crickets are *the* most musical of all insects. The Snowy Cricket, however, of the United States, and no doubt elsewhere in America, is a very striking performer, especially at night, when it emits sounds which Nathaniel Hawthorne has likened to "audible stillness," and

of which he says: “If moonlight could be heard it would sound like that.” Thoreau describes it as a “slumbrous breathing,” but according to the State Entomologist of the United States, this “slumbrous breathing,” or “audible stillness,” consists of “a shrill re-teat, re-teat, re-teat,” which Mr. Leland Howard,^[26] indeed, thinks the best description, but is not quite my idea—nor probably Hawthorne’s—of how moonlight would sound. Harrington—who I suppose is another entomologist—does not interfere with any of these opinions, but describes something which he has seen, and can find nothing about in books. “While the male,” he says, “is energetically shuffling together his wings, raised almost vertically, the female may be seen standing just behind him, and with her head applied to the base of the wings, evidently eager to get the full benefit of every note produced.”^[26] No doubt the female likes the notes—that, indeed, is the *rationale* of their utterance—but what they are really like it is impossible to make out from these various descriptions, another of which, by the way, is “a rhythmic beat.” Possibly they are no more extraordinary (at any rate, “re-teat” is not) than those of our own, and cheerful, house-cricket, which to my ear have always sounded very pretty, but which Cowper evidently did not care about except as a matter of association, since he thus alludes to them in the *Task*:

“Sounds inharmonious in themselves and harsh,
Yet heard in scenes where peace for ever (*sic*) reigns,
And only there, please highly for their sake.”

No doubt there are associations, though these, belonging to the kitchen, appear to me to be of another and blither description, but the “sounds” themselves, in my opinion, are neither harsh nor inharmonious, as far as any unpleasantness to the ear is conveyed by the last word.

One interesting point about the song of crickets is that the number of notes uttered in any given space of time—per minute, say—varies according to the temperature, the two rising together. Professor Dolbeare was the first, as far as I know, to call attention to this fact, and he is thus confirmed by a lady: “One cool evening a cricket was caught and brought into a warm room. In a few minutes it began to chirp nearly twice as rapidly as the out-of-door crickets. Its rate very nearly conformed to the observed rate maintained on other evenings under the same temperature conditions (as now indoors). From this series of observations we found that the rate of

chirping was, as Professor Dolbeare says, very closely dependent on the temperature.”^[27] So the crickets are little thermometers—sixty-three degrees Fahrenheit to one hundred chirps per minute.

As we have seen, the Katydids give concerts, and we may therefore infer that they like their own music in a musically appreciative way; that they listen to each other as critical connoisseurs, whether they have other feelings or not, and that it is not a mere matter of the female alone admiring the sounds made by the male, just because *he* makes them. In all this, however, the admiration is confined—at least, as far as we know—to one species—that to which the musician belongs. Katydids appreciate the performances of Katydids. But there is one group of performers whose music gives satisfaction, not merely to individuals of other species than their own, but to such as are not even included in the same order with them, so that racial pride or family prejudice cannot be the reason of it. Towards these stars we will now turn our gaze.

All who have lived in the more southern parts of the world, including the southern countries of Europe, must have made the acquaintance of the cicadas, for in these regions they are large insects, conspicuous by their appearance when once seen, and by their song long before they are noticed. There is something very uncouth—one might almost say grotesquely humorous—yet at the same time pleasing and lovable about the broad flat heads and great goggle eyes of these insects, in the which it is easy to imagine some quaint sort of expression that seems to mean or suggest something for which the language supplies no word. Their wings, both long and broad, which, when folded, project far beyond the extremity of the abdomen, concealing everything save the great head and the wide shield or boss of the thorax, help also in giving them a most salient and characteristic appearance, and make them look more aerial than they really are. Their legs, whilst they retain their ordinary resting attitude, are entirely hidden, and so too are the organs of the mouth, which combine to make a sharp-pointed beak. Thus their appearance is typical of air and sunshine, and anything so gross as mere feeding or terrestrial locomotion seems foreign to their nature. The ancients, who loved and admired the cicadas extremely, thinking them the most fortunate of creatures, supposed that they lived entirely on dew.

“Oh Tettix, drunk with sipping dew,

What musician equals you?"

sings Anacreon, or someone who imitated him and wrote very gracefully, for Tettix was a common Greek name for the cicada. Really they live on the sap of the trees on which they sit, and there may even be two opinions about their music. To me it is pleasant enough—full of the joy of the sunshine, as it were, and its loudness and the continuous way in which it goes on excites one's wonder. In regard to the way in which it is produced, Darwin says, at page 351 of his immortal work, *The Descent of Man*: "The sound, according to Laudois, who has recently studied the subject, is produced by the vibration of the lips of the spiracles, which are set in motion by a current of air emitted by the tracheæ. It is increased by a wonderfully complex resounding apparatus, consisting of two cavities covered with scales. Hence the sound may truly be called a voice. In the female the musical apparatus is present, but very much less developed than in the male, and is never used for producing sound." As the Greeks, who must have had their observers, used to say—

"Happy the cicadas' lives,
Since they all have voiceless wives."

This sounds all right—I mean the account of the apparatus—but according to Dr. Powell, of New Zealand, it is all wrong. Writing in the *Transactions of the New Zealand Institute*,^[28] Dr. Powell, after quoting the above passage, says, "I am, of course, ignorant of the details of his description; but unless the cicada which he describes differs essentially in the nature of its musical organs from those found in New Zealand, and also from those described more or less correctly by other authors, especially Réaumur, he is most certainly in error." Dr. Powell, then, after telling us that the stridulating organs of the cicada are constructed on a principle unique in nature, viz. a vibrating membrane, continues: "In the male, on the upper surface of the first ring of the abdomen, on either side, may be seen a crescent-shaped opening, and on examining this opening with a magnifying-glass it will be seen to lead into a shallow cavity, closed in by a horny membrane. This membrane is highly elastic, and the sound is produced by the contraction of the muscle straightening out the folds of the membrane; this produces a click and, on the muscle relaxing, the membrane, from its elasticity, springs back with another click." That this is really the way in which the sounds are

produced seems proved by the fact that “if a live insect be caught, and these membranes be observed during the act of stridulation, they will be seen to be vibrating rapidly in time with the beats of the shrill sound.”

But what about the “wonderfully complex resounding apparatus, consisting of two cavities covered with scales”? After a full examination and various experiments, Dr. Powell arrives at the unexpected conclusion that the sound is in no way dependent upon these “large transparent, drum-like membranes,” as he calls the cavities in question. I was “much surprised,” he says, “to find that the large drums seemed to take no part in the production of the sound, and the idea occurred to me that they might be hearing organs; but on examining the females, which are dumb and do not possess the stridulating organs, I found that the drums exist, indeed, but are quite rudimentary instead of being large, as we should expect to find them, were they subservient to the sense of hearing.” If, however, the drums did answer the purpose of a resounding apparatus in the male, we should expect to find them exactly as they are in the female, and so strong does the evidence of their suppression in her appear to me, that I cannot help thinking that, in spite of all Dr. Powell’s observations and experiments, he was somehow mistaken, and that in nature they do act in this way.

As to the quality of the sound produced by the cicada—of its song, as we may call it—this varies greatly in the different species, for there are many cicadas. Speaking of that of the largest—the great *Pomponia imperatoria* of Borneo—as big as a mouse, one may almost say, Mr. Annandale remarks, “The sound produced by this species is, at the beginning of the song, like the winding up of a large clock, and ends by being comparable to the notes of a penny whistle. Between these extremes it rises in a series of trills, each of which concludes with a kind of click. Each section of the song is faster, louder, and clearer than the one which preceded it, until, almost five minutes after the cicada’s settling, the noise suddenly comes to an end as the insect flies off to another tree, where it commences again.”^[29] This great pompous imperial insect—to give it a free rendering of its Latin name—sits shrouded in the mysteries of the deeper jungle, while smaller and less majestic babblers haunt its skirtings and the village groves. “Another species, commonly heard at night in the jungle, has a clear, loud, clarion-like call, which can be heard for a great distance.”^[29]

Of the three New Zealand species of cicada—or those found in Canterbury—a large and small green, and a black one, the two first, Dr. Powell tells us, say “crrrk-crrrk-crrrk,” the second “r-r-r-r-r,” and the third “crrrk-rrrrr,” *ad infinitum*. “Many persons,” he adds, “are totally unable to hear the voice of the small green cicada, or any very acute sounds, and inasmuch as the entire range of the human ear is, according to Helmholtz, eleven octaves, it has been justly remarked that the air may be filled with shrill insect sounds, which may be perfectly audible to the insects themselves, but absolutely inaudible to our grosser senses.”^[30]

It is in Natal—at least, the fact has been observed there—that the cicadas, as they sing, are listened to by admiring groups of other insects. These appear to be beautiful creatures, having wings of a soft, gauzy texture, but iridescent, and shot with the colours of the rainbow. A band of these radiant attendants, consisting sometimes of a dozen or fifteen, fly to the tree where a cicada is sitting and arrange themselves in a semicircle around it, facing its head. They are “all ear” evidently, and, as the sweet sounds continue, one or other of the listeners will advance and touch the antennæ or legs of the object of its admiration. Such marks of appreciation, however, though flattering in proportion to their undoubted sincerity, are not to the taste of the cicada, who will sometimes, whilst in the midst of its song, strike out vigorously with a foot or so—for, of course, it has six—causing its too obtrusive admirers to retreat to a more respectful distance, where they continue to listen with every sign of being extremely pleased.^[31] Some years ago we did not even know the name of these musical-connoisseur-like, and withal very beautiful insects, but now they have been identified by Mr. Kirby, at the British Museum, as *Nothochrysa gigantea*, so we are all much the wiser, and have a weight lifted from our minds.

CHAPTER IX

A Greek mistake—Nature vindicated—Cicadas provided for—A difficult feat—Perseverance rewarded—Cicadas in story—Dear to Apollo—Men before the Muses—Plato and Socrates—Athenian views—A mausoleum for pets—The Greek ploughman—Apollo's judgment—Hercules' bad taste—Modern survivals—A beneficent insect—Elementary education in Tuscany.

THE Greeks thought that the life of the cicadas was all joy, but modern research has been successful in removing the reproach of inconsistency from the general scheme of creation. All is in order, as it now appears: the cicada's case has been considered, and a very handsome wasp provided for it. At least, I think it is handsome. It is large and strong, I know, as is necessary for the part it has to perform, but I cannot quite remember the colours it flies under; an expression which, though metaphorical, may be pardoned, since flags have much to do with such dramas as that now to be described. For as the joyous, sun-loving creature sits in its accustomed place, chirruping forth those shrill yet musical notes which I, at least, have never wearied of, the destroyer is at hand, and settling on its broad back, curves its abdomen beneath that of the poor blithe singer, and in a moment has done its work. As the sting enters, the happy note that has been sounding regularly for the last hour, perhaps, is changed to a discordant scream of pain, and with a spasmodic spring or flutter—the last, or near the last, that it will ever make—the cicada, with the wasp still clinging to it, falls to the ground. This is awkward for the wasp, who doubtless considers herself aggrieved in the matter, since the cicada is so bulky that, powerful as she is, she can neither lift it from the ground in flight, nor is she prepared to drag it all the way to her burrow. What, then, is she to do, or of what use to her is the prize she has obtained with such adroitness? But she has her plan, and though the captious behaviour of the cicada has, for the moment, a little deranged it, it is not permanently frustrated. Slowly, but with firm insistence, she drags her victim to the tree on which a moment before it was so happily seated, and then exerting all her force, begins to mount the trunk

with it. Often she has to pause and rest, often it seems as though the task would be beyond her, but she continues the laborious ascent, sometimes for upwards of an hour, until at last a height has been reached at which it is possible for her to put her great project into execution. This is no other than to fly down obliquely, with her victim clasped in her arms, to the pleasant little sarcophagus which she has previously prepared for it, for though flight upwards, or in a straight line, with such a burden, is out of the question, her strength is equal to this. It is necessary, however, that she should balance the body nicely, and make a fair and uninterrupted start, in order not to be overweighted and again fall. Her enterprise is “full of poise and difficult weight,” and cannot be successfully carried out in face of the rude struggles of a tiresome obstructive not “in tune with the infinite.” These struggles, however, have now ceased; the cicada is in a comatose condition, and, having adjusted it properly, and assumed the requisite attitude and position, our wasp—whose scientific name, by the way, is *Sphecius speciosus*—launches herself, with “the white man’s burden” she has “taken up,” from her coign of vantage, and reaches home with it in safety. How high she has previously ascended the tree I cannot say, since my informant does not, but it would be interesting to ascertain both this and the average distance which she has to fly to her nest, and to compare the one with the other. Unless the latter is very much greater than the former—and as the journey is constantly downwards it cannot, one would think, be very far—then we must see in the wasp’s choice of a toilsome ascent up a perpendicular tree-trunk, in preference to a horizontal journey along the ground, a triumph of instinct over intelligence, and it is, indeed, quite possible that, having always been accustomed to fly back with her prize, which perhaps was not always so heavy, she should go through as much labour to enable her to do this as, differently directed, would attain the end for which it is employed.



A WASP BEARING OFF A CICADA.

After the wasp has killed the cicada, they both fall to the ground. Strong as the wasp is it is not easy for her to carry such a heavy insect to her nest. But she has her plan. Slowly but persistently she drags her victim to a tree-trunk and up it, though it may take her an hour to reach the requisite height. Then she sails off for her nest on an inclined plane, with wings extended, and her victim clasped in her arms.

The burrow of this wasp consists, we are told, “of a gently sloping entrance, extending for about six inches, when, ordinarily, a turn is made at right angles, and the excavation is continued for six or eight inches farther, ending in a globular cell an inch and a half in diameter. Frequently a number of branches leave the main burrow at about the same point, each terminating in a round cell.”^[32] In each of these cells either one or two cicadas are deposited, and it would seem that when there are two, only one of these is provided with an egg, so that some of the wasp-larvæ have double rations. As the female *speciosus* (her arguments, I think, would need to be specious to make one in love with a scheme in which she plays such a part) is very much larger than the male, it seems more than probable that the female eggs are laid in the chambers which contain two cicadæ, and the male ones in those which accommodate a single one only. If so, then these solitary wasps must have the same control over the sex of the eggs laid by them as the queen bee has. The social ones, should this be the case, no doubt have, too, but as the former must have preceded the latter, it would appear that this power has not been developed to meet the needs of a complex state of society—as has been generally supposed—but in accordance with much more simple conditions. The fact, however, if it be one, has not yet been demonstrated.

“The delicate white, elongate egg of the wasp is laid under the middle leg of the cicada, and when it hatches, the larva protrudes its head and begins at once to draw nourishment from between the segments of its victim. The egg hatches in two or three days, and the larva attains full growth in a week, or a little more. It feeds entirely from the outside, and, when full-grown, spins a white silken cocoon (mixed with much earth, however), which is finished at the expiration of two days. It remains in the cocoon, unchanged, through the winter, transforming to pupa only in the following spring, and shortly before the appearance of the true insect. When the adult hatches it gnaws its way out of the cocoon, and so on up through the burrow to the surface of the ground, thus completing its life-round in a full year.”^[32] How long, exactly, the life of the cicada lasts after it has entered into hospitable relations with the *speciosus* I am unable to say.

Such, then, is the end of the cicada, in spite of the love of Apollo, who, according to the Anacreontic ode, bestowed upon it its shrill song. Thus it dies, though “cherished by the Muses, painless and fleshless, almost equal

to the gods.” Whether it be fleshless *speciosus*, in the larval state, best knows (on the latter point there will have been no means of comparison), that it is painless one can only hope. It is something, however, to be so known to fame. Homer himself alludes to the cicada in terms of respect, calling its shrill song “delicate music,” whilst Hesiod tells of “the dark-winged Tettix, when he begins to sing to men of the coming summer; he whose meat and drink is of the refreshing dew, and who all day long and at break of day pours forth his voice.”

There was no end, apparently, to the love of the ancients—especially the Greeks—for the cicadas, or tettiges—for they were known by both names—or to the graceful things they said of them. From poets and philosophers down to ploughmen, all were equally fond of them. “We bless thee, Tettix,” says a poet whose name has been merged in that of one who is now a name only, though a great one—Anacreon, namely—“We bless thee for that seated on the tree-tops, sipping the dew, thou singest royally.... Oh, sweetest of summer prophets! honoured by mortals, thou art cherished by the Muses. Phœbus himself loves thee, and gave thee thy shrill song”; and Plato tells us that “as music soothes the mind and dissipates fatigue, so the ploughman loves and cherishes the cicada for its song.” The Greek ploughman, apparently, was a less gross embodiment than the one of the present day, after twenty-five centuries or so of improvement. To Apollo the cicadas were sacred, because they “everlastingly sang to the sun,”^[33] and, for the Muses, they had once supplied their place. “As the story goes,” says Plato, “before the Muses lived the cicadas were men on earth, and so loved song and singing that, to lose no time from it, they left off eating, and so died of that dear delight. But, in death, they became cicadas, and this boon was granted them by the Muses, lately born, that on earth they should eat no more, but only sing until they died again, and that then they should return to the Muses to tell them who, amongst mortals, loved and worshipped them most.” “A lover of music like yourself,” says Socrates in the “Phædrus” of Plato, addressing one of *his* worshippers, “ought surely to have heard this story of the cicadas, how they were once human beings, but died through forgetting to eat. But now, dear to the Muses, they hunger no more, thirst no more, but sing only, from their birth. And in heaven they tell Terpsichore of the dancers, Erato of the lovers, Calliope, eldest of the nine, and Urania, of those whose heart is in philosophy—and thus they whisper to them all.”

So established were these and similar stories that, in Greece, a cicada perched on a harp was often engraved upon gems as the symbol of the Muses, and, were there a musical contest, one had only to settle on the lyre or pipe of the competitor it favoured, for the prize to be instantly adjudged to that one—since Apollo was then held to have spoken. Only in the absence of such indication were other methods of forming a conclusion resorted to. In common with other graceful creatures, cicadas were often kept as pets by the Greeks, and that mausoleums were sometimes raised to these favourites we know from the following epigram of the poetess Anytie—written probably for the friend it celebrates:—“For a grasshopper, a nightingale of the fields, and for an oak-haunting cicada Myro has built one common tomb. There the maiden sits and weeps for three pets, torn from her by unrelenting Hades.”

Amongst the Athenians the cicadas were looked upon as children of the soil of Attica, and those only who, like them, had been born upon it, were permitted to twist the golden tettix, or bodkin, amidst their flowing locks, thus forming the knot in which they were accustomed to wear them. This privileged bodkin received its name through being surmounted with the head, in gold, of a cicada, or tettix, and the wearers—or bearers—of these insignia—which were strictly forbidden to strangers—were known for this reason as Tettigophori. They were most proud of the distinction, and, indeed, as it showed them to be Athenians, they had a somewhat better right to be than is common in such cases. Yet, amidst all this praise, we meet, here and there, with a dissentient note. Hercules, for instance, feeling inclined to sleep, once, on the banks of the river, opposite where the town of Locris stood, and not being able to, on account of the perpetual singing of the cicadas, took it so seriously that he prayed to the gods to put a stop to their disturbing him. The gods, with whom Hercules was always a favourite, heard his prayer, and cicadas, from that time, ceased to sing opposite Locris, though they swarmed all round about that town. Here it seems just to be hinted that Hercules was not very fond of the cicadas' song, and Virgil—but he was a Roman—has called it (*infandum!*) a *creaking* note. On the whole, however, when he mentions these insects, he gives us a pleasing picture.

“*Sole sub ardenti, resonant arbusta cicadis,*”

he sings; a line which seems bathed in sunlight, and makes one see the green lizards too. On the whole I cannot help thinking that Virgil loved the cicadas.

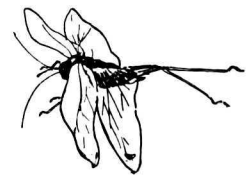
It is interesting to find that in modern Italy, generally, but especially in Tuscany, the old ideas and legends in regard to the cicadas have not yet died out. Still, according to the Tuscan peasant, they were maids—not men—before the Muses, till Apollo, as a mark of his favour, promoted them into insects. Now, however, but little distinction seems to be drawn between cicadas and crickets, or grasshoppers, and, indeed, this was to some extent the case in classical times—the three often figuring together on ancient coins or rings. Amongst all of these—and together they supply a number of species—the greatest favourite with the Tuscan peasant of to-day—as perhaps it was in days long gone by—is a beautiful grey-green grasshopper, which the Americans would call a Katydid, but is, here, the cavalletta. This insect is looked upon as the special patron of children, upon whom it has the power of conferring musical and poetic genius, as well as more general mental endowments. To perform this properly, however, it must enter the room where its little favourite lies asleep, and this it seems often to do. The mother, should she see it, has her own part to play in the matter, which she does by tying the beneficent insect, by a long thread, to the bed-post, and chanting the following verses, with the idea, probably, that “then the charm is firm and good.”

“Cavalletta, good and fair,
You bring good fortune everywhere,
Then since into this house you’ve come,
Oh, bring good fortune to our home,
Unto me and everyone,
But bring it mostly to my son.
Cavalletta, this I pray,
Bring, and do not take away.
In life you were a lady, full
Of talent, good, and beautiful,
Let me pray, as this is true,
You’ll give my child some talent too.
And when you fly from east to west,
May you, in turn, be truly blest,
For though an insect form you bear,
You’re still a spirit good and fair.”



A LUCK-BRINGING GRASSHOPPER

In Tuscany, if this insect comes into a child's room whilst asleep, it is the mother's duty to attach the grasshopper by a thread to the child's bed to bring good fortune. The grasshopper is shown in the right-hand corner.



As the child grows older, and learns to talk, he is instructed in the truth of the matter, and taught by heart the following verses, which he must repeat whenever he sees a Cavalletta:—

“I am but little, as you see,
But yet I may a genius be;
And if, when grown, I shall be great
And make a name in Church and State,
I’ll not forget that one fine day,
As I in cradle sleeping lay,
A Cavalletta blessed me there,
In answer to my mother’s prayer.”[\[34\]](#)

We are not told what happens to the Cavalletta that has been tied up, after “the charm’s wound up.” The proper thing for the mother to do would certainly be to let it go, but I can’t help thinking that what she really does do is to put her foot on it, under the idea that only that can make the thing quite certain. That would be so like the peasantry—of any country.

CHAPTER X

Cicadas in England—A blower of bubbles—The prolific Aphis—A nice calculation—
Scientific curiosity—Dragon-fly armies—The son of the south-west wind.

IT is generally understood that there are no cicadas or tettixes in England, and this—with a reservation in favour of a single species residing in the New Forest—is roundly asserted in various entomological works of authority. Since, however, Mr. George Bowdler Buckton, F.R.S., has written a monograph of the British *Cicadæ*, or *Tettigidæ*, in two volumes, each of which has a number of plates giving figures of the various species, all with their Latin names, there would seem to be a conflict of learned opinion; and I, for my part—since one of these species has relations with a nice little parasite which I should like to describe—am of opinion, after profound investigation and impartial weighing of the evidence on both sides, that Mr. Buckton is right. What strikes one at first sight as curious is that numbers of creatures, as large sometimes as humble bees, or larger, and of very striking appearance—often quite brilliantly coloured—should for so long have escaped observation; for certainly one has never seen them oneself, and, on making inquiries, one soon finds that nobody else has. But there is an explanation of this seeming miracle, and that of a not very satisfactory nature. One may have noticed, whilst going through the plates, that in the neighbourhood of each striking figure there are two little irrelevant-looking black lines, drawn soft and fine, very unobtrusive, looking as though they wished to elude observation; and gradually it begins to dawn upon you that these lines represent the real size in linear measurement of the very salient, *outré*-looking creature you are looking at. This, then, is the key to the mystery. England is full of cicadas, but they are all so small that nobody can see them—at least without taking some trouble. So our poets have been silent, our philosophers have made no reflections, and our ploughmen, to this day, are without a proper objective for those appreciative perceptions of life around them which, if it only existed, there might be some evidence of

their possessing. Our aristocracy too, or old county families, have never been able to “think gold of themselves,” as the saying is, on account of their golden tettix-pins, though the feeling itself has not been entirely denied them. In a word, our national character has been uninfluenced by cicadas, and, on this, two questions arise: first—for it is no use to start on an assumption—whether faults exist in it, and then, if they do, whether all or any of them are due to this cause. But such matters are for the historian to deal with, and would be out of place in the pages of a work like this.

Though cicadas are so small in England—whilst their voices, if they have any, as there seems no particular reason to doubt, are too attenuated to be audible to our human ears—yet they are not quite invisible. When seen, however, they are known by some other name, such as frog-hoppers, tree-hoppers, or the like. Some of these, in their larval stage, which much resembles the adult, take a great deal of pains to conceal themselves, though in this they have another reason than that of wishing to elude observation. Our common cuckoo-spit is a good instance of this, and also of how a wrong explanation of a common and easily observed phenomenon may for a long time be given, not only in popular works, but also in scientific textbooks or monographs, or within the supposedly up-to-date pages of various encyclopædias. The cuckoo-spit, as everyone knows, sits in the midst of a little bower of froth (allied to that other of bliss perhaps) which, on being examined, resolves itself into an accumulation of bubbles, having a somewhat sticky consistency. We had always been told—and still are now very often, though the contrary has been well made out—that these bubbles proceeded from the insect itself, after the manner of any other secretion. But this is not the case. The secretion here is only a clear fluid, and into this the insect afterwards blows bubbles by a mechanical process, and through the addition of air. It is Professor E. S. Morse who, in the pages of *Appleton's Popular Scientific Monthly*,^[35] has thus revolutionised all our ideas on this subject. His account is as follows: “The so-called frog-spittle or cuckoo-spit appears as little flecks of froth on grass, buttercups, and many other plants during the early summer. Immersed in this froth is found a little green insect, sometimes two or three of them concealed by the same moist covering. This little creature represents the early stage of an insect which, in its full growth, still lives upon grass, and is easily recognised by its triangular shape and its ability of jumping like a grasshopper.”

“If the insect is cleared from the mass of froth it will crawl quite rapidly along the stem of the plant, stopping, at times, to pierce the stem for the purpose of sucking the juices within, and finally settling down in earnest, clutching the stem with its legs. After sucking for some time, a clear fluid is seen to exude from the end of the abdomen, flowing over the body first, and gradually filling up the spaces between the legs and the lower part of the body and the stem upon which it rests. During all this time not a trace of an air-bubble appears; simply a clear, slightly viscid fluid is exuded, and this is the only matter that escapes from the insect. This state of partial immersion continues for an hour or more. During this time, and even when walking, the posterior segments of the insect’s abdomen are extended at intervals, the abdomen turning upwards at the same time. It is a kind of reaching-up movement, but whether this action accompanies a discharge of fluid, or is an attempt at reaching for air, I have not ascertained. Suddenly the insect begins to make bubbles by turning its tail out of the fluid, opening the posterior segment, and grasping, as with a pair of clasps, a moiety of air, then turning the tail down into the fluid, again, and instantly allowing the enclosed air to escape. These movements go on at the rate of seventy or eighty times a minute. The tail is moved alternately to the right and left in perfect rhythm, so that the bubbles are distributed on both sides of the body, and these are crowded towards the head, till the entire fluid is filled with bubbles, and the froth thus made runs over the back and around the stem. In half a minute some thirty or forty bubbles are made in this way—a bulk of air two or three times exceeding that of the body—without the slightest diminution in the size of the body.”

It seems clear, therefore, that the air which is put to this purpose is abstracted directly from the atmosphere, and that neither it nor the bubbles manufactured through it have ever been within the body of the insect. Moreover, if the little bubble-maker be thoroughly dried—which, according to Professor Morse, is a matter of difficulty—it will continue to secrete such spare fluid as it still has, but not the tiniest bubble is seen to issue with this. If set in a drop of water it struggles to the surface, and then goes through the same process of blowing bubbles as it has done when immersed in fluid of its own distilling. The result, however, is not the same, for the water will not hold the bubbles, which constantly disappear. Such, then, is the manner in which the frothy pool is made. What purposes does it answer? That of a pond, apparently, for it would appear that in their larval state these little

frog-, or tree-hoppers, are to some extent aquatic insects. If kept dry and not allowed to renew their supply of fluid, their body shrivels, and before long they die. This is not through suffocation, since they can breathe air, by means of spiracles, in the ordinary way. If, however, they are examined closely, certain leaf-like appendages may be detected upon each side of the seventh and eighth segments of the abdomen, and Professor Morse suggests that these may be of the nature of branchiæ, or gills, enabling the insect to breathe, also, in water or fluid, by abstracting the air from it, after the manner of a fish, as some other aquatic larvæ do. “As many of these,” he says, “respire in two ways, either inhaling air through the spiracles, or by means of branchial leaflets, so *Aphrophora* (for that is the classic name of our insect) may likewise utilise its branchial tufts for the same purpose. Thus we may see the reason for this bubble-blowing, since each fresh bubble added to the mass may aerate the fluid, so to speak, and thus secure at intervals a fresh supply of oxygen.”^[36]

In early spring, if one examines the leaf-buds of rose trees, which now begin to swell, one may often see tiny little black specks, like grains of gunpowder, scattered over their surface, especially within any fold or crevice which it presents. These are the eggs of the Aphides, insects which, if not cicadas, are not so very far removed from them, and which, looked at from various points of view, are extremely interesting little creatures. One of these points of view, which we may conveniently start from, is their extraordinary rate of increase, which exceeds even that of the Chinese. “A single insect,” says Mr. Buckton, “hatched from one of these shining black ova may be the mother of many billions of young, even during her lifetime. Réaumur calculated that one Aphis may be the mother of the enormous number of 5,904,900,000 individuals during the month or six weeks of her existence. But neither Tongard nor Morren is satisfied with this estimate, both declaring that quintillions are within the capabilities of a single mother’s efforts. Professor Huxley (who, by the way, was not interested in the alleged phenomena of spiritualism, *even if true*) makes a curious calculation which, at any rate, affords some approximate idea of what a quintillion of Aphides might mean. Assuming that an Aphis weighs as little as one-thousandth of a grain (which is less than I should ever have thought), and that it requires a man to be very stout to weigh more than two million grains, he shows that the tenth brood of Aphides alone, without adding the

product of all the generations which precede the tenth, if all the members survive the perils to which they are exposed, contains more ponderable substance than 500,000,000 of stout men: that is, more than the whole population of China.”^[37] This, it appears, is an under-estimate, which is rather annoying, for one would like to call it a gross exaggeration. But facts are facts—in whatever degree they may interest one—and it is impossible not to feel respect for an insect like this, especially in these days, when the diminished returns of the census are beginning to cause alarm as to the future destinies even of our own once proudly fecund race. It is a wonderful record for a single individual—to have weighed down China—and when Mr. Buckton remarks that facts like these regarding the prolific nature of Aphides “afford sufficient explanation of the occurrence of the extraordinary swarms so often noticed by authors,”^[37] nobody is likely to disagree with him. With billions a certainty, and quintillions in the air, swarms seem amply accounted for.

One of the authors here alluded to is our homely immortal, White of Selborne. “I shall here mention,” he says, “an emigration of small Aphides, which was observed in the village of Selborne no longer ago than August 1st, 1785. At about three o’clock in the afternoon of that day, which was very hot, the people of this village were surprised by a shower of Aphides, or smother-flies, which fell in these parts. Those that were walking in the streets at that juncture found themselves covered with these insects, which settled, also, on the hedges and gardens, blackening all the vegetables where they alighted. My annuals were discoloured with them, and the stalks of a bed of onions were quite coated over for six days after. These armies were then, no doubt, in a state of emigration and shifting their quarters; and might have come, as far as we know, from the great hop-plantations of Kent or Sussex, the wind being, all that day, in the easterly quarter. They were observed, at the same time, in great clouds about Farnham, and all along the vale from Farnham to Alton.”^[38]

Other great migrations of Aphides have at various times been observed. In the autumn of 1834 the city of Gand was invaded, and, one may almost say, taken by a vast army of them, and at Bruges and Antwerp the same swarm is said to have darkened the sun,^[39] a result of such gatherings more noticeable elsewhere than in England, since our sun usually is darkened. Insects, though their movements are not so regular, nor, as a rule, so

noticeable as those of birds, yet often migrate—how often or how regularly it is difficult to say. Locusts are, of course, the stock example as well as the most terrific one, but perhaps dragon-flies, were they as destructive, would have been as much noted in this connexion. Their migrations seem to be tolerably frequent, and a record of them between 1494 and 1868 has been published by Koppen, a German entomologist. In 1881 a great flight of them took place in Illinois. “The air,” we are told, “for miles around seemed literally alive with these dragon-flies, from a foot above ground to as far as eye could reach, all flying in the same direction, a south-westerly course, and the few that would occasionally cross the track of the majority could all the more easily be noticed from the very regular and swift course they generally pursued; but even these few stray ones would soon fall in with the rest again. Very few were seen alighting and all carefully avoided any movable obstacles.”^[40] This migration took place during a very dry season, and may have been caused by it owing to the drying up of swamps, ponds, etc., in which the insects would otherwise have laid their eggs, obliging them to seek other suitable places.

In the spring of 1900 a great migration of dragon-flies was observed in Belgium. “All the observers agree that the insects flew rather low, with astonishing regularity, and without resting; that they kept close to the earth, where there were no obstacles, but that they mounted to a height of 10 to 12 mètres when houses or trees were in the way. They did not go round obstacles in their line of route, but surmounted them, and descended on the other side. According to some observers, their flight was very slow, others again asserting that it was very swift. When the velocity could be estimated, however, it was found to be at 5 mètres per second or 18 kilometres (11¼ miles) per hour (so that the slows have it). In general they went in groups, more or less isolated, and more or less dense.” The writer of the above account—a Belgian—concludes thus: “All the facts point to the following conclusions: The dragon-flies of the 5th came from regions situated to the east of the country, which they entered in several columns, flying at a great altitude: between 7 and 8 a.m. they descended towards the earth, continuing their route towards the west. But we remain in ignorance of their point of departure. The swarm probably quitted its usual habitation early in the morning, and immediately flew to a great height. It was only on arriving near the earth that they flew against the wind”^[41] (which, however, they then continued to do).

What Mr. Hudson calls “dragon-fly storms” are a special phenomenon of the Pampas. In this case the cause of the migration—for such movements seem to come under this heading—is a special wind called the *pampero*, that blows south-west from the interior of the Pampas. It is very violent, cold, and dry, and the dragon-flies evidently fear it. The “storm” is thus described by Mr. Hudson: “It is in summer and autumn that the large dragon-flies appear; not *with* the wind, but—and this is the most curious part of the matter—in advance of it; and inasmuch as these insects are not seen in the country at other times, and frequently appear in seasons of prolonged drought, when all the marshes and watercourses for many hundreds of miles are dry, they must, of course, traverse immense distances, flying before the wind at a speed of 70 or 80 miles an hour. On some occasions they appear almost simultaneously with the wind, going by like a flash, and instantly disappearing from sight. You have scarcely time to see them before the wind strikes you. As a rule, however, they make their appearance from 5 to 15 minutes before the wind strikes; and when they are in great numbers, the air, to a height of 10 or 12 feet above the surface of the ground, is all at once seen to be full of them, rushing past with extraordinary velocity in a north-easterly direction. In very oppressive weather, and when the swiftly advancing *pampero* brings no moving mountains of mingled cloud and dust, and is, consequently, not expected, the sudden apparition of the dragon-fly is a most welcome one, for then an immediate burst of cold wind is confidently looked for. In the expressive vernacular of the gauchos the large dragon-fly is called ‘*hijo del pampero*,’ son of the south-west wind.”^[42]

CHAPTER XI

Aphides and their enemies—Curious interrelations—The biter bit—Altruistic development—Bread and beer protectors—Saved by ladybirds.

WITH prolific powers which have been successful in arousing the interest even of the late Professor Huxley, it is a comfort to think that the numbers of the Aphides are always being kept down by the operation of certain well-contrived causes, most of which take the shape of various insect enemies. Were all of these, or perhaps were any one of them, entirely removed, the whole world apparently might find itself deep buried beneath a “star-y-pointing pyramid” of insect organisms, for what cannot quintillions, in the hands of a competent mathematician, achieve? The wonder certainly seems that any kind of check should be sufficient. We owe our safety, in part, as might have been surmised, to a small Ichneumon Fly, the traces of whose work may generally be seen in a number of brown lifeless corpses, which are dotted about like so many skeletons at the feast, amongst any collection of living Aphides. These, if examined more closely, are seen to be mere empty sacs, each one having at some part of it a quite circular aperture, through which the issuing guest has escaped. But the amount of the good thus effected is not to be estimated by the number of these shells, these nests from which the bird has flown. Such are only in the last stage of things, whilst almost all, including the healthiest-looking of the living Aphides, are probably travelling along the same road, to arrive at the same goal. All, or almost all, have within them a guest whose energies are unceasingly devoted to absorbing the whole of their interior arrangements into itself, and gradually taking their place. “In June,” says Mr. Buckton, “during the hot weather, I have seen, at the same time, as many as three of these flies on one rose sprig, each poised on the back of an Aphis, which throws itself into many contortions for the purpose of throwing off its enemy. The Ichneumon, however, remains fixed on the back for ten or more minutes (as though enjoying the situation) before the ovipositor is thrust under the skin of the victim and the egg is laid. The Aphis appears to suffer

at first but little, since it soon resumes its occupation of pumping up the sap. A worm-like, or more commonly, a maggot-like creature, according to the species of the parasite, hatches from this egg, which revels in the organised nutritious fluid elaborated by the Aphis. The greater part of the abdomen is occupied by this maggot when it becomes full fed, and then it may often be seen through the transparent integument, as a grub curled into a semicircle. Finally, the Aphis dies, the grub ceases to feed, and after a certain period of rest cuts out of the roof of its prison a circular plate, like a trap-door, as regular in form as if a carpenter's centre-bit had been used. The emerging fly has four wings, long antennæ composed of numerous joints, a wasp-like body and legs, and is in every way suited for its marauding expeditions.”^[43]

Other and more interesting dramas can take place within the body of an Aphis; wheels within wheels, one masterpiece of economic contrivance enclosing another, perfection more perfected. Along what path, indeed, can the beauty and wisdom of Nature—those endless steps from endless seeming halting-places that become, when reached, but so many points of fresh departure—be run to earth, so to speak? The brain becomes, at last, almost weary in the pursuit of wisdom's ways, and even the delighted spirit would fain cry, “Hold! Enough!” Thoughts like these are powerfully excited by the following picture.

It is spring, early spring, and already the young Aphides have begun, with gladness born of the opening year, to absorb the sweet sap from the stems of the immature wheat-crop. Later, however, when July's sun shines brightly behind its cooling screen of clouds—for in England all climatic extremes are tempered—they ascend in “numbers numberless” to attack the ear itself. What, then, can save the hope of the husbandman? What but *Ephedrus plagiator*, a small black-winged Ichneumon Fly that even now is at work? As each individual Aphis clings to the wheat, it becomes, in spite of its efforts to fill itself, hollower and hollower; its appetite flags, and ere it can fatally affect the plant on which man's life and the machinations of the protectionist depend, it has become a mere brown pupa-case for a body other than its own to develop in. A day or two, and almost on every grain of the wheat hangs an insect, lifeless, but—oh, floweret springing from the tomb!—life-filled. Hardly a living Aphis is to be seen feeding amongst them. The wheat-crop has been saved. But the march of events, thus unfolded, does not end here. Another drama has to be played out ere the full

life issues from the once living cradle whose contents it has absorbed and become, nor will it bear the image of that particular Ichneumon Fly that laid its little egg, some weeks ago, in the body that seemed so designed for it. Like the Aphis, the Ichneumon, too, must learn to live for others, thus rising through selfishness to a purer and higher embodiment. If we pass a little later through the same cornfield, another insect, differing from, yet of the same general type as the prior parasite, may be seen running to and fro over the wheat-ears, tapping each tenanted abode with tremulously quivering antennæ, which, as well as the whole body, seem to vibrate with excitement. It is looking for lodgings, but not every house so touched can be utilised, for *Ceraphrus Carpenteri*—such is the new tenant's name—must first be satisfied that none of its own species have already taken possession. Once assured on this point, however, its duty lies plain before it, and bending its antennæ against the wheat-ear, so as to form a fulcrum, it turns the tip of its abdomen towards that of the dead Aphis, and with its ovipositor commences to saw through the skin. As much as ten minutes may be occupied in the accomplishment of this task, for the sarcophagus that has thus to be pierced is hard, and the ovipositor, though short, is not stout, but slenderly formed. But there is no flagging of energy, and at length, when efforts steadily continued have been crowned with success, the same deft instrument is again employed to pierce the sleeping *ephedrus*, and a second egg is deposited in this second cradle. To this new tenant the former one must now yield up the juices of its body, even as those of the Aphis were freely rendered unto it. It must die in its turn, but by its death another lives, and thus the physical act of aggression, which we call selfishness, becomes the seed-bed, as it were, or forcing-house—the food-plant, to use an entomological simile—of a moral altruism. True, the Aphis may at first struggle, the maggot, pierced by the ovipositor, may flinch for a moment, but after that there is complete passivity, without which there can be no complete acquiescence. Self-absorption, that is the moral of it all; for the true self of the Aphis, which is not represented by the outer husk, is absorbed into the Ichneumon, and so in regard to the latter. Thus, throughout the animal kingdom we must look to the inner, and not to the outer, significance. What matters it, though Nature be “red in tooth and claw,” if the fierce rendings of the outer integuments are but as the first gropings towards interior rest and calm? And should we not, in the lower walks of life, look to the soul through the body, and see in processes which,

with ourselves, might seem to represent the flesh only, a blending which but anticipates the more complete separation? Thus, and thus only, as it appears to me, can we impart beauty to a scheme which, without this key, must appear selfish and unsatisfactory. The key may be hard to find, but when we once hold it we need no longer repine.

Besides the Wheat Aphis, which but for such arrangements as have here been glanced at, would almost deprive us of bread, we have the Hop Aphis, a species the dread of which is still more strongly disseminated amongst the masses of this country, inasmuch as its interference would be with the supply of beer. No wonder, then, that the little ladybird is beloved by all, since, but for its efforts, many a poor man might have to live in a state of enforced sobriety, which, in its turn, must deleteriously affect that position of respect and esteem which many illustrious and highly placed individuals now hold in the hearts of the people, so that a general disturbal of habits and ideas, amounting almost to moral chaos, would attend any serious diminution in the numbers of this insect, England's true guardian angel. But it were unjust to claim an undue share in the merit of recognising work like this. Appreciation of such services is, as one might expect, widely spread, and is expressed in such popular names as, for instance, in Lombardy "Bestioline del Signore," in Tuscany "Madonnine" or "Marioline," in France "Bête" or "Vache à Dieu," and in Germany "Sonnenkäfer." The first-named countries, indeed, are not, or used not to be, beer-drinking, so that unless this little madonna is a patroness of the vine too, they are not so easy to understand. It may, however, be incidentally mentioned that the ladybirds are good friends to the orchards, and destroy many thousands of apple- or plum-eating Aphides.

Aphides, in fact, of some kind or other, are what they have come into the world to destroy. It is their mission, their epos.

The following account of the habits of these beneficent creatures, principally in the above connection, is given by Mr. Buckton, a profound student of insect life-histories: "The food of Coccinella (the ladybird) consists almost exclusively of Aphides. Their marvellous voracity is shown equally in their larval and their winged condition. The former stage may be commonly seen throughout early summer as slaty-grey or brown six-footed creatures, covered with tufted tubercles, and provided with mandibles efficient both for holding and sucking out the juices of their victims. In

some years the imagos (or grown insects) are wonderfully numerous, and when they take wing form vast swarms which travel great distances. In the year 1869 such a cloud passed over a large part of Kent, Sussex, and Surrey,” and their effect (of the same genial nature as that of Bacchus wandering through the earth) was soon seen in the good hop crop of the following year. “Although the Coccinella is not restricted to the Hop Aphis for its food, it frequently follows its migrations, and travels on the same winds. Whilst feeding, the Aphis is held and manipulated by the jaws or palpi of the Coccinella, and the devouring operation proceeds amidst the struggles of the victims from the apex of the abdomen to the thorax, which parts, together with the head and legs, are finally rejected.”^[43] This process—which is by no means confined to insect life, but extends upwards from it even into the highly organised mammalia—does not seem to be a pleasant one to witness, for Mr. Buckton remarks upon it: “We may express some hope, in sympathy with the Aphis, that the automatic theory of animal life may here find some place, and that reflex action may explain the fact that, under the microscope, the mutilated remains of the Aphis, without stomach and without internal organs, have been seen to walk away and live after the operation for a considerable time! Automatically the Coccinella furbishes up its jaws and antennæ in readiness for another meal. From thirty to forty Aphides may thus be consumed in one hour.”^[43]

Automatically perhaps—that is to say, between the lines—we may gather Mr. Buckton’s opinion of the automatic theory. There are some theories which seem held, like dykes or barriers, to prevent the sea from getting in. One doesn’t want the sea to get in, because it would swamp such a lot of things, which, although quite artificial, one is not prepared to part with, but one doesn’t believe in the barrier except for that particular purpose to which it is applied. The automatic theory in regard to animals is a case in point. Scientific men make use of it in order to keep out another, which they don’t want to have to admit, though they do, as a fact, believe in it. This, again, one can read between the lines whenever they give any account of their observations on this or that animal, whether it be dog or elephant, ants or something much lower down in the scale—rotifers, for instance, or amœba in the ocean of a watch-glass. One sees what they really mean very well then, though they may not themselves be aware of it, but they are never in the least *convincing* when they air their automatic theory. Aphides, as may

have been gathered incidentally, feed wholly upon the juices or sap of plants. Active *ab ovo* (which means from the egg), “their occupation,” says Mr. Buckton, “is to grow as fast as possible,” and with a view to this end the rostrum or beak, with its enclosed sucking or pumping apparatus, is fully developed from the very commencement, “often, indeed, to such an abnormal extent that it forms an awkward appendage, trailing behind the body whilst walking.”^[43] The insect does not, however, walk much, but, settling itself down on the twig or stem where, perhaps, it first saw the light, pierces the bark with the instrument thus provided for it, and commences to suck up the sap into its mouth. This is not a process which can be indulged in with impunity to the plant, especially since Aphides reside in great societies upon the same one, and turn their attention to every part of it, not even excepting the roots. Troops of small Aphides, in fact, have sometimes been found in the pips of large codling apples. In consequence of this excessive drain upon their fluids, which is as though our own blood were to be sucked, plants thus invaded by Aphides become greatly weakened, and their young shoots and leaves have a distorted appearance in consequence. Others, either through this cause alone, or in consequence of some poison or acid injected by the Aphis, have gall-like excrescences produced upon them. These have a hollow interior, into which the Aphides penetrate, and there take up their residence. Such swellings thus become their houses, and therefore, since it is a great advantage to the Aphides to be sheltered in this way, it is possible that some special instinct through the exercise of which the tree is thus affected, may have been implanted in them by the action of natural selection.

Aphides are often spoken of by entomologists as if a very high degree of interest attached to them, and, no doubt, in many respects this is the case. As we have seen, they exhibit certain phenomena of *corporealism* (which *did* interest Professor Huxley) to a greater extent perhaps than any other creature, though of this I am not at all sure; but after all one soon gets over the wonder of that, especially since there is no realising it, and then it does not seem to raise a creature to a very high level of interest. Again, to quote authority, “there is a most curious alternation of broods in these insects, some forms being winged and with separate sexes, and others wingless or apterous and capable of producing their kind for an indefinite number of generations before a sexual brood is again developed. In fact, the anomalies

of members of this family are endless, and it would require volumes to epitomise even the comparatively little which has already been discovered with reference to their habits and transformations.”^[44] Still, for all this, it is difficult to look long at an Aphis, or a collection of Aphides upon a rose tree or any other of the plants they affect, without getting heartily tired of them, and for me, as perhaps for most people, the principal interest about these sluggish creatures lies in the relations which have become established, without any intelligent efforts on their own part, between them and ants—but it will be best to reserve the discussion of this subject for the following chapter.

CHAPTER XII

Ants and their honey-cows—A mutual benefit—Unity of motive—The end and the means—Two ways of getting honey—Insect cattle—Wasps as cow-milkers—A cow-keeping bee—Ant cow-sheds—Aphides in ants' nests—Children of light and darkness—Forethought extraordinary.

A drop of honey, or something like it, is the connecting bond between the ant and the Aphis. It is exuded by the latter through certain tubercles which are situated at the end of the abdomen, and is, of course, the product of the endless quantities of sap, which, so long as it lasts, these insects are for ever pumping up from the plant they inhabit, and swallowing. This honey, or honey-dew, to use the more special name bestowed on it, the ants want, but they are not content with drinking it whenever it issues from its manufacturers, in natural course. This is not sufficient, and they have learned to increase the flow of so valued a beverage by their own efforts—in other words, they milk the Aphides, which thus become their cows. To do this they tap them with their antennæ, softly and gently, on the sides of the abdomen—a quick little shower of touches. Under the influence of this probably pleasant sensation the Aphis becomes willing to part, and, raising the abdomen, “teems her refreshing dew” in a drop from the tip of it. This action of the ants cannot, in Europe, be successfully imitated, at least it has not been, and if an ant is not forthcoming the fluid is contained in the body of the Aphis until necessity compels its being ejected. Probably the ants, if delayed in their visits, are missed by the Aphides, as a cow misses her milker, and long before they do excrete, as the process is called, they would perhaps have done so had they felt able. The sensation no doubt of the ant’s antennæ on the abdomen has become, through usage, the almost necessary stimulus to the act produced by it.

The above remarks are best illustrated by a quotation from Darwin, which, in my opinion, should always be given in any general account of the relations of ants and Aphides. “I removed,” says Darwin, “all the ants from a group of about a dozen Aphides on a dock plant, and prevented their

attendance during several hours. After this interval I felt sure that the Aphides would want to excrete. I watched them for some time through a lens, but not one excreted. I then tickled and stroked them with a hair in the same manner, as well as I could, as the ants do with their antennæ; but not one excreted. Afterwards I allowed an ant to visit them, and it immediately seemed, by its eager way of running about, to be well aware what a rich flock it had discovered; it then began to play with its antennæ on the abdomen first of one Aphis and then of another; and each, as soon as it felt the antennæ, lifted up its abdomen and excreted a limpid drop of sweet juice, which was eagerly devoured by the ant. Even the quite young Aphides behaved in this manner, showing that the action was instinctive, and not the result of experience. It is certain, from the observations of Huber, that the Aphides show no dislike to the ants: if the latter be not present, they are at last compelled to eject their excretion. But, as the excretion is extremely viscid it is no doubt a convenience to the Aphides to have it removed; therefore, probably, they do not excrete solely for the good of the ants.”^[45]

If the reverse of this were the case, if the Aphides did excrete for the sole benefit of the ants, then, in Darwin’s own opinion, the case for natural selection would be broken down, and with this there would be some better ground of reason for those who would see in relations of this sort a set-off, as it were, against the never-ending bloodshed and rapine, accompanied with suffering in varied—often in an intense—degree, which is the very stuff out of which Nature has woven her mantle. But there can be no essential difference where the principle at work is precisely the same. So long as a creature does benefit itself, the way in which it does it, and the incidental effects of its doing so, are of no consequence; it is the motive power that the philosopher has to consider, and there is little comfort—if comfort be needed—in knowing that an animal, to do itself good, is doing good to some other, when one also knows that, governed by the same incentive, it would as cheerfully prey upon that other’s eye. As Hamlet says, in such a case “the readiness is all.”

As an illustration of this truth here is another picture of how ants procure honey from a weaker creature that may happen to have swallowed it, when it is not to be obtained by the soft methods of persuasion. “Once upon a time,” says Dr. Lincecum, “there dwelt in my yard a flourishing colony of

the very smallest species of black ant,” and having described how these Lilliputians found and ate some syrup belonging to the household, and were in consequence attacked by a larger and stronger species, he continues, “They”—that is the attacking party—“grabbed up the heavily burdened little fellows, doubled them, and biting open the abdomen, drew out the full sac, and seemed to swallow it. Then, casting the lacerated carcase aside, they furiously sprung upon another of the panic-stricken crowd and repeated the horrid operation.”^[46] Clearly, then, Nature, so long as she can attain her end, cares not by what means she attains it.

Independently of any feeling of comfort which the Aphides may experience in being milked by the ants, observation at once shows that they benefit largely, in a general way, by the attentions of the latter. It is not enough for the ants to milk their cows when they happen to meet them. They go very much farther than this, and cow-keeping is of as much importance with them as with us. Lucky the Aphis who has a guard of ants round it, fiery warriors prepared to defend their property against all foes. None need be feared now. Let but an Ichneumon buzz, and a dozen stalwarts start to the rescue.

“I dare thee but to breathe upon my love.”

“Fear not, sweet wench, they shall not touch thee, Kate,
I’ll buckler thee against a million.”

And so they do indeed, or against any reasonable number. But there is no combination amongst these banditti. Each comes but to eat his own Aphis, and no one thinks of helping a friend. All therefore are powerless before the organised attack of so fierce a bodyguard. Whilst the ants are with them the Aphides are quite safe, and they are often permanently guarded in this way. Other ants take even more elaborate precautions for the safety of their property, placing them in stalls, where they stand, by plastering earth, etc., about the plant on which they are feeding. Lastly, others still conduct them into their own nest, where they keep them, sometimes, in a chamber specially prepared for their reception, every necessary measure being taken for their proper nourishment, and, as one may say, comfort. Nay, the very eggs of the Aphides are tended by the ants, and hatch out in their own nurseries. Nor is it little for which they do all this, since, taking their size

into consideration, the yield of these ant-cows each day must be much greater than that of our own—at least, I should imagine so.

It is not all ants who do these things, nor do any do all of them, but where there are Aphides and also ants, it would seem to be the exception rather than the rule for the latter to neglect them.

But Aphides, though the principal ant-cows of Europe, are not the only ones even there, whilst elsewhere various other species that have this honey-excreting property become their substitutes. “In the tropics,” says Belt, in his much-observing work, “their place is taken in a great measure by species of *coccidæ* and genera of *Homoptera*, such as *Membracis* and its allies. My pineapples were greatly subject to the attacks of a small, soft-bodied, brown coccus, that was always guarded by a little black stinging ant (*Solenopsis*). This ant took great care of the scale-insects, and attacked savagely any one interfering with them, as I often found to my cost when trying to clear my pines by being stung severely by them. Not content with watching over their cattle, the ants brought up grains of damp earth, and built domed galleries over them, in which, under the vigilant guard of their savage little attendants, the scale insects must, I think, have been secure from the attacks of all enemies.”^[47] And again, the same naturalist tells us, “The pawpaw trees growing in my garden were infested by a small brown species of *Membracis*—one of the leaf-hoppers that laid its eggs in a cottony nest on the under part of the leaves. The hopper would stand covering the nest until the young were hatched. These were little soft-bodied, dark-coloured insects, looking like Aphides, but more robust, and with the hind segments turned up. From the end of these the little larvæ exuded drops of honey, and were assiduously attended by small ants belonging to two species of the genus *Pheidole*. A third ant—a species of *Hypoclinea*—which I have mentioned before as a cowardly species, whenever it found any young hoppers unattended, would relieve them of their honey, but would scamper away on the approach of any of the *Pheidole*. The latter do not sting, but they attack and bite the hand if the young hoppers are interfered with.”^[47] The latter “are, when young, so soft-bodied and sluggish in their movements, and there are so many enemies ready to prey upon them, that I imagine that in the tropics many species would be exterminated if it were not for the protection of the ants.”^[47]

But these leaf-hoppers had not only ants, but wasps to protect them, and there were constant skirmishes and bickerings on their account between the two. The wasp obtained the honey just in the same way as the ants—namely, by stroking the hoppers with its antennæ, and its possession of wings, more than its greater size, gave it a clear advantage over its rival. It did not grapple with the latter, even when there was only a single one to dispute its right, but, rising on the wing, and hovering about till a good opportunity presented itself, it would dart down suddenly on the impertinent little dwarf, and strike it from the leaf or stem. So quick was this action that Mr. Belt could not determine whether it was with the feet or the mandibles that the wasp delivered its blow, but he thinks it was with the former; that is to say, the front pair of them. But in spite of its superiority in single combat, the wasp could not prevail against the numbers of the ants. If, indeed, it was first in the field, there was not much difficulty, for though the leaf would before long be found by some or other of the ants, yet the first arrivals were only pioneers, and when once they were knocked off it had to be found again, only for a similar fate to befall the new discoverers. Often, however, the wasp would try to clear a leaf already in possession of the ants, and the way to which was known. But in this it was never successful, for though many fell, streams of others came rushing up, so that the wasp had no time to enjoy the fruits of its labours, but was obliged to keep constantly fighting, and before long was tired out. Though a giant amongst pigmies, and having wings—a sort of flying-dragon contending with an army of knights—yet it did not despise its small enemies, and evidently dreaded lest any of them should succeed in fastening on it. No doubt it knew—from inheritance, or experience, or both—that an ant clinging to a leg was a difficult thing to get rid of, and to avoid being placed in this position it never fought upon the ground—that is, the leaf—but only on the wing, in the manner described. Had it used its mandibles to bite with, the ants would have seized them, and some might have got on its body. Its sting played no part, doubtless because the small size and hard bodies of the ants would have rendered it ineffective.

We see from the above account that ants are not the only insects that can make discreet use of honey-yielding creatures, though they excel all others in this respect. Wasps have also learnt to milk, if not to stall, their kine, and to wasps, it would seem, must be added—which need not surprise us—at least one species of bee. A correspondent, whose name and date of

communication I cannot now remember, says, writing to *Nature*: “Fritz Muller has observed in Brazil a larva of a leaf-hopper—*Umbonia indicator*—which is used, like the Aphides by the ants, as milch cattle by a species of stingless bee—*Trigonia Cagafogo*. This bee is fond of oily matters, and feeds on carrion, old stinking cheese, and oil secreted by various plants. Although stingless, it possesses a very intense venom, which causes a most lively irritation of the skin.” I wish I could give the details of a fact so interesting, but have not had the opportunity of reading the original account from which this bald statement is taken. The ants, therefore, have rivals in this industry, and possibly such rivalry may exist to an extent hitherto unsuspected.

Though the protection of these insects by the ants architecturally—by moist earth placed round them, that is to say—is mentioned in the above account, it is not dwelt upon, and seems to play but a small part in the general drama. Some ants, however, rely solely on this method. Mr. Gaudie, writing in the *Victorian Naturalist*,^[48] gives the following account of one of these: “A small species of ant, commonly distributed in the Mallee, has a curious habit of keeping in close confinement a rather large mealy Aphis, which feeds on the stems of young eucalyptus gum trees. Round and over these Aphides the ants construct a domed covering of particles of bark, grass, etc., which serves the double purpose of imprisoning the aphides and excluding other ants. Some of these coverings appear to be entirely closed, whilst others have an opening left in the edges. This doorway is, however, constantly guarded by a pair of ants, which continually move about in the open space, and seem much impressed with the importance of the duty assigned to them. Each enclosure contains generally from three to a dozen Aphides, and about the same number of ants. Upon making a breach in some of these structures for the purpose of observation, I have noticed that many of the live stock were immediately seized by the ants and forcibly removed to a place of safety. The ant under notice is about a quarter of an inch in length, and is of a uniform dark, reddish-brown colour, and forms its ordinary habitation under logs, or in old rotten stumps, and sometimes in the ground. Several other species of ants are very assiduous in their attendance on the various aphides, *tettigonidæ*, and coccids, but the above is the only kind I have noticed that uses such extraordinary means to secure a monopoly of the much-prized ‘honey-dew.’”

For ants that keep and rear Aphides in their nests we need not go farther than our own little yellow one—*Lasius flavus*. They guard and look after the eggs of their protégés, which form little black shiny clusters, with the same care that they bestow on their own, and when they are hatched set about providing food for the young aphides. This, it would appear, does not consist of the roots of various plants penetrating the nest itself, for Sir John Lubbock found that the first business of the ants, after the young aphides had appeared, was to conduct or carry them out of the nest, evidently in order that they should find their natural food. None being at hand under these artificial conditions, and the plants required not being known, the poor aphides all died, and this happened again the following year. In the year succeeding to this, however, Sir John was more fortunate, and this is the account he gives of his interesting discovery: “The eggs commenced to hatch the first week in March. Near one of my nests of *Lasius flavus*, in which I had placed some of the eggs in question, was a glass containing living specimens of several species of plants commonly found on or around ants’ nests. To this some of the aphides were brought by the ants. Shortly afterwards I observed on a plant of daisy, in the axils of the leaves, some small aphides very much resembling those from my nest, though we had not actually traced them continuously. They seemed thriving, and remained stationary on the daisy. Moreover, whether they had sprung from the black eggs or not, the ants evidently valued them, for they built up a wall of earth round and over them. So things remained throughout the summer, but on the 9th of October I found that the aphides had laid some eggs exactly resembling those found in the ants’ nests; and on examining daisy-plants from outside I found on many of them similar aphides and more or less of the same eggs.”^[49]

As the young aphides had been brought by the ants to the daisies, and as they had subsequently laid their eggs there, it would certainly seem that the ants are accustomed to collect these eggs from without, and that the aphides do not lay them in the nest. When they are hatched the young aphides, as we have seen, are taken out to feed and lay, and these new eggs laid by them are, in their turn, brought in and tended by the ants. This, as Sir John Lubbock remarks, is a much more remarkable thing than if the aphides, living in the nest with the ants, simply laid their eggs there. In that case they would probably hatch out whether they were tended or not, and it could not

be long before the ants would become aware of their value. Here, however, we see this knowledge—how first obtained we know not—exhibited in a more striking manner, and also a great degree of foresight displayed, since as the eggs, except for accidents, would hatch where they were, it can only be with the idea of providing against these that the ants bring them into their nest. There they are safe from many dangers which threaten them above ground, and are not exposed to the rigours of winter or other climatic vicissitudes.

What are we to say of this act? I think there might be one or two things to say, but Sir John Lubbock says this: “Our ants may not, perhaps, lay up food for the winter; but they do more, for they keep, during six months, the eggs which will enable them to procure food during the following summer—a case of prudence unexampled in the animal kingdom.”^[49] There is a slight national note here which should, perhaps, make us suspicious. At least, I am always suspicious when a Frenchman praises anything French, an Englishman anything English—even ants or the climate—a Tierra del Fuegian anything in Tierra del Fuego, and so on. No doubt if prudence really induces the act, it is very great, but if we could imagine any other cause through which the habit might have begun, natural selection would have brought about the rest, since those ants which stored aphides’ eggs would have had more aphides, and consequently more honey-dew to nourish them than those which did not. Now the eggs might at first have been eaten, and so carried down, as provisions, or aphides, brought into the nest, might have laid before they got out again. However the act originated, it is probably a prudential one now, but if the growth of prudence has been aided by that of an inherited habit, having nothing to do with this, it is not quite so remarkable. But what, exactly, does “our” in the above passage mean? *Lasius flavus* is not confined to England—at least, I suppose not—and if other countries have a claim on its mental powers, our cue should rather be to undervalue them—at least, the note of national vanity should be held in check by the all as powerful one of national prejudice.

Besides these particular aphides, which leave the nest directly after leaving the egg, there are four or five other species which live in it altogether, and feed on the roots of various growing plants. Some nests which I had contained a few, but under natural conditions they are to be found, I believe, in abundance. Special chambers, it would seem, are given up to them, and

in Kirby's *Marvels of Ant Life* there is a picture of such a "subterranean cow-house." The question arises, where do these aphides lay their eggs, and, if in the nest, does not it largely discount the intelligence, or prudence, attributed to *Lasius flavus* in bringing the other ones into it? In that case, since the eggs of the various species probably resemble one another, any found outside would be brought in by the ants, just as their own larvæ or pupæ would be—or anything else which they value—nor need we ascribe greater foresight to the one act than to the other.

Ants, however, do more wonderful things in relation to aphides than this that Sir John Lubbock has recorded, and if that act is unexampled, as an exhibition of prudence, elsewhere in the animal kingdom, it is not, I think, in this particular branch of it. First it must be remarked that amongst the aphides we have what is called the "alternation of generations," that is to say a light-loving generation that feeds on the stems and leaves of upper earth, produces one that loves darkness, whose food is only the underground roots of the plants their parents lived on. This brood in its turn gives birth to another, which forthwith seeks the sun, and so the round goes on. There is this difference in the two broods, that the light-lovers, nevertheless, seek out darkness when the time comes to lay their eggs, whilst the children of darkness lay theirs in the caves where they have, all their lives, lived. That ants should be aware of all this, and habitually adapt their cow-keeping economy to circumstances so recondite, seems very extraordinary, but it would certainly appear to be the case. Thus when *Lasius fuliginosus* (another Franco-Britannico, etc., species) sees *Schizoneura venusta*—its particular Aphis—seated on a grass stem, and evidently wishing to lay her eggs, it knows at once what to do. Soft and large, with voluminous wings, such an insect is not well fitted for burrowing. She could hardly do it, in fact, so the ants, recognising this, begin to do it for her, and soon drive a tunnel leading down to the roots of the grass, through which they lead her, first, however, having clipped off her wings, which are now but a useless encumbrance.

Arrived at the terminus, the ants make a proper apartment for their cow Aphis, and here, in the midst of warm sympathisers, and with every comfort and luxury about her, she no longer hesitates to lay her eggs. In due time they hatch, producing wingless aphides, and from the brood thus raised the ants obtain their honey. When, however, this crawling generation have in

turn produced another winged one, the ants, far from seeking to detain these in a place where they would only die, again set to work to make tunnels, through which they conduct them successively to the upper air. One tunnel, one would think, would be sufficient for the purpose; but Lichtenstein, who observed these facts in the south of France,^[50] states that each Aphis, as it issues from the egg, has a separate one made for it by the ants. Having reached the surface, these cave-born Ariels spread their wings and fly away. Where they will settle no ant knows, but to the community that has freed them they are lost, probably—they and their eggs—for ever. Do the ants know this? If they do, they do not repine at it, for they know also that the perpetuation of the species, through which alone they can hope for fresh honey, has been provided for. This seems to me altogether to outdo the prudential feat of *Lasius flavus*, and since *Lasius fuliginosus* is distributed probably throughout the greater part of Europe, all the nations that do honour to that portion of the earth's surface are at equal liberty to think of it with patriotic complacency as “our ant.” For my part, I will only say this, that, whether it is or not, I think it deserves to be a Japanese ant—or that the Japanese, nowadays, much more deserve to have it than we do: that perhaps is the better way of putting it.

CHAPTER XIII

Cow caterpillars—The adventures of Theophrastus—Cave-born Ariels—Led to the sky—A strange attraction—Ant slaves and slave-holders—Slave-making raids—Feeble masters—An ant mystery—Effects of slavery—The decadent's reply.

AS we have seen, both in this chapter and a former one, aphides are not the only insects which yield the ants honey—or something honey-sweet—and are cherished by them in consequence. There are, for instance, the coccidæ, or scale insects, as mentioned by Belt; but whilst some of this family are milked in the same way as the aphides, to which, indeed, they bear a strong resemblance, others are simply eaten, as though they were sweets. To them might be said in warning, “Make yourself all honey, and the *ants* will swallow you,” but who can modify the nature of his own juices? Then there are the ants’-nest beetles, many of which have a sweet downiness which the ants enjoy licking, and are for this reason carried about with them when they move from one place to another. Not that they are always carried, for one little beetle, at any rate, whose name—it must be a diminutive—is *Formicoxenus nitidulus*, is accustomed to ride on the backs of its protectors, like the little cockroaches discovered by Professor Wheeler.

But perhaps the most interesting parallel to the aphides, as cows, is to be found in certain caterpillars, which are as soft and defenceless as they are, and represent a class of creatures which ants habitually prey upon. A certain family of butterflies, however, commonly known as the Blues, but entitled to the scholarly name of *Lycænidæ*, produce caterpillars which bear, upon the twelfth segment of the body, a certain honey-holding reservoir which, when full or nearly so, may be made to yield its contents through the same treatment which is so effectual in the case of the aphides. The ants tap or titillate the body of the caterpillar, near where the gland is situated, with their antennæ, and the caterpillars, charmed with such affability, overflow in return. This interesting fact has been observed in various parts of the Old World, and also in North America; but the most detailed account which we

have of it comes from India. In this case, as in all the others, the caterpillar is a quite small one, and feeds on the leaves of a certain tree, bearing both “an astringent yellow fruit” and the name of *Zizyphus jujuba*, though, by the way, jujubes are not, as a rule, astringent. The name of this little caterpillar—it would scorn to be behind the tree it feeds on in such a matter—is *Tarucus theophrastus*, so now we have something to fix it in the memory. The ant that patronises it is a large black one—its name I cannot give—and here, too, as in the case of the aphidean relations, we have, in the most noteworthy of the actions recorded, a very remarkable instance of what looks like foresight, and foresight, too, of a very large and general kind. In the first place, the ants make a nest at the foot of the trees in which the caterpillars reside, and here, during the period of their growth and nourishment, they avail themselves of their services. But when this period is over, and the caterpillars are about to change into chrysalids, then a strange scene takes place. All over the tree, ants are now to be seen running about in a state of the greatest excitement, and whenever they meet a caterpillar descending, or preparing to descend, the trunk, in order to burrow into the earth at its base, and there pass its pupal stage of existence, they conduct it down themselves and relieve it from the labour of digging, just in the same way as our English ants do with the aphides.

Still stranger is the scene which reveals itself if the earth at the base of the tree be removed, for then it is seen that chrysalids, and caterpillars that are about to turn into chrysalids, are clinging all round the trunk, whilst all amongst them are the ants, helping to place this one or that one in position. The band thus formed round the tree may be several inches broad, and it is always remarkably even, as though arranged on æsthetic principles. As the light shines in, the ants become agitated, and seizing hold of their property—for in this light they consider the caterpillars—begin to rebury them, so that in time, if the annoyance continues, they will form a fresh circle of bodies lower down the tree. Here, then, is an ants’ nest, described as temporary by Mrs. Wyllie, from whose interesting account^[51] the above facts are taken, full of butterfly chrysalids, and in about a week it becomes full of butterflies themselves, and amidst the rough, black bodies of hosts of earth-working Calibans, colours born of the rainbow gleam and flash from the fairy wings of delicate insect Ariels. Each one of these was helped from its cradle, thus strangely situated, by a little group of these gnomes, who

then assisted it to unfold its wings, and guided its uncertain steps. Later, when strength has come to it, and something—it knows not what—like an upward desire, these same gnomes will lead it to the portals of their gloomy Hades, where it will spread its wings and fly to meet the light. In so strange a way, led by such uncouth guides, does Ariel find the sky. Yet, as though the place of their new birth—gloomy though it be and opposed to their light-loving natures—had yet some nameless attraction for them, crowds of these butterflies may be seen, for some time after their exodus, hovering over the nest, before they leave it for ever to dwell in the courts of the sun.

Just as in the case of the aphides released by *Lasius fuliginosus*, these ants will never see their butterflies again, nor will they gain any after advantage that can with certainty be traced to the particular individuals thus set free. But they gain in such a manner as, if the reflection really occurred to them, would make ants not much below men. The process of reasoning would be this: “Though we may very likely not get any caterpillars from the eggs which these butterflies will lay, yet we ought not to kill them, because then there would be so many butterflies less in the world, to lay eggs, and if we did this every year, and other ants too, caterpillars, as well as butterflies, would become scarce, and at last we should not get enough.” For myself I doubt if ants really do reason like this, but by what steps this habit of releasing butterflies so as to ensure the perpetuation of the species has come about, I don’t quite see. In the case of the aphides, perhaps it has been through actual observation of their habits, and here, too, this hypothesis may not be excluded, since the butterflies might well be seen laying their eggs, nor is it unlikely that these are watched, and the issuing caterpillars tended from the beginning. For the purposes of the ants, indeed, all aphides, and every theophrastus, would be the same, and they might very well think that those which they found laying, or about to lay their eggs, were the very ones previously liberated by them from the nest. Thus the difficulty involved in supposing that they must reason in a general and not merely in a particular way does not really exist.

Some of these black caterpillar-tending ants of India are not always so lucky as to secure stock. They may live far from a *jujube* tree, and so never meet the right species; but if ever they do, even though it be in the most unexpected manner, they are not taken by surprise, notwithstanding that other caterpillars are habitually devoured by them. Mrs. Wyllie proved this

by an experiment. "I took Theophrastus," she tells us, "from a tree, and introduced him on the pathway of another company of the same species of ant, which lived in our verandah, but kept no farm, and it was odd to see the ants come tumbling out headlong to fight the intruder, and the sudden way in which they cooled down on investigation of the foe. None attempted to harm him, and he was politely escorted across the boundary, the ants running alongside and feeling him all over with their antennæ. This must have been instinctive, as they could have had no former knowledge of him as a 'milk-giver.'" Mrs. Wyllie adds that "the dead chrysalids in an ants' nest are carefully removed and thrown away outside; the ants also distinguish between the dead and the living."

Anyone observing or reading about ants might exclaim, on finding that they utilised the natural product of other insects and kept them in captivity in order to do so, "Where will this end? May they not, then, also keep slaves?" And in very fact, as all the world now knows, we do find what is called slavery amongst ants, though to me it hardly seems the right word, since there is perfect willingness on the part of the slave, and no power of punishment lies with the master. There is equality, moreover, since this is not a matter of the kind of things which one class of a community does and another does not do, but of the spirit in which each does them. With the ants we have the Japanese spirit—or rather the Japanese seem getting nearer to the ants—and so there is real equality. However, the first act which makes these creatures slaves—for I will use terms as I find them—is one of violent and deadly hostility, and through it they are, of set purpose, taken possession of and carried off to the nest. At that period, however, they are yet in the cradle, have yet to be born into their last and most perfect state of life. From the moment they are so born they grow up as a part, and indeed the most important part, of the body politic, and of such pleasure and consideration as obtains in ant-life they have their full share.

From the above it may be gathered that these ant-slaves are ants themselves, and this, indeed, is the case. One species of ant raids the nest of another, overpowers the able-bodied inhabitants, slays or incapacitates a certain number, and carries away with it to its own nest as many of the helpless pupæ as it is able to. For a great many years—thousands probably—these combats and carrying off of spoil had been observed, but it had always been imagined that the pupæ—or ants' eggs, as they are commonly called—were

taken as provisions, merely to be stored in the nest of the victor, and there eaten at leisure. The discovery of the real truth was an era in the study of ant-history, and it was made by a Frenchman—Pierre Huber—a man of whom Darwin says that he was a “better observer even than his celebrated father,” for Pierre was the son of François Huber, the blind man, who yet found out all about bees. I hardly see how he can have been *better* myself; but the son was not blind, and, of course, eyesight is an advantage in observation.

The particular species of ant concerning which this discovery was made is *Formica rufescens*, or *Polyergus rufescens*—the reader may take his choice—and Darwin, who impresses the facts of ant-slavery upon the mind better than a dozen books specially devoted to ants or insects, says of it: “This ant is absolutely dependent on its slaves, and without their aid the species would certainly become extinct in a single year. The males and fertile females do no work of any kind, and the workers, or sterile females, though most energetic and courageous in capturing slaves, do no other work. They are incapable of making their own nests or of feeding their own larvæ. When the old nest is found inconvenient, and they have to migrate, it is the slaves which determine the migration, and actually carry their masters (one might just as well call *them* *their* slaves) in their jaws. So utterly helpless are the masters, that when Huber shut up thirty of them without a slave, but with plenty of the food which they like best, and with their own larvæ and pupæ to stimulate them to work, they did nothing; they could not even feed themselves, and many perished of hunger. Huber then introduced a single slave, and she instantly set to work, fed and saved the survivors, made some cells, and tended the larvæ, and put all to rights. What can be more extraordinary than these well-ascertained facts?”^[52] The slave-ant in this case is *Formica fusca*, and it is also held in bondage by another species of slave-maker, viz. *Formica sanguinea*—or the Blood-red Ant—as was likewise a discovery of Pierre Huber. This last species is found in the south of England, and its slave-making habits have been observed by Darwin, who opened fourteen nests and found a few slaves in all of them. “The slaves,” he tells us, “are black, and not above half the size of their red masters, so that the contrast in their appearance is great.”^[52] The black ants were not often seen by Darwin to leave the nest, and others who have observed their habits in England have considered them as “strictly

household slaves.”^[52] Huber, however, whose observations were carried on in Switzerland, says that “their chief office is to search for aphides,”^[52] and this would take them far afield. In Switzerland, however, slaves seem to be more numerous in the nests of the Blood-red Ant, and Darwin attributes the difference in their habits to this account. Huber also tells us that the Swiss slaves “habitually work with their masters in making the nest, and they alone open and close the doors in the morning and evening.”^[52] This is done, I suppose, by placing pellets of earth in the mouth of the entrance-tunnel and removing them again; but there is one species of ant which would have only to place or remove itself, for this purpose, since its large head, by being wedged into the passage, stops it up, and thus fulfils the office of a front door. The ant that does this must be one belonging to a certain caste of workers having very large heads, for the heads of the other ones would not be large enough. The nest of this species is made in decaying wood, and there is always some worker who thus uses his large head as a stopper, removing it when a fellow-townsmen wishes to enter the nest, but presenting its smooth, impenetrable surface, guarded with jaws, to all unauthorised intruders. It is Forel, one of the best ant-observers of to-day, who tells us this, and the ant apparently is *Lasius fuliginosus*, which is a British species, and, according to an account which I have already referred to, does not seem to be always a wood-borer. *Formica sanguinea*, however, does things—or has things done for it—after a more ordinary fashion.

Darwin was the witness of a slave-raid on the part of *F. sanguinea* which was not, in this instance, very successful. He says: “They approached, and were vigorously repulsed by an independent community of the slave-species (*F. fusca*), sometimes as many as three of these ants clinging to the legs of the slave-making *sanguinea*. The latter ruthlessly killed their small opponents, and carried their dead bodies as food to their nest, twenty-nine yards distant; but they were prevented from getting any pupæ to rear as slaves. I then dug up a small parcel of the pupæ of *Formica fusca* from another nest, and put them down on a bare spot near the place of combat; they were eagerly seized and carried off by the tyrants, who perhaps fancied that, after all, they had been victorious in their late combat.”^[52] In his work, *Ants and their Ways*,^[53] the Rev. Farren White describes a similar raid which was—or rather which had been, for it was nearly over when he arrived on the scene—wholly successful. Here, however, the oppressed species seems

to have made a very poor resistance, though very likely it had been more vigorous in the earlier stages of the raid. “I watched a *fusca*,” says Mr. White, “carrying off a pupa from behind the entrance whence the *sanguineæ* were issuing forth. Immediately it saw one of the enemy approaching, it dropped its charge and left it to its fate. The *sanguinea* then gave it a push, and drove it off in double-quick time”; and, again, “I noticed a *sanguinea* coming up out of the nest with a pupa, and a *fusca*, observing it, went up a fern-frond with the utmost expedition.”^[53] Other observations of a similar nature were made, and the conclusion arrived at by Mr. White is “that between the *fuscæ* and the *sanguineæ* there is a well-defined and clearly pronounced antagonism. In presence of the *sanguineæ* the *fuscæ* were terror-stricken. In fact the depredators had it all their own way, and were able in this instance, at least, to carry out their marvellous instincts without destroying a single life.”^[53] It will be seen how ill this accords with the account given by Darwin. My own way of accounting for the discrepancy is that, in the first instance, the little *fuscæ* were flushed with success, and, in the second, demoralised through defeat. The same effects would follow the same causes in all but the most splendid human armies.

The raids made by the first-mentioned species, *Polyergus rufescens*, or, as Huber calls them, the Amazon ants, are of an even more determined description, for none are braver, or perhaps so brave. If one of these should find herself alone and in the midst of enemies, she makes no effort to escape, as many, though not all, other species would, but fights on to the end, making constant agile leaps to this side or that, at every one of which she transfixes an enemy, and dies at last biting hard. To fight, indeed, is the whole end, aim, and business of life for an Amazon, and we have already seen how they do no work, and are washed, fed, and carried by their servants. It is not quite true, however, that they cannot feed themselves, as Pierre Huber thought, and had good reason to think, for a well-known living observer—Herr Wasmann—has discovered that their mandibles are so constructed as to enable them “to absorb nourishment from eggs or pupæ.” Possibly the mandibles are hollow, and communicate thus with the mouth, as is the case with some other insects, but I have not Wasmann’s account at hand, and his exponent says only this. Wasmann tells us also that these Amazons will “absorb nourishment,” however they do it, even from the eggs of their own species. They cannot, however, feed on liquid food, and

as they had no other when shut up without servants, that is why they died, or would have died, had these not been brought them in time. So too, though their slaves wash and brush them, yet they are always brushing themselves and attending generally to their own toilette, and this they do even amidst

“All the currents of a heady fight,”

so that Wasmann has compared them to the Spartans combing their long hair before the battle of Thermopylæ, though we are not told that they combed it after the fight had begun.

Still, it seems plain that the habit of keeping slaves has exercised a degrading influence on these ants, and this tendency is much more markedly apparent in several other species. One of these with a really dreadful name, *Strongylognathus*—we might call them Strong Ants, but they seem to be weak ones—is described by Forel as “*une triste caricature*” of the Amazons, and the extraordinary thing is that, though themselves feeble and enervated, they manage to make, or by some means obtain as slaves, the workers of a much more robust species—*Tetramorium*, to wit; workers, by the way, are the only class of ants ever enslaved. These weak ants fight in the same way as the Amazons themselves; but, though spirited enough, they are so much inferior in bodily vigour to the *Tetramoriums*, “a courageous species living in large communities, that in a battle between the two, artificially instigated by Forel, almost all the slave-holders were killed, without being able to avenge their deaths even on a single one of those whom they aspired to rule.” Yet they won the day, or rather the already enslaved *Tetramoriums*, who marched to do battle for them, won it for the few survivors in their ranks. From this we can see how, when these decadents once have slaves, they may get more. The difficulty is how they are obtained in the first instance—when a nest is first founded by a queen of the slave-making species, for example. It might be supposed, finding two kinds of ants living together, one weaker and much more helpless than the other, that the former lived a parasitic existence in the nest of the latter, and was not a slave-owner at all; but this theory is disproved by the fact that no males or females of the *Tetramoriums* are ever to be found, showing that it is not they but the others who are the true founders of the nest.

A still more extraordinary instance of a slave-holding species of ant than the one just mentioned is *Anergates atratulus*, for in this there are no workers at all, only kings and queens, who are waited on, and their eggs and larvæ fed and tended, by the slave species—*Tetramorium*, in this case also—just as though these latter were their true-born subjects. Here too the slave species is only represented by workers. These male and female *Anergates*—a worker of the species has never been known—are both few in number and weak in themselves. When a pair of them (or a fertilised queen) go off to found a new colony, how do they, or how do their few weak descendants, impress a strong fierce species into their service, by whom the nest is built, and every other service performed? The question remains unanswered. Nobody knows. Several theories have been advanced, one by Sir John Lubbock, who supposes that the king and queen of *Anergates* assassinate the queen of *Tetramorium* and reign in her stead,^[54] and another, more recently, by Wasmann, whose idea is that fertile queens of *Anergates* are sometimes adopted by a colony of *Tetramoriums* who have lost their own queen. This last is the newest suggestion, and is considered just at present, perhaps for that reason, the most probable. To me Sir John Lubbock's view seems likelier to be correct, since it is more usual in nature for the weak to prey, as parasites, upon the strong, than for the strong to seek assistance of the weak. True, I can form no idea as to how the assassination of the rightful queen takes place, but Nature is full of resources, and will do much to promote a really worthy end.

I will conclude this chapter by quoting some remarks of Sir John Lubbock as to the ill effects which the institution of slavery exercises, with ants as with men, upon the character of the slave-holder. "These four genera," he says, "offer us every gradation from lawless violence to contemptible parasitism. *Formica sanguinea*, which may be assumed to have comparatively recently taken to slave-making, has not, as yet, been materially affected. *Polyergus*, on the contrary, already illustrates the lowering tendency of slavery. They have lost their knowledge of art and their natural affection for their young! They are, however, bold and powerful marauders. In *Strongylognathus* the enervating influence of slavery has gone further, and told even on the bodily strength. They are no longer able to capture their slaves in fair and open warfare. Still, they retain a semblance of authority, and, when roused, will fight bravely, though in

vain. In *Anergates*, finally, we come to the last scene of this sad history. We may safely conclude that in distant times their ancestors lived, as so many ants do now, partly by hunting, partly on honey; that by degrees they became bold marauders, and gradually took to keeping slaves; that for a time they maintained their strength and agility, though losing, by degrees, their real independence, their arts, and even many of their instincts; that gradually even their bodily force dwindled away under the enervating influence to which they had subjected themselves, until they sank to their present degraded condition—weak in body and mind, few in numbers, and, apparently, nearly extinct, the miserable representatives of far superior ancestors, maintaining a precarious existence as contemptible parasites of their former slaves.”^[55]

Since, however, in all these cases the masters are still truly served by their slaves, who make them comfortable, and have no more sense of their degradation than they themselves have, an answer might be made to these moralisings. However various the masks behind which true motives lie hid, happiness, diversely conceived of, is the one end and aim of all. Does it, then, really much matter by what means it is attained? Till we can show that these slave-holding ants have become less and less happy, we are only tilting at shadows, and an *Anergates* might very well say, in regard to the above view, “Tut, prut, drop your heroics. I am very comfortable; these strong fellows work for me. I like not working, and what I am I wish to be.”

CHAPTER XIV

Ant partnerships—How some ants feed—Persuasive methods—An *imperium in imperio*—Amusement by instinct—Begging the question—Nest within nest—Ant errors v. human perfection—Distorted arguments—How partnerships begin—Housing an enemy—Ant ogres.

THE relation of slave and slave-master—to use the received terminology—is not the only one of a social and friendly nature in which ants of different species stand towards one another; for as will have been gathered in the previous chapter, slavery amongst ants is a quite friendly institution, conducted, in fact, upon the “liberty-equality-fraternity” principle. Some species of ants, however, inhabit the nests of other species, or build their own amidst theirs in such a way as almost to make them one, and thus they live as perpetual guests, not only without paying for such accommodation by rendering their hosts any services, but often forcing these latter to be of service to them in other ways also. Thus, a small species of Texan ant whose first or Christian name is *Leptothorax*, but whose surname has not yet been fixed upon, lives on these terms in the nests of a larger one, the celebrated *Myrmica brevinodis*. Whether Professor Wheeler was the discoverer of the little ant I am not quite sure, but he was the first, I think, to observe its relations with the big one and those of the big one with it, and his account of them is excessively interesting. “A small dish,” he says, “containing a syrup of sugar and water was placed near the nest (an artificial one under close observation). This was soon found by two of the *Myrmica* workers, which at once gorged themselves with the liquid and returned into the nest.”^[56] Soon afterwards a *Leptothorax* worker entered it also, and having run or tracked down one of the two honey-gorged creatures, forthwith got up on to its back, and, seated there, began to lick its head, an attention which it supplemented with a soft, persuasive titillation with its antennæ, whilst at the same time communicating a motion to its abdomen, which Professor Wheeler is so convinced must have been accompanied with certain sounds—known to the learned as stridulations—

that he does not hesitate to affirm that it was thrown “into stridulatory oscillation.” Nor was the *Myrmica* deaf to such an appeal. It slackened its pace, hesitated, then paused, and as though unable longer to resist the influence, folded its antennæ and appeared to give itself up to the full pleasure of the thing. The tempter, now, still making soft play with the antennæ, lowered its own head, and began to lick the *Myrmica* first on one cheek and then the other, including also the mandibles and parts adjoining. Thus fostered, a dewy moisture, drawn evidently from the reservoir of lately swallowed nectar, began to glisten on the lips of the large ant, and, increasing rapidly to a droplet, was re-imbibed by the expectant little one. “The latter,” says Professor Wheeler, “then dismounted, ran to another *Myrmica*, climbed on its back, and repeated the very same performance. Again it took toll, and passed on to still another *Myrmica*.”^[56] Up to the present the attention of Professor Wheeler had been concentrated on the doings of this one individual, but now, turning his attention to other parts of the nest, he “observed that nearly all the *Leptothorax* workers were similarly employed. In one corner a number of *Myrmica* workers had formed a circle about a few of their small larvæ, which they were cleansing and feeding. A *Leptothorax* soon found its way to this cluster, and stepped from the back of one ant to that of another, lavishing a shampoo on each in turn, and apparently filling its crop with the liquid contributions thus solicited.”

The above method of obtaining food appears to be peculiar to these ant parasites. Beetles, for example, solicit it either by taps or touches with the antennæ—which is a similar one indeed, but does not go so far nor involve a ride—or else by stroking the face of their host with their fore-feet. Other species of ants, when soliciting food from one another or demanding it from their slaves, employ a more or less similar method, whilst the *Lepismid* that we have before spoken of is a thief pure and simple. Licking seems to be the personal discovery of *Leptothorax*, and being licked the peculiar privilege of *Myrmica brevinodis*. That it is a valued one is clear, but the price asked for it is not always forthcoming, possibly because there is not always anything to forthcome. On such barren occasions *Leptothorax* makes the best of a bad job, and dismounting from its first love, runs about looking for another.

Sometimes, after having licked the head and face of its patron, the poor petitioner turns round and proceeds to do the same by its abdomen. This, perhaps, is a last effort of persuasion, but Professor Wheeler rather supposes the surface of *Myrmica*'s body to be "covered with some agreeable secretion." Queen *Myrmicas*, however, seem to be very rarely treated to any sort of licking, and males apparently never. The reason of this, probably, is that both queens and males are themselves accustomed to receive their food from the workers by a similar process of regurgitation, and are probably therefore not in the habit of regurgitating it. They are therefore neglected by the little parasites, who console themselves by being all the more insistent with those who have something to give. These—that is to say, the workers—are waylaid whenever they enter the nest, as having presumably found something to eat outside it, and, in order to be on the spot, at once their importunate lickers, who seem to live in a perpetual state of crying, "Give! give!" keep in the more or less immediate proximity of the entrance, or entrances, should there be more than one. Professor Wheeler, indeed, doubts if the *Leptos* ever feed themselves in the ordinary way, but inasmuch as they were on one occasion seen by him to do so, such doubt appears to me to be uncalled for.

These little ants make, in regard to the big ones within whose nest they live, a sort of *imperium in imperio*. In a small chamber surrounded by the large galleries of the *Myrmicas*, and communicating with these by a passage too narrow for the latter to pass through, lives the queen with a small number of workers; eight of them in the nest observed by Professor Wheeler, together with a few larvæ, almost filling the cavity. They appear to be on affectionate terms with one another, the workers feeding their queen in the most assiduous manner, and she often playing with them like a cat with her kittens, throwing them on to their backs, and then "hugging and kissing them" (as Professor Wheeler describes it) *con amore*. Not that the Professor himself takes this view of it, for after hesitating whether to ascribe such behaviour to maternal affection, "the play instinct," or hunger, he decides for the latter—on what grounds, since there was a continual passage of viands from one ant to another, the queen especially being "assiduously fed," I am unable to see. What, too, is "the play instinct," except a mere term made use of in order to suggest the idea of automatism in regard to an act which hardly seems to admit of such an interpretation? Instincts

represent imperious necessities which, if not attended to, the species must fail or perish. Such, at any rate, are the grounds on which they must be supposed to have been originally built up. But what creature has had to play in order to survive? Not ants, surely, who work so hard that they cannot stand in need of more exercise than their daily life affords them. Nor, it would seem, is such an instinct developed amongst other insects, which again seems to show that it cannot be of any great importance. When, therefore, we find that ants, the most wonderful of all insects, do play, this strongly suggests their possession of an intelligence analogous to that of the higher animals. Instinct, however, is largely independent of intelligence, such as we understand it, and therefore, to allude to “the play instinct” in ants before the instinctive character of the act has been made out, is to prejudge the question whether ants are automatic or reasoning beings.

The smallness of the passages leading from the interior chamber of *Leptothorax* to the broad galleries of the *Myrmicas* suggests that the latter were not intended to pass through them; but we cannot really draw this inference, since an ant in tunnelling would allow for the size of its own body, but not for that of another species. Certain it is that the big ants constantly force their way through the narrow passages, thus partly breaking down the wall, and that they are then received by the little ones in a quite friendly manner, and persuaded to part with some of their interior stores. Still, when this has been effected, their friends seem mildly desirous that they should go, and, as soon as they have gone, set to work to repair the breaches made by their entrance. No sooner has this been done, however, than they are broken down again, and so it may continue, apparently, for an indefinite period, at any rate in nests constructed for observational purposes, and where the conditions are, therefore, more or less artificial. Whether it is so to anything like the same extent under nature may well be doubted, for that any creature should live in a state of never-ending useless labour does not seem likely; and, moreover, unless the one ant could have made itself comfortable within the nest of the other, why should it have become established there at all? But whatever it may be outside the study, this is Professor Wheeler’s account of what fell under his observation: “At one p.m.,” he tells us, “the *Myrmica* workers discovered the hiding-place of their little companions, and two of them, in single file, shouldered their way through the narrow passage, enlarging it as they proceeded. As soon as the

head of the first *Myrmica* appeared in the chamber, the *Leptothoraxes* which had been attending to their morning toilet and that of their larvæ, and to the careful arrangement of their eggs, turned to meet the intruders.”

“Now dreadful deeds
Might have ensued,”

and for such an upshot, indeed, upon the first occasion, Professor Wheeler was prepared. “For an instant,” he says, “I fully expected to see a fierce battle, but I had misjudged the *Leptothorax* character. To my surprise the *Myrmicas* on entering were received with a profusion of shampooing, and, though sadly crowding the occupants of the little chamber, they let themselves down comfortably, and appeared to experience all the sensuous satisfaction of a couple of *roués* who have dropped into a Turkish bath for the night. Yet the little *Leptos*, though behaving in this friendly manner” (their conduct indeed was not more disinterested than upon other occasions), “seemed to have some dim desire to remove the *Myrmicas* from their nest, for from time to time one was seen to pull with her mandibles at the fore leg or antenna of one of the intruders, as if to remind her that there are limits to polite hospitality.” Professor Wheeler adds that “this was the only act even approaching hostility witnessed between the two species. The *Myrmicas* never showed the slightest irritation towards the *Leptos*, never seized them in their mandibles or even menaced them. They seemed rather to look upon the little creatures with gentle benevolence, much as human adults regard little children. They never passed their little guests without the antennal greeting, and the *Leptos* shampooed their hosts with comical zeal.”^[56] The continued breaking down and repair of the dividing wall is then described, with the conclusion that “in their natural environment the *Leptothoraxes* would not be cramped for space, and would probably dig their cell where they would not so frequently be disturbed by their inquisitive hosts.”

As regards the possible effects upon the *Myrmicas* of having thus frequently to render up food swallowed for their own nourishment, it must be remembered that amongst most ants this is a thing of custom; and, again, it seems probable that there would be an internal sense on the part of the regurgitating individual as to concessions of this nature having gone as far as it was healthy that they should go. As we have seen, they are not

infrequently refused. Professor Wheeler, however, came to the conclusion that *Myrmica* colonies suffer very considerably from this cause, and on this he makes the following comment: “If I have correctly estimated the influences which may tend to diminish the fecundity and prosperity of the *Myrmicas*, we have in this double nest another striking demonstration of the complete absence in ants of any faculty of reason. For if the *Myrmicas* possessed a glimmer of this faculty they could easily annihilate the gluttonous little nest-mates that are for ever roaming about their galleries like so many animated stomach-pumps.”^[56] Yes, truly a *most* “striking demonstration,” seeing that we human ants can annihilate, all in a moment, any evil that has insensibly gained a footing amongst us, and with which we have been familiarised from birth. Custom, growing gradually from unnoticed beginnings, plays no part at all amongst us—never affects our views in the very slightest degree. In Europe we hang up all the brewers and distillers; whilst mobs of infuriated Chinamen rend in pieces the vendors of opium and crushers of their women’s feet. There is no such thing in *human* nature as tolerating an evil for the pleasure that lives in it; no *man* ruins his health, and sinks into an early grave, through being a slave to sensual pleasures. Nor can what is manifestly wrong seem right to *us*; there is no pernicious, obstinate, wilful shutting of *our* eyes. What a contrast does all this present with such a state of affairs as we are here considering! And how plain it is that there can be no reasoning power in the ant, since reason and right conduct are synonymous with man!

What is the origin of these strange co-partnerships—for there are others—which we find existing between ants of two different species living in the same or in one double nest? As we know, the different species of ants are commonly very hostile to one another; and for any to enter the nest of some other one is to court destruction, if they be not the stronger party. Nay, they dare not even enter a strange nest of their own species. It seems probable, therefore—this, at least, is my own view of it—that such friendly cohabitation has come about through the channel, not of peace, but of war—through successful encroachments which, from being unavoidable, have come gradually to be less and less resented, till the two parties, mingling freely, have learnt to live on other terms. Now there are ants which live, like ogres, in the nests of other species, preying upon their eggs and young. Such a one is *Solenopsis fugax*—to whom we will come presently—but it is

perhaps even more interesting to find in species between whom relations of a similar nature to those which we have been considering exist, occasional slight traces of a mutual hostility. As between the *Leptos* and *Myrmicas* indeed this has only as yet been noticed, very faintly, on the part of the former; but in regard to another pair who live together—our great wood-ant, namely, *Formica rufa* and tiny little *Formicoxenus nitidulus*—Professor Wheeler remarks: “On one occasion in one of my artificial nests, in which the ants had previously lived on good terms with one another, I saw a *Formica* touching a *Formicoxenus* with her antennæ and menacing her with her mandibles, but she departed without even attempting to seize her. In the same nest I found a *Formicoxenus* which had seized the leg of a *Formica* in its mandibles and had died in this position.”^[57] Other observers, too, have from time to time—but only very occasionally—noticed facts of the same sort. Through such exceptional slight indications we may perhaps see, “as in a glass darkly,” what things were at the beginning.

Let us now look at the beginning. *Solenopsis fugax*, Sir John Lubbock tells us, “makes its chambers and galleries in the walls of the nests of larger species, and is the bitter enemy of its hosts. The latter cannot get at them because they are too large to enter the galleries. The little *Solenopses*, therefore, are quite safe, and, as it appears, make incursions into the nurseries of the larger ant and carry off the larvæ as food. It is as if we had small dwarfs about eighteen inches to two feet long harbouring in the walls of our houses and every now and then carrying off some of our children into their horrid dens.”^[58] This is the general proposition. Monsieur Janet can add a few particulars. “The *Solenopsis*,” he says, “may establish itself near almost any other ants of our country, and is found especially with....” Here follows a list all in Latin, but our common Wood-ant—the large one that makes those great heaps of pine-needles—and the Amazon, or slave-making ant, are contained in it.^[59] “The *Solenopsis* nest,” continues M. Janet, “may partially surround that of its neighbour’s, or it may even be partly excavated in the masses of earth which separate the galleries of the latter. In each case—and probably, too, when, as is frequent, there is not such close contiguity—fine connecting galleries enable the *Solenopsis* to make incursions into the nests of their neighbours, where, as we shall see, they find an abundance of food. The actual nest consists of a number of small circular chambers about 8-20 mm. in diameter and only 6-8 mm. in height. Most of these

chambers are separated from one another by several centimetres, and are connected by slender galleries, often less than two millimetres in diameter, entering the chambers at their walls, ceilings, or floors, which latter are remarkably clean, smooth, and hard.”^[59]

The food of which these horrid little ants find such an abundance is, of course, the cocoons and larvæ of their unfortunate neighbours, and M. Janet gives the following account of the way in which they dispose of them: “From ten to thirty of them,” he says, “climb up on to a cocoon and cover it with little perforations which, finally, becoming confluent, enable them to reach its contents. If it contains a pupa, the legs and antennæ fall an easy prey to the mandibles of the *Solenopses*. In this case the victim is cut into, sucked, and torn into very small pieces, which the ants hasten to carry away into the interior of the nest. The operation is much more difficult in the case of a larva which has just spun its cocoon. Such a one I have seen the *Solenopses* drag into the interior of the nest and keep working at for twenty-four hours. At the expiration of this period the larva began to look flaccid (as may be believed), and was covered with little black dots which were, sometimes, double, corresponding with the little wounds made by the mandibles of its assassins. Numbers of the latter were busy lapping up the liquid which exuded from the wounds, but it was not until thirty-six hours had elapsed that the larva was entirely devoured.”^[59] This is certainly not a pleasant picture, yet, if our surmise is correct, the remote descendants of these murderous *Solenopses* may become as harmless and as pretty in their ways as the little *Leptos*, a reflection which goes far to discount any uncomfortable feelings we might otherwise have been inclined to have in regard to the general plan or scheme of things. Thus, in nature, though occasionally a slight shadow may seem to rest upon the landscape, the next moment the very memory of it is lost in a blaze of sunlight glory.

Forel believes that when a *Solenopsis*, and one of the larger species of ants that it plagues, meet, the latter are unable to see it on account of its small size, so that, practically, it is invisible. This seems a strange doctrine, since the same ants can see smaller things; and yet, from their behaviour under such circumstances, M. Janet is inclined to think so too. It can hardly be that they shun combat, though the *Solenopses*, in spite of their small size, are able, even here, by virtue of their numbers, and being armed with stings, to meet their victims, as one may almost call them, upon equal terms. M.

Janet, indeed, once saw so strong and warlike a species as the slave-making *Formica sanguinea* killed by some half-dozen *Solenopses*, but he adds that on such occasions a considerable number of the latter were, generally, killed also. This, however, is not sufficient to abate the evil, so perhaps the molested species, finding that fighting is of no use, accustomed to see *Solenopses* from their birth, recognising, too, as a part of their own atmosphere, the distinctive smell which they, no doubt, possess, accept them like some disagreeable part of their lives, and try to make the best of it.

CHAPTER XV

Ant wonders—Leaves cut for mushroom-growing—How ants plant mushrooms—A nest in a mushroom-bed—“Psychic plasticity”—Two opinions—Ant stupidity—Unfair comparisons—The ant and the servant-maid—Mushroom-growing beetles—Choked by ambrosia—Intelligent uselessness—Automatic phraseology—A curious insect.

ANTS, as everybody knows, have a special faculty for doing extraordinary things. Only a few of these have been mentioned in the last and preceding chapters, and only a few more can be touched upon in this. To do the subject anything like justice, a whole large book would be required, not a few chapters merely of a quite small one. What ants do, indeed, reminds me of the refrain, constantly repeated, of a certain old ballad lately brought to my notice, viz.—

“Wonders, notable wonders! never the like was heard.”

For instance, they grow mushrooms (rice, or some cereals, they used to grow and reap, but lately they have not been allowed to); they use their own larvæ as an implement to sew or stick things together with, thus making little shuttles of them; they make bridges of their own bodies, by which they pass over rivers—even wide ones, it would seem, at least for them—which otherwise would be impassable; they allow themselves to be made into honey-pots and kept full for the good of the general community, who take a little of them when they want it; they have cemeteries, and would appear even to feel something like awe or respect in the presence of their own dead; they cause certain plants to grow and come to maturity, which would otherwise die, in order to make a house in them, and so on and so on, many other wonders equally notable, to say nothing of those which have already been recounted.

To take the first on the list—I hardly believe in a classification of wonders—Belt, who was an engineer, but ought to have given up his whole life to observations of this sort, was the first, I believe, to find out that ants were

mushroom-growers. Like others, when he came to Nicaragua he saw the leaf-cutting ants passing in long, double columns backwards and forwards between their nests and the trees, the homeward-bound column laden with their little crescent-shaped bits of green leaf, the outgoing one empty-handed. “The first acquaintance a stranger generally makes with them,” says Belt, “is on encountering their paths on the outskirts of the forest crowded with the ants; one lot carrying off the pieces of leaves, each piece about the size of a sixpence, and held up vertically between the jaws of the ant; another lot hurrying along in an opposite direction empty-handed, but eager to get loaded with their leafy burdens. If he follows this last division, it will lead him to some young trees or shrubs, up which the ants mount; and where each one, stationing itself on the edge of a leaf, commences to make a circular cut, with its scissor-like jaws, from the edge, its hinder feet being the centre on which it turns. When the piece is nearly cut off it is still stationed upon it, and it looks as though it would fall to the ground with it, but on being finally detached the ant is generally found to have hold of the leaf with one foot, and soon righting itself, and arranging its burden to its satisfaction, it sets off at once on its return. Following it again, it is seen to join a throng of others, each laden like itself, and, without a moment’s delay, it hurries along the well-worn path. As it proceeds, other paths, each thronged with busy workers, come in from the sides, until the main road often gets to be seven or eight inches broad, and more thronged than the streets of the city of London. Standing near the mounds, one sees from every point of the compass ant-paths leading to them, all thronged with the busy workers carrying their leafy burdens. As far as the eye can distinguish their tiny forms, troops upon troops of leaves are moving up towards the central point and disappearing down the numerous tunnelled passages. The outgoing empty-handed hosts are partly concealed amongst the bulky burdens of the incomers, and can only be distinguished by looking closely amongst them.”^[60]

It used to be supposed that these leaves themselves, in a decaying state, were the food of the ants, whilst another theory was that they were used to make a sort of underground roof to the nest with. Belt’s discovery took everybody—including himself—completely by surprise. “I believe,” he says, “the real use they make of them is as a manure, on which grows a minute species of fungus, on which they feed: that they are in reality

mushroom growers and eaters”;^[60] and he thus narrates the circumstances which led him to this conclusion:—

“When I first began my warfare against the ants that attacked my garden, I dug down deeply into some of their nests. In our mining operations we also, on two occasions, carried our excavations from below up through very large formicariums, so that all their underground workings were exposed to observation. I found their nests below to consist of numerous rounded chambers, about as large as a man’s head, connected together by tunnelled passages leading from one chamber to another. Notwithstanding that many columns of the ants were continually carrying in the cut leaves, I could never find any quantity of these in the burrows, and it was evident that they were used up in some way immediately they were brought in. The chambers were always about three-parts filled with a speckled brown flocculent, spongy-looking mass of a light and loosely connected substance. This mass, which I have called the ant-food, proved on examination to be composed of minutely subdivided pieces of leaves, withered to a brown colour and overgrown and lightly connected together by a minute white fungus that ramified in every direction throughout it.”^[60] Belt assured himself in many ways, but not through actually seeing them do so, that this fungus was what the ants fed on, and he adds, “that they do not eat the leaves themselves I convinced myself; for I found near the tenanted chambers deserted ones filled with the refuse particles of leaves that had been exhausted as manure for the fungus, and were now left, and served as food for larvæ of *Staphylinidæ* and other beetles.”^[60]

Belt’s conclusions have been since amply verified, and the actual process of preparing the leaves and laying down the mushroom-beds, as well as the clipping and—if I mistake not—eating of the mushrooms, has been observed. Herr Möller—a German observer who resided for some years in tropical America—is usually referred to in this connection; but such extracts from his writings as I have come across are to me less *convincing* than the following account of Mr. Edward Tanner, which is contained in the *Journal of the Trinidad Field Club*.^[61] The observations were made with ants in confinement, as were Herr Möller’s also, I believe. “Each forager,” says Mr. Tanner, “drops her portion of leaf in the nest, which is taken up as required by the small workers, and carried to a clear space in the nest to be cleaned. This is done with their mandibles, and if considered too large, it is cut into

smaller pieces. It is then taken in hand by the larger workers, who lick it with their tongues. Then comes the most important part, which is almost always done by the larger workers, who manipulate it between their mandibles, the ant using her palpi, tongue, three of her legs, and her antennæ while doing so. It now becomes a small, almost black ball, varying in size from a mustard-seed to the finest dust-shot, according to the size of the piece of leaf that has been manipulated, which varies from $\frac{1}{8}$ by $\frac{1}{8}$ to $\frac{1}{4}$ by $\frac{1}{4}$ of an inch. These balls, really pulp, are then built on to an edge of the fungus-bed by the larger workers, and are slightly smoothed down as the work proceeds. The new surface is then planted by the smaller workers with slips of the fungus brought from the older part of the nest. Each plant is planted separately, and they know exactly how far apart the plants should be. It sometimes looks as if the plants had been put in too scantily in places, yet in about forty hours, if the humidity is regulated, it is all evenly covered with a mantle as if of very fine snow. It is this fungus they eat, and with small portions of it the workers feed the larvæ.”

The statement herein contained that the ants plant the new portion of their mushroom-bed with slips or plants taken from the already growing fungus is, as far as I know, new. I do not remember it in Herr Möller’s paper,^[62] who speaks of the *hyphæ* of the fungus growing through and round the little leaf-balls within a few hours, but without reference to their being planted, nor is it alluded to by Professor Wheeler, who has studied the mushroom-growing ant—whether the same or a similar species I know not—in Texas. Forel, again, speaking of an allied form in Colombia, says, “The largest workers triturate the leaves”; and again, “the medium-sized workers of the minim caste are for ever clipping the threads of the fungus, which then develops the ‘Kohlrabi’ (the little round swellings, that is to say), on which the ants feed.”^[62] Possibly this last may allude to the planting, but if so, it is the reverse of clearly put. Professor Wheeler also alludes to this constant clipping of the fungus, and sees in it the probable cause of the mutilation of the antennæ of the little blind cockroaches that live with these ants and take toll of their mushrooms.^[63] But as these constitute the sole food of their insect cultivators, it is natural that the latter should frequently clip in order to eat them, and the clipping would, no doubt, stimulate their growth. All this, however, is different from the actual deliberate planting of the fungus on newly laid-down portions of the bed—an act which would imply a very

clear intention, and make the ants farmers in the same way that we are. This, however, need not be the case if they only lay down the beds, for these at one time probably constituted their actual food, the crop of fungus being merely incidental. But if the ants deliberately plant the fungus, then, indeed, they must know precisely, in a human way, what they are about.

As we have seen, the leaves, from which, in their state of pulp, the mushrooms spring, are stored up by the ants in large underground chambers; but these mushroom-beds, or gardens, as they are often called, are themselves a sort of nest, containing tunnels and chambers, and not merely unformed heaps. It is in one or other of these chambers that the queen ant of the nest resides, a majestic creature, almost an inch long, but inflated both with pride and eggs to a disproportionate extent. Her sons and virgin daughters, who will some day be queens themselves, keep her company, whilst all about in the galleries and all over the broad, flat surface of the garden, which resembles a large flattened sponge, walk the different castes of workers, some large, some small, some medium-sized, with a few big-headed soldiers here and there amongst them, as though to keep the crowd in order. Whether they have really any such duty assigned them we do not know, but they do not appear to do any work, whilst the others are all busy at something, and the smaller workers particularly keep threading the stalks and filaments of the fungus in order to weed out any extraneous useless growth from amongst it.^[63]

It is a sad reflection—thus sighs Professor Wheeler—that so much ordered energy, such apparent intelligence, should all be really due to—*what* he does not seem to be quite certain about, not automatism entirely perhaps, but if not, then semi- or demi-semi-automatism, tempered with “psychic plasticity.” Against this view of the matter we have that of Belt, who, after giving two instances, which came under his own observation, of intelligent adaptation, on the part of ants, to meet particular circumstances, exclaims, “Can it be contended that such insects are not able to determine by reasoning powers which is the best way of doing a thing, or that their actions are not guided by thought and reflection?”^[64] But then Belt was not provided with the term “psychic plasticity,” and without it he could only infer intelligence from any intelligent act.

Still it cannot be denied that a great many instances have been given—noticeably in the case of “*our* ants” by Sir John Lubbock^[65]—in which these

paragons of insects have behaved very stupidly, or shall we say—for why should a creature that cannot be intelligent be stupid either?—with great “psychic *rigidity*”? Certainly such contradictions are very puzzling, but I would suggest one way of trying to estimate better the rigid type of ant intellect, which I believe to be absolutely new, and that is to compare it not with one’s own brain—or Darwin’s—but with that of a rigid type of person. It is wonderful what a difference this might make in our conclusions. An ant, for instance, that is unable, under some special circumstances, to get a thing down into its nest, because it persists in holding or pulling it, in the way it has always been accustomed to, or another that would rather be blown into the water along a known road than leave it for a new one, makes a poor figure in presence of the seven sages, or amidst a circle of senior wranglers mentally called up for its confusion; but we should think, rather, of some pin-headed servant-maid, setting an article of furniture each morning in the place that, with evident intention, you have removed it from overnight, or making up a larger and larger fire as the weather gets warmer and warmer. One should think of the obstinacy with which many people cling to old habits which changed times have made useless, or even harmful, and of how numbers not only prefer inferior things they are used to, to the most decisive improvements, but hate and revile such improvements as though they were undeniable evils. Instances will occur to everyone. I would rather not mention any for fear of alienating nine out of every ten of my readers. We should think, also, of savages or primitive, slow-moving peoples. What a great unadaptability, for instance, did the Matabele show in their methods of encountering our countrymen during the war, and throughout the rising; as also in that rising itself, since it was against all those well-known blessings which our empire confers upon savages.^[66] It is to these less exalted levels of human faculty that we should look when we seek to compare an ant’s mind—when out of its usual set track—with our own, if we wish to do the ant any justice. That we pursue an opposite plan is my own explanation of many a partial verdict. To every experimenter in these directions (who should happen to ask my advice) I would say, first, “Do you know, or have you ever known, a really silly person?” and on his beginning, at once, with “Yes, Mrs.” or “Miss” (as the case may be), I would strike in peremptorily thus: “Then keep *her*—not Newton—in your mind as a standard of comparison.”

That ants should intentionally cultivate mushrooms will appear wonderful to everybody, and some will see in it the high-water mark of their mental development, by whatever path it has been arrived at. It seems natural to connect such doings with the fact that “in ants the cerebral ganglia are of extraordinary dimensions, and in all the Hymenoptera these ganglia are many times larger than in the less intelligent orders, such as beetles.”^[67] Yet the brain of an ant—“one of the most marvellous atoms of matter in the world, perhaps more so than the brain of a man”—is “not so large as the quarter of a small pin’s head.”^[67] Of what size, then, can a beetle’s be?—especially that family of beetles which grow and cultivate mushrooms, just in the same way that ants do. It would be suggestive—though I hardly know of what—should it be found that, comparatively speaking, *they* have no brain at all.

The beetles alluded to have been named, with reference to the particular kind of mushrooms they grow, ambrosia beetles, though in what the great superiority of these over those raised by the ants lies I do not know, for no one appears to have tasted them. It has been agreed, however, to call them ambrosia. “One of the most remarkable facts,” says Mr. Froggatt, “is that each group of these beetles is associated with a certain kind of ambrosia or fungus, notwithstanding that they are found in different timbers. This substance is actually cultivated by the mother beetle upon a carefully prepared layer or bed of wood-débris, generally at the end of the gallery; but in others the ambrosia is grown only in certain brood chambers of peculiar construction, whilst in others again it is propagated in beds near the cradles of the larvæ!”^[68] When the latter hatch, they find a supply of celestial food awaiting them, and can walk about the various galleries, feeding upon it to their hearts’ content.

In other cases, however—that is to say, with other species—social development has gone further, and, besides boring galleries, the mother-beetle excavates a number of cells in their walls, like rows of bedrooms opening out of either side of a passage. She does not, however, quite finish her bedrooms, but, whilst they are still incomplete, lays an egg in each, and when this hatches, the young beetle, then in its larval state, takes up the task where she left off, and in time completes it. All the while they are growing up the mother feeds the young ones, and, between the intervals of doing so, stops up the entrance to the cell with a “plug”—such is the word employed;

“to what base uses we may return, Horatio!”—of ambrosia. In time, when they have acquired the full imago form, each female beetle flies away to make a burrow and rear a family of her own, and in some species she is accompanied in this marriage flight, as it may be called, by the male. In others, however, the males are wingless, and remain in the burrow, till, when their appointed time comes, they die. Whether the male, when winged, assists the female in her mining operations I am not quite sure, inasmuch as that point seems to be avoided in the accounts which I have been able to consult, but the wingless male would not be able to do so, as he would be left behind in the burrow when the female flew away to found another colony.

The fungus, when it has once commenced to grow, increases very rapidly, so that if the number of beetles in the nest is much diminished, as, say, by some accident, the rest cannot eat enough to keep it down, and so, it would appear, are suffocated. It is asserted, however, that when the wingless males are deserted by the females, and would otherwise perish in this way, they all collect together in a few of the galleries and feast on the ambrosia there growing. By this means, we are told, they “prolong for a time their useless existence”—an ungrateful way of putting it, so it seems to me, as the poor things have already been useful in a very indispensable manner, so that their existence as a whole is anything but useless, and to separate a part of it from the rest and carp at that is silly as well as ill-natured. But it is the fashion to speak in this harsh way of the male insect, beginning with the drone bee; whereas when the female has done all that *she* can do—which is often just to lay her eggs—nobody talks of *her* useless existence. Fashion is a curious thing, and ants, even if they be automatons, are not the only creatures that do things automatically.

It is certainly very curious, if it be true, that the wingless male beetles should, by thus congregating together in this way, and so saving their lives, show more intelligence than the winged females, who, under similar circumstances, are choked with their ambrosia, as the Duke of Clarence was with his nectar in a malmsey-butt. It is true that with the males the thing happens every year, whereas with the females it is only accidental; but, in the particular circumstances, it is difficult to see how inheritance can have had anything to do with it. Here, then, are a particular family of beetles who live the same sort of social life that ants and bees do, which discovery

appears to have been made by a Mr. Hubbard not so many years ago, and from whose paper on the subject all the above particulars have been taken, though only through the medium of various magazines, since even at the British Museum I was unable to get the paper itself. So long ago, however, as 1844 a certain Herr Theo Hartig “published an article on the ambrosia of *Xyleborus* (*Bostrichus*) *dispar*, in which he showed that it was a fungus growth (*pilzrasen*), and he named the fungus *Monilia candida*.” This statement is made by Mr. Hubbard in his much more recent account. Not feeling perfectly certain from it whether the origin, as well as the nature, of the strange-named substance was not also divined by the German investigator, I quote the reference in order not to do him a possible injustice, for to me it seems that there have been few more interesting discoveries than this of these ant-like, ambrosia-growing beetles. But why the ants only grow mushrooms, thus allowing themselves to be enormously outdone by an inferior insect, is more than I can understand.

And now a word of justice to these beetles. It might be supposed that, by burrowing into trees, they caused the death of the latter, but this is not really the case. Writing in *The Agricultural Gazette* of New South Wales for August, 1900, Mr. Froggatt, the Government Entomologist, makes the following statement absolving *Xleborus*: “This curious little beetle (*X. solidus*) is rather plentiful about Sydney, and is frequently sent to us taken out of the trunks of fruit trees, which it is supposed to have killed; but in all cases that have come under my notice it has had nothing to do with the tree dying, but is attracted to the tree as soon as it becomes sick, the bark begins to wither, and the first symptoms of decay set in.” Mr. Froggatt adds: “The instinct that leads these and other wood-boring beetles to a tree as soon as it is sick is something marvellous; in the tropics I have collected many fine, rare species upon the freshly cut tent-poles in our camp, attracted to the wood, but otherwise seldom found in the bush.” This instinct would seem to be a remarkably developed scent, though why a severed branch should smell differently from the tree of which it but a moment before made a part it is not easy to imagine. However, we cannot, without evidence, attribute clairvoyance to beetles, and perhaps it is the cut from which the scent emanates.

Another example of an insect which is neither an ant, bee, wasp, nor white ant, but which yet may be said to live a true social life, is the little creature

which, under the name of *Psocus venosus* and as belonging to the order *Carrodentia*, will be familiar to everyone. It is nearly related to the so-called book-lice, but lives in the open air, “being seen,” says Mr. Leland Howard, “upon the trunks of trees, in flocks numbering from twelve to forty or fifty individuals.”^[69] These browse together like a herd of miniature cattle on the various lichens that embrace the bark, and these they nibble so closely that wherever they move they leave a bare track behind them. Sometimes one family and sometimes several are included in the herd, all ages and stages being represented, from the wingless but free-moving larvæ to the winged imago form. The latter, however, though they be thus provided, will not readily forsake their young, but the whole of them, when alarmed, first run all together, and then, if the cause of disquietude continue, suddenly scatter as though in panic, and run hither and thither, in all and every direction. When the danger seems over, they close up their ranks again, and go on browsing as before.

The female *Psocus* lays her eggs in little clusters of from fifteen to twenty, and protects each cluster under a sort of dome or shield of gnawed wood which she presses upon them so that they stick to it. She is said to brood over the eggs, but this does not appear to mean that she actually incubates them. Rather, she remains about, keeping watchful guard till they are hatched, and then takes the young to find pasture, walking at their head like a hen in front of her chickens. From such beginnings as these it seems possible that the social life of ants has been, in the course of ages, evolved and developed.

CHAPTER XVI

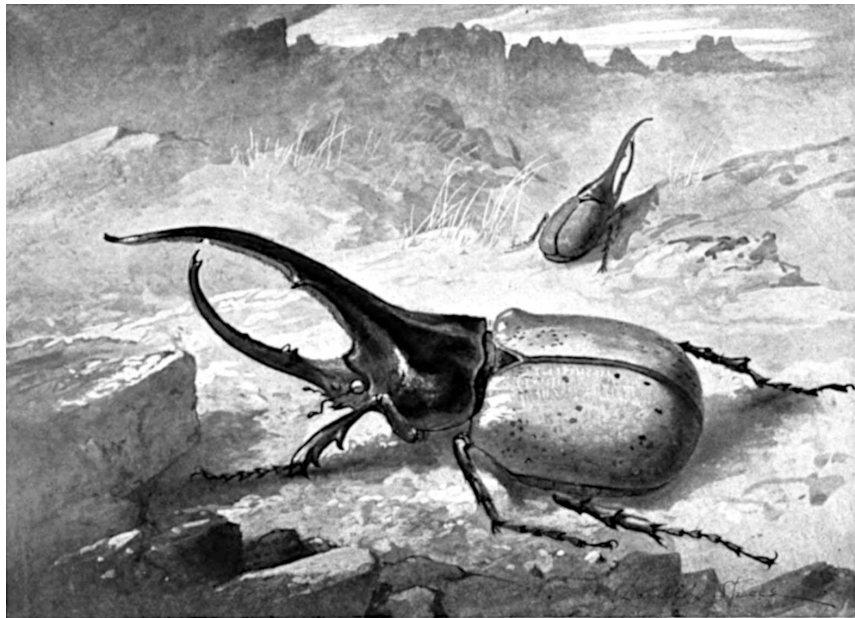
From wood to ambrosia—Wood-boring beetles—Rival claimants—Stag and other beetles—Metempsychosis—Flies with horns—Comical combatants—Female encouragement—The sacred Scarabæus—A beetle with a profession—Table companions—Old and new fallacies—From theft to partnership.

IT is, no doubt, through feeding on wood that the beetles we have been considering came in time to feed on ambrosia. The particular fungus, that is to say, which for some unaccountable reason has received this name, appeared as a natural growth upon the walls of their tunnels, and in time it came to be thought necessary, and its coming was arranged for. By similar steps, probably, the leaves once carried as food to their nests by the sauba ants, or cooshies, have become the soil merely on which that food is grown; and so I have no doubt myself that even if the present agricultural ants of Texas do not purposely sow and afterwards reap the rice that springs up around the circular mound of their domicile, their descendants will do so. Indeed, it seems rather curious that, with such facilities for a gradual development, the habit has not yet been acquired; and this is the chief reason which inclines me to suspend judgment on the question, and wait for further observations. Far from thinking the thing too wonderful, I wonder if it be not the case. Such wonder, however, is for ants and not for beetles, except, indeed, ambrosia beetles, who certainly merit it, though in an opposite way. No other wood-borers of the order are anything but wood-eaters, or, at any rate, if a few feed on fungus, as would not be improbable, should it happen to appear, they have nothing to do with the cultivation of it. The words "as far as we know," however, must be added to the foregoing statement. Numbers of beetles pass the larval and pupal stage of their existence within the trunk of a decaying or even of a perfectly sound tree, from which they issue after the final metamorphosis has been made. Amongst these is our own stag-beetle, in whom, that is to say in whose caterpillar, some suppose themselves to see the *Cossus* of the ancient Romans, which was as much appreciated by them, and, no doubt, justly, as

are earthworms by the Chinese. Others, however, believe this to have been the large red meaty-looking caterpillar of the goat-moth; and as the one conjecture is quite as plausible as the other, the only, or, at any rate, the best way of arriving at a conclusion would be to try them both—a simple plan which, as far as I know, has not yet been adopted.

The stag-beetle—when of maximum size, that is to say, for it varies amazingly in this respect—is much the largest beetle which this our island possesses, but though, with its huge, antler-like jaws, it makes a good perennial illustration for all books of popular entomology, its merits seem to end there, for either there is nothing or nothing has yet been observed particularly interesting about it. No doubt, if we look at the matter from an absolute rather than a relative point of view, the second of these two explanations is the correct one, for a creature has only to be studied in order to become interesting; but as compared with ants, bees, wasps, and many other insects, beetles, or, at any rate, the vast majority of them, are not so very entertaining in their habits, and the stag-beetle has no superiority in this respect to correspond with its size and uncommon appearance. This appearance, however, is confined to the male, who alone possesses the great branching mandibles on which its greater size also is largely dependent. It would be natural to suppose that these formidable weapons, as they certainly appear, stood in relation to the combats of the males for the possession of the females, yet it is often stated that the short, sharp pincers of the latter, which can be made, it would seem—anyone who doubts may try—to meet in the flesh, are really the more efficient of the two. Be this as it may, it is not improbable that the stag-beetle's jaws, since they are very handsome, may have been developed less as weapons than as ornaments, under the laws of sexual selection. Darwin, if I remember, was doubtfully of this opinion, and he attributes many strange projections and processes on the head or thorax of other beetles—as notably that huge one with a snout like a weaver's beam, called the Hercules beetle—to the same agency. No use for this extraordinary trunk, as one may term it, has as yet been discovered, but as the under portion is covered with a thick matting of soft brown hairs, it would seem as though it had some office to perform, unless indeed we suppose this *chevelure* to be likewise admired. A lesser, though still tremendous, projection, starting from the head, as the other one does from the thorax, is likewise unaccounted for, for though the two together make in appearance a pair of uncouth and irregular pincers, they neither are,

nor apparently can be used in this way. Nothing appears to be known of this strange creature's habits, and the same may be said in regard to most of the more remarkable-looking beetles of the world, as well as those which are not so extraordinary in their appearance. The ways of beetles, in fact, have been but little studied, and it is perhaps not too much to say that if for every thousand that fill the show-cases in museums we could know the life-history of one, we might with infinite advantage, in exchange for this knowledge, throw the whole pin-forest of them into the sea. In what light this fact, if true, exhibits the labours of those naturalists—as the world calls them—who, living for years amidst the life-teeming regions of the earth, have spent their whole time in constantly killing and killing, coming home, at last, with an acre of carcasses, to write a book containing hardly anything of first-hand observation—the soul of natural history—I will not pause to inquire.



THE HERCULES BEETLE.

The enormous beetle from which this illustration was drawn, though not a particularly large specimen, is six inches long, and the upper jaw measures three inches and a half.

There are many other species of beetles, the males of which are ornamented about the head and thorax with all sorts of knobs and projections, so that, with some, one might think that a one-horned or two-horned rhinoceros had undergone metempsychosis, as it is called, that its soul, that is to say, had transmigrated into the body of an insect, which latter had been fashioned so as fancifully to resemble its old one. However, as this would be a downward journey, it is more satisfactory to imagine that certain beetles have been “translated” into rhinoceroses. As to these kinds of excrescences, Darwin believed them to be of the nature of adornments, and since their owners—the males—have not been seen to use them in warfare, and indeed do not appear to fight, it is difficult to imagine any other *raison d’être* for them. This seems all the more likely because certain flies found in the Malay Archipelago have likewise excrescences, which we have to call horns, and these too are confined to the males, though it is hardly to be imagined that they would fight in a manner to make them of service. These flies must be most extraordinary creatures to look at. They have long legs, which they

draw together underneath them, so as to stand very high, and their horns are not only conspicuous by their size and shape, but also by being brightly coloured. Thus in one species they are a beautiful pink with a light stripe down the centre, and bordered on each side with black. In another the colours are yellow, black, and brown, and though *Elaphomia cervicornis* has to be contented with black, and pale tips, yet his are the finest pair of all, being nearly as long as his body, and branched so as to look like a pair of slender and delicate stag's horns. The other pairs are not like this, one of them being rather club-shaped, and therefore less horn-like, whilst another has an extraordinary resemblance to the antlers of an elk, which are broad and palmated, so that it is the *Elaphomia alcicornis*. Here, therefore, are both horned beetles and horned flies who yet do not fight with their horns, so that unless they serve as ornaments it would be a puzzle to say what they do serve as; for as the male beetles do not fight, which is the principal way in which male creatures, including man, show their vigour, why should we suppose them to be more vigorous than the females?

In some other beetles, which do fight, the sexes do not differ conspicuously, nor do their facial or other peculiarities appear to bear any special relation to warfare. Thus “those curious little beetles, the *Brenthidæ*” of the Malay Archipelago, have an extraordinarily long snout—or rostrum, to talk entomologically—at the end of which come the jaws and antennæ, and this rostrum is used by the female to bore holes in decaying wood, where she afterwards deposits her eggs. The males, however, do what they can with them as weapons, and Dr. Wallace has seen two of them fighting together in a very comic manner. “Each,” he tells us, “had a fore-leg laid across the neck of the other, and the rostrum bent quite in an attitude of defiance, and looking most ridiculous.”^[70] On another occasion “two were fighting for a female, who stood close by busy at her boring. They pushed at each other with their rostra, and clawed and thumped, apparently in the greatest rage, although their coats of mail must have saved both from injury. The small one, however, soon ran away, acknowledging himself vanquished.”^[70] *Lethrus cephalotes* is another fighting beetle, and here the males, instead of horns or anything extraordinary, have merely somewhat larger mandibles than the females. “The two sexes,” says Darwin, “inhabit the same burrow. If, during the breeding season, a strange male attempts to enter the burrow he is attacked; the female does not remain passive, but closes the mouth of

the burrow and encourages her mate by continually pushing him on from behind; and the battle lasts until the aggressor is killed or runs away.”^[71]

Of yet another species, the *Ateuchus cicatricosus*, the sexes “live in pairs, and seem much attached to each other; the male excites the female to roll the balls of dung in which the ova are deposited, and if she is removed he becomes much agitated. If the male is removed the female ceases all work, and, as M. Brulerie believes, would remain on the same spot until she died.”^[72] But M. Brulerie was reckoning apparently without M. Fabre, since whose investigations in this last department it may be said that “*nous avons changé tout cela.*” For this *Ateuchus* is none other than the celebrated Scarabæus, or sacred beetle, and, in the first place, M. Fabre has shown that the balls of dung, which are rolled about by them with so much perseverance and energy, do not contain the ova, as it was always thought that they did, but are merely provender and nothing more, and though sometimes they are rolled by two beetles together, these are not the male and female, or, at any rate, they need not be. They are just as likely to be two males or two females, and in any case, though the two may be of opposite sexes, they do not represent a mated pair. Simply when the two—if, as is by no means always the case, more than one take part in the rolling—have pulled and pushed the ball to a suitable place, they make a hole in the ground, into which they drag it, and, having closed the aperture, sit and feast at their leisure.

According to Fabre the vital principle contained in the egg would be destroyed were it rolled about in this fashion, so when the mother Scarabæus, who, it would appear, works in this matter alone, is ready to lay her eggs, she first makes an excavation, and then brings the dung down into it in pellets, till there is a heap of it, which fills the whole concern. Then “the first thing to do is to select very carefully, taking what is most delicate for the inner layers, upon which the larva will feed, and the coarser for the outer ones, which merely serve as a protecting shell. There around a central hollow which receives the egg the materials must be arranged layer after layer, according to their decreasing fineness and nutritive value; the strata must be made consistent, and adhere one to another; and finally the bits of fibre in the outside crust, which has to protect the whole thing, must be felted together.”^[73]

Thus, when the grub first issues from the egg, it finds light digestible food ready to hand, which becomes coarser and more fibrous with its growth and increased capacity of assimilating such stronger diet. As more and more is eaten, the ball, which is about the size of an ordinary apple, becomes hollower and hollower, till at last, when only the outer crust remains, the grub is ready to enter upon that wonderful series of changes—called its metamorphoses—which will bring it forth into this larger ball of dirt, a complete beetle, with a useful profession, that of scavenger, immediately open to it.

Thus a fallacy which, according to Fabre, dates from the time of the Pharaohs, viz. that every ball of dung which one might at any time see a Scarabæus beetle rolling and trundling along contained its egg, has been finally disposed of, nor is this the only one. It used to be thought, not only that any two beetles rolling a ball between them were male and female, but also that any single one that happened to be in difficulties would immediately fly off and summon a comrade or two to its aid. Fabre denies this altogether, and maintains that in this rolling away of provisions each individual beetle is purely a self-seeker. It is true, as we have seen, that the *bonne bouche* will often be eaten by two Scarabæi—never more—in the cavern prepared beforehand for its reception, but, according to Fabre, this is only because it is to the mutual interest of both to act in this way, since neither can succeed in appropriating the ball to itself, in spite of efforts—which in other cases, however, may be successful—to do so.

In all such cases the one beetle is the real owner of the ball, whilst the other is only there with the intention of stealing it if he can. Thus the thief will often let himself be pushed along by the honest worker, lying flat on the ball, and doing no work whatever, though at other times, when a rise in the ground makes it difficult for a single beetle to roll it, he will assist with all his power. Again, whilst the one Scarabæus is hollowing out a cave for the approaching banquet to take place in, the other, left with the ball, will, after some time, begin to go off with it alone, and unless pursued by the owner before he has gone too far, he accomplishes his purpose, and eats it all himself. Thus he has associated himself to the maker of the ball with the distinct idea of stealing it if he can. He has this plot in his mind, to pretend partnership, to even give real assistance, but to watch his opportunity and decamp when it occurs. That, at least, is the view suggested for our

adoption, but I cannot say that it recommends itself to me. Fabre, in my opinion, has disposed of one error only to fall into another of precisely the same kind. He says very justly in regard to the idea that one beetle would deliberately fly away and summon others to its assistance, "It is no slight thing to admit that an insect has a truly surprising grasp of the situation, and a facility for communicating its ideas to others of its kind more surprising still. Are we to suppose that a Scarabæus in distress conceives the idea of begging for help, flies off, explores the country round, etc.?" Very true; but if we are not to suppose this, I certainly will not suppose, either, that this same Scarabæus can conceive the idea of pretending to assist another in order to rob him of his property. This would be as deep a laid scheme as the other, and the facts of the case, as given by Fabre himself, do not appear to me to lend themselves to such an explanation.

The point of these interesting relations has been, in my opinion, entirely missed. What we really see in them, or what, at least, is there for us to see, is the beginnings of order and social polity, evolving themselves out of lawlessness and the strong hand. Further, it has already gone some little way, for the fact that two Scarabæi do, as a matter of fact, assist each other very materially in rolling the ball, and that they do sit and eat it together in the same chamber, are not to be got over by any such amusing fancy picture as this brilliant writer, as well as keen observer, has given us. It is no use fixing our eyes upon that part of the conduct of the beetle which we are invited to call the thief, in contradistinction to the owner—I doubt myself if Fabre has always kept the two distinct from beginning to end—if we pass over the other and much more interesting parts of it. Why does this beetle help to get the ball up a hill, and why does he sit for some considerable time with it outside the cave that the other is making, before he begins to go off with it? Does he say to himself, in the first case, "If we don't get it to the right place, to begin with, he'll never dig a hole and leave me outside with it," and, in the second, "I'll wait till he has come out and found it all right, several times, so that his suspicions may be put to sleep"? This would be scheming with a vengeance; but *serviteur Monsieur Fabre*, I must refer you to your own incredulity in another matter. I will never accept such an explanation, and the view which I take of the whole affair is this. The beetle which Fabre calls the thief is under the sway probably of precisely the same feeling as the other one—the rightful owner. He has seized a piece of dung, and as he seizes it, whether another has it at the time or not, it appears to be

his, that being the only idea of rightful ownership which is not too large for his comprehension. Finding, however, that another beetle has exactly the same idea as himself, he is forced, willy nilly—and the experience is being constantly repeated—to accommodate himself to this circumstance and make the best of it. The ball—this is the one great advantage—will continue to roll even if he does not push it. Therefore he can afford to be lazy sometimes, and be pushed along with it. The ball stops: in that case he must push it, and, even without this incentive, it would appear from Fabre's account that the two often work together. Thus, from the very necessities of the case, it is evident that a sense of partnership—at least a feeling of doing work in combination with another—has begun to dawn in the mind of the *Scarabæus*. The fact that when the one beetle is left alone with the ball, whilst the other excavates, it does not immediately go off, but stays a little, as though waiting to be rejoined, suggests to my mind that this feeling, fostered by custom, has already gone some way, though it is not wonderful that, after a time, the primitive individualistic tendency should again assert itself. But when the fugitive is overtaken, it assists the other to roll the ball back, and the end of it all is a meal shared peacefully between the two, in one common apartment. If we suppose that the instinct, or capacity, of working together for some common end has had any beginning, surely we might expect to find it in some such state of affairs as this. The result of primitive conceptions is that two *Scarabæi* are often obliged to roll one ball between them, and if there be any advantage in this arrangement, natural selection will no doubt do the rest. That it has already begun to do it is, I think, very probable; but Fabre was not an evolutionist.

CHAPTER XVII

Do ants sow and reap?—Rival observers—The Texan v. Macaulay's schoolboy—More evidence wanted—How ants cross rivers—Tubular bridges—Ant armies—A world in flight—Living nests—Ants and plants—Mutual dependence—Nests in thorns and tubers—Ant honey-pots—Business humanity—Burial customs—A strange observation—Two views of ants.

MUSHROOM-GROWING, especially if the ants plant the mushrooms in the way stated by Mr. Tanner, is just as extraordinary, I think, as their habit of planting a field with ant-rice and reaping it at the proper time would be, did they really practise it. Up to a little while ago it certainly seemed as though they did, for there was Dr. Lincecum's definite statement based upon twelve years' observation, and this, if not confirmed by Mr. McCook, was, at any rate, not contradicted by him. On the contrary, McCook mentioned a good many facts pointing in the direction of Lincecum's assertion, and though he did not consider them decisive, he could see no reason why the ants should not act in this way, as indeed there is none: so that as he had only stayed a few months where Lincecum had lived for twelve years, he seemed like a weaker witness supporting, according to his opportunities of observation, a much stronger one. Now, however, comes another witness, whose opportunities have also been great, and in a somewhat heavy-handed way, in a spirit of myth-slaying and irrelevant reference to supposed schoolboy knowledge, hardly required in face of all that ants are known to do, denies the whole thing.

First, however, let us have the assertion as originally made by Lincecum, which is, that on the summit of the mound of their nests, from which they carefully clear away all other vegetation, the harvesting ants sow the seed of a certain plant called ant-rice for the purpose of subsequently reaping a harvest of the grain. It is sown in time for the autumnal rains to bring up, and at the beginning of November a green row or ring of ant-rice, about four inches wide, is seen springing up round the circumference of the *disk* (as the circular top of the mound is, for some reason, always called). In the

vicinity of this circular ring the ants do not permit a single spire of any other grass or weed to remain a day, but leave the aristida or ant-rice untouched until it ripens, which occurs in June of the next year. After the maturing and harvesting of the seed, the dry stubble is cut away and removed from the disk, which is thus left unencumbered until the ensuing autumn, when the same species of grass again appears as before, and so on. ^[74] After stating in a letter to Darwin that he has seen all this taking place year after year, Dr. Lincecum adds:—"There can be no doubt of the fact that the particular species of grain-bearing grass mentioned above is intentionally planted. In farmer-like manner the ground upon which it stands is carefully divested of all other grasses and weeds during the time it is growing. When it is ripe the grain is taken care of, the dry stubble cut away and carried off, the paved area being left unencumbered until the ensuing autumn, when the same ant-rice reappears within the same circle and receives the same agricultural attention as was bestowed upon the previous crop, and so on year after year, as I *know* to be the case in all situations where the ants' settlements are protected from graminivorous animals." ^[75] Lincecum also believed that the ants were able in some way to prevent the seed stored in their nests from germinating. This same fact has been asserted, and apparently proved, by Moggridge, in regard to the harvesting ant of southern Europe, and he also states that, if in spite of the precaution any seeds begin to sprout, the ants by gnawing off the tips of the radicles would prevent the germination from proceeding.

This, then, is the case *for* the harvesting ant, as we may say; for if these things be true they are certainly much to its credit, whereas, if not, the scandal is so great that it ought to change its name. Let us now hear the case against, as stated by Professor Wheeler, after which readers may make up their minds, if they can, for I have not quite done so yet. I quote in full, so that the two statements may be balanced against each other, and this, I hope, will be more interesting than the usual "Mr. So and So, however, disputes this and thinks, etc."—another line or two in which the contrary proposition of the one before is stated at about the same length. This is what Professor Wheeler, who "speaks home—you may relish him more (or at least as much) in the soldier as the scholar"—has to say: "It may not be altogether out of place in this paper to record a few other observations on *Pogonomyrmex molifacicus*, inasmuch as this form has been singled out

among all the known members of the genus as presenting certain remarkable instincts. Lincecum is responsible for the myth that this *Pogonomyrmex* sows a certain species of grass, the 'ant-rice' (*Aristida oligantha*), protects it from harm and frees it from weeds while it is growing, for the purpose of reaping the grain. This notion, which even the Texan schoolboy (not Macaulay's, who probably knew as much about it) has come to regard as a joke, has been widely cited, largely because the great Darwin stood sponsor for its publication in the Journal of the Linnean Society. McCook, after spending a few weeks in Texas observing the ant in question and recording his observations in a book of 310 pages, failed to obtain any evidence either for or against the Lincecum myth and merely succeeded in extending its vogue by admitting its plausibility. Two years of nearly continuous observation enable me to suggest the probable source of Lincecum's and McCook's misconceptions. In either case the observer has started with a few facts, and has then stopped short to draw inferences before gathering more facts. If the nests of *Molifacicus* be studied during the cool winter months—and this is the only time to study them leisurely and comfortably, since the cold subdues the fiery stings of their inhabitants—the seeds which the ants have garnered in many of their chambers will often be found to have sprouted. It is, therefore, certain that these ants are not able to prevent the seed from germinating, as Moggridge claims for the European species of Messar, except by conveying them to drier chambers; and in protracted spells of wet weather even this precaution seems to be of no avail. On sunny days the ants may often be seen removing these seeds when they have sprouted too far to be left for food, and carrying them to the refuse heap, which is always at the extremity of the cleared earthen disk or mound. In this place the seeds thus cast away as inedible often take root, and, somewhat later, form an arc of tall grass more or less closely approximating to a complete circle round the nest. Since these ants feed largely, though by no means exclusively, on grass-seeds, and since these particular seeds are a very common and favourite article of food, it is easy to see how their grass should often predominate in the circle. In reality, however, only a small percentage of the nests, and only those situated in certain localities, present such circles. Now to state that the ant, like a provident farmer, sows this cereal, and guards and weeds it for the sake of garnering its grain, is as absurd as to say that the family cook is planting and maintaining an orchard when some of the peach-stones which she has

carelessly thrown into the backyard, with the other kitchen refuse, chance to grow into peach trees.”^[75] Certainly such a thing should have been observed before the statement was made, and, if it has not been, the facts seem more probably accounted for on the above explanation.

Professor Wheeler goes on to say that “there are several other facts which show that the special ring of grass about the nest is an unintentional and inconstant result of the activities of the ant colony. For instance, one often finds very flourishing ant-colonies that have existed for years in the midst of much-travelled roads, or in stone side-walls, often a hundred or more feet from any vegetation whatever (without any ant-ridge on their mounds therefore). Again, it is very evident that even a complete circle of grass like those described by Lincecum and McCook would be entirely inadequate to supply more than a very small fraction of the grain necessary for the support of a flourishing colony of these ants. Hence they are always obliged to make long trips into the surrounding vegetation, and thereby wear out regular paths, which radiate in different directions, often to a distance of forty to sixty feet from the entrance of the nest. The existence of these paths, which are often found in connection with grass-encircled nests, is alone sufficient to disprove Lincecum’s statements.”^[75] It certainly seems easier to suppose that Lincecum misinterpreted certain facts, not themselves in dispute, than that an explanation on which so many considerations seem to throw doubt is the correct one. One thing, at least, seems certain—if some of these ant communities grow grain of set purpose, all of them do not. This may be possible, but more proof of it than Lincecum has brought is demanded. If the ants really sow and reap the grain that grows upon their mounds, and, more especially, if they carefully keep the patch clear, it ought not to be difficult to see them doing so. This last would be decisive, whereas the other two are by no means so.

That ants should use their own larvæ like a shuttle, and for the same purpose, seems as strange a thing as one can well imagine, but there is no doubt at all about it, the act having been witnessed on various occasions by competent observers, whose evidence is mutually corroborated. The species in question is common in Eastern Asia, and is accustomed to make little houses or arbours for itself by bending leaves round so that the edges meet, and then fixing them together, as some caterpillars do. Now the larva can do something which the grown ant cannot, which is to spin a cocoon from a

sort of gummy, thread-like substance which issues from the mouth. Whilst one group of ants therefore join to keep the leaf bent in the proper position, another take each a larva in their jaws, and pass it from edge to edge of the leaf, applying its mouth to each edge, until the two are bound firmly together.^[76] Whether this is a more or less remarkable habit than growing mushrooms it would be difficult, perhaps, to decide, nor is there any need to try, since such questions are more interesting left uncertain.

It is well known, or at least credibly asserted, that ants cross rivers by clinging one to another from the branch of a tree overhanging the water, till the end of this living chain, as it becomes longer and longer, is carried by the force of the current to the opposite bank, where a bridge is formed, over which the main body marches.^[77] According to Du Chaillu the ants in Africa make, not only a bridge, but a tunnel—"a high, safe tubular bridge through which the whole vast regiment marches in regular order."^[78] These are the celebrated driver or bashikouay ants, who, when upon their terrible marauding marches, put every living creature, including man, to flight, though for many flight is in vain. Size and strength are here no protection. "The elephant and gorilla fly before them; the black men run for their lives." So says Du Chaillu, and, sure enough, when the skins of some of the poor gorillas he shot arrived in England, several of these ants were found amongst the hair.^[79] In the forests of equatorial Africa, abounding—if they have not all been shot by this time—with large animals, these hunting-raids must give rise to some stirring scenes. What crashings through the trees and undergrowth! What uncouth sounds, perhaps, of mingled pain and rage! How a bitten gorilla would express himself! What a subject for a picture if a herd of elephants, a few families of gorillas, a score or so of lions, with a few leopards, and baboons, perhaps a rhinoceros, and any number of antelopes, were all to come rushing down together to where an artist stood ready for them! I should like to see the picture he would draw.



PURSUED BY DRIVER ANTS

The greatest beasts of the forest will fly before these terrible little insects, one of which is shown in the left-hand corner.



A more remarkable sight even than an ant-bridge is perhaps an ant-nest, by which I mean, not an ants' nest in the ordinary sense of the term, but a nest made of ants. The following quotation from the much-containing *Naturalist in Nicaragua*, page 25, will explain this hard saying. "They make their temporary habitations in hollow trees, and sometimes underneath large fallen trunks that offer suitable hollows. A nest that I came across in the latter situation was open at one side. The ants were clustered together in a dense mass like a great swarm of bees hanging from the roof, but reaching to the ground below. Their innumerable long legs looked like brown threads binding together the mass, which must have been at least a cubic yard in bulk, and contained hundreds of thousands of individuals, although many columns were outside, some bringing in the pupæ of ants, others the legs and dissected bodies of various insects. I was surprised to see in this living nest tubular passages leading down to the centre of the mass, kept open just as if it had been formed of inorganic materials. Down these holes the ants who were bringing in booty passed with their prey." Of the many curiously

constructed or strangely produced dwellings of ants, this made out of their own bodies is amongst the most remarkable.

Many ants live in the interior of various plants. The plant generally benefits as much as the insect by this arrangement, so that there is a mutual dependence between the two, which in some cases is carried to such an extent that the life of one or both seems a necessary part of that of the other. In Borneo, for instance, a certain large tuber which grows on the branches of aged trees is always found inhabited by a certain red ant, of small size, but fierce disposition, which rushes out and attacks anyone who ventures at all near its dwelling. The seed of this tuber is disseminated in the same way as is our own mistletoe, through the agency of birds, that is to say, the seed being surrounded by a similar pulpy mass, which adheres to the branch on which it falls. Soon after germination the tuber, which is shaped something like a carrot, begins to develop, but whilst still quite small its growth ceases and in this state it would remain, and before long, die, if it should not happen to be found by the ants in question. If it should be, however, its life is assured. They immediately bore a hole at the base of the stem, upon which this enlarges to a great degree, so that soon there is room for them to excavate galleries in the cellular tissue of the interior, and to form a populous colony. The whole tuber is soon perforated in all directions, and becomes a living and growing formicarium, the great accretion of cellular tissue which has made this possible having been caused by the poison—if we may call it so—of the ant's bite, in the same way as the sting of the gall-fly raises galls upon the oak.^[80] Of course, from the moment that the ants appear the tuber is safe from any other insect, or small bird, or mammal that might otherwise do it harm. The ants in defending their nest would defend it, and it is on this principle of mutual advantage that such ant and plant alliances have been brought about.

Thus the dry, arid plains, called savannahs, of tropical America support a species of acacia of which the thorns, characteristic of the family, grow in pairs and are shaped exactly like the horns of some oxen. Every pair of these horns becomes in time an ants' nest, and if the tree be touched or shaken, the ants rush out full of fury in defence of their habitations. Thus every tree is tenanted by a large army of retainers, who almost more than the thorns themselves, which have been developed for the same purpose, protect it against browsing quadrupeds. Its thorns, however, would be no

protection against the leaf-cutting ants in search of materials for their mushroom-beds, whereas these are kept at bay by a hostile species, smaller indeed, but armed with a powerful sting. "For these services," says Belt, "the ants are not only securely housed by the plant, but are provided with a bountiful supply of food; and to secure their attendance at the right time and place, this food is so arranged and distributed as to effect that object with wonderful perfection. The leaves are bi-pinnate (double, that is to say), and at the base of each pair of leaflets, on the mid-rib, is a crater-formed gland, which, when the leaves are young, secretes a honey-like liquid. Of this the ants are very fond; and they are constantly running about from one gland to another to sip up the honey as it is secreted. But this is not all; there is a still more wonderful provision of solid food. At the end of each of the small divisions of the compound leaflet there is, when the leaf first unfolds, a little yellow, fruit-like body, united to it by a point at its base. Examined through a microscope, this little appendage looks like a golden pear. When the leaf first unfolds the little pears are not quite ripe, and the ants are continually employed going from one to another examining them. When an ant finds one sufficiently advanced it bites the small point of attachment; then, bending down the fruit-like body, it breaks it off and bears it away in triumph to the nest. All the fruit-like bodies do not ripen at once, but successively, so that the ants are kept about the young leaf for some time after it unfolds. Thus the young leaves are always guarded by the ants; and no caterpillar or larger animal could attempt to injure them without being attacked by the little warriors." Thus, as Mr. Belt very aptly puts it, "the ants are really kept by the acacia as a standing army to protect its leaves from the attacks of herbivorous mammals and insects."^[81]

As for the honey or honey-pot ants, they were first heard of in America, and various floating stories, which seemed more or less hard to credit, having got into circulation about them, without there being any positive knowledge to check them, Dr. McCook, to remove this grave reproach to transatlantic entomology, started off one day to observe them. He soon found that the main fact which had been stated was correct, viz. that a certain sect or caste of these ants, disregarding the Italian warning, were in the habit of making themselves all honey, to be swallowed in consequence by the rest of the community. These are the so-called honey-pots, and so well do they deserve their name, that when full the abdomen becomes almost perfectly circular, like a glass globe, and so enormously swollen that the body in proportion to

it is like a grain of wheat stuck into a cherry or gooseberry.^[82] The legs dangle towards the ground, but hardly, or only by a great effort, reach it, and in this last state of distension the insect may find it impossible to get about, though as a rule by dragging or pushing herself along sideways, she is able to do so to a certain extent. These honey-jars have special chambers for their accommodation, and here they hang in clusters from the roof, awaiting the visit of any worker, who upon signifying his wants—it would seem after climbing up to them—is fed, after the ordinary ant manner, by regurgitation. In the same way the honey-bearers are themselves filled, or more properly speaking, feed themselves, since the mouth arrangement, in spite of the direction in which things seem hastening, has not yet become so simple as in the case of a real jar.

The honey which the rotunds, as McCook calls them, receive from the workers is gathered at night, and is obtained almost entirely from the galls of oak trees, which, when pierced by the ant's mandibles, exude a white transparent liquid in minute globules. This is greedily licked up by the ants and distributed by them after the return home, not only to the rotunds, but to such of their fellow-workers as may not have taken part in the expedition.^[83] The honey thus obtained is pleasant to ant and human taste alike, and the Indians of New Mexico, as no doubt elsewhere, obtain it by the simple process of squeezing the insect—breaking the honey-jar, as one may say. They also make from it a fermented liquor having intoxicating powers, so that one need not wonder that the idea of farming the honey-ant, like the honey-bee, has been seriously discussed in the United States. McCook, however, has pointed out that “the limited quantity of the product would prevent a profitable industry,” and he adds: “Besides, the sentiment against the use of honey thus taken from living insects, which is worthy of all respect, would not be overcome.”^[83] Personally I think it would be overcome, and pretty quickly, too, as are most other sentiments that stand in the way of pleasure or profit. Women would get it under first, as in the case of birds, seals, etc., and the world would soon follow, with “woman's influence” upon its lips. But let me not be unjust. I do not believe in sentiment as a working force in the case at all. If the ants are not to be squeezed it will be on commercial considerations.

That the worker-ants—and for that matter the others also—are extremely fond of the honey so curiously stored by them, will be easily believed, and

an unpleasant illustration of their greediness in this respect was often observed by McCook when capturing a nest. The swollen bodies of the rotunds, on these occasions, were sometimes unavoidably ruptured, whereupon such workers as happened to be near these unfortunates, forgetting their alarm, which had hitherto been great, and the ruin and confusion all around them, paused in their flight, or aimless movements, and greedily lapped up the overflowing honey.^[84] It is all the more interesting, therefore, to learn that when the “little life” of these poor honeypots is at length “rounded with a sleep,” their contained treasure, though so easily obtainable, goes with them to the grave, the idea of opening the full crop, and imbibing the contents, never seeming to occur to any ant. This is all the more remarkable in that the workers, when they recognise that life is extinct, carefully separate the abdomen from the thorax by sawing through, with their mandibles, the little connecting stalk called the petiole. The two parts are then removed separately, that representing the head half being carried, whilst the “golden bowl” of the body “unbroken,” though with “the spirit fled for ever,” is rolled along the various chambers and galleries of the nest, till it finally finds a resting-place in the cemetery just beyond its precincts.^[84] To what are we to attribute the non-utilisation of the honey in the dead body? Even were it possible that the ants could forget that it was there, they cannot be unconscious of what must be smelt, as well as seen, through the semi-transparent walls of the abdomen. Some feeling must restrain them—what, I am not prepared to say in a work which does not aim at being scientific.

Here, then, we have one most suggestive illustration—“suggestive,” I think, is a very useful word—of the funeral habits of ants. Many others could be instanced, but I will end this chapter, and small account of ant doings, generally, with the following extract from the *Proceedings of the Linnæan Society* (1861). The observer was a Mrs. Hutton, of Sydney; and Romanes, who quotes her account in his *Animal Intelligence*, remarks that though she is not a well-known observer the facts reported were such as scarcely to admit of a mistake. Personally, I attach no weight whatever to anybody’s not being known as an observer. Want of leisure, or unpropitious circumstances generally, must prevent large numbers of people from seeing what they would be very well able to note accurately if they did, or from recording what they do see; whilst, on the other hand, leisure, joined to taste

in a certain direction, makes many a quite average observer known as a good one. A good observer, in fact, is rather one who is always keeping on, and does not weary, than one who can see a single salient thing more plainly than most other people; and, again, it is easy to set a fictitious value merely on being before the public.

Having thus defended Mrs. Hutton, I proceed now to quote her account: “I saw,” she says, “a large number of ants surrounding the dead ones” (soldier ants which she had herself killed and left lying on the ground some half-hour previously), “and determined to watch their proceedings closely. I followed four or five that started off from the rest towards a hillock a short distance off, in which was an ants’ nest. This they entered, and in about five minutes they reappeared, followed by others. All fell into rank, walking regularly and slowly, two by two, until they arrived at the spot where lay the dead bodies of the soldier ants. In a few minutes two of the ants advanced and took up the dead body of one of their comrades; then two others, and so on, until all were ready to march. First walked two ants bearing a body, then two without a burden; then two others with another dead ant, and so on, until the line was extended to about forty pairs, and the procession now moved slowly onwards, followed by an irregular body of about two hundred ants. Occasionally the two laden ants stopped, and laying down the dead one, it was taken up by the two walking unburdened behind them, and thus, by occasionally relieving each other, they arrived at a sandy spot near the sea. The body of ants now commenced digging with their jaws a number of holes in the ground, into each of which a dead ant was laid, where they now laboured on until they had filled up the graves. This did not quite finish the remarkable circumstances attending this funeral of the ants. Some six or seven individuals had attempted to run off without performing their share of the task of digging; these were caught and brought back, when they were at once attacked by the body of ants and killed upon the spot. A single grave was quickly dug, and they were all dropped into it.”

“Prodigious!” as Dominie Sampson would have said, and certainly I think this is one of the most remarkable observations upon ants that has ever been made. As far as the burying is concerned, it has been corroborated by the Rev. W. Farrar White, who, at the same time, corroborates Pliny; but how strange are all the circumstances! What was it, one wonders, that made just a few of the crowd shirk their share of the labour—for this is not like ants.

Some strange, uncanny feeling in connection with the dead bodies may be suspected; but seeing that, as the Russian proverb truly says, “Another man’s soul is darkness,” it is not very likely that we shall ever know what ants feel.

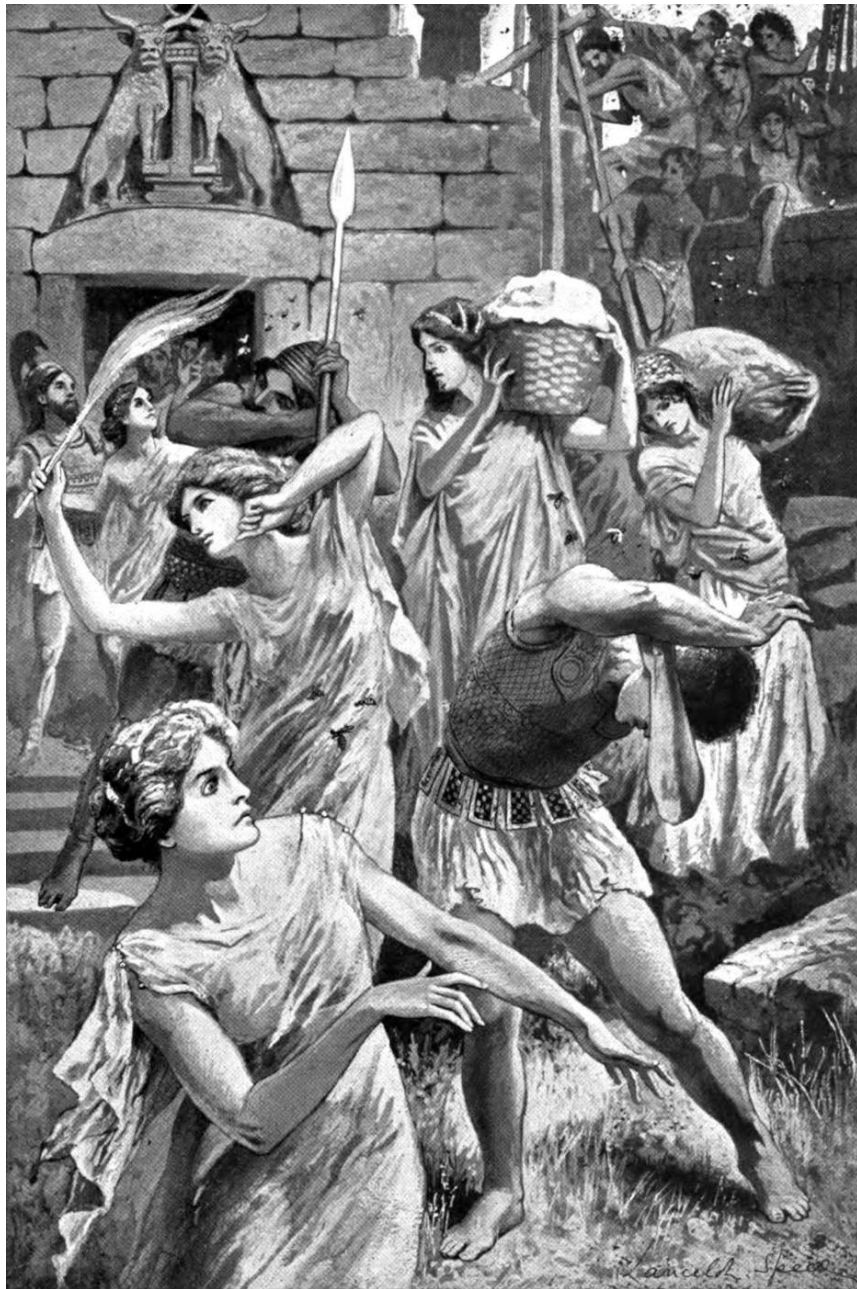
One interesting question is suggested in this connection, though I have never known it raised yet. Two views of what ants are, excluding compromises, may be taken—the automatic one, tempered with “psychic plasticity,” of Professor Wheeler, and that formed by Mr. Belt, who, having fully satisfied himself—from the keenest observation, be it remembered—of their reasoning powers and capacities, remarks, “When we see these intelligent insects dwelling together in orderly communities of many thousands of individuals, their social instincts developed to a high degree of perfection, making their marches with the regularity of disciplined troops, showing ingenuity in the crossing of difficult places, assisting each other in danger, defending their nests at the risk of their own lives, communicating information rapidly to a great distance, making a regular division of work, the whole community taking charge of the rearing of the young, and all imbued with the strongest sense of industry, each individual labouring not for itself alone, but for all its fellows, we may imagine that Sir Thomas More’s description of Utopia might have been applied with greater justice to such a community than to any human society.”^[85] Now, if Belt’s view be the correct one, or if the evidence in favour of it be at all strong, is it not time for us to ask ourselves, merely as a moral problem, how far we, in our clumsy and imperfect human state, have a right to kill ants and tumble Utopia to pieces, simply for our amusement, intellectual or otherwise? Ought we to do this? Or ought we, like a lady who lives in America and writes to very scientific papers, to imprison queens who do no harm, and make ourselves learned at the expense of one, or both of their antennæ, during the term of their natural lives? However simply and sweetly we may talk of this, however much true womanly feeling may enter into the narrative, nay, even though we give the queens pet names, is it really right?

CHAPTER XVIII

Bees and wasps—A bee's masonry—What happens to caterpillars—Living food—Variations in instinct—A wasp's implement—Unreal distinctions—A cautious observer—Bees that make tunnels—A wonderful instinct—Leaf-cutting bees—Nests made of poppy-leaves—Born in the purple—Commercial philosophy—The appreciative white man—Economy of labour—Bees and rats—Busy shadows—A bee double.

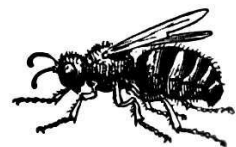
THE consideration of ants naturally leads to that of bees, but of the life and doings of the hive-bee—made common now in a hundred practical treatises and bee-keeper's manuals—it is not the design of this little book to treat. Wasps are less written about, but even here, in a work which can only deal with a very few insects out of a very great many, a choice may be permitted one, so I will merely observe, in regard to the common species, that in my opinion wasps are much less irascible than bees—in fact, quite good-natured compared to them—but at the same time, owing to their room-entering, table-pillaging propensities, much greater nuisances, so that they deserve stern treatment, but a more charitable estimate of their character. Hornets, again—which seldom offend in this way—appear to me to be very peaceable insects, as though, wielding a mighty weapon, they felt that they had no need to use it except on “a striking emergency.” Such a definition would apply to the running of a stage-coach, diligence, omnibus, waggon, etc.—in fact, any large vehicle—into their nest on the highway, in which case the consequences, one may well believe, would be appalling. Never having been in such a position myself, and being without trustworthy information on the subject, my powers of description are useless here, but there is a way of dealing with this emergency also. This reminds me, however, of an account which I have read somewhere or other of hornets having once stopped a Roman army. This may seem surprising nowadays, but we must remember that in classical times armies did not possess artillery. There is therefore nothing invidious in the opinion which I here express, that however much they may have stopped the Romans, they would never stop the Japanese.

In both ants and bees we find solitary and social species, so as in ants we have been considering the latter only, we will now reverse the process with bees. There are many interesting species of solitary bees, but it must be premised that the word “solitary” is to be understood here in a special rather than in a general sense. As far as mere numbers are concerned, there is often a large community of bees building their cells in close proximity all at the same time, but each builds its cell for itself alone, or rather for its family—no one thinks of helping its neighbour. There is no co-operation, in fact, and that makes all the difference. It would be all the same to every one of the bees that are building so close together if all the rest went away and left it to work alone. And yet we cannot even quite say this, because, in one case, at any rate, though every individual bee makes its own cell and thinks only of that and of its own family, yet, when all the cells are finished, the whole community join in making one mud roof over the whole of them. By this we see how difficult it is to find quite separate places for allied animals, and how the habits of one are apt to slide gradually into those of another. Still, we must do the best we can, and take words as we find them, remembering that the locusts, as already explained, do *not* belong to the *locustidæ*.



DRIVEN OUT BY HORNETS.

Aelian, in his "Natural History," says that a city in Crete was attacked by such a plague of hornets that the inhabitants were driven to abandon it, and build a new city on another site. A hornet is shown to the right of this inscription.



Amongst the best known of the solitary species of bees are the Carpenter Bees, the Carding and Tapestry Bees, and the Mason Bees. Of the latter a great French observer, who, though he lives now, belongs really to the days of Réaumur and Swammerdam, has something to tell us. Speaking not of Réaumur's maison bee—that “splendid Hymenopteron with its dark violet wings and costume of black velvet”^[86]—but of a smaller species—*Chalicodoma sicula*—he says: “You should see the active bee at work when the road is dazzling white in the hot sunshine. Between the neighbouring farm where she is building and the road where the mortar is prepared there is a deep hum of the bees perpetually crossing each other as they come and go. The air seems traversed by constant trails of smoke, so rapid and direct is their flight. Those who go carry away a pellet of mortar as big as small shot: those who come settle on the hardest and driest spots. Their whole body vibrates as they scratch with the tips of their mandibles and rake with their forefeet to extract atoms of earth and grains of sand, which, being rolled between their teeth, become moist with saliva, and unite. They work with such ardour that they will let themselves be crushed under the foot of a passer-by rather than move.”^[86] Then comes the making of the actual nest, or little collection of cells. “After choosing a boulder,” says Fabre, “she comes with a pellet of mortar in her mandibles, and arranges it in a ring on the surface of the pebble. The forefeet, and, above all, the mandibles, which are her most important tools, work the material, which is kept plastic by the gradually disgorged saliva. To consolidate the unbaked clay, angular pieces of gravel as large as a small bean are worked in singly on the outside of the still soft mass. This is the foundation of the edifice. Other layers are added, until the cell has the required height of three or four centimetres. The masonry is formed by stones laid on one another and cemented with lime, and can stand comparison with our own. Layers of mortar sparingly used hold them together. The cell completed, the bee sets to work at once to store it. The neighbouring flowers, especially those of *Genista scorpius*, which in May turn the alluviums of the torrents golden, furnish sugared liquids and pollen. She comes with her crop swelled with honey, and all yellow underneath with pollen dust, and plunges head first into the cell, where for some moments one may see her work her body in a way which tells that she is disgorging honey. Her crop emptied, she comes out, but only to go in again at once, this time backwards. With her two hind feet she now frees herself from her load of pollen by brushing herself underneath. Again she

goes out, and returns head first. She must stir the materials with her mandibles for a spoon, and mix all thoroughly together. When the cell is half full it is stored; an egg must be laid on the honey paste, and the door has to be closed. This is all done without delay. The orifice is closed by a cover of undiluted mortar, worked from the circumference to the centre. Two days, at most, seem required for the whole work.”^[86] Afterwards several more cells—making a continuous group of from six to ten—are added, and when all is completed, the mason bee “builds a thick cover over the whole group, which, being of a material impermeable to water, and almost a non-conductor, is at once a defence against heat and cold and damp. This material is the usual mortar, made of earth and saliva, only with no small stones in it. The nest is now a rude dome, about as big as half an orange; one would take it for a clod of mud flung against a stone, where it had dried. Nothing outside betrays its contents—no suggestion of cells, none of labour. To the ordinary eye it is only a chance splash of mud.”

Of course, when the eggs are hatched, the bee larvæ feed on the stored pollen and honey, a pleasing picture which suggests another something like it, though not altogether the same. I allude to certain species of solitary wasps, which, urged by the same feelings of maternal solicitude, choose a living caterpillar, grasshopper, spider, etc., for the future sustenance of their young. Take, for instance, *Ammophila urnaria* of North America, whose habits in this respect have been carefully studied. This wasp is about an inch long, with very long legs, and a waist even exaggeratedly wasp-like. It is black in colour, but with a red mark running round the fore part of the abdomen. At the proper time she—for, of course, we are dealing with the female—may be seen running about the ground, and eagerly searching the various plants and grasses that come in her way. Occasionally, as though in lightness both of heart and body, she gives a leap off the ground, and at other times will fly up from it more deliberately, to make an examination of some overhanging leaf. At last, as a result of these little aerial excursions, let us say, she knocks down a certain green caterpillar of the kind wanted, and with maternal devotion full upon her, at once sets to work. The caterpillar, however, though taken by surprise, and assaulted the instant it has touched the ground, resists strenuously, as though instinctively knowing, and highly disapproving of, the fate in store for it. It is larger and more bulky than the wasp, and its contortions are so powerful that the latter

is several times repulsed in her assaults. She is not discouraged, however, but continues perseveringly to fly at the caterpillar, till at last she takes it at a disadvantage, possibly in a moment of weariness, and alighting with her long legs on each side of the large, soft body, seizes it by the neck with her mandibles, and holds it fast. Now the caterpillar, stimulated doubtless by the painful, or at least unwelcome nip, struggles with redoubled energy; but it is beneath its oppressor, who, straddling over it and never relaxing her grasp, lifts it at last, with an effort, a little from the ground, and inserting her curved abdomen like a fish hook beneath it, strikes in a more effective and certain way than did ever the most benevolently contemplative member of all the fishing fraternity. The result is instantly apparent, for with the entry of that deadly sting into its body, all struggles on the part of the caterpillar cease, and it lies a living corpse at the feet of its cruel oppressor. The latter, after remaining still for some moments as though to give her victim time to realise and appreciate its situation, stings it again and then again, each time choosing, as she has done before, for the locality of the operation, the junction of two out of the dozen or so segments into which the long length of the caterpillar is divided. Then she flies up, but after circling a little above the scene of her triumph, she descends again, and gives her victim, though now helpless and paralysed, a taste or two more of her quality. The first part of her business is now done, and well done. She has earned a rest, or rather she may exchange one form of activity for another. Accordingly she proceeds to indulge in the pleasures of the toilette, and it is not till this is completely finished that she flies with, or drags, her victim to the neat little burial-place, representing also her future nursery, which she has already provided for it.^[87]

The above illustration is taken from the account of a particular case which fell under the keen observation of G. W. and E. G. Peckham, two well-known American entomologists. On other occasions, however, this wasp—that is to say, various individuals of the same species—besides stinging the caterpillar, went through another and more curious process. This consisted in biting and squeezing the anterior upper portion—the neck as we may call it—of their victim.^[87] The same operation was also observed by Fabre when he watched his good mothers, but though I have called it biting and squeezing, that is not the right term for a *savant* to employ. *He* calls it malaxation, which, perhaps, means doing both at the same time. Biting,

however, would seem to imply no less, but, perhaps in order to bite scientifically, it is necessary to take a piece out, or at least to make the blood come, though in common parlance this does not, or did not, hold good, since Sampson bit his thumb at Abram and Balthasar, in the first scene of *Romeo and Juliet*, but it cannot be supposed—nor does the context support such a view—that he bit it so hard as that. Malaxation, however, let it be; but why such a process on the part of the wasp should be necessary it is not easy to see, since the mandibles are not poisonous like the sting, and the latter is all in all sufficient to produce the paralysis required, as is apparent in the instance already given, where the sting alone was employed. To me it seems possible that this malaxation may be a happiness to the wasp merely, as the shaking of a rat certainly is to a terrier, whatever other advantages accrue from it. That insects, like other animals, including man—who, indeed, is the crowning instance—take a savage pleasure in overpowering and killing their prey, I have myself very little doubt.

We have seen that this wasp stung the caterpillar between the segments of its body, and, as we will assume—for the result seems to warrant the inference—in the central part of it, so that the sting, entering the great nervous cord or ganglion, which is situated in this region, with little swellings at each of the segmental rings, produced the described paralysis. It was Fabre's view that this must always be the case, and he thought likewise, in accordance with his own observation, that the caterpillar received a sting at the junction of all or nearly all the segments of its body. Otherwise it would be imperfectly stung, and in consequence not sufficiently paralysed to prevent its struggling, and so detaching the young larva, or perhaps the egg, which, as it would seem, is laid on, and not inside, the body of its living provisions. On the other hand, were the caterpillar stung too severely, so as to be killed outright, the grub when hatched would only have putrid meat to feed upon, and this again, it was assumed, would be fatal to its existence. On these grounds Fabre concluded that we had here an instinct which must have been perfect from the beginning, since as anything short of such perfection would be followed by the death of the larva, those gradual steps by which, on the theory of natural selection, all excellence either of structure or instinct has been attained, could not in a case like this have had any existence.

But all this has been exploded by subsequent observation. What Fabre saw he knew, but in all that he inferred without seeing he was entirely mistaken. As observed by the Peckhams, a caterpillar may be either stung so slightly as to be quite lively, and yet not succeed in shaking off the wasp larva hatched on its body, or so severely as to die almost immediately, yet without detriment to the larva who feeds on its discoloured and more or less putrified body, with the same gusto, and apparent benefit, as though it were warm with life.^[87] Thus the question seems not so much to be, how can such perfection of instinct as was observed by Fabre have been attained through the process of natural selection, as why it should have been attained; or perhaps we may even go further and ask if this supposed perfection exists at all, and whether Fabre did not deceive himself. A wasp having secured a caterpillar is, of course, at liberty to sting it as often as it pleases. Why, then, should one wasp behave quite like another one in this respect? Here, as elsewhere, there would be some amount—perhaps a considerable amount—of variation in individual disposition, and wasps of milder or less savage mood would sting less frequently than their fiercer fellows. There might, therefore, as it appears to me, be a large amount of fluctuation both in the number and degree of severity of the stings—if indeed there is any regulation in this respect—and also in the consequent injury to the caterpillar or other insect, without any particular scope being offered for natural selection to play a part. What room, indeed, for such a force can there be if it makes no difference to the wasp-grub whether the caterpillar which is to be its food, is stung badly or slightly, or whether it lives or dies?

That this is really the case seems to be implied in the following paragraph which I quote from the same interesting work that I have before referred to^[87] “The conclusions that we draw from the study of this genus differ in the most striking manner from those of Fabre. The one pre-eminent, unmistakable, and ever-present fact” (the invariable fact, as one might say) “is variability. Variability in every particular—in the shape of the nest and the manner of digging it, in the condition of the nest (whether closed or open) when left temporarily, in the method of stinging the prey, in the degree of malaxation, in the manner of carrying the victim, in the way of closing the nest, and last and most important of all, in the condition produced in the victims of the stinging, some of them dying and becoming ‘veritable cadavers,’ to use an expressive term of Fabre’s, long before the

larva is ready to begin on them, while others live long past the time at which they would have been attacked and destroyed if we had not interfered with the natural course of events. And all this variability we get from the study of nine wasps and fifteen caterpillars”!^[87] Fabre’s ideas therefore seem totally disproved, but though natural selection—the counter-theory to his own—has no doubt produced the *Sphex* and *Ammophila*, with their habit of stinging and storing caterpillars, to serve as food for their young, it does not follow that it has done anything more than this; for though variation be the stuff in which natural selection works, it need not always work in it—any more than a tailor need always make clothes because there is an abundance of cloth.



SOLITARY WASPS

In the upper part of the picture a solitary wasp is seen attacking a caterpillar on a leaf. Beneath is another of the same species busy pounding the entrance to its burrow with a pebble.

In the digging and closing of her burrow—her nesting-habits, as we may call them—our *Ammophila* is almost as interesting to watch as in her mode of proceeding with caterpillars, though here a certain well-known stimulus

to human enjoyment which I need not enlarge upon is wanting. Having found a convenient spot for her nursery, she digs, with her mandibles and front pair of legs, a little tunnel in the ground, to about the length of her own body, and at the end of it hollows out a round chamber or cavern just large enough to make comfortable quarters for a pair of invalid caterpillars—a hospital for incurables, we may call it to begin with, but soon to become their tomb. Having dug to about her middle, the wasp backs out, with a little pellet of collected earth held firmly in her mandibles. With this she flies to a little distance and then, letting it drop, alights on the ground and takes a little rest before returning to continue her work. She may either fly or run back, for her legs are as highly developed as her wings—she is in fact a very perfect athlete. The process of excavation is now continued, there is more burrowing, more flying away with the earth dug out, and before long the nursery-vault is completed. The next thing is to find caterpillars for it, but before flying away to look for one, *Ammophila* carefully conceals the entrance to her tunnel with pellets of earth, which she often brings from a distance, and will not be satisfied with unless they seem well adapted for their purpose. At last, when the aperture is both blocked and hidden, she starts off upon the still more important undertaking which has been already described, and after a longer or shorter interval—if her quest is successful—returns with a nicely stung caterpillar. As two are required there must be another journey and another stopping up of the burrow, before the final one, which is of a more solid nature, occupying, sometimes, as much as twenty minutes. In thus bringing her labours to a conclusion, *Ammophila* often shows a wonderful degree of intelligent foresight—foresight we must term it if we admit the intelligence; for sometimes she will drag a leaf over the entrance to the tunnel, though now filled in, or taking a stone as large as her head in her mandibles, will pound down the earth with it to make it firm and compact.^[87]

It used to be said—and may be still by that large class of people who are for ever making false parallels and artificial distinctions—that man was the only animal that made intelligent use of an instrument, but Darwin instanced a monkey cracking a nut with a stone, and an elephant breaking off a bough to fan itself with. Here, in an insect, we have a case which is perhaps even more to the point—more extraordinary, that is to say; for certainly the idea of flattening and pressing down earth over a general

surface, and of taking something to do it with, seems a little less obvious than that of cracking a nut, in a similar manner, and therefore to require more thought in the planner of such a process. No wonder that the delighted witnesses of this interesting fact flung themselves on the ground on each side of the unconscious inspirer of their wonder, in order to have a better sight of it; but that a previous observer of the same thing should have waited a year before publishing what he had seen, because he feared such a statement would not be believed,^[87] is to my mind a display of prudence almost as wonderful, though not nearly so edifying, as that of *Ammophila* herself. If we are not to make known what we see, because people who believe in their own and nobody else's eyesight are not likely to credit it, how is evidence to accumulate for the benefit of the more intelligent part of the community? It is only of this small minority that we should think, or, rather, we should not think of anything but the truth, where truth is concerned.

Returning to bees of the solitary kind, "the operations of the wood-piercers," says Bingley, "merit our careful attention." They shall have it for a moment, but space is against them. However, the female of the species, *Xylocopa violacea*, which for some reason is disliked by householders, bores in the springtime, by the aid of her strong mandibles alone, neat little circular tunnels in such objects as garden-seats, gates, front doors, arbours, window-shutters, rustic tables, and the like. At first, it is stated, she "bores perpendicularly, but when she has advanced about half an inch she changes her direction, and then proceeds nearly parallel with its sides for twelve or fifteen inches. If the wood of the seat, door, table, etc., be sufficiently thick, she sometimes forms three or four of these long holes in its interior, a labour which for a single insect seems prodigious, and in the execution of it some weeks are sometimes employed. On the ground, for about a foot from the place in which one of these bees is working, little heaps of timber-dust are to be seen. These heaps daily increase in size, and the particles that compose them are almost as large as those produced by a hand-saw."^[88] When the tunnels are finished, the mother bee divides them into some dozen little rooms about an inch deep, making the divisions of wood-dust, which she cements together by aid of a glutinous secretion with which she is furnished. Before each cell is closed it is filled with a paste composed of the farina of flowers, mixed with honey (it makes one envy the grub), and

an egg is deposited in it. As each cell takes some time to make and provision, it is obvious that the egg in the lower one will hatch a little sooner than the one above it, and so on right up to the top. If, therefore, any one of the larvæ “were to force its way upwards, which it could easily do, it would not only disturb, but would infallibly destroy all those lodged in the superior cells.”^[88] Here, however, natural selection (called Providence in Bingley’s time) steps in, and “has wisely prevented this devastation, for the head of the nymph (chrysalis), and consequently of the emerging bee, is always placed in a downward direction.”^[88] Of course, therefore, the insect moves forward in the direction towards which it looks at birth—its new birth, that is to say; and, moreover—this is the really astonishing thing —“the mother digs a hole at the bottom of the long tube, which makes a communication between the undermost cell and the open air. By this contrivance, as all the bees instinctively endeavour to cut their way downward, they find an easy and convenient passage, for they have only to pierce the floor of their cells in order to make their escape, and this they do with their teeth very readily.” As regards this communicating passage, however, it is presumably a mere continuation of the tunnel, and was probably once occupied with cells, like the upper portion. Why the lower end of the tunnel should have come in time to be left empty—what advantage, that is to say, its being so represents—it is not easy to see, but possibly it stood in danger of being reached by a bird’s beak, through what has now become the exit of escape merely.

As there is a leaf-cutting ant, so, too, there is a leaf-cutting bee, but here the resemblance ends, since no thought of food or fungus enters the mind of the latter insect. Her more simple and direct object is to make the severed leaves into cells, and this she does with wonderful skill and ingenuity. The cells, which are made by rolling the pieces of leaves round within a tunnel or gallery, previously excavated in the earth, are separated from one another by a circular piece, which fits into the tube with extreme nicety, making at once the ceiling of each lower compartment and the floor of that above it. As each is finished the bee, as in the other instances, fills it with a mixture of honey and pollen, upon which she then lays an egg, and finally closes the mouth of the tunnel. The leaf principally used for the manufacture of these pretty cradles is the rose leaf, and the paste which fills them is of a rose-red colour, owing to the pollen having been collected for the most part from thistles.

The children of the leaf-cutting bee, therefore, are delicately housed, but what are they, in this respect, to those of the poppy or tapestry bee, who are born like the Byzantine princes in a “purple^[89] chamber” made of the rich leaves of poppies? Here they lie, or crawl, in state, one to each royal apartment, which is filled, almost to the brim, with the sweet food of bees. Yet when they come forth, at last, it is not as gorgeous imperial creatures clothed in “purple and pall,” but only little ordinary-looking black bees covered all over with dirty grey hair. And their beautiful purple poppy chamber has been seen by no one—not even by themselves probably—buried as it is full three inches deep in the earth. Wallace, somewhere in his *Malay Archipelago*, moralises over the beautiful little kingbird of paradise, sparkling out its life amidst the forest solitudes of a remote island, unseen by human eyes, save those of savages, till once or twice, perhaps, in a century, some wandering white man, who alone is capable of appreciating its beauty, comes to bang its life out and bear away its skin. The thought of this dainty little crimson-tapestried bower lying in black darkness, like a grave beneath one’s feet, rouses a similar train of reflection in the mind, but perhaps it would do so more strongly were it associated in the same degree with ideas of sport or profitable collecting.

The carding-bee is interesting, not so much for the nest which it makes, as for the wonderful way in which it makes it—or, to express it more justly, it is more especially interesting on this account. Having either made or found a suitable cavity, these bees under-roof it with a thick thatching of moss. To carry this, bit by bit, to the place, would take them a very long time, so, instead of doing so, they stand one behind another, with their backs toward the nest, in a line that reaches from the moss to its entrance. The furthest bee then pulls out a piece with her mandibles, cards it with her fore feet, and, with the others, passes it on, beneath her body, to the second bee, who passes it to the third, the third to the fourth, and so on, all down the line, the last bee entering the nest with it. Thus these bees do with moss exactly what rats have been seen to do with eggs, when transporting them to their burrows. A most interesting anecdote of this is quoted by Romanes,^[90] from Jesse’s *Gleanings*, but it does not appear to be in my edition, which I had thought was a complete one. All I can find is a bare reference to their having been known to “convey” eggs from a box, in this way—“convey the wise it call.” This is an annoying discovery, for I detest all selections, not

made by myself, from Palgrave's *Golden Treasury* to any man's "Hundred Rest Books." But if my edition is a complete one, then the thing should be looked into, for the anecdote quoted by Romanes is not there—at least it is untraceable through the index.

Like ants, bees are subject to parasites, and as some belonging to the former are ants themselves, so with bees we have the same thing, but developed to a still more striking extent; for there is no ant that I know of that lives with another which it so closely resembles that the latter is unable to distinguish it from itself. Such bees, however, there are. Some of our humble bees, for instance, go through life thus attended by a double whose existence it never for one moment suspects. The two, indeed, are linked in the closest bonds of social intimacy. Together they leave the nest, together they fly from flower to flower, together they re-enter it. Together, too, they seem to glow in industry—to emulate each other's toil. But all the while that the true industrious bee is collecting pollen and nectar the double is only pretending to do so—or rather, let us say, seeming—and whilst the former bustles about, feeding the larvæ and making the cells, the latter only bustles about, like the shadow of a busy person on the wall. But when the true bee lays an egg, the double lays one too, almost at the same moment, and in the very same cell. Both are then hatched, together, and the two larvæ grow up in the same cradle, nourished by the same food, make their transformation side by side, and so creep forth into the nest. The two, as I say, are hardly to be distinguished one from another, yet all the while one is a real true-hearted humble bee, and the other a mere show, a stage make-up, an outer shell without any of the proper qualities inside. And the best of the joke is, that, probably, the false or cuckoo bee, as it is called, is as much deceived as its foster relatives, and imagines itself a good honest sterling member of the community. It is forced by nature to cheat, but the fraud is unconscious, and the impostor is imposed on in its turn.

Thus in the insect world we have something which can only be brought about, amongst ourselves, through a conscious disguise, by means of wigs, false moustaches, etc.—what we call an impersonation—but here is a life-long impersonation which costs the "born actor" no trouble. Why is this? What is the meaning of it? Why should one bee—or any other insect or creature—look just like another one, and yet have a Latin name of its own, which the other has no right to? Why should the individuals of one species

be hardly more like each other than they are like the individuals of another species, even though—as is often the case—these two species are widely separated in the system of nature? Such are the questions to which a consideration of these cuckoo-bees, as they are called, give rise. They will be answered, if at all, in the following chapter.

CHAPTER XIX

Natural selection—Protective resemblances—A locust's stratagem—Mock leaf-cutting ants—Flowery dissemblers—A Malay explanation—Snake-suggesting caterpillars—A prudent lizard—Inconclusive experiments—A bogus ant—Flies that live with bees—A caterpillar that dresses up—A portrait-modelling caterpillar.

EVERYBODY knows nowadays how all the different species of animals and plants, living and extinct, have come into existence. It was quite simple. All they had to do was to keep on varying. Some of them varied in a way that was good for them, some in a way that was bad. The latter died, but the others increased and multiplied, and as the process was always going on, and it is impossible to vary long without becoming changed, it happened that creatures which had started with a certain appearance got in time to have quite another one, so that they would not have been recognised by the people who used to know them, if these same people had kept alive. However, as the process was so slow that it took millions of years, and is still going on, awkward things of this sort never happened, and so, as nobody had ever seen one species of animal change into another before their eyes, they found it difficult to believe that they ever had done so; for the ordinary person says "seeing's believing," though he believes in all sorts of things that neither he nor anyone else ever has seen, or is ever likely to. Still, for all that, he thinks his own eyesight must be better than anyone else's.

This process of eternal change, with the changes for the better surviving, and those for the worse dying out, is what is called natural selection, and if we understand it—as there are few now who do not—we can understand this, that if any kind of creature is so strong and formidable that it would be an advantage for weaker creatures to be mistaken for it, then it is not at all unlikely that some of these weaker creatures will get more and more to resemble it, until at last they are so mistaken. For instance, our common wasps, who are armed with a formidable sting, and are very skilful in using it, are not attacked by any other insect, excepting hornets, which are not

common. Any fly or moth, therefore, that resembles a wasp will be generally left alone, and the more so the more it resembles it. Accordingly we do find flies and moths that look very like wasps, and live safely in consequence. Still more would it be of advantage to look like a hornet, and there is a moth so like one that it is called the Hornet-clear-wing.

On this same principle of being mistaken for something that is safe from attack or annoyance, all sorts of animals, and in a special degree insects, have come to look like various objects around them, and amidst which they live, such as stems of grass, pieces of moss or stick, leaves, flowers, and so on—some of the resemblances being more special and extraordinary. These are things which, though the eye may see, it does not as a rule dwell upon, because there are so many others round about. Who, for instance, would look at any particular blade of grass? So a bird that would pounce down upon an insect that it saw moving amongst the leaves of a tree, if there was no doubt that it was an insect, would not take any pains to examine what only looked like one leaf amongst many.

Thus, throughout nature we have these curious resemblances of certain creatures to certain other creatures, or to the plants or inanimate objects around them, but it is principally amongst insects that the phenomenon is met with, probably because they increase and multiply so quickly that there has been more time both for the laws of inheritance and for the great controlling one of natural selection to have come into play. Whatever is the reason, there is no doubt about the fact, which will be best illustrated by one or two salient instances.

Ants, though they fall a prey to various animals larger than themselves—such as birds or ant-eaters—yet in their relations with other insects occupy a position of comparative safety, on account of their weapons and pugnacity, and, still more, of their numbers. The driver ants of equatorial Africa, and their South American representatives, the *ecitons*, are indeed, when they set out on their foraging expeditions, the terror, not only of insects, but of all animal life. “Wherever they move,” says Bates, referring to the latter, “the whole animal world is set in commotion, and every creature tries to get out of their way.”^[91] This, however, as they climb trees, and send out encircling columns which enclose a considerable extent of ground, is difficult, or rather, impossible, for all such as cannot fly some distance without alighting; for if an ant or two once seize upon them all is over. An insect,

therefore, that cannot evade the onset of such an enemy is lucky if it has some such means of ensuring its safety as has been above referred to. One at least thus specially favoured inhabits Nicaragua. “I was much surprised,” says Mr. Belt, “with the behaviour of a green leaf-like locust. This insect stood immovably amongst a host of ants, many of which ran over its legs, without ever discovering there was food within their reach. So fixed was its instinctive knowledge that its safety depended on its immovability, that it allowed me to pick it up and replace it amongst the ants without making a single effort to escape. It might easily have escaped from the ants by using its wings, but it would only have fallen into as great a danger, for the numerous birds that accompany the army of ants are ever on the outlook for any insect that may fly up, and the heavy flying locusts, grasshoppers, and cockroaches have no chance of escape.”^[92] This locust resembled a green leaf which, as we have seen, was a very protective resemblance indeed. It might, however, had it been a smaller insect, have resembled one of the ants themselves, and in that case could have run about with them, pretending or appearing to forage, also with perfect impunity. Whether the *Eciton* has such a double I know not, but various ants have. With some it is a spider that assumes their form. With others, as we have seen to a partial extent, a caterpillar, but the *Sauba*, or leaf-cutting ant—which is also the mushroom-growing one—is understudied, leaf and all, by an insect which, though the order to which it belongs has been determined, has not yet apparently received a name. “An example,” says Professor Poulton, “of protective mimicry, which I believe to be more wonderful in its detail and complexity than any which has been hitherto described, was observed and interpreted by my friend Mr. W. L. Sclater, in 1886, during his investigations in British Guiana. Mr. Sclater and his native servant had been collecting insects by shaking the branches of a tree over a sheet. The servant, although described as a very acute observer, saw an insect on the sheet which he mistook for one of the abundant Cooshie ants (perhaps the native name), carrying its little jagged segment of leaf over its back. Mr. Sclater looked more closely, and saw that it was an entirely different insect belonging to the order *Homoptera*. Its length was about that of an ant carrying its leaf. The leaf was represented by the thin flattened body of the insect which in its dorsal part is so compressed laterally that it is no thicker than a leaf” (or as we would say, which along the back is no thicker than a leaf), “and terminates in a sharp, jagged edge. The head and legs were brown, and suggested the

appearance of that part of an ant which is uncovered by the piece of leaf. The jagged dorsal line, when seen in profile, evidently corresponds to the roughly gnawed edge of the fragment of leaf, for Mr. Sclater tells me that the contour of the latter is generally shaped by the mandibles of the ant rather than due to the natural margin.”^[93]

The above-mentioned insect is a dweller in trees, and one might have supposed that a general resemblance to the leaves among which it moves would have been a sufficient protection for it. This probably was the beginning of the deception, which became more complex as time went on. In the leaf-like back of the insect we see probably the original disguise, but as the eyes of birds became more acute they began to pierce through it, more especially when the creature walked. Round about the would-be leaf, however, the leaf-cutting ants—distasteful to the birds that so affected it—were constantly moving and walking. If only it could get to resemble one of these it might be as active as it pleased, and especially if its motions, as well as its appearance, became ant- or ant and leaf-like. And this, indeed, was what gradually began to take place. Variation was always going on, and natural selection was always at hand to mould and shape its results. The two insects were, to begin with, of much the same size, and the general leaf-like appearance of the one was a good basis on which the more particular resemblance to the cut piece of leaf, carried by the other, might be founded. A few deeper washes of brown, some not very profound modifications of contour, and an ant-suggesting legs and body began to appear beneath it. Meanwhile, however, hundreds of thousands—nay, millions—of bad or mediocre copies were swept away, the species became rarer and rarer—trembled, perhaps, on the verge of extinction; but just when it might have appeared to the birds, who were no longer able to obtain a once much-enjoyed morsel, that it really was extinct, it was saved; nature’s object had been gained. A certain number of individuals were left, were close at hand even; even now, at that very moment, one might be crawling on the same twig where a despondent bird sat, only it was not to be distinguished from a leaf-cutting ant. Such are the ways of nature, such the slaughter that attends her victories.

In Borneo, and the Malay Archipelago generally, there is a pretty pink flower known as the “Straits Rhododendron.” Once a gentleman was looking at one of these flowers and admiring it, when all at once it turned

round and stared him in the face. It was not a flower, but a mantis; its flattened legs—pink like them—made the petals; its abdomen, turned up over the back and held thus motionless, resembled an opening bud. “When I held the branch on which the insect had established itself in my hand I could not tell exactly where animal tissue commenced and where flower ended, so perfectly was the one assimilated to the other, both in colour and surface-texture.”^[94] When once established on a flower this mantis would remain there quite motionless, if undisturbed, until it had occasion to leave it; and of course, in nature, had any insect settled on or near it, it would have instantly been seized. The ways of the mantis are well known. “Under a most sanctimonious aspect,” says Fabre, speaking of the little green one of Provence, “it hides the morals of a cannibal”;^[95] and, indeed, the female, which is larger and stronger than the male, will often turn upon the latter and devour it in the very midst of a love-passage. This it does, as in all other cases, by suddenly launching forward one or both of its fore-arms—which have been previously held in an attitude of prayer—and enclosing the body of the victim between their first and second segments, each of which is toothed along the edge like a saw. The double row of teeth meet in the body, which, held aloft, and writhing on either side of the trap, is devoured piecemeal by the mantis, who, with its sharp jaws, tears little mouthfuls out of it as long as it, or its appetite, lasts. This process, made more interesting by the way in which it was brought about, was witnessed in the case of the above-mentioned species. Small flies frequently settled upon it as it sat motionless, flower-like amidst flowers. “These it made no attempt either to drive off or to capture; its motions seeming rather to attract than repel them. After a short time a larger *Dipteron*, as big as a common house-fly, alighted on the inflorescence within reach of the predatory limbs. Then the mantis became active immediately; the fly was seized, torn in pieces, and devoured.”^[96] Such are the real propensities of the seeming flower, and such, too, it may be observed, are those of some actual flowers—to wit, insectivorous ones.

To the Malays, however, whose minds are not yet open to the doctrines of protective or aggressive resemblance, or to evolution generally, this mantis is a flower, they “know not seems.” The blossoms of “the *sendudok*” have become alive, and perhaps some analogies suggested by their own life-experience temper their surprise at such an apparent change of disposition.

They say, too, that few men ever see more than one flower-mantis in the whole course of their lives, so rare a creature is it. In this, no doubt, they are right; yet it would be possible, perhaps, even for a Malay to see several without knowing anything about it. Native eyes are almost always sharper and better than those of the Europeans who come amongst them; but, on the other hand, no native goes about like a modern entomologist, with his eyes specially open in one direction and the possibilities of protective resemblance in his mind.

The same naturalist, during the same expedition, was singularly delighted to secure a larva, whose resemblance to a snake was “so startlingly accurate that I was, for a moment, completely deceived.”^[96] A description follows which, as it is of that kind which deals longly and learnedly in details without producing any particular general effect, may be left out. It would seem, however, that this caterpillar, like many others, has the power of withdrawing its actual head into a fold or two of its skin, which is here so marked that it performs the office of a mask, obscuring and taking the place of the real head thus obliterated. The mask is furnished with two spots, which at once become the creature’s eyes, and both in colour, shape, and general appearance bear a remarkable resemblance to those of a snake; whilst a wrinkled fold, running back on either side from what appears to be the snout, suggests the mouth, and the flattened head with its characteristic arrangement of broad, flattened scales is also indicated by certain markings and colours on the required part of the caterpillar’s body. An apparent head like this, thrown suddenly up as though threatening to dart forward with a hiss and distended jaws, might alarm anyone, and such a mock demonstration is evidently required to give full effect to the disguise. Thus we are told that “when the larva was moving about with the anterior segments well expanded the resemblance to a snake was not so startling; but directly it was touched the terrifying attitude was assumed, the anterior segments being drawn in and the front of the body turned towards the aggressor. When, at the same time, the hinder part of the body was hidden by leaves the deception became complete, and if effective enough to deceive, even temporarily, a human being, it must surely be equally effective in deterring less highly organised and timid foes.”

For the “timid” certainly, but for the “less highly organised” the conclusion does not seem so plain. No sight is better than a bird’s, and it is practice that

makes perfect in any particular direction. Still, unless we suppose the disguise to be accidental merely—and this no one with a knowledge of the whole subject can do—the object of it seems clearly apparent, and we may, therefore, assume that, on the whole, it is successful—to the extent, at any rate, of keeping the species in existence. In such matters, however, there is nothing like practical experiments, if one has the chance of making them, as the finder of the caterpillar in question must have had, since he says, “Unfortunately I was unable to test the efficacy of the disguise, for fear of losing the larva, which I was anxious to rear for the purpose of identification.”^[96] To me this appears a false judgment. Such a test would have been much more interesting, surely, especially if resulting in the way anticipated, than a dry pinned specimen and a Latin name.

Another large snake-resembling caterpillar was found by Bates in the forests of Brazil, and the likeness was sufficiently striking to alarm several people to whom he showed it. But it is not necessary to go so far afield, for here in England, according to Professor Poulton, we have an excellent example of this kind of protective resemblance. This is no other than the caterpillar of the elephant hawk-moth, which by withdrawing its head into its body—just as does the Bornean species—produces a similar false face, with a pair, or, indeed, two pairs of fierce-looking eyes.^[97] This caterpillar feeds on the great willow herb, and when at rest keeps amongst the dead brown leaves at the base of the stem. “As soon,” says Professor Poulton, “as the leaves are rustled by an approaching enemy, the caterpillar swiftly draws its head and the three first body-rings into the two next rings, bearing the eye-like marks. These two rings are thus swollen, and look like the head of the animal, upon which four enormous, terrible-looking eyes are prominent. The effect is greatly heightened by the suddenness of the transformation, which endows an innocent-looking and inconspicuous animal with a terrifying and serpent-like appearance.”^[98]

With this caterpillar, since naturalists know what to call it, and there is no chance of its handing down any of their names in Latin to posterity, it has been possible to make experiments, and on the whole perhaps they have been in favour of the protective resemblance theory. The most interesting one—that I have read, that is to say—was made by Professor Poulton with a full-sized green lizard, and is thus described by him: “The lizard was evidently suspicious, and yet afraid to attack the caterpillar, which

maintained the terrifying attitude in the most complete manner throughout. The lizard kept boldly advancing, and then retreating in fright; but at each advance it approached rather nearer to the caterpillar. After this had taken place many times and nothing had happened, the lizard grew bolder and ventured to gently bite what appeared to be the head of the caterpillar; it then swiftly retired, but finding that there was no retaliation it again advanced and gave it a rather harder bite. After a few bites had been given in this cautious manner, the lizard appeared satisfied that the whole thing was a fraud, and devoured the caterpillar in the ordinary manner.”^[98] Professor Poulton has no doubt as to the lizard having been alarmed at first by the appearance of the caterpillar, and adds that he has never seen one act in the same way on any other occasion; other large hawk-moth caterpillars being eaten at once with entire *sang-froid*. It may be observed, however, that if every lizard were to act in the way recorded, under natural conditions, the advantage to the caterpillar would be nil, since though a species may survive through not being eaten, it certainly will not through being eaten with hesitation. And why should a lizard be more timid in the open air than in a box or a fern-case? Unless we assume, therefore, that this particular one was bolder than most others would be, the result of the experiment was not for, but against, the theory it was designed to test; and since we have no business to assume, the only thing to do is to get more caterpillars, and give them to more lizards. Small birds, however—and this in a country like England is more to the point—seem really to fear these pseudo-snakes to the extent of flying away from them.^[99] But would an ordinary large caterpillar of the *Sphingidæ*—say, of a privet hawk or death’s-head moth—frighten them in the same way? If so, then again we are nowhere.

Perhaps a still more extraordinary instance of protective resemblance than any of the foregoing is that of a caterpillar which pretends to be an ant—one provided with an efficient sting, and of an irritable disposition. Here, as in the snake cases, it is by one portion of the body only that the fraud is perpetrated, but this, instead of being the front, is the hind part, in which, perhaps, it offers a unique example of the sort. The colour of the caterpillar is exactly that of the ant, and whilst its extremity represents the latter’s head, two black spots which are there situated bear an equally close resemblance to the eyes. The jaws are represented by the last pair of false

legs or claspers, which are of disproportionate size, and can upon occasion be stretched widely apart, whilst a number of thin, tentacle-like processes, attached in pairs to the segments of the body, have all the appearance of an ant's legs and antennæ. Armed with these properties, which, however, in a state of quiescence are not very recognisable, the caterpillar waits, as one may say, to have its feelings ruffled, when, by flinging the hinder part of its body into the air, each separate appurtenance begins at once to act the part assigned it, and the whole becomes a startling make-up. The head, with eyes, is jerked from side to side, the jaws gape, the legs move, the antennæ quiver, and an angry, threatening ant starts, as by magic, into being. "When," says Mr. Annandale, "the caterpillar is seen in an end-on position, or when the anterior two-thirds of the body are hidden, the resemblance is positively startling," so that "it is difficult to imagine how a lizard or a frog with a previous experience of the ant could fail to be deterred."^[100]

In the light of the above cases, that of the cuckoo-bees does not seem so very wonderful, since both the species are bees, and all or most of the members of any group or family of animals as a rule bear some resemblance to one another, since they descend from a common and not very remote ancestor. Many flies, however, have almost as close a resemblance to various bees and wasps, whilst one of the latter is even the model for a species of cricket, which would otherwise fall a victim to it and others of its family. There is a beetle, too, so like a wasp, not only in its appearance, but in the way in which it runs about and moves its antennæ, that anyone almost would be taken in. Whether, under this disguise, it enters wasps' nests and preys upon the larvæ, as the bee-like *Volucella* flies enter the nests of the humble-bees they imitate, I do not know, nor, I think, does anyone, but this might very well be the case. These flies, however, now I come to think of it, do not really injure the bees. It used to be the idea that they did, but lately it has been discovered that they are only scavengers, feeding on all the waste products of their hosts, and even on their dead bodies should such opportunities arise.^[101] The humble-bees, on their part, seem to appreciate these services, though we are not entitled to say that they admit the flies into their nests on this account, since they probably do so owing to their likeness to themselves.

Of the walking-stick insects, which are hardly to be distinguished, even with close attention, from the grass or twigs on which they cling, everyone

has heard or read, and the caterpillars, common enough in England, which remain motionless, projecting like a twig from its stem, and looking just the same as one, are almost as good instances of unconscious deception. But neither these caterpillars, nor any of the other insects that have been mentioned, do anything, except through the attitudes they assume, to produce their wonderful disguises. They have nothing to do with the cutting out of the material. They do not dress up for the part themselves. That, however, is what some caterpillars do. There is one, for instance, in Borneo, that has a number of spines arranged in pairs down its back, and on each of these spines it fixes several little buds of the plant on which it is feeding, such buds, and not the leaves of the plant, being the actual food it eats. Consequently the caterpillar, which is quite a small one, looks like a spray of tiny buds itself, and can hardly, by possibility, be noticed amidst its flowery chaplet. The buds are not impaled on the spines, as might be supposed, but are attached to them with silk, which the caterpillar weaves for the purpose, and the whole process of the thing has been observed by the gentleman who gives the account, and who is no less competent a person than the curator of the Sarawak Museum. This is what he says: “A bud would be shorn off with the mandibles, then held in the two front pairs of legs, and covered all over with silk issuing from the mouth of the caterpillar. The caterpillar then twisted the front part of its body round, and attached with silk the bud to one of the spinous processes, and another bud would then be attached to this, and so on until a sufficiently long string—generally three or four buds—was made, when operations on another spine would be commenced. The caterpillar fed on these buds, scooping out the interior, and when not hurried, using the empty shells in preference to whole buds for its covering. When irritated it curled up, and remained thus for fifteen or twenty minutes. At other times it would sway about, looking like a branchlet blown by the breeze.”^[102]

In time this caterpillar made “a silk cocoon covered with buds,” but it never turned into a butterfly, for ants attacked it, and its life was nipped in the bud. It appears to be a very rare caterpillar, and nobody knows what butterfly it belongs to, or what is its full Latin name. Since it is a *Geometer*, however, why not *Geometer ignota* under a sketch (as given in *Nature*, June 25th, 1903), in the cabinet—which would, in all cases, be the better plan?

I really do not know whether this or another caterpillar of South America be the more extraordinary, for if the one makes itself like something, the other makes something like itself. *Anæa* (*sp?*)—I give the name as I find it—is a little green caterpillar having a very funny nondescript sort of shape—as much like a little piece of gnawed-out leaf, left hanging to the midrib, as anything else. Such an object, however, is not one of the common ones of nature, and if it stood alone might be unrecognised or misinterpreted. The caterpillar, therefore, feeding along the midrib of the leaf, gnaws out a number of such little pieces, more or less like itself, and leaves them sticking upright along it, attached by a point or two. All the rest of the leaf at that part of the midrib, it apparently eats, or bites away, so that there remains only the slender, bare stalk, with several bits of leaf upon it, one of which is the caterpillar. To say which bit is he is now very difficult, and it looks as if none of them were. This caterpillar is, of course, green, like the leaf he feeds on, but he is not the same colour all over. He is light above and dark below, and this exactly suits—I have it on authority—the chiaroscuro of the situation, so that, both in light and shadow, he looks for all the world like a little elongated bit of green leaf attached to the midrib by a couple of stalks.^[103] One would say, “Some caterpillars must have been eating that leaf”; but one would never think the caterpillar that had been eating it was still there.

CHAPTER XX

Butterfly resemblances—A living leaf—How spiders trap butterflies—Butterfly doubles—Suggested explanation—More evidence wanted—Warning coloration—A theory on trust—A straightforward test—Advice to naturalists—A strange omission.

SOME of the most remarkable instances of protective resemblances amongst insects are exhibited by butterflies, one, perhaps, being the most perfect existing under nature; however, I only say perhaps. This is the world-renowned leaf butterfly of Sumatra, and elsewhere in the Malay Archipelago. Of the great purple emperor family, it is purple on the upper surface, and gleams like a meteor as it shoots about in the rich, sun-bathed atmosphere of the tropics, its conspicuousness being enhanced by a sort of miniature, sharp-pointed swallow-tail, in which the hinder pair of wings end, and a broad, orange bar, like a sash or scarf of honour, running right across the anterior wings. It flies boldly and strongly, and when it descends upon a bush or shrub it is as though a little purple torch had shot through the foliage; but all at once, even though you see it come down just in front of you, it has vanished utterly—the torch has gone out. You may look and look, but unless you know the trick, and have seen the settling, and never taken your eyes off the exact spot, you will never find the butterfly, or see anything more of it until, all at once, it gleams in the air again. For the under part of the leaf butterfly's lovely purple wings is like the leaf indeed—"the sere, the yellow leaf"—with a midrib running down the centre veinings on either side, a curled tip at the top, a stalk at the bottom, and everything proper to leaves, but not as a rule to butterflies. All four wings join in this effect, for being thrown up in the usual way when the insect settles, the leaf-like shape is thus brought about, one-half of the under surface being seen on each side in clear profile, whilst the purple now lies hid within, like the pictures on a folded screen. As for the body of the butterfly, that is hidden inside the wings too; the legs are all but invisible, and the two little pointed swallow-tails, just touching the plant's stem with their mutual tip, make the stalk of the leaf. Even on the wall of a room or a

curtain it would seem as though a dead leaf were sticking there; how much more when, as is always the case, the butterfly flies into some bush or thicket crowded with dry, brown leaves, and settles all amongst them. It is not that you don't see it there that makes you miss it, but that you see it and scores of brown leaves all about it, every one of which looks just the same as itself.



PROTECTIVE MIMICRY

The picture at the top shows birds pursuing butterflies, while in the one below the same birds have lost their prey, as the butterflies have alighted and show only the underside of their wings, which are practically indistinguishable from the neighbouring leaves.

To make the matter plainer, in case this is not a very accurate description, here is the account of an eye-witness: “This species,” says Dr. Wallace, “was not uncommon in dry woods and thickets, and I often endeavoured to capture it without success, for after flying a short distance it would enter a bush among dry or dead leaves, and however carefully I crept up to the spot, I could never discover it till it would suddenly dart out again, and then disappear in a similar place. At length I was fortunate enough to see the exact spot where the butterfly settled, and though I lost sight of it for some time, I at length discovered that it was close before my eyes, but that in its position of repose it so closely resembled a dead leaf attached to a twig as almost certainly to deceive the eye, even when gazing full upon it.” Then follows a minute explanation of the imposture. “The end of the upper wings terminates in a fine point, just as the leaves of many tropical shrubs and trees are pointed, while the lower wings are somewhat more obtuse, and are lengthened out into a short, thick tail. Between these two points there runs a dark, curved line exactly representing the midrib of a leaf, and from this radiate on each side a few oblique marks, which well imitate the lateral veins. The tint of the under surface varies much, but it is always some ashy brown or reddish colour, which matches with those of dead leaves. The habit of the species is always to rest on a twig and among dead or dry leaves, and in this position, with the wings closely pressed together, their outline is exactly that of a moderately-sized leaf, slightly curved or shrivelled. The tail of the hind wings forms a perfect stalk, and touches the stick, while the insect is supported by the middle pair of legs, which are not noticed among the twigs and fibres that surround it. The head and antennæ are drawn back between the wings, so as to be quite concealed, and there is a little notch hollowed out at the very base of the wings, which allows the head to be retracted sufficiently. All these varied details combine to produce a disguise that is so complete and marvellous as to astonish everyone who observes it, and the habits of the insect are such as to utilise all these peculiarities, and render them available in such a manner as to remove all doubt of the purpose of this singular case of mimicry, which is undoubtedly a protection to the insect. Its strong and swift flight is sufficient to save it from its enemies when on the wing, but if it were equally conspicuous when at rest it could not long escape extinction, owing to the attacks of the insectivorous birds and reptiles that abound in the tropical forests.”^[104]

Dr. Wallace then speaks of another closely allied species which is common in India, on the under surface of whose wings there are sometimes, to the boot of all that has been described, in the way of disguise, “patches and spots formed of small black dots, so closely resembling the way in which minute fungi grow on leaves that it is almost impossible, at first, not to believe that fungi have grown on the butterflies themselves.”^[104] The minuteness of a resemblance like this is really very surprising, for it seems as though the butterfly-hunting bird or insect—some powerful wasp may represent the latter—was capable of minutely examining the object in question, and saying to itself, as it were, “I don’t think that can be a leaf, because there are no black spots upon it,” or *vice versâ*. In reality, however, it is no doubt the general effect, to which every detail contributes, that tells. What such resemblances do seem to me to show—and this, I think, is a new idea—is the accuracy and precision of some insects’ sight. How insects see things has long been a question, and many, I suppose, think it quite uncertain whether a leaf, for instance, throws the same picture on their retina that it does on ours. But if, to deceive them, the copy must be such that it also deceives us, is it not clear that it does? Otherwise the effect of the original could probably be reproduced by a less accurate copy. How little, after all, does the finest painting really resemble nature! The effect alone does so, not the means by which it is arrived at. Surely, then, if an insect, looking at a leaf or any other object, received but a general impression of colour, with an outline more or less blurred, or ill-defined, these copies of nature by nature—made to deceive—would bear witness to the fact. A study of protective resemblances is perhaps the best way of forming an idea as to how creatures, other than ourselves, see the world. It is even possible that such resemblances exist, which we, because we see things differently, are totally incapable of detecting.

I do not know if any other striking case of resemblance to an inanimate object (if plant life can be included under this term) is offered by the butterfly world, though there are several more of the same kind, but I cannot remember one just now. No doubt there are many which have not yet been discovered. We have, however, various instances of concealment even here in England, as, for instance, the peacock butterfly; but these, as well as special resemblances, are, for the most part, more marked in moths. The lappet moth, indeed, though it does not quite get the shape, looks very like a

dead brown leaf, whilst in the buff-tip moth we are supposed to have a special resemblance to a piece of rotting wood, clothed with moss or lichen, and broken at each end. Personally, I have never received the impression of such a definite object, but only a general one of rot and decay. Even here, however, I do not believe I could ever be taken in, for the yellow head and tips of the wings, which are supposed to offer a perfect resemblance to the two broken ends of the piece of wood, are to me the tell-tale parts, and instantly cry out, “Moth!” In fact, soft as is the colouring of the buff-tip, it still seems to me a salient object, and I do not think very much of that bird’s eyesight who fails to detect it under anything like favourable circumstances.

Another moth that flies by day, and is not uncommon in the United States, bears, when sitting on a leaf, a much stronger resemblance to a bird-dropping, but in this not uncommon form of imitation moths, and all other insects, are outdone by spiders, who use it aggressively against them, and particularly, it would seem, against butterflies, as the following instances will show. Mr. Forbes once, whilst travelling in Java, saw a butterfly settled upon a bird-dropping. He watched it for some time, and then, wondering at its long stay, approached cautiously, and, slowly extending his hand, actually caught it by its wings, between his finger and thumb—no mean feat, as it seemed, yet there was nothing to boast of. As he lifted the butterfly only the wings came away, the rest of it staying with the supposed bird-dropping, which was now seen to be a spider, who, having caught the butterfly by means of this shameful imposture, was quietly occupied in eating it. The disguise in this case was of the most wonderful perfection. “Such excreta,” says Mr. Forbes, who discovered this one, “consist of a central and denser portion of a pure white, chalk-like colour, streaked here and there with black, and surrounded by a thin border of the dried-up, more fluid part.”^[105] The appearance of each of these constituent parts was successfully counterfeited by the spider in question, who, in its own person, represented the more solid material, and spun the rest with its web.

As I know from early experience, when a naturalist makes a prize, all at once, of some interesting specimen, for some time afterwards he expects, or, rather, feels as if he would see some other on every leaf or twig; but time went by and no more of these “vain, delusive” spiders presented themselves. At length, years afterwards, the same naturalist found himself by the banks of the Moesi river, in Sumatra (which sounds much more

interesting than the Thames, for instance), and this was his second experience. “I was,” he says, “rather dreamily looking on the shrubs before me, when I became conscious of my eyes resting on a bird-excreta-marked leaf. How strange, I thought, it is that I have never got another specimen of that curious spider I found in Java, which simulated a patch just like this! I plucked the leaf by the petiole while so cogitating and looked at it half-listlessly for some moments, mentally remarking how closely that other spider had copied nature, when, to my delighted surprise, I discovered that I had actually secured a second specimen, but the imitation was so exquisite that I really did not perceive how matters stood for several moments. The spider never moved while I was plucking or twirling the leaf, and it was only when I placed the tip of my little finger on it that I observed that it was a spider, when it, without any displacement of itself, flashed its falces into my flesh.”^[105] (He means it bit him.)

Not all butterflies are entrapped by the kind of simulacrum here noticed. Nature can adapt herself to every taste, and in South Africa there are spiders who make themselves attractive by appearing to be flowers. Of some of these and their *modus operandi* Mr. Rowland Trimen, who was curator of the Cape Town Museum, gives the following interesting account: “Many species of spiders,” he says, “are well adapted to succeed by being coloured in resemblance to the flowers in or on which they await the arrival of their victims. One that inhabits Cape Town is of the exact rose-red of the flowers of the oleander, and, to more effectually conceal it, the palpi, top of the cephalothorax, and four lateral stripes on the abdomen, are white, according remarkably with the irregular white markings so frequent on the petals of *Nerium*.”^[106] These, indeed, must be beautiful spiders, and one would like to hear a little more of them, but Mr. Trimen goes at once from red to yellow. “I was led,” he continues, “to notice a yellow spider of the same group in consequence of seeing that two of a number of butterflies on the flowers of *Senecio pulugera* did not, on my approach, fly off with their companions. Each of these unfortunates turned out to be in the clutches of a spider, and when I released them I observed their captors very narrowly, and I found that the latter’s close resemblance to the *Senecio* flowers was not one of colour alone, but due also to attitude. This spider, holding on to the flower stalk by the two hinder pairs of legs, extended the two long front pairs upward and laterally. In this position it was scarcely possible to believe that

it was not a flower seen in profile, the rounded abdomen representing the central mass of florets and the extended legs the ray florets, while to complete the illusion the femora of the front pair of legs, adressed to the thorax, have each a longitudinal red stripe, which represents the ferruginous stripe on the sepals of the flower.”^[106]

Later on, Mr. Trimen was so fortunate as actually to see a butterfly caught by another flowery impostor:—“The butterfly,” he tells us, “was engaged in honey-sucking on a white flower-head of *Lantana*, and explored each individual flower with its proboscis. While I was watching it, the butterfly touched and partly walked over what looked like a slightly folded or crumpled flower about the middle of the cluster. This turned out to be a spider, which instantly seized the butterfly, throwing forward its front legs, somewhat after the fashion of a mantis. In this spider the effect of the little depressions on the limb of the corolla was given by some depressed lines on the back of its smooth white abdomen.”^[106]

Other spiders resemble snail-shells, others ants, and one, at least, is like a small scorpion, but we will return to the butterflies. As I have said, except for that wonderful copy of a leaf, already described, I cannot think of any very extraordinary resemblances amongst them, belonging to that class, but there are others which form a little class of their own. In the last chapter we have seen bees imitating bees, and in this we will make the acquaintance of certain butterflies which, as it were, pretend to be of a species which they do not really belong to. Thus in Brazil, by the great River Amazons, a number of large showy butterflies are found which belong to the family of the *Heliconea*, and wherever these fly they are accompanied by various other butterflies, belonging to quite different families, which are nevertheless so extremely like them that even Mr. Bates, who, for eleven years, ran up and down the Amazons with a butterfly-net in his hand, could never be quite sure which kind it was that he was going to catch. Often, when he thought he had got a *Heliconea* he was perfectly thunderstruck to find it was really a *Papilio*, *Pieris*, *Euterpe*, *Leptalis*, *Protogonius*, *Ithoneis*, *Diopis*, *Pericopis*, *Hyelosia*, or something of that sort; or again, when it was one of these he was after, and at last he thought he had it in the net, he would be petrified, on looking more closely, to find that what he had really caught was a *Heliconea*.

But now, as all these butterflies were alike or nearly alike, how could Mr. Bates tell—or how had anybody been able to tell before him—that they were really not all the same species—that a *Heliconea* was not a *Papilio*, or that a *Papilio*, *Pieris*, *Euterpe*, etc., were not all of them *Heliconeas*? This, at first sight, seems a difficult question to answer, but really it is not, because, in all these families of butterflies, the various species composing them bear a kind of generalised resemblance to one another: there is a family likeness, in fact, and this is not only the case in regard to their outward appearance—the shape and colour of their wings, etc.—but it applies, in a still greater degree, to their structure and internal economy. Thus, however strongly a *Pieris*, or one of those others, might resemble a *Heliconea*, the trained eye of an entomologist could easily see that it really was a member of another family, and since, in resembling the *Heliconea*, it departed from the general type of pattern and colouring exhibited by the family to which it belonged, whilst this one species of *Heliconea* it resembled was like the others, it might be inferred that the latter was the imitated and not the imitating form. Again—and this is still more decisive evidence in the cases where it applies—the resemblance is often confined to one sex of the copying species, viz. the female, so that whilst she is hardly to be distinguished from the model on which she has founded herself, the male retains the appearance, together with all the other characteristics, of the race to which both he and she belong.

But now came a further question, the most puzzling or, at any rate, the most important one of all, viz. Why should the one butterfly imitate, or rather resemble the other, in such an extraordinary degree—a degree seeming to preclude the possibility of mere chance having brought it about? This question Mr. Bates is supposed to have been the first to answer, though I cannot help thinking, myself, that he has only extended an explanation, which, in some cases, was so obvious that no one had thought of pointing it out, to these other cases where it was not nearly so easy to see. For what can be plainer, as Mr. Bates himself remarks, than that a moth, for instance, by closely resembling a hornet, would escape the attacks of birds that might otherwise have devoured it? I cannot think but that so patent an explanation had been in the minds of many long before 1862, and though no one previous to that date may have applied the principle of natural selection to

such cases, it must be remembered that natural selection had been established by Darwin some ten or twelve years before.

Bates, however, besides making an ingenious application of the above principle to a special case, gave a real reason for something which was not at all obvious, viz. why one butterfly should be a gainer by closely resembling another; and this no one had hitherto been able to do. His surmise, which has since in many instances been confirmed, is as follows. Having first pointed out that the *Heliconea* butterflies are a numerous, flourishing race, whilst those species that imitate them are poor in numbers, he says, “What advantages the *Heliconidæ* possess to make them so flourishing a group, and, consequently, the object of so much mimetic resemblance, it is not easy to discover. There is nothing apparent in their structure or habits which could render them safe from persecution by the numerous insectivorous animals which are constantly on the watch in the same parts of the forest which they inhabit. It is probable they are unpalatable to insect enemies. Some of them have glands near the end of the abdomen which they protrude when roughly handled; it is well known that similar organs in other families secrete fetid liquids or gases and that these serve as a protection to the species. They have all a peculiar smell. I never saw the flocks of slow-flying *Heliconidæ* in the woods persecuted by birds or dragon-flies, to which they would have been easy prey; nor when at rest on leaves did they appear to be molested by lizards, or the predaceous flies which were very often seen pouncing on butterflies of other families. If they owe their flourishing existence to this cause it would be intelligible why species whose scanty number of individuals reveals a less protected condition, should be disguised in their dress, and thus share their immunity. Is it not probable, seeing the excessive abundance of the one species and the fewness of individuals of the other, that the *Heliconea* is free from the persecution to which the *Leptalis* is subjected?”^[107]

No sooner was this suggestion made than naturalists all over the world began to test it, or rather to say that it ought to be tested. Some experiments have been made, but they have not been very numerous, and it can hardly be said that they entirely support Bates’s view. Sometimes they do and sometimes they do not, so as there is no reason to suppose that every butterfly is relished by every kind of insect-eating creature, this is not conclusive, till the same tests are employed in regard to butterflies that are

not imitated in this way; for if the latter have not been imitated on that account, it need not be on that account that others have been imitated. Thus Belt says, as the result of his observations, “The *Heliconidæ* are distasteful to most animals; I have seen even spiders drop them out of their webs again; and small monkeys, which are extremely fond of insects, will not eat them, as I have proved over and over again.”^[108] He also “observed a pair of birds that were bringing butterflies and dragon-flies to their young, and although the *Heliconidæ* swarmed in the neighbourhood, and are of weak flight so as to be easily caught, the birds never brought one to their nest.”^[109] This seems very good evidence of the truth of Bates’s theory, but then, as against it, we learn from the same observer that “another spider that frequented flowers seemed to be fond of these very same butterflies,” and as to the spiders which were seen to drop them out of their webs, they may resolve themselves into one, since farther on Belt says, “A large species of spider also used to drop them out of its web when I put them into it.”^[109] Then we are told that “there is, however, a yellow and black-banded wasp that catches them to store his nest with”; and which, having done so, “would quietly bite off its wings, roll it up into a ball, and fly off with it.”^[110] Professor Poulton calls these cases “interesting exceptions,” and easily accounts for them. But might not further observation keep adding to the number of exceptions, until at last, they become so numerous that all one could say would be this: “There is a great choice of insects in tropical America, and some creatures may prefer one kind and some another, to whatever species they belong.” In India, again, where there is another family of butterflies having doubles, or understudies, only one species was refused by all the mantids which a French naturalist gave them to. Others were eaten by all of them.

Has any man tried eating one of these butterflies? That was what Professor Wheeler did to test another supposed case of the same sort. Here the insect was a large and very conspicuously coloured day-flying moth. This moth has not an understudy, as far as is known, but it was supposed, then, to be an example of what is called “warning coloration,” that is to say, its bright colours were believed to be flourished in the face of any and every animal it might meet with, in order to warn them that it was not good to eat. Otherwise some bird, or lizard, or other creature might kill it before it had time to find out that it wasn’t. For instance, had it been just a brown moth—

there are so many of these and most of them good to eat—how was it to be distinguished from others? But such a get-up as *that*—black and white wings and a black and orange body—once seen it was not to be forgotten. It was like the red flag at a rifle range, warning one off, and this is the theory of warning coloration. So Professor Wheeler, as he rode through the deserts of Wyoming, with the moths all about him, resolved to test this theory which had lived for a long time, and still goes on living a good deal on trust. “He dismounted from his horse and proceeded to masticate the body of one of the moths. To his astonishment the little flavour that it contained was mild and pleasant, one may almost say nut-like.”^[111] Perhaps it may be thought that, on the “*de gustibus*” principle, what is pleasant to a human being might be disagreeable to a bird or a lizard; but Professor Wheeler tried another experiment. “Another day-flying moth, common in our eastern States, has deep black wings, each adorned with a pair of large yellow spots, and there is a dash of orange on its legs. It certainly cannot be a mimetic species (if it were, of course, one would not expect it to be nasty) as there is no other day-flying moth which could serve as its model. Several of these moths were given to some lizards that had previously been well fed on house-flies and could not, therefore, be very hungry. The moths were seized at once, and devoured, with evident signs of relish.”^[111]

As a result of these experiments Professor Wheeler concludes that “naturalists should be more careful in imputing nauseous or disagreeable qualities to some conspicuously coloured animals,” and he suggests that “if every field entomologist could only bring himself to repeat the writer’s experiment on one of many cases of ‘flaunted nauseousness’ and place his taste impressions on record, we should in the course of time have a really valuable body of evidence, for we can hardly assume that beasts, birds, and reptiles can find things ‘nauseous’ which are quite tasteless, or even pleasant, to the human palate.”^[111] “*Il n’y a pas de réplique à cela,*” and how it is that so simple a plan did not occur to Mr. Bates during all the eleven years he was on the Amazon it is not very easy to imagine. On the whole, perhaps, it may be said that the reason why certain butterflies are imitated by other butterflies has not been so satisfactorily settled as the fact that they are so imitated. But, on the other hand, there is some—perhaps much—evidence of the truth of Bates’s theory, and, moreover, that theory is

in itself so plausible that it seems to require a good deal of evidence to overthrow it.

It is not only in South America that butterflies dress up like one another. Instances similar to those here given occur also in Africa and the Malay Archipelago, as well as in other parts of the world. There is even one doubtful case in England, both the copy and original being moths. Moths, especially the day-flying ones, are represented in these phenomena as well as butterflies, which are sometimes imitated by them.

CHAPTER XXI

Sights of the forest—A butterfly bridge—Bird-winged butterflies—“What’s in a name?”—Scientific sensibility—Resemblance v. mimicry—A convenient wrong word—Beauty in nature—Nuptial display—Strange counter-theory—*Lucus a non lucendo*—Reasoning by contraries—True in Topsy-turvydom—Butterfly courtship—Form and colour—A curious suggestion—Powers of defective eyesight.

THOUGH the principle of protective resemblance, as explained in the last chapter, will account for the colours and markings of those butterflies which imitate the *Heliconidæ*, it does not explain how the *Heliconidæ* themselves came to be as they are; for in nature every mark and line and shade has a meaning, and has come into existence by virtue of some law or another. Beauty itself, independently of any *arrière pensée*, as we may call it—remembering those flower-resembling spiders—requires to be explained.

Nor are the *Heliconidæ* themselves, though gaudily dressed showy butterflies, anything like so beautiful as many others; for instance, as the *Morphos*, those giants of their kind, who sweep like stars through the tropical forests on wings whose whole broad surface is blue, but a blue that flashes like a drawn sword and has a hundred glints and gleamings of ever-varying light. High-fliers they are for the most part, keeping to the tops of the trees, but every morning, and always at the same time, they make a descent into the glades and alleys of the forest, where for a little they flap lazily backwards and forwards, now in eclipse, now flashing forth into sunlight, as though to flaunt their beauties in the face of the lower world. Such openings in the primeval forest are often made by the fall of great trees—for even where the axe is not there is death in the midst of life—and as these majestic insects sail high above them in a world of air and light, their shadows fall upon the place beneath, and trace their course along the ground. When the sun’s rays strike into them, such clearings become the gathering grounds of various butterflies. Besides the great *Morphos*, the flashing of whose wings in the sunlight can be seen sometimes a quarter of a mile away,^[112] various species of *Heliconeas*—the one we have just been

reading about—whose black wings are in some species spotted with scarlet, in others with white or blue, waltz about the bushes or undergrowth, “dancing in the chequered shade”; fritillaries somewhat like our own, but of larger size and more effective colouring, fly higher up, about the tree-trunks, whilst over the ground itself, carpeted as it often is with flowers fallen from the leafy world above, and scenting the air, ghost-like butterflies, whose clear, transparent wings are without any colouring matter, ceaselessly hover and flit.

Wherever there is a river, many-coloured armies, bivouacked amidst its various shoals and reefs, sit sucking water through, the moist particles of the sand, whilst others, in even greater numbers, pass and repass from one bank to another, making, however wide the waterway, an aerial fluttering bridge. Other butterflies, also denizens of the great Brazilian or Central American forests, have broad white wings, shot with a satiny lustre, whilst those of yet another are like glass, with one opaque spot of violet and blue, in the midst of each of them. In flight this spot is the only part that can be seen, and it looks, Bates tells us, “like the wandering petal of a flower.”^[113] There are swallow-tailed butterflies, too, whose livery of deep, soft green, and deeper velvet black, set off with roseate hues, is amongst the richest of all—“rich, not gaudy,” so at least I should term it, for I have seen it, putting flowers to shame, on the lower slopes of the Himalayas. Here these butterflies—double the size of our *Machaon*, and of another shape, with racquet-rather than swallow-tails, flew on the open hillside, courting the sun, but in Brazil they keep to the forest depths, where, like Una, they “make a sunshine in the shady place.”

The butterflies of South America are almost rivalled—quite they cannot be—by those of the Malay Archipelago. Here we have the great Bird-winged Butterfly, discovered by Dr. Wallace, who calls it “elegant,” and bestows upon it a name (*Ornithoptera Brookeana*) which is not quite that.^[114] However, “What’s in a name? That which we call a *Brookeana*,” etc.—we must reverse the proposition. There is no describing such a creature—at least, not convincingly. Suffice it to say that its wings “almost resemble those of a sphinx moth in shape,” and are deep, velvety black, but lit up with a band of green feather-like markings, so brilliant and lovely that they reminded its discoverer “of the wing-coverts of the Mexican trogon laid upon velvet”^[114]—and that for anyone who has seen a trogon, even stuffed

and dried, is to say enough. Besides this adornment the great *Brookeana* has “a broad neck-collar of vivid crimson and a few delicate white touches on the outer margins of the hind wings.”^[114] Another *Ornithoptera* without the *Brookeana* is “the largest, the most perfect, and the most beautiful of butterflies.” The two last, however, are matter of opinion, and I should think myself that a *Morpho* with its azure wings, sometimes seven inches across, and flashing a quarter of a mile away, would run—or fly—it hard. Then there is a yet more gorgeous species of “Bird-winged Butterfly,” with wings equalling or even exceeding the *Morpho*’s in expanse, whose ample surface is divided between flame-like orange and a black so deep, rich, and velvety that it seems to glow—“the pride of the Eastern tropics, one of the most gorgeously coloured butterflies in the world.”^[115]

This was the butterfly that gave Dr. Wallace a headache, “so great was the excitement produced by what will appear to most people a very inadequate cause.”^[115] Surely not; for the cause here alluded to was not the insect’s beauty—which had been seen before, free and untrammelled, without any such ill-consequence—but its capture and anticipated transference to the cyanide bottle. It was not mere æsthetic emotion, therefore, that produced the headache, but scientific enthusiasm, of which no man need feel ashamed. It is easy for us on such occasions to mistake our feelings, but the clue to them, I think, is this, that however beautiful a creature may be, and however appreciative we may think ourselves of such beauty, yet if we resolve, in the true interests of science, to take that creature’s life, then the scientific spirit must be far stronger in us than mere admiration of its beauty. This test I would apply to another account which Dr. Wallace gives us of the capture of “one of the most magnificent insects that the world contains.” “I trembled,” he says, “with excitement as I saw it coming majestically towards me, and could hardly believe I had really succeeded in my stroke till I had taken it out of the net and was gazing, lost in admiration, at the velvet black and brilliant green of its wings—seven inches across—its golden body, and crimson breast. It is true I had seen similar insects in cabinets at home, but it is quite another thing to capture such oneself—to feel it struggling between one’s fingers, and to gaze upon its fresh and living beauty, a bright gem shining out amid the silent gloom of a dark and tangled forest. The village of Dobbo held that evening at least one contented man.”^[116] When we consider that the “fresh and living

beauty” was caught for the very sake of being made dead and mouldy, and that the “bright gem” which would otherwise have continued to flash in the forest was about to become one of those very same specimens that had been looked at with such an inferior degree of interest, we must admit, I think, that the higher of two passions was predominant here, and that the author, in dwelling only upon the other and lower one—mere delight of the eye—has done himself less than justice.

Mr. Bates, without headaches, has given us some very pleasing pictures of butterfly-life in the tropics, and in doing so he has instinctively, as it were, kept the killing and capturing in the background. “The number and variety of gaily-tinted butterflies,” he tells us, “sporting about in this grove on sunny days, were so great that the bright moving flakes of colour gave quite a character to the physiognomy of the place. It was impossible to walk far without disturbing flocks of them from the damp sand at the edge of the water, where they congregate to imbibe the moisture. They were of almost all colours, sizes, and shapes. I noticed here altogether eighty species belonging to twenty-two different genera. It is a singular fact that, with very few exceptions, all the individuals of these various species thus sporting in sunny places were of the male sex; their partners, which are much more soberly dressed and immensely less numerous than the males, being confined to the shades of the woods. Every afternoon, as the sun was getting low, I used to notice these gaudy, sunshine-loving swains trooping off to the forest, where I suppose they would find their sweethearts and wives.”^[117] What a delightful scene! Here, “next to the very common sulphur-yellow and orange-coloured kinds, the most abundant were about a dozen species of *Eunica*, of large size, conspicuous from their liveries of glossy dark blue and purple. A superbly adorned creature, the *Callithea Markii*, having wings of a thick texture, coloured sapphire blue and orange, was only an occasional visitor. On certain days, when the weather was very calm, two small gilded green species literally swarmed on the sands, their glittering wings lying wide open on the flat surface.”^[117]

Such, then, are the colours of butterflies, and, as may be imagined, comparatively few of these gorgeous liveries have been acquired through the principle of protective resemblance—using the term in its widest sense, to include those cases where one species becomes, as it were, the double or wraith of another. Mimicry is the word which, by a ludicrous process of

false reasoning, naturalists have convinced themselves it is right to apply to this particular kind of resemblance, and no other one; though why a butterfly should *mimic* another butterfly and only *resemble* a leaf, “*quien sabe?*” as the Spaniards say. The principal reason adduced for this misuse of language, viz. that the wrong word is more convenient than any right one, providing us with the useful series, *mimic*, *mimicry*, *mimetic*, *mimicker*, *mimicked*, *mimicking*, obviously applies to the one case as well as the other, and if it is an advantage to be absurd in one way, surely it is a double one to be absurd in two. On these grounds I would suggest to naturalists that, having broken down the proper and natural confinements of the word in question, they should rather extend its use than limit it, to the extent even of calling their children “mimetic forms,” should they happen to resemble them, and thinking twice before punishing a son for merely “mimicking” his father.

But, leaving this, how—to take the jargon as one finds it—are the glorious colours of butterflies to be explained when they are due neither to protective resemblance, which is not mimicry, though it very much resembles it, nor to mimicry, which is distinct from protective resemblance, though mimicking it exactly? Certainly neither of these will do to account for uncopied hues and patterns, which are like nothing in the world but their own loveliness, unless, indeed, it be the rival glories of the most resplendent birds. Still less will aggressive resemblance—though, as we see, it can make spiders look like flowers—explain them. The great governing cause which produces such effects as these, as well as most others in nature, is natural selection; but we must look beyond natural selection even, if we wish to understand all the beauty that we see in the animal world, and especially the higher developments of it.

Darwin, as we know, was the great demonstrator—though not the first conceiver—of the law of natural selection,^{[\[118\]](#)} and on this he might have cried “*Nunc dimittis*” and retired, so to speak, leaving someone else to find out the other law; but instead of that he went on and demonstrated that too. This other law, the evidence for which is really overwhelming, and has never been met by anything better than a conceited over-estimation of human superiority, wrapped up in a cloud of wrong reasoning, is that of sexual selection, which implies that, in the choice of their mates, animals, like ourselves, are guided by some sort of preference; and as this with them

—again like ourselves—is usually determined by the element of beauty, the most beautiful partners are being constantly selected, and species in consequence become more and more beautiful. This process, however, is usually confined to the males, they being the eager wooers, whilst the females only wait to be courted, and then shyly and modestly choose—such, at least, is the supposition. This masculine beauty is often inherited by that sex only to which it is so useful, but in other cases it is transmitted to the female also. Thus, to take birds, where the results of the law of sexual selection are on the whole most pronounced, we have, on the one hand, the pheasants and birds of paradise, where the male alone is resplendent, whilst in the trogons, parrots, and many other species the beauty is common to both the sexes. As is well known, the males of various highly ornate birds are accustomed to make the most extraordinary display of their beauty before the females, making the most of the parts most richly decorated, assuming just such attitudes as are required in order to give these their full advantage, and, in fact, taking pains and trouble in a high degree and of a very special and peculiar kind, which must either be directed to an end which seems perfectly plain and apparent, or else so much waste of time—due to no special cause and without any particular meaning—an alternative which the opponents of sexual selection do not in the least mind accepting.

To give one instance of what is called nuptial display in birds—for it will serve to illustrate what Darwin supposes to take place with some insects also, as well as forming a basis of comparison with what has been more carefully observed in the case of spiders—Belt in his often quoted work gives us the following pretty picture of humming-bird courtship. Speaking of a beautiful blue, green, and white species (*Florisuga mellivora*), he says: “I have seen the female sitting quietly on a branch, and two males displaying their charms in front of her. One would shoot up like a rocket, then suddenly expanding the snow-white tail like an inverted parachute, slowly descend in front of her, turning round gradually to show off both back and front. The effect was heightened by the wings being invisible from a distance of a few yards, both from their great velocity of movement and from not having the metallic lustre of the rest of the body. The expanded white tail covered more space than all the rest of the bird, and was evidently the grand feature in the performance. Whilst one was descending, the other would shoot up and come slowly down expanded. The entertainment would

end in a fight between the two performers; but whether the most beautiful or the most pugnacious was the accepted suitor I know not.”^[119]

Here the display, as well as the intention, seems evident enough, and it is not a whit more so than in hundreds of other cases collected by Darwin during his lifetime, and which have been largely added to since his death. As the hen is constantly present during these performances, and as she has been known on various occasions to show a strong partiality, or the reverse, to this or that male bird, we have here a solid basis of observed fact on which to raise an hypothesis. On what facts the counter one rests, as propounded by Dr. Wallace, viz. that colour and antics are produced by superior vigour resident in the male, it is less easy to see, unless, indeed, such as point in a quite opposite direction may, on a sort of *lucus a non lucendo* principle, be held to support it. If this be conceded, then, indeed, we have a plentiful crop, nor need we any longer feel sceptical because the eagle, say—that bird of fierce energy—does not flash out like all the crown jewels together as it descends on its prey, or because the swift, whose vital force is, perhaps, even greater, leaves no train of jewelled light to die all day, behind it, on the air. Nor need we wonder that the trogons, though as resplendent as the swifts, swallows^[120] and eagles are dull-coloured, should be as lazy and sluggish as these are energetic: nor that, whereas the females of some humming-birds are sober-suited, those of others, though their vigour would seem to be in no way superior, are as gem-like as their mates: or that the males as well as the females of some wholly dull-coloured ones, and of many other plain birds, seem bursting with vigour, and indulge in all sorts of strange antics and dances: that a cock partridge, for instance, seems as vigorous as a cock pheasant, and that bright colours and pugnacity are dissociated in such tremendous fighters as the ruff, the coot, and the blackcock. Multiply such instances by the score or the hundred—as can easily be done—and if only the above-stated principle be granted, we get more and more proof of the correctness of a theory to which facts, if dealt with in the more usual way, would be almost instantly fatal. After all, this would be a more satisfactory mode of procedure than that of tolerating a travesty on the strength of a high reputation. There is such a principle in nature as the *lucus a non lucendo* one, so, as we admit the word mimicry in a false sense—because it is convenient—why not admit that? It would be not one whit less convenient—for the theory.

But, handling facts in another way, can we explain the beautiful colours of butterflies as we explain the brilliant plumage of birds—by sexual selection, that is to say? Of this there is not so much direct evidence as could be wished, for butterfly courtship is a long affair, and, for various reasons, is not so easy to watch under natural conditions, as in the case of birds, though this, too, is often beset with difficulties. We know, however, that the male is often much more beautiful than the female, that he pirouettes around her, and that she remains often “icy insensible”—in fact refuses him—which certainly implies a power of choice. Rival males, too, will “whirl round each other with the greatest rapidity, and appear to be incited by the greatest ferocity.”^[121] That butterflies, like bees, perceive and are attracted by colours is well known, and it would be strange, indeed, if they were not alive to the many very beautiful and complex patterns on their own wings, when these cannot have been evolved through any principle of protection—since they resemble or suggest nothing—when, in fact, if not beauty, it is difficult to see the object aimed at. Yet the strange suggestion has been made that, though butterflies see colours, they cannot see form, that their sight is defective in some peculiar kind of way. But if form is outline—and if not, what is it?—where is the distinction, seeing that the beginning and leaving off of a colour or of two or more colours must make an outline, and therefore a form? If we see the colours of a pattern where the one ends and the other begins, we see that pattern, and on the other hand, if we could not distinguish one colour from another, or colour from something uncoloured—as, say, the air—we should be blind to colour, as well as to form. Form can hardly be called a thing of itself. It is rather the line of demarcation between two or more things, so that, if each of these is clearly perceived, the form or outline which their juxtaposition makes must be also perceived. Assuming that butterflies see the beautiful arrangements of colour—eyes, spots, bands, lines, etc.—in such a way as can alone account for their being there to see—as well as we do, that is to say—then it is absurd to imagine that they have no perception of form.

On what is this assertion based? Mr. Scudder relies on the following facts: “Christy,” he tells us, “observed in Manitoba one of the swallow-tails fluttering over the bushes, evidently in search of flowers. As he watched it, it settled momentarily, and exactly as if it had mistaken it for a yellow flower, on a twig of *Betula glandulosa*, bearing withered leaves of a bright

yellow colour.”^[122] But might not the association of ideas raised by a familiar colour in an insect’s mind overpower for a moment its judgment? Might it not do so in the case of a man also? And should we think a person very stupid who, for a moment, mistaking a yellow leaf for a yellow flower, stretched out his hand to pick it? Pooh! once again,^[123] let us think of people who do foolish things—kings, generals, cabinet ministers, servant-maids, etc.—not of infallible persons. We should not be too severe—not “break a butterfly on a wheel.”

Again—this is Mr. Scudder’s second instance: “Albert Müller records seeing the blue Alexis of Europe fly towards a very small bit of pale blue paper lying upon the grass, and stop within an inch or two of it, as if to settle, doubtless mistaking it for another of its own kind.”^[124] Surely this is rather in favour of the butterfly’s sight than otherwise, since it discovered its mistake and did not settle. Who, too, can tell the precise moment at which the mistake was discovered, since the piece of blue paper might have puzzled the butterfly—piqued its curiosity to know what it was—even after it knew what it was not? Thirdly, “Plateau has observed the small tortoise-shell butterfly fly rapidly towards a cluster of artificial flowers.”^[124] And who cannot be taken in by artificial flowers? “Such examples as these,” says Mr. Scudder, “seem to indicate that butterflies may perceive colour in mass, but in no case indicate any further visual powers.”^[124] To me they indicate that butterflies can make mistakes. Mistakes rarely show one’s perfections, but other indications of further visual powers are not wanting. For instance, Mr. Scudder himself says: “One of my favourite modes of showing this characteristic (inquisitiveness) to unbelieving friends has been to toss my cap high in the air, when these butterflies will often dart, dash at, and play around it as it begins again to descend.”^[124] How do they play around this moving object in the air if it represents to them only “colour in mass,” and not a defined shape and outline? Were it otherwise, they would fly right into it, and be carried down with it sometimes on top of them. But if they see all parts of the colour so that they can nicely avoid it, and sport about its periphery, then they see the shape of the cap. Then, again, Mr. Scudder tells us: “Many kinds are of a lively and even pugnacious disposition, and perch themselves upon the tip of a twig, or on a stone, or some such outlook, and dash at the first butterfly that passes, *especially if it be one of their own species*,”^[125] then the two advance and retreat, forwards

and backwards, time and again, circle round each other with amazing celerity, all the while, perchance, mounting skywards, until suddenly they part, dash to the ground, and the now quiet pursuer again stations himself on the very spot he quitted for the fray.”^[126] How does he do *that* without accurate eyesight, with good defining power?—to which, indeed, the whole performance bears witness. Elsewhere, too, this pronounced characteristic of returning to the exact spot, left some little time ago, is dwelt upon. To me it seems a complete upsettal of the defective eyesight theory, or, since good eyesight could do no more, what does such defectiveness matter?

The following description also, which Bates gives us, of butterfly-life by the Amazons, does not suggest that any of these bright-day-lovers, these children of the sun, need write an “*Apologia pro oculis suis*.” “The fine showy *Heliconii*,” he says, “often assemble in small parties, or by twos and threes, apparently to sport together or perform a kind of dance” (my “dancing in the chequered shade,” therefore, was no inapt quotation). “I believe the parties are composed chiefly of males. The sport begins generally between a single pair. They advance, retire, glide right and left in face of each other, wheel round to a considerable distance, again approach, and so on; a third joins in, then a fourth, or more. *They never touch*;^[127] when too many are congregated a general flutter takes place, and they all fly off, to fall in again by pairs shortly afterwards.”^[128] Lastly, Belt tells us this: “Here a large spider built strong, yellow, silken webs joined one on to the other, so as to make a complete curtain of web, in which were entangled many large butterflies, generally forest species, caught when flying across the clearing. I was at first surprised to find that the kinds that frequent open places were not caught, although they abounded on low white-flowered shrubs close to the webs; but on getting behind them and trying to frighten them within the silken curtain, their instinct taught them to avoid it, for, although startled, they threaded their way through open spaces and between the webs with the greatest ease.”^[129] If a butterfly with defective eyesight can thread its way between spiders’ webs, so as never to be caught, “with the greatest ease,” “why, then, say an old man can do somewhat”—but it must be without spectacles.

CHAPTER XXII

Beautiful spiders—The “Peckham paper”—Spider courtship—Male antics and love-dances—Occasional accidents—Strength of the evidence—The one explanation—Darwin’s last words—His theory established.

SPIDERS, as we have seen, may attain beauty by getting more and more like flowers, but beauty is not the attribute with which they are principally connected in our minds. Rather they are a synonym of something uncouth and horrid-looking, as well as of skill and persevering industry. For those of us, however, who have lived in the tropics they have other associations, for here, side by side with the most hideous of monsters, huge, dark, and hairy, are found others, small and gem-like, flashing indeed with beauty, the representatives in their order of the humming-birds, those “living sunbeams” of the Indians, amongst birds. These lovely little spiders belong to a particular family, the *Attidæ*, which has been placed by common consent at the head of all the others, since, whilst structurally, and in other respects, it is inferior to none, “it contains among its 1,500 species the greatest amount of sexual differentiation and the highest development of ornamentation.” Dr. Wallace, after noticing “their immense numbers, variety, and beauty,” in tropical South America, says, “Many of them are so exquisitely coloured as to resemble jewels rather than spiders”;^[130] and again, in his work on the Malay Archipelago, he alludes to them as “perfect gems of beauty.”^[131]

These little radiant spiders live amongst flowers and foliage, and here they chase such small insects as their size allows them to cope with. Besides running, they make little leaps into the air, and so, if they can manage it, come down on their prey, for which reason they are often called “jumping spiders.” This is a very different mode of action from that of remaining perfectly still till a butterfly or other insect happens to settle on one, and it is accordingly instructive to find that, great as is the beauty of these flower-haunting spiders, yet it does not resemble that of the flowers amongst which

it is displayed. The iridescent flashes and sparkles more resemble those of the mineral than of the vegetable world—where, indeed, they hardly exist—and must serve, as well as their active movements, to point them out to their enemies even amidst a background of flowers. It is not upon principles of protection, therefore, or to acquire a dissembling resemblance that such bright brilliancy has been developed in these little creatures.

Since, therefore, these spiders could not have become beautiful on any principle of protective or aggressive resemblance, nor yet of warning coloration, for which there would here be no opening, and had yet become beautiful in a high degree, they seemed to Professor and Mrs. Peckham to offer a good subject for the testing of the theory of sexual selection, and deciding as to whether Darwin or Wallace was right in that matter. After several months of careful, and often very laborious observation—rewarded, however, by the most interesting results—they have given their answer, and this answer, resting as it does on the most irrefragable evidence, should be decisive for all time. It may safely be asserted that anyone who, after reading the “Peckham paper,” as it may well be called, is not convinced both that the male spiders of this beautiful family woo the females by displaying their beauty before them, and that the females carefully watch the display, accepting only such as please them sufficiently and rejecting the others, never will be convinced, since only by the spiders actually speaking, which is not likely to happen, could the evidence be bettered. If, indeed, the female had been heard to say “Pretty i’ faith,” or “You are a fine young man,” just before her actions gave clear, or still clearer indication that this was in her mind, had she murmured “Take me” as she let herself be taken, and had the male asked, after the way common in novels, “Was it my abdomen or the stripes on my palpi that made you first fall in love with me?” then, perhaps, even those who believe that the higher spiritual love is for man alone would have been converted—and yet I know not, since assertions so unlikely in themselves might have flung doubt on the whole paper.

But, however this may be, the evidence now offered us in favour of Darwin’s views can never be strengthened except in this way, so that, as far as proof is possible in such a matter, sexual selection as a law and principal agent of beauty in nature is now proved, though, at the same time, several more facts are added to those upon which the counter hypothesis seems

based, and which would certainly prove it in Topsy-turvydom. To take these first, the authors of the paper in question have sought to apply to spiders “the hypothesis that the brighter colour of the male is due to his greater activity and vital force.” “Beginning,” they say, “with the most brilliant family—the *Attidæ*—we find that the females are, with few exceptions, larger, stronger, and more pugnacious than the males. Thus we placed two females of *Phidippus morsitans* together in a glass jar. No sooner did they observe each other than both prepared for battle. Eyeing each other with a firm glance, they slowly advanced, and in a moment were locked in deadly combat. Within a few seconds the cephalothorax of one was pierced by the fang of the other, and with a convulsive tremor it relaxed its hold and fell dead. We placed together eight pairs in all, and in each instance the fight was short and even to the death. Subsequently we put in a well-developed male, which, though smaller, was compactly built and apparently strong enough to bring the virago to terms, but to our surprise he seemed alarmed and retreated, trying to avoid her. She, however, followed him up and finally killed him.”^[132]

So much for *Phidippus morsitans*. Coming to *Dendryphantes elegans*, the authors, who kept a number together in a large box, “were much struck by the greater quarrelsomeness of the females. They would frequently go out of their way to chase each other, and they were much more circumspect in approaching each other than were the males.”^[133] Again they say, “Valkenaer, Menge, Hentz, and others give numerous instances where the male meets his death through the fierceness of his mate. In fact the danger is so imminent that after a successful courtship it is the habit in several genera (*e.g. Epeira* and *Tegnaria*) for the male to retire with precipitation from the web of the female as a reasonable precaution; yet the rule is for the male to be more ‘beautified’ than the female.”^[133]

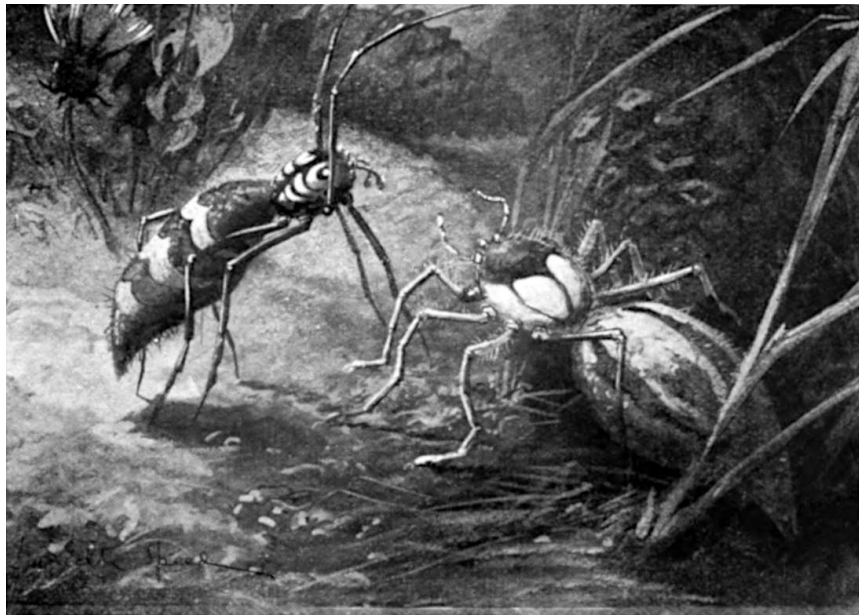
Coming now to the actual courtship of these brilliant spiders, the authors placed pairs of several species in square wooden boxes, having a cloth bottom, on which they could easily move about. One of the species experimented on was *Dendryphantes elegans* mentioned only a moment ago—such a name is not to be forgotten—whose beauty is thus described: “The male is covered with iridescent scales, his general colour being green. In the female the colouring is dark but iridescent, and in certain lights has lovely rosy tints. In the sunlight both shine with the metallic splendour of

humming-birds. The male alone has a superciliary fringe of hairs on either side of his head, his first legs being also larger and more adorned than those of his mate.”^[133]

Yet the extra vigour from which this special growth is supposed to have sprung has not, as we shall see, affected his growth in general. “The female is much larger, and her loveliness is accompanied by an extreme irritability of temper, which the male seems to regard as a constant menace to his safety; but his eagerness being great and his manner devoted and tender, he gradually overcomes her opposition. Her change of mood is only brought about after much patient courting on his part”.^[133] And now comes the minutely interesting description of this iridescent, *couleur de rose* courtship. “While from three to five inches distant from her he begins to wave his plummy legs in a way that reminds one of a windmill. She eyes him fiercely, and he keeps at the proper distance for a long time. If he comes close she dashes at him and he quickly retreats. Sometimes he becomes bolder, and when within an inch pauses with the first legs outstretched before him, not raised, as is common in other species; the palpi also (in insects it would be the antennæ) are held stiffly out in front, with the points together. Again she drives him off, and so the play continues. Now the male grows excited, as he approaches her, and while still several inches away whirls completely around and around; pausing, he runs closer, and begins to make his abdomen quiver as he stands on tiptoe in front of her. Prancing from side to side, he grows bolder and bolder, while she seems less fierce, and yielding to the excitement, lifts up her magnificently coloured abdomen, holding it at one time vertically and at another sideways to him. She no longer rushes at him, but retreats a little as he approaches. At last he comes close to her, lying flat, with his first legs stretched out and quivering. With the tips of his front legs he gently pats her; this seems to arouse the old demon of resistance, and she drives him back. Again and again he pats her, with a caressing movement, gradually creeping nearer and nearer, which she now permits without resistance,”^[133] and so on,

“Till the happy ‘yes’ falters from her mouth,”

almost as exciting, though not quite so detailed, as the climax scene of a latter-day novel.



1. A solitary spider dancing before its mate.



2. A cockroach attacking an astonished scorpion. Its weapons are the spines on its powerful hind legs.

Of the courtship of another species—*Habrocestum splendens*—we have the following account: “The male, a magnificent fellow when we first caught him, displayed for a long time before the female. He began by advancing a few inches before her, and then backing off again, this being repeated many times. After a while he settled down under a little web in the corner. The female, troubled by this indifferent treatment, advanced towards him; he came out and she fell back. This play was kept up for some time, but at length the male began his courting in earnest. When within a few inches of her he began a rapid dance from side to side, raising the whole body high on the tips of the legs, the first pair being directed forward and the palpi clasped together, with the abdomen turned to one side and lifted up. After a short dance he stood motionless, striking an attitude and remaining quiet for half a minute. Then he turned his back on her, moving irregularly about, with his legs forward and his palpi vibrating. Again he dances sideways before her, strutting and showing off like a peacock, or whirling around and around.”^[133]

On such occasions the female would “commonly move nearer to him and appear much excited herself. We at first supposed that this turning around was accidental, but it happened so regularly at a certain stage of the courtship that we concluded it was an important part of the display, serving to better show off his brilliant abdomen.”^[133] Of this there can hardly be a doubt, since on every occasion the male spider, whatever his species, assumed such attitudes as displayed his best points to the best advantage—a fact which recalls the following passage in one of Darwin’s letters: “I am very glad to hear of your cases of the two sets of *Hesperiadæ* (a butterfly), which display their wings differently, according to which surface is coloured. I cannot believe that such display is accidental or purposeless.”

How glad, and more than glad, would Darwin have been to have read the tale of these spiders! It is, indeed, one of those ironies of fate, of which the world is so full, that he did not live to see this demonstration—for it is no less—of the truth of his most original and elevating views; elevating they may be well called, since they allow to the animal world an æsthetic faculty, the power, once thought exclusively human, of appreciating beauty. It is curious how willing many are to exalt humanity at the expense of all other beings. The higher faculties they like, and perception of the beautiful they like, and spirituality—especially in love—they like very much indeed; but they only like these things in their own species. That is to say, conceit lies at the bottom of all this exaltation. Such man-worshippers would not have more of a good thing in the world, but less, so that they may have all there is of it. On such grounds the war against evolution was waged, and its last struggles are against sexual selection. The body has been given up, but the spirit, which touches us yet more nearly, is still fiercely defended.

In *Hasarius Hoyi* “the sexes are very different, the male being the more conspicuous of the two. In his dances, the male has several movements. Most commonly he goes from side to side, with his first legs obliquely up. At other times he twists the abdomen to one side, and, bending low on the other, goes first in one direction for about two inches, and then, reversing, circles to the opposite point. The females are very savage, especially with each other, and even the members of the sterner sex are not always free from danger when paying their preparatory addresses. Once we saw a female eagerly watching a prancing male, and, as he slowly approached her, she raised her legs as if to strike him, but he, nothing daunted by her

unkindly reception of his attentions, advanced even nearer, when she seized him and seemed to hold him by the head for a minute—he struggling. At last he freed himself and ran away.”^[133] Yet “this same male, after a time, courted her successfully.” That so much savagery has to be overcome in the female, and finally is overcome by these dances, shows how powerfully she must be affected by them. Of another and previously undescribed species, “a dozen or more males, and about half as many females,” were found by the authors “assembled together” under natural conditions. “The males were rushing hither and thither, dancing opposite now one female and now another. Often two males met each other, when a short passage of arms followed. The males were very quarrelsome, and had frequent fights, but we never found that they were injured. Indeed, after having watched hundreds of seemingly terrible battles between the males of this and other species, the conclusion has been forced upon us that they are all sham affairs, gotten up for the purpose of displaying before the females, who commonly stand by, interested spectators.”^[133]

Then there is a small ant-like species, who, “unlike most of the *Attid* males, keeps all his feet on the ground during his courtship. Raising himself on the tips of the posterior six, he slightly inclines his head downwards by bending his front legs, their convex surface being always turned forward. His abdomen is lifted vertically, so that it is at a right angle to the rest of his body. In this position he sways from side to side. After a moment he drops the abdomen, runs a few steps nearer the female, and then tips his body and begins to sway again. Now he runs in one direction, now in another, pausing every few moments to rock from side to side and to bend his brilliant legs, so that she may look full at them.”^[133] What can be clearer than this? And here, indeed, the authors remark: “We were much impressed by the fact that the attitude taken by the males served perfectly to show off their fine points to the female. We had never known the male of this species until the day that we caught this one and put him into the mating-box, and it was while studying his courtship that we noticed how he differed from the female in his iridescent first legs. He could not have chosen a better position than the one he took to make a display.”^[133]

Elsewhere, in another experiment with the same species, the authors, after remarking that if these specially modified front legs were held in any other way the effect of the flattened and iridescent surface would be lost, go on to

say: “This is a good example of what we have again and again observed in the courtship of the *Attidæ*: that whatever fine points of colour or structure the male possesses, his actions before the female display them to the very best advantage. In whatever part the special merit may lie, he sedulously strives to bring it to the notice and impress its beauty upon the mind of the female to whom he is paying his addresses.”^[133] As for the female, she is throughout described as watching the male eagerly and with the greatest interest, and that this interest is not always felt from the first, but is aroused by degrees, becoming, at last, so strong as to suspend for a time the natural inclination to assault and eat the wooer, is all the more significant. That there are dangers in these courtships there has been some indication, “but worse remains behind.” *Phidippus rufus* was caught once and eaten in an unguarded moment, and whilst *Phidippus morsitans* was waving his particularly handsome first pair of legs, “thickly adorned with white hairs,” precisely the same thing happened to him. Still, on the whole, such incidents are exceptional.

Particularly interesting is the account given of the courtship of *Saitis pulex*, a male of which species was introduced into a box already occupied by a female. “He saw her as she stood perfectly still, twelve inches away; the glance seemed to excite him, and he moved towards her; when some four inches from her he stood still, and then began the most remarkable performances that an amorous male could offer to an admiring female. She eyed him eagerly, changing her position from time to time, so that he might be always in view. He, raising his whole body on one side, by straightening out the legs, and lowering it on the other by folding the first two pairs of legs up and under, leant so far over as to be in danger of losing his balance, which he only maintained by sidling rapidly towards the lowered side. The palpus, too, on this side, was turned back to correspond to the direction of the legs nearest it. He moved in a semicircle for about two inches, and then instantly reversed the position of the legs and circled in the opposite direction, gradually approaching nearer and nearer to the female. Now she dashes towards him, while he, raising his first pair of legs, extends them upward and forward as if to hold her off, but withal slowly retreats. Again and again he circles from side to side, she gazing towards him in a softer mood, evidently admiring the grace of his antics. This is repeated until we have counted one hundred and eleven circles made by the ardent little male.

Now he approaches nearer and nearer, and when almost within reach, whirls madly around and around her, she joining and whirling with him in a giddy maze. Again he falls back and resumes his semicircular motions with his body tilted over; she, all excitement, lowers her head and raises her body so that it is almost vertical; both draw nearer”^[133]—and the male, now, for some short period is in no danger of being eaten.

Lastly—for this must be the last example—we have a species—*Astia vittata*—in which the male is represented by two differing forms, each of which dances before the female in its own particular way. One of these forms is red, like the female, which he resembles in other respects, so that this must be taken as the original specific type. The other, which has evidently been developed from it, in deference to the æsthetic preferences of the female, is black, with the special adornment of three tufts of hair on his head, or thereabouts, that part of a spider which is termed the cephalothorax. These tufts stick bolt upright, rising together, but separating about half-way up, and give to their fortunate possessor—for, as we shall see, he is fortunate—a very spruce and dapper appearance. Looked at dispassionately, if one can do that, they are certainly as handsome as moustaches, and there is no reason in the nature of things why they should not be admired as much. So, indeed, they are, and that the admiration bestowed upon each is of an equally high nature I, at any rate, see no reason to doubt.

The following description will show what a spider with moustaches can achieve: “The *vittata* form, which is quite like the female, when he approaches her raises his first legs either so that they point forward or upward, keeping his palpi stiffly outstretched, while the tip of his abdomen is bent to the ground. This position he commonly takes when three or four inches away. While he retains this attitude he keeps curving and waving his legs in a very curious manner. Frequently he raises only one of the legs of the first pair, running all the time from side to side. As he draws nearer to the female he lowers his body to the ground, and, dropping his legs also, places the two anterior pairs so that the tips touch in front, the proximal joints being turned almost at a right angle to the body. Now he glides in a semicircle before the female, sometimes advancing, sometimes receding, until at last she accepts his addresses. The *niger* form, evidently a later development, is much the more lively of the two, and whenever the two

varieties were seen to compete for a female the black one was successful.”^[133]

Here, surely, is a final answer to those assertions as to indifference on the part of the female, which, though made in the teeth of probability, are often, on account of the difficulties of observation, almost impossible to disprove. Here are two kinds of males, one lively and with moustaches, the other not so lively and without them; as the first is always, or even, say, generally chosen, his appearance must be preferred. Were it only his liveliness, as Dr. Wallace has suggested, why should he have acquired another dress as well as another dance? or, if the female can have a choice as between liveliness and slowness, as between a jig and a minuet, why, in Heaven’s name, should she not have one as between one get-up and another? Sexual selection might, I think, be put to the test in this one species with its two male forms. Let but a sufficient number of courtships be observed and reported on, and if *niger*, in a large percentage of them, wins the day, choice on the part of the female—the only link in the chain of evidence which it is at all possible to deny—is a proved thing.

But to continue: “He—*niger*—is bolder in his manners (no wonder he prevails), and we have never seen him assume the prone position, as the red form did, when close to the female. He always held one or both of the first legs high in the air, waving them wildly to and fro; or when the female became excited, he stood perfectly motionless before her, sometimes for a whole minute, seeming to fascinate her by the power of his glance”^[133]—greatly aided probably by the three tufts of hair showing through the archway of the uplifted legs. Here, again, too, as in some of the other species—perhaps all—“although the males were continually waving their first legs at each other, their quarrels were harmless. It was quite otherwise with the females, since they not only kept the other sex in awe of them, but not infrequently in their battles killed each other.”^[133] As the males cannot win the females by fighting, what have they to contend with effectively except these curious, elaborate, and most interesting displays, the purpose of which is so excessively obvious? On the other hand, the fact that the females yield, almost against their nature, to these displays, that they are slowly and gradually won through their means, is proof positive that they like them, and if so, how is it possible that they should not like one more or less than another? What, in fact, is choice but a greater or less reaction to

this stimulus or to that? The initial absurdity of laying claim to a monopoly of such a capacity as this, either in our matrimonial affairs, or any other matter in which animals participate, has not been sufficiently dwelt upon.

Professor Poulton, in considering this case of *Astra vittata* with its two male forms, one of which is always chosen by the female in preference to the other, remarks (with his own italics), “It must be admitted that these facts afford the *strongest support* to the theory of Sexual Selection.”^[134] He thus endorses—as anyone, I think, not hard-set the other way, must endorse—the opinion of the authors of the paper that “in the *Attidæ* we have conclusive evidence that the females pay close attention to the love dances of the males, and also that they have not only the power, but the will, to exercise a choice among the suitors for their favour,”^[135] to which he adds this rider: “Remembering that this conclusion has only been reached in the *Attidæ* by the closest study, I think we may safely explain the smaller confidence with which we can speak of other animals by the want of sufficiently careful and systematic investigation.”

The process of the narrative having led, in the last chapter or two, to a discussion of some of the ways in which insects become shaped and coloured through natural selection, sexual selection seemed marked out as the subject for this one. The reason why I have filled it with extracts from a certain very interesting paper has been a better one than that of saving myself trouble. That paper—the most important one perhaps that has ever been written on the subject—is a wonderful confirmation of Darwin’s views, but Darwin, as it appears to me, has not benefited by it in the way that he ought to do in the popular mind. There is no work that I know of, written upon merely popular lines, that brings these facts forward, and yet I feel sure that to large numbers of people, who yet do not care to read books avowedly scientific, they must be extremely interesting, not only in themselves, but as allowing them both to form a judgment on the subject, and on the correctness or otherwise of Darwin’s views—for Darwin is an interesting and picturesque figure far beyond the close borough of science.

Now the general more intelligent public who read, perhaps, widely, but not very deeply or very specially, know that Darwin believed in two forces—natural and sexual selection—by the joint action of which, species, as he held, had been gradually modified and evolved, and they know that the

former of these two has been accepted by science, but that to the latter there has been much more opposition, and that it is not—or is not supposed to be—established like the other. Many, perhaps, may have read Dr. Wallace's *Darwinism*, a work in which Darwin's most distinctive and original view—that one whose conception, apparently, he shared with nobody and on which he based much of the argument contained in his *Descent of Man*—is considered and rejected in a way which makes the title of the book misleading, surely, if not a somewhat comically ludicrous misnomer. All those who have read it, as well as many who have not, will be interested—they cannot fail to be—in the wonderful record of spider courtships contained in these extracts, and having reflected on them, they will, if I mistake not, be much more impressed with the arguments *for* this part of Darwinism than they were with those brought against it in the book of that name.

All these latter arguments, by the way—the languor of swallows as against the vitality of parrots, trogons, etc.—were well known to Darwin himself; and as no one was, at the same time, more impartial in considering, and more capable of correctly estimating, facts hostile to his own theories, or which, at first sight, might seem to be so, it may not be out of place to end this chapter with a reference to what he thought of them. This we may gather from a statement contained in a paper—the last, presumably, ever written by him—which was read before the Zoological Society but a few hours before his death, and which is as follows: “I may, perhaps, be here permitted to say that, after having carefully weighed, to the best of my ability, the various arguments which have been advanced against the principle of sexual selection, I remain firmly convinced of its truth.”

CHAPTER XXIII

Web making spiders—Dangerous wooings—An unkind lady-love—Lizard-eating spiders—Enlightened curiosity—Rival entomologists—Instinct of resignation—A worm-eating spider—Alternative explanation—The dangers of patriotism—Trap-door spiders—Web-flying spiders—Spiders that nearly fly—Spider navigators—The raft and the diving-bell.

NONE of the spiders mentioned in the last chapter are web-makers. These latter are not dancers; that is to say, the males do not dance before the females when they wish to recommend themselves as husbands. Instead, they pull at the strands of the web, whilst stationed at its circumference, in a manner which has a distinct meaning for the female, who sits in the centre, and who replies by other twitches. These may be either of an encouraging or repellent nature, and it is only in the former case that the lover ventures to approach. This, however, he must do with extreme caution, and prepared at any moment to drop and hang suspended by a thread should the object of his attentions, who greatly exceeds him in size, change her mind or conceive some cause of displeasure against him. Should he not be sufficiently quick on such occasions, he is liable to be spun up between the long legs of his lady-love as though he were a fly, and disposed of accordingly. This was observed in 1798 by Raymond Maria de Termayer, who remarks upon it: "Perhaps overpowering hunger compelled her to do it, but the act was very ferocious."

The most curious thing in these *webepathic* courtships, as one may call them, is that the female spider seems to know the particular jerk or twitch of any strand of the web which is made by a male, and to distinguish it, perfectly, from the vibrations set in motion by a fly or other insect that enters it, for upon these occasions, though her back may be turned towards her admirer, she does not trouble to look round, whereas in the latter case she would not only do so, but come rushing down to secure her victim—if she were hungry, perhaps it should be added. On the other hand, as has been already mentioned, the male can interpret the wishes of the female from the

movement she imparts to the thread, and regulates his conduct accordingly. Webepathy, therefore, does not seem a name ill-chosen to describe this system of intercommunication of ideas.

The spider mentioned above as devouring her lover was the common garden or geometric one, as it is sometimes called, which in England is the largest example of a web-spinner. In other parts of the world, however, web-spinning spiders attain to a much larger size, and their webs, of course, are in proportion. The largest, perhaps, are found in Madagascar, and the gigantic fabrics which many of these weave are curiously utilised by smaller spiders of a parasitic disposition, who spin their own little webs between the thick strands of those of their hosts. Here they live in perfect amity with the latter, in whose presence they find a protection against the attacks of small birds—for these, it would seem, stand in awe of these huge spiders, in whose toils they are sometimes accidentally caught, and by whom they are then devoured. So, at least, Vinson, the historian of the spiders of Madagascar, would seem to imply, if he does not actually make the statement, of which I will not be quite sure.^[136] That the great *Mygale* of South America eats birds is now an established fact, Bates having given an account of it in his well-known work, *The Naturalist on the River Amazon*.^[137] In this case also the birds—for there were two of them—were caught in a web, but it was not a geometric one, in which the spider sat, but a much denser and more closely woven fabric stretched across a crevice, or irregularity, in the trunk of a tree, the spider—which was of much larger size than the largest *Epeira*—keeping watch behind it in the recess.



A BIRD-CATCHING SPIDER'S WEB.

This enormous Madagascar spider spins webs so strong that birds are caught and held in them. In one of the large meshes will be seen a small parasitic spider's web for catching flies and other insects. The smaller spider is not only permitted to do this, but is protected by its host from the attacks of the smaller birds.

Other spiders—as doubtless the *Mygale* if he can get them—will eat lizards, as the following account by Mr. Frederick Pollock will show: “Having procured from the Deserta Grande some fine specimens of this large and handsome spider (*Lycosa*—a kind of tarantula) in the early part of this year, and having provided suitable cages with glass lids for them, I was anxious to ascertain how large an animal the largest spider would take; and for this purpose I obtained some lizards about three inches long, including the tail. Three of these lizards were killed and devoured by one spider during the time I kept it. They were eaten bones and head and claws and all, the only remnant of the feast being a small ball about a quarter of an inch in diameter, which was cast aside at the bottom of the cage.”^[138] But why were not some larger lizards tried, since there was no difficulty about three inches? Every inch would have increased the fun—I mean have added to the scientific interest. But perhaps there were none larger.

Mr. Pollock goes on to say that “the islands of Madeira, Porto Santo, and Deserta Grande all lie within an area about fifty miles across. They have each its own peculiar large *Lycosa*, no two being alike; and it is a very remarkable fact that these *Lycosæ* vary in size inversely with the magnitude of the island on which they are found—Madeira, the largest island, having the smallest *Lycosa*, and Deserta Grande, the smallest island, having by far the largest spider. The mode of defence of all these varieties of *Lycosæ* is precisely the same. They elevate the thorax, raise the first pair of legs high up, and opening wide asunder their falces, strike at and seize any object, such as the end of a pencil” (or the tail of a lizard) “in a most formidable manner.”^[138]

There is another lizard-eating spider, or at least a spider that will eat lizards when formally introduced to them, and that in a very scientifically interesting manner, the lizard showing such a lively sense of its situation, and the jaws of the spider working in a way which is very curious. These jaws, it appears, are double, also “cheliform,” “denticulate,” and several other things, from which I gather that there are two pairs, each pair working something like the claws of a crab, but with a sawing action, adapted to their toothed surface. By an extremely beautiful adjustment, when the spider in question seizes its prey, one pair of jaws holds on to it, whilst the other saws into it, and then the pair which has been sawing, holds, and the

pair which has been holding, saws, and so on alternately, a division, and yet, at the same time, a combination of labour.

“Like to a double cherry, seeming parted,
But yet an union in partition.”

The efficacy of the arrangement was well tested by an Anglo-Indian scientist upon a lizard three inches long, *exclusive* this time of the tail. “The spider sprang upon it, and made a seizure immediately behind the shoulder. The poor lizard struggled violently at first, rolling over and over in its agony, but the spider kept firm hold, and gradually sawed away with its double jaws into the very entrails of its victim.”^[139] There was an interesting variation between this case and the last, where, it will be remembered, the lizards were eaten “bones and head and claws and all,” whereas here “the only parts *uneaten* were the jaws and part of the skin.” This lizard, however, was “at least five inches long from nose to extremity of tail”; but then, again, the spider must have been larger too, though clearly its meal was something in the nature of a feat, since after it “it remained gorged and motionless for about a fortnight, being much swollen and distended.”^[139] There is no mention of this in the other case, which would seem to imply that the result was different. If so, we have here a fact of great interest—what fact, scientifically elicited, is not?—but in order to establish it upon a really firm basis, further experiments should be made, and, once more, as the limit of size has evidently not yet been reached, I would recommend a lizard of six inches long.

This spider, however, now I come to remember, is not really one, but a *solpugid*, and a *solpugid* is a creature so like a spider that it used to be thought one some years ago, but now belongs, not only to a family, but to an order of its own, which comes somewhere between the scorpions and the true spiders. They are large creatures, and their bite is very severe, though it does not appear to be poisonous. Some of the species are nocturnal, as is the case with the one above mentioned, which was christened *Galeodes vorax* by its discoverer, Captain Hutton, if he indeed discovered it. Captain Hutton, being a great entomologist—to attain which title one has only to put pins through insects—used to lay a sheet on the ground at night, and stand a lantern upon it. Numerous insects were attracted to the light, and this

brought *Galeodes*—who is a great entomologist too, though without the pins—upon the *tapis* or *draps*.^[139]

Often there would be fights between two rivals, and of these, or, rather, of the general fighting, and one may also say yielding, habits of the species, Captain Hutton gives the following somewhat curious account. “They plant their true feet” (for these *Galeodes* have a pair which look exactly like feet, but are really their palpi) “firmly on the ground, the body at the same time being elevated and the two pairs of palpi held out in front, to ward off the attack. In this attitude they advance and retire, according as either gains a slight advantage, endeavouring to throw each other to one side, so as to expose some vulnerable part or form an opening for attack; and when this is once effected the fortunate wrestler instantly takes advantage of it, and rushing in, seizes his adversary behind the thorax, and the combat is ended, the vanquished victim yielding himself, without further struggle, to his inevitable fate.” Similarly, “if, in their efforts to get away, they are brought into contact, the one instantly seizes the other and devours him, the victim making no struggle whatever; but if they meet face to face they both enter into a wrestling match for life or death.”

This habit of yielding as soon as there is no more use in struggling seems a very strange one, since it is opposed to the primary instinct of self-preservation, and it is not easy to see how the species can benefit by certain individuals dying in a passive manner, unless, indeed, by refusing to do so they might injure the victor, who, by dying afterwards, would add to the tribal mortality. If this be really the explanation, we are reminded of Huber’s statement as to two queen bees, when each has it in its power to sting the other, being seized with a sort of horror, under the influence of which they separate, thus avoiding the catastrophe of leaving the hive queenless.

Though, as we have seen, there are some spiders which eat birds, and others which eat lizards, yet both these interesting things take place abroad. Here, however, in England, it would seem that we have a spider which eats worms, catching them at the end of a long woven bag which descends into the earth, and into which the worm somehow manages to get. How it does so, however, is not at all clear, since the bag, which is sometimes a foot long, is described as having no opening at either end, the spider living enclosed in it, apparently a permanent prisoner. Still a sac like this would

seem as difficult of entry for flies as for worms, and the spider, which is three-quarters of an inch in length, and armed with very large mandibles, or falces—to use the approved word—must, it is plain, live on something. As a matter of fact, it was living on a worm when found by Mr. Brown, who gives the following account of the matter. “On drawing out one of the sacs,” he says, “I observed a worm at the lower end, partially within the sac and partially outside, and it was evident that the spider had been eating a considerable portion of its anterior extremity.” One would have thought that a careful examination of the way in which the worm had got into the sac would have thrown light on the problem, but of this we hear nothing more.

When taken out of the ground the sac was limp and flaccid, but afterwards the spider inflated it, and it was then seen to have some minute valves—“openings,” that is to say, “protected or covered by a little valve or door.” They were not, however, to be detected in every nest—possibly on account of their very small size. That the object of these valvular openings is to admit air seems obvious, for spiders breathe through lungs and require a good supply of oxygen. It was now concluded, both by Mr. Brown and Mr. Newman—a well-known entomologist—that this particular spider lived on worms and resided permanently in a long subterranean sac or bag, which it had the power of inflating with air. A different view, however, was propounded by Mr. Meade—an authority on spiders—who suggested that the worm had only got into the sac by accident, and that the spider, like other subterranean nest-makers, probably came out at night and fed abroad, returning to rest at home during the daytime. To do this it would, indeed, have to unweave one end of its sac—probably the upper one—and then do it up again, but there was no reason, in Mr. Meade’s opinion, why it should not act in this way. To me, however, it seems unlikely that the minute valves, made with such care, should be destroyed in this manner and made afresh every day; and moreover, when Mr. Brown looked again at the sac, in order to test this theory, he could find no evidence of its having been dealt with in this Penelopean manner. There were no traces of fresh silk. The evidence, therefore, seems to be more in favour of *Attipus sulzeri*—for that is its name—being a genuine worm-eating spider. If so, it is worthy of all respect as a curiously aberrant form.^[140]

These spiders are of a deep brown colour, with a very soft abdomen and a generally half-baked appearance, but with hard, black, shining mandibles.

There was only a single individual—evidently the female—in the sacs taken by Mr. Brown, but I myself was the finder of one such sac—for I feel sure it must have been the same—in the New Forest, and in this a pair were amicably settled, one being about twice the size of the other. This, if I mistake not, was in May, but I also remember, or seem to, that the bag was quite open at one end. Thus, then, stand the facts. Upon them I think we are justified in believing that there is a worm-eating spider in England, but of course it would be a very high honour for any country to have such a creature, so that there is a danger of letting one's patriotic feelings run away with one.

In these sac-making spiders we see, perhaps, the ancestors of, or rather travellers towards, those which crown a silk-lined perpendicular tunnel with a skilfully made trap-door. The latter is furnished with a hinge, and should it be discovered, the spider, seizing it from within, endeavours with might and main to prevent its being raised. As is well known, the upper surface of these trap-doors, or, as we may call them, lids, are covered by the spider with such materials—leaves, grass, moss, etc.—as surround the site of its nest, so that when shut down they are indistinguishable from the general surface of the ground.

Another use to which the webs of spiders are put is that of a parachute, on which the little creature—for small species alone may enjoy this luxury—sails delicately through the air. This, however is not the nearest approach made by any of the tribe to actual flight, though in practice it almost surpasses that power, even as possessed by many winged insects, who do indeed cleave the air, but cannot ride upon it in a filmy chariot, twinkling in the rays of the sun. Still there is one spider that, though it has not yet achieved wings, is in process of developing them. This little semi-Ariel—but the subject will be best done justice to in the glowing language of the Rev. O. P. Cambridge: “Adult male, length rather above two lines. The abdomen is of an elongated oval form, and rather flattened; its upper side is furnished with an epidermis, which is continued, laterally, on either side to an extent considerably exceeding the width of the abdomen, and of a semi-oval or elliptical form; the outer portion of this epidermis, on either side, is capable of being depressed and folded round the abdomen, or elevated and expanded to its full width, after the manner of wings. Mr. H. H. B. Bradley, of Sydney, New South Wales, to whom I am indebted for examples of this

exceedingly interesting and remarkable spider, tells me that he has observed them elevating and depressing the flaps, and also actually using them as wings or supporters to sustain the length of their leaps. That this, as with an analogous appendage in the flying squirrel, should be intended for such sustentation one could have but little doubt, after examining it even in the preserved specimens. The three examples were all found on one spot near Sydney, in the month of October, running and jumping on low plants and flowers.”^[141]

It is delightful to think of a little delicate spider-body like this, rising gracefully from the petal of one flower, expanding its thin, filmy fringes, and descending in a long slanting line through the air, like a flying squirrel or a galeopithecus, onto the petal of another. Even were its appearance no more than elegant, this would be a most pleasing sight. But it is much more than this. Various hues meet in its diminutive body, and so harmonious and pleasing is the general effect produced by them that the first captor of so much loveliness was enraptured as he gazed on his prize, whilst even Mr. Cambridge, with only dried specimens to fire his imagination, yet cannot choose but exclaim: “It is difficult to describe adequately the great beauty of the colouring of this spider.”^[141]

Spiders, then, either through gossamer or their own structural modification, seem engaged in the conquest of the air. There remains but the water, and this element also they have partly subdued. There are raft-spiders and diving-bell spiders. The first, having woven a few dried leaves, stalks, grasses, etc., launches out as courageously from the shore as the first navigator, whose heart, according to Horace, was thrice bound in brass, but who probably was timid and cautious. Our spider, however, has no fears, nor need it to have any, since no sort of capsizal can affect a structure which answers its purpose as well one side up as another; whilst even if it were to sink—though that hardly lies in its nature—there is always the water to run on. The raft, in fact, is only like the nest or web ashore—a place to have a comfortable meal in. The prey—some aquatic insect—is caught generally on the surface of the stream, and the spider, after each successful raid, skims back with her booty to the little self-guiding boat which it has temporarily left. There, when no longer hungry, she sits and scuds about, careless and pleasure-loving, like another little Phædria in her “flitt barck” over the waters of the Idle Lake.^[142]

“Thus,” says Büchner, “everywhere in nature are battle, craft, and ingenuity, all following the merciless law of egoism, in order to maintain their own lives and to destroy those of others.” In man, indeed, there is some counterpoise to all this in the mind and *façon de parler*; but the lower animals do not think so much, and, having no proper language, cannot even talk altruistically.

Lastly, we have the water-spider, whose little spun nest, against the submerged stem of some aquatic plant, is open at the bottom like a diving-bell, and filled with air which its owner carries down from the surface in successive bubbles, each one looking “like a globe of quicksilver.” To collect them she swims on her back, and, in some manner, entangles them amongst the numerous hairs with which her abdomen is covered, where they cling safely all through the journey,

“Proud of that delicate lodgment.”

There may be other bubbles as pretty, perhaps, but few, by bursting, do such good to those who have cherished them. In winter, it would seem, the spider closes the entrance to the diving-bell, and sleeps, dry and soft, in a well-aired bed, in spite of the damp situation.

CHAPTER XXIV

Aquatic insects—Lyonnet's water-beetle—A floating cradle—Larva and pupa—An ingenious contrivance—Nothing useless—The imaginary philosopher—How the cradle is made—The mysterious "mast"—Later observation—The giant water-bug—An oppressed husband.

SPIDERS having brought us to the water, it may be as well, or even better, in view of the title of this work, to say something about water-insects. Of these, so long as the water be fresh, and not salt, there are many, and the largest, perhaps, if he exceeds some of the dragon-fly larvæ and the Giant Water-Bug of America, must be the Great Water-Beetle—*Hydrophilus piceus*—which is larger even than the much commoner one—*Dytiscus*—which everybody knows, and which is *the* water-beetle to most people.

It is the fate of some animals to become associated for all time in our minds with the name of some particular man, as, for instance, the bee is with that of Huber, and the ephemera with that of Swammerdam. Again, the fame of Lyonnet, though he was skilled in eight languages, and became cypher secretary and confidential translator to the United Provinces of Holland, is principally bound up with a certain caterpillar, viz. that of the goat moth, of which creature, though only an amateur in such matters, he made dissections and executed plates, which have never yet been surpassed, and are supposed to be entirely unsurpassable. In a lesser degree his memory is associated with this particular water-beetle—the great one, into the heart of whose mystery he was the first to pierce: "In the beginning of July," he tells us, "I had noticed in the ditches a kind of cocoon which I did not recognise. It was whitish, of the size of the end of the finger, nearly spherical, but rather oval and flattened. The surface, which looked like tow, was not quite smooth. One of the two ends was flatter than the other, and furnished with a raised rim. From the space within this rim projected a sort of little tapering mast about as long as the cocoon."^[143]

These cocoons, when opened, were found to contain about a hundred eggs. Lyonnet kept them in water till the eggs hatched. “The larvæ,” he says, when this had taken place, “remained one day enclosed in the cocoon before escaping. Then they made an oval aperture in the lower part of the flattened end of the cocoon, and escaped through this into the water.”^[143] Here they fed upon snails, their manner of eating which is thus described: “The larva seizes the snail with its mandibles, then bends its body backwards and rests the snail upon the broad back, which serves as a table (as with the larva of the grain-eating ant of Texas). In this position, holding the snail in its legs, the larva breaks the shell, and devours it.”^[143]

When full-fed the larvæ left the water, and one of them was placed by Lyonnet in a box full of moistened earth. This it entered, and, some days afterwards, changed into a large white pupa or chrysalis, about which there was one curious feature, viz. that “on each side of the head”—or, as an entomologist would say nowadays, “on the fore part of the prothorax—were three brown, strong hooks. Two others of the same kind were found at the hinder end of the body.”^[143] These hooks were solid, so that they could contain no part of the perfect insect, and Lyonnet points to them as good examples of *apparently* useless structures. Their office, indeed, he himself knows, but he does not reveal it till the usual philosopher has been imagined who denies that they can have any. Then, of course, comes the anticipated discomfiture of this unwary person—so frequent in the eighteenth century—who, unwarned by experience, has walked quietly into the trap. “In the damp earth which the pupa requires the above-described hooks fulfil a purpose, unexpected by us, but, at the same time, of great importance. The skin of the pupa is very delicate. Lying on damp earth, it could hardly escape injury, and the weight of the body might easily give it a distorted shape. But (*Monsieur le Philosophe*) the pupa protects itself from these dangers by assuming an unusual attitude. It extends itself back downwards in a horizontal position, and supports the weight of its body by the three sets of hooks, as upon a tripod. In this attitude, though surrounded on all sides by moist earth, it keeps its body from actual contact with any object until it has assumed its final shape. Thus,” continues Lyonnet, turning full upon the stupefied philosopher, “we see how necessary are those hooks, which at first sight appeared so useless. To decide that this or that structure is superfluous because we cannot guess its use is truly ridiculous in beings

whose information is so limited as ours.”^[143] Applauding shouts (“*Mais certainement!*” “*C’est vrai cela!*”) rend the air, and the imaginary philosopher goes out in a state of painful confusion.

The above facts, first made known by Lyonnet, have been confirmed by subsequent observers, such as Miger, and the pupa of another and much smaller water-beetle is now known to support itself in the same manner, or, rather, on the same principle, since the place of hooks at either extremity is taken by spiny projections, with which the back is covered.

Lyonnet now turned his attention to the mature beetle, and especially to the female, whom he was anxious to see make her cocoon. Having put a few in a large wooden trough and supplied them with some floating weed, “I had,” he says, “before long, the pleasure of seeing the female *Hydrophilus* betake herself to work under my eyes. I found, to my surprise, that, like the spider, she had her spinneret at the hinder end of the body. Two small brown prominences enclosed each a delicate conical tube, from each of which a separate thread proceeded, and with these the cocoon was woven in the following way. At first, lying upside-down near the surface of the water, the beetle buried the hinder part of her body, and the two hindermost pairs of legs, in the weed, whilst with the first pair, which were free, she drew and pressed the weed around the end of her body, moulding it to its shape. She then began to weave what seemed the under half of her cocoon, but having finished this part she turned over with it so that it became the upper half, and then wove the real under one. The two curved surfaces were then woven together, and in about an hour and a quarter the body of the cocoon was finished. For about two hours after this the beetle remained still, her back being uppermost. At first her body was buried in the cocoon up to the thorax, but one could see that she was gradually withdrawing it. During these two hours of apparent rest she laid her eggs, not at hazard, but in regular order, side by side, the pointed ends uppermost. This work accomplished, she closed the mouth of the cocoon, and then began to spin the little mast, which gradually rose above the surface of the water till it had attained the requisite height, and the cocoon was then finished.”^[143]

Lyonnet was unable to discover the use of the so-called mast, and it remains a mystery to this day, so that the imaginary philosopher might have a better chance here, were it not his *métier* to be put to confusion. It is hollow, and as the cocoon contains air, with which the beetle supplies it—just as the

water-spider does her diving-bell—Miger, whose observations were made in 1807, some fifty years after those of Lyonnet, supposed it might serve as the channel of entry. But, although hollow, it has no orifice, but is closed at the end, and this does not seem to accord with the above view. Mr. G. A. Laker, a modern observer, does not think that the spike can serve as a balance to the cocoon, since this is usually attached to some weed, or other supporting substance. He, however, cut the spike off two of the cocoons, and the eggs in both of these remained unhatched. Moreover, these cocoons subsequently sank, whereas in their normal state they “are so constructed that when floating loose the spike retains its proper position, and even if the cocoon be held so that the spike is parallel with the water and then suddenly released, it immediately rights itself.”^[144] The balance theory, therefore, certainly seems to have something in its favour. Lyonnet’s own conjecture was that the mast, as he calls it—a designation against which Miger protests—might merely represent the waste silk which the beetle felt impelled to get rid of. The time taken by the beetle in making the whole cocoon is about five hours, whilst the mast, spike, or turned-up point, as Miger severely calls it, takes it half an hour. It is curious that whereas Lyonnet’s cocoons held “about a hundred” eggs, Mr. Laker gives the number as “usually between fifty and sixty.”^[144]

As space has its exigencies—and long may it continue to have—I will here merely mention such names as *Gyrinus*, *Dytiscus*, *Hydrobius*, *Donacia*, etc., “and let *them* speak for me,” but having paid some attention to the great water-beetle, silence in regard to the giant water-bug would be hardly gracious, and might be ill taken. This terrific creature is like a monstrous exaggeration of our own water-scorpion, to which it bears a distorted, but real resemblance, minus, however, the long ovipositor—the so-called tail behind. Its appearance is not to be described. Like other bugs, and as are the aphides and cicadas for less cruel purposes, it is armed with a long, sharp-pointed beak, through which, having plunged it into the body of its prey, on whose back it has previously leaped, it sucks the life-juices, holding on, all the while, with its two curved, claw-like front legs. Its strength is in accordance with its size, and both are such that it finds no one in its own circle, so to speak, at all capable of contending with it. “It is the facile master of the ponds and estuaries of the tidal creeks and rivers of the Atlantic States,” says Uhler. “Developing in the quiet pools, secreting itself

beneath stones or rubbish, it watches the approach of a *Pomotis*, mud-minnow, frog, or other small-sized tenant of the water, when it darts with sudden rapidity upon its unprepared victim, grasps the creature with its strong, clasping fore-legs, plunges its deadly beak deep into the flesh, and proceeds with the utmost coolness to leisurely suck its blood. A copious supply of saliva is poured into the wound, and no doubt aids in producing the paralysis which so speedily follows its puncture in small creatures.”^[145]

Another American water-bug of similar build, but much smaller size, has the same general habits, to which it adds the more special one of carrying about its eggs on its back, where, in time, they are hatched, but do not, it would appear—though this seems somewhat out of harmony with the practical spirit of nature—proceed at once to suck their parent’s blood, an omission which, as it would be a most moving instance of unselfish surrender on the part of the latter, is, perhaps, to be regretted. Possibly the reason is that the eggs are not fixed upon the right back, so that even were this *dénouement* to take place, we should not have an instance of maternal, but only of paternal affection. This, for some reason, is not so effective as the other, and therefore Nature, who, as we know, is a consummate artist, may not care to waste her materials on an inferior situation.

Be this as it may, the fact that the domestic economy of these water-bugs did not proceed, throughout, upon the lines that might have been expected may first have led a German observer—Schmidt—to suspect something unusual, in consequence of which misgiving he looked more closely into the matter, and found—what had not before been imagined—that the male and not the female was the egg-bearer. He was not, however, able to determine how this arrangement was brought about, or with what feelings the male received and bore his burden. This was left for Miss Slater, who found that the females in her aquarium insisted upon laying their eggs on the backs of the males, that the latter objected to their doing so, which led to a struggle between the two, often lasting for two or three hours, but ending invariably in the victory of the female. The male has, then, to bow to necessity, but he does not do so in a cheerful spirit, nor even without some further efforts to escape his destiny. “That he chafes under the burden,” says Miss Slater, “is unmistakable; in fact my suspicions as to the sex of the egg-carrier were first aroused by watching one in an aquarium which was trying to free itself from its load of eggs, an exhibition of a lack of maternal

interest not to be expected in a female carrying her own eggs. Generally the *Zaithas* are very active, darting about with great rapidity, but an egg-bearer remains quietly clinging to a leaf, with the end of the abdomen just out of the water. If attacked he meekly receives the blows, seemingly preferring death (which, in several cases, was the result) to the indignity of carrying and caring for the eggs.”^[145] This last, however, is not very explicit, so that, the whole account not being to hand, I cannot say what precisely happened.

It is curious that the male should be so spiritless, after receiving the eggs, for this would seem to nullify such advantages as the arrangement might otherwise offer. The eggs must be laid somewhere, and might be supposed safer on the back of the male than elsewhere—in which fact, perhaps, we may see the origin of the instinct. But if the male, sinking under his burden, is able neither to defend himself nor it, this advantage seems nullified.

Estuaries and tidal creeks, which, as we have seen, are included in the habitat of these water-bugs—at least, of the giant one—bring us gradually to the sea. That there are marine insects we know, but they do not appear to extend beyond the tidal beach, on the sands of which they expatiate, when the sea is out, and burrow into them on its return. All are small, and still smaller is the amount said about them, even in such works as are precisely those where all that is known on the subject ought to be stated—systematic works of natural history, for instance, which take “*Arthropoda* (Insects, etc.)” in their due order, but do not so much as tell you whether marine ones exist or not. Yet the date of such works is after 1895. For these reasons, and another which has been once or twice before alluded to, I have but one remark to make about marine insects, and I will make that in the next chapter.

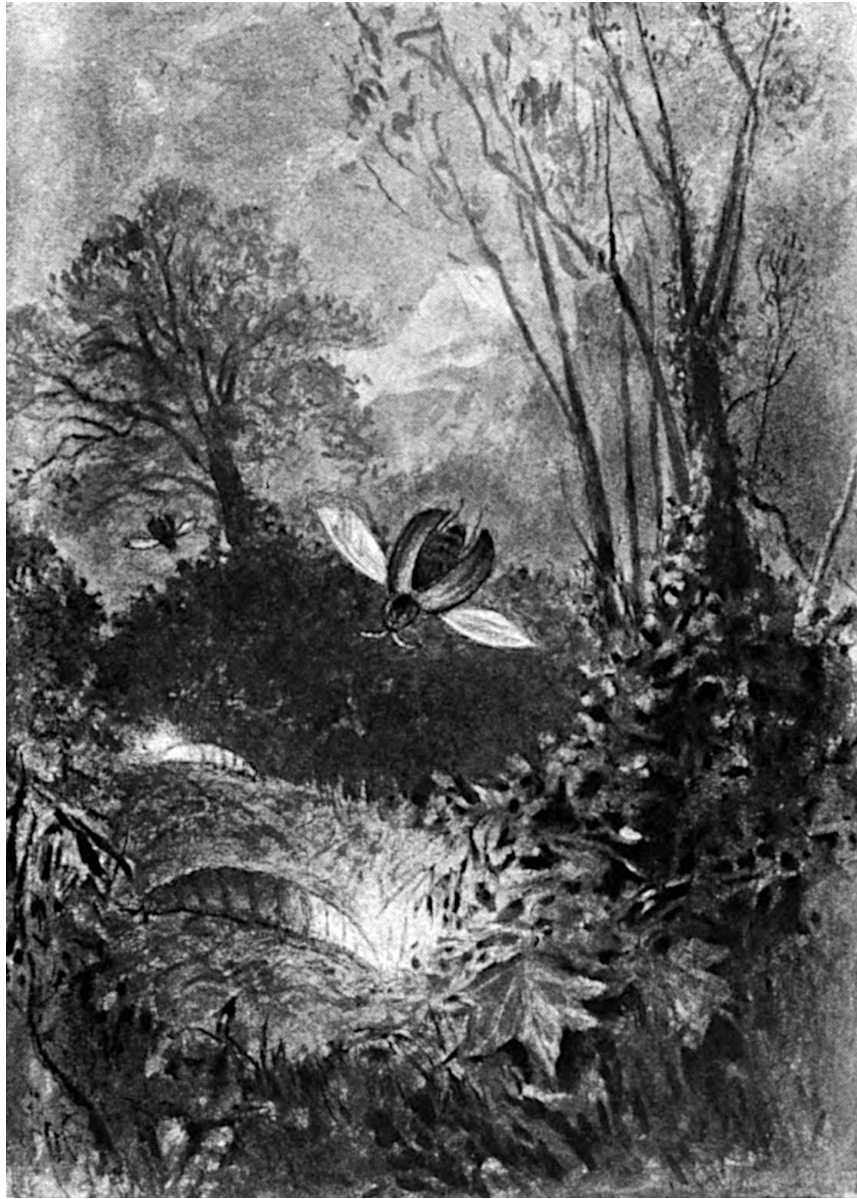
CHAPTER XXV

One remark—Phosphorescent insects—Glow-worms and fire-flies—Fiery courtship—
A beetle with three lamps—Travelling by beetle-light—The great lantern-fly
controversy—Is it luminous?—Madame Merian’s statement—Contradictory
evidence—A Chinese edict—Suggested use of the “lantern”—Confirmation
required—Luminous centipedes.

NO marine insect—this is the remark—is phosphorescent—that is to say, as far as I know, which is a very saving clause indeed. This seems curious, because, as everyone knows, other sea-dwelling creatures are, producing most wonderful and beautiful effects, and, moreover, the luminous property is active in many terrestrial insects. Of these the glow-worm is a familiar and, though, perhaps, the humblest, a very beautiful example. At any rate, there are insects of the glow-worm family whose fires are far less “ineffectual,” or, to speak more truly, far outglow those of our own species. What, for instance, can be more gorgeous than the green or orange lights—for they differ in colour according to the sexes—with which the nights and the rich vegetation of the West Indies are brilliantly, yet softly, lit up? Nothing, surely, if it be not the name of the creature producing such splendour, which is *Pygolampis xanthophotis*^[146]—not one syllable less.

Whether it is the male or the female that gives out the green or the orange light, I do not know, nor in my opinion do various monographists in various encyclopædias and text-books, though they make no such avowal, but content themselves with not saying. However, it is not a matter of importance except to the insect producing it, in whose breast the one or the other colour arouses very different sensations—rivalry or love. For there is no doubt now that these lovely illuminations, as well as those of our own glow-worm and of every other light-bearing creature, have relation to the needs and wants of their producers, to whose æsthetic sense, and not to ours, they are intended to appeal. That they appeal also to our own is a mere irrelevant side-issue, not considered, so to speak, by the force under whose pressure these beauties were called forth, and not of the smallest

consequence. It was not always thought so, and were the pride of man reachable by such considerations it might humiliate us to reflect that displays, which in real beauty immeasurably surpass our clumsy illuminations and fireworks, are made nightly, not for our eyes, but for those of a beetle.



INSECTS THAT CARRY LAMPS

The glow-worms in this picture are rather larger than life. The male insects have wings; it is the females chiefly, if not solely, that emit the soft, beautiful light.

Gilbert White, however, in the eighteenth century, exclaims amidst some very pleasing verses:

“For see, the glow-worm lights her amorous fire!”[\[148\]](#)

on which one of his editors of the nineteenth remarks: “This is still the generally received notion, but the fact is that both sexes of the glow-worm are phosphorescent, not only in the perfect insect, but also in the larva and even pupa state.”^[147] But this does not affect White’s statement, which is the simple fact, as well as “the generally received notion,” and, moreover, though our own male glow-worm is phosphorescent, it is not so brilliantly so as the female. Indeed, in the ninth edition of the *Encyclopædia Britannica*—which is later than this editorial note—it is stated not to be so at all, so that even if White believed this—which is not very clear—he has been supported by learned authority for a very long time.

In other species the male is the more brilliant, or the sexes do not differ greatly in this respect, each one lighting its “amorous fire” in the degree that nature allows it to—as no doubt our own male does too. Of this fact, which, in the light of Darwinism, might have been boldly assumed, there is no longer any doubt after Professor Emery’s interesting observations^[149] on the Italian species *Luciola Italica*. These were made in the meadows around Bologna, where, having caught some females, the Professor imprisoned them in glass tubes and laid them down amidst the grass. In this situation, though smell as an attractive agent was excluded, males would come flashing to the glass, and, on the other hand, as soon as the lamp of any of these became visible, the female would kindle her own, if it had previously been unlighted. Arrived on the spot, the male would dash madly about the unapproachable female, who continued to light her lamp at him till another, and then others, arrived, when it is to be supposed that her favours were distributed. In the end there would sometimes be a dozen fiery rivals glowing and flashing round the tube. But though the female shot out her attractive beams with evident intent to please, it does not appear that she was the seeker in the business, since we hear only of males flying to the imprisoned females, and not of females pursuing these males. To such modest merit, therefore, as a nice distinction between different ways of attaining the same end may entitle her, the female glow-worm also is entitled.

The light of the two sexes in the Italian glow-worm is described by Professor Emery as being the same in colour and intensity, but differing in some other respects. The flashes of the male, for instance, are more quickly recurrent, whilst those of the female gleam out at longer intervals, but last

for a longer time. They are, also, more tremulous, as well as more restricted, though what is meant by this last expression, since the brightness is said to be equal, is not quite apparent. Possibly it may imply that the light proceeds from a lesser area of the body, but, if so, this should be clearly stated, even in a *résumé*. I can find no reference to such a fact, if it be one, in the text-books.

From the above it is evident that the glow-worm's fires are anything but "uneffectual" from the point of view of the insect, but Shakespeare was no doubt thinking of something very different—their paling, namely, before the light of dawn. According to Gilbert White, however, they should have been out long ago—the glow-worm being too wise to afford opportunities of comparison in this respect. Thus subtly does the naturalist of Selborne impugn the accuracy of the Bard of Avon: "By observing," he says, "two glow-worms which were brought from the field to the bank in the garden, it appeared to us that these little creatures put out their lamps between eleven and twelve, and shine no more for the rest of the night."^[149] The intention here, though cleverly disguised, is not sufficiently so to escape detection. It was possibly seen through by the late Charles and Mary Cowden Clarke, who in one of the million or so notes to their edition of Shakespeare, say, without distinct reference to the passage in question:—"Uneffectual. There is double signification included in this word; it means the glow-worm's light, which shines without giving heat, and which no longer shows when morning appears."^[150] Thus whilst not committing themselves to White's opinion they provide a safe refuge for their author, in case it should prove in time to be correct; according to the sound principle contained in a Russian proverb which says, "Had he known where he was going to fall, he would have laid down straw."

In tropical countries fire-flies take the place of glow-worms with us, and though the light which these give out is not so soft and poetic as the lovely green or golden green one of the latter, yet it is more effectively beautiful, owing to the way in which it wanders through the night, appearing and disappearing in successive brilliant flashes. For here the beetle that carries the lamp is a flier, and flashes it about at pleasure through the air, having the power, it would seem, either of showing or concealing its light. The effect of a number of these points of brilliancy, gleaming out, now here, now there, on the soft night air of the tropics, is inexpressibly beautiful, as

though, in a smaller firmament, innumerable miniature stars had ceaseless birth and death.

Women, who like to emphasise their own beauty, or the want of it, by placing themselves in juxtaposition with every lovely thing in nature, and care not if a thousand deaths go to help one smile or glance, have not forgotten the fire-flies. They put them in their hair, or wire them onto their dresses, threading them together, sometimes, in long bands, which they wind about their fair—or otherwise—persons; they do this, more especially, when going out to parties, fancy-dress balls, or other social entertainments. The advantages are obvious, for the homeliest features may be thus lighted up, and the dullest woman become brilliant. No wonder that in some South American cities—Vera Cruz for example—these fire-fly beetles form quite an important article of trade, all for toilette purposes.^[151] The natives catch them by waving sticks with burning coals tied to their ends through the air, by the light of which they are attracted, and so come within reach of a long-handled butterfly net. When caught, they are put into a box covered with a little netting of wire, and there kept till wanted. They are fed upon sugar-cane, and twice a day must be bathed in tepid water.^[151]

What is done with the poor beetles after they have contributed to the night's amusement we are never told—whether those that have been all wired together are unwired and let go, or pulled off in two or more pieces to save trouble, as seems more likely. It is likelier still perhaps, in the houses of the rich, that the whole thing is flung aside, and the poor living lamps left to struggle till they die—unprovided with sugar-cane. But such details are not thought worth mentioning. The charming effect is the one thing dwelt upon, and charming it may very well be, though to gain it through a mass of even insect discomfort is, to my mind, a contemptible thing. Fancy fifty or a hundred uncomfortable, writhing, struggling things on the dress that a lady is dancing in, every one of which, if let go, would make a wandering star in the air more really worth looking at than the whole ball-room together! By substituting flowers for women, however, effects far more beautiful are gained through less reprehensible means. The fire-beetles—why should they be called flies?—are in this case confined in small globes of delicate glass, set amidst clusters of flowers, or flowering shrubs, and thus they softly illuminate the garden. Give them some sugar-cane whilst the party is in progress, and let them go next morning, and they will have had very little

to complain of—a strange experience for any lower creature that gets into the clutches of the highest one.

The most wonderful of all the fire-beetles is the large one of near two inches long—quite, or more, if we count the antennæ—that inhabits Mexico, where in ancient times it was used as a lantern by the Aztecs in their night-journeys, as it still is by their modern descendants. It is wonderful, not by reason of its size merely, or, in any special degree, of the light it emits—though this is brilliant in proportion to it—but because it carries three separate lamps: two above, situated on either side of the thorax, and one on the under side, just in front of the abdomen. Thus, as it turns or varies in its flight, one flash of the most intense brilliancy follows another, like the revolving light of a lighthouse. The colour of the light is described as a rich green—richest, however, or at least brightest, on the under surface.^[151] The beauty and dazzling effect of this upon a dark night can be imagined, and is thus described by Dr. Kidder: “Before retracing my steps I stood for a few moments looking down into the Cimmerian blackness of the gulf before me; and while thus gazing a luminous mass seemed to start from the very centre. I watched it as it floated up, revealing in its slow flight the long leaves of the palm *Euterpe edulis*, and the minuter foliage of other trees. It came directly towards me, lighting up the gloom around with its three luminosities, which I could distinctly see.”^[151] There is something wonderfully poetical in the thought of winged beings like this pursuing each other through the night, by the light of these glorious flashes—the “light of their own loveliness,” it may well be called, since it is, indeed, their beauty. It seems curious and a waste that where there is the greatest capacity of poetic imagination we should find the least, or almost the least, realisation of it in habit and structure.

We know from Oviedo that the Mexican Indians, when they travelled at night, were accustomed to fasten these great refulgent beetles on their hands and feet, and thus pass flaming through the country. They danced, too, by their light, and even wove or painted by it. Why, therefore, could not lamps of great power, as well as beauty, be evolved from such insects by bringing the selective agency of man to bear upon them? The phosphorescent principle in living nature has not perhaps been made the most of by us. Was more made of it by the Aztecs? and did they turn their attention to the systematic rearing of these living lamps?—for, from hearing so little about

them one would not think that these insects were so useful now, as, from the above account and what other contemporary Spanish writers tell us, it would seem that they were, at the time of this old and cruelly destroyed civilisation.

Holder, in his work on phosphorescent animals, either quotes or refers to Prescott as saying that “when the Spaniards visited the country”—that is, Mexico, “the air was filled with the *cucujo*, a species of large beetle which emits an intense phosphoric light from its body strong enough to enable one to read by. These wandering flies, seen in the darkness of the night, were converted by the excited imagination of the besieged into an army of matchlocks.” Surely, from such a foundation, something as superior to it as are our cultivated fruits, or domestic breeds, to the wild stocks from which they sprung, might in no long time be produced, since it is not to be supposed but that some individuals of the *Pyrophorus* give a stronger light than others. The above passage, by the way, if it be from the *Conquest of Mexico*, as one might suppose it to be, is most carefully concealed in the index, which, however, it might very well be, and yet exist, as I know from much teasing experience. As to the matchlocks, would to Heaven the old Mexicans, as well as the Peruvians, had had them, or, still better, 11-inch Howitzers. I might then have something more to say about these wonderful beetles. All I can add now is that the light appears to be used by the insect as a guide to its own movements, since when the celebrated Dr. Dubois covered one of the side ones with wax, this caused the individual so treated to walk in a curve, and when “both spots were covered it soon stopped, and then moved in an uncertain manner, carefully feeling the ground with its antennæ.”^[151] But I do not know if “both” here means all three of the lamps, or only the two upper ones.

If there be any luminous insect that eclipses the *Pyrophorus* it must be the great lantern-fly—also of South America—provided only that the great lantern-fly is luminous. That is a most essential point, and it does not appear yet to have been satisfactorily made out. The principal evidence on the affirmative side is that of Madame Merian, who was right about the *Mygale*—the great bird-killing spider—and who here speaks as an actual eye-witness. Her account is as follows: “The Indians,” she says, “once brought me, before I knew that they shone at night, a large number of these lantern-flies, which I shut up in a large wooden box. In the night they made such a

noise that I awoke in a fright, and ordered a light to be brought, not knowing from whence the noise proceeded. As soon as we found that it came from the box we opened it, but were still more alarmed, and let it fall to the ground in a fright at seeing a flame of fire come out of it; and as many animals as came out, so many flames of fire appeared. When we found this to be the case we recovered from our fright, and again collected the insects, highly admiring their splendid appearance.”^[151]

Here, then, is a definite statement, from which all possibility of mistake seems excluded, if, as I suppose is the case, there is no doubt as to the specific identity of the insect which was the subject of it, and which is thus described by Mr. Holder in the work already mentioned: “The *Fulgora lanternaria* of South America,” he tells us, “is nearly three and a half inches long from tip of head to extremity of tail (*i.e.* abdomen), and almost five and a half inches broad with its wings expanded.” Truly a goodly insect, of right portly dimensions, and if it be not really luminous—upon occasions, at any rate, for it certainly is not so generally—it is so much the greater pity. But to continue: “The body is of a lengthened oval shape, while the head is distinguished by a singular prolongation, which sometimes equals the rest of the body in size.” This is a most remarkable appendage, if it may be called so, hollow and with a blown-up, inflated sort of look. It does, indeed, to some extent resemble a Chinese lantern, and seems made to be lighted up. The colour, too, suggests this, since it is striped longitudinally with red and yellow, presenting quite a gala appearance. Accordingly, it is said to be here that the luminous property of this strange insect exists. This is its lantern, and, by reason of it, it has received its name of lantern-fly.

And yet, since that night when Madame Merian had her interesting experience, we meet with no one, apparently, who can unequivocally say that he has seen the Great Lantern-Fly with its lantern alight. On the other hand, we have some second-hand statements which have almost the value of first, such as that of M. Westmael, who “assures us that a friend of his observed the luminosity”;^[151] whilst “John C. Branner, PH.D., states that when in South America he was often informed that it was luminous, but never could find anyone who had personally seen the light.”^[151] The curious thing is that there are other lantern-flies belonging to other parts of the world, and in regard to them too we have the same doubt and discrepancy, the same assurances and general belief, the same categorical denials. Thus a

distinguished authority on the subject of phosphorescence— Dr. Phipson—in referring to the smaller Chinese species, *Fulgora candelaria*—the candle-fly—says: “It is from these appendages, the sides of which are transparent, that the phosphoric light appears.” And again: “It is said also that the trunk of a tree covered with numerous individuals of *Fulgora candelaria*, some in movement, others in repose, presents a very grand spectacle, impossible to describe, but which may be witnessed sometimes in China.”^[151] It would seem, too, that there exists a Chinese edict which forbids young women to keep these candle-flies; and if this is not with the idea of preventing their use as signals, or of checking vanity, it is difficult to see what the object of such an enactment can be.

Lastly, we are told by Packard, in his *Guide to Insects*, that “Mr. Caleb Cooke, of Salem, who resided several years in Zanzibar, Africa, told me that the lantern-fly is said by the native to be luminous. They state that the long snout lights up in the night, and in describing it say its head is like a lamp (*keetchwa kand-tah*).”

All this evidence appears to me to point in one way, and one way only—I mean, of course, in its entirety, since otherwise it points in two ways. But even if it is possible that in one country alone an insect—well known and conspicuous—can have got the reputation of being luminous without really being so, at least occasionally, this can hardly have come about in regard to the same, or some allied insect, in three or four countries. Added to this we have Madame Merian’s direct evidence, but, on the other hand, it is perfectly clear that these insects are not always, or even generally, luminous. The conclusion, then, seems irresistible that they occasionally are so, that, for some reason or other, the phosphorescent principle is active in them only at certain times or seasons. Why this should be so we do not know, but there is nothing inconceivable in it; and some other animals—for instance, centipedes—would seem to be luminous at some times and not at others.

The so-called lantern or snout being a very remarkable organ, for which some use must be assumed, the likelihood of its sometimes becoming a lamp would be increased considerably, if, so far as we knew, it performed no other office. This was how the case stood till lately; but in 1899 there was the Skeat Expedition for scientific purposes to the Malay Archipelago,

and on its return Mr. Nelson Annandale propounded a theory in regard to the more ordinary use, at least, of the organ in question, which was based on his own observation. His account is as follows: “The curious anterior prolongation of the head in many genera of the *Fulgoridæ* has long puzzled entomologists. At Biserat, in Jalor, I was fortunate enough to observe the real use of this peculiar structural modification. On the morning of May 30th I noticed a specimen of *Hotinus spinola* seated on the trunk of a Durian tree in the village, and incautiously attempted to catch it in my hand. The insect remained almost still, merely drawing in its legs towards its body and pressing the claws firmly against the bark, until I had almost touched it. Then it lowered its head with very great rapidity, flew up into the air without spreading its wings, and alighted on the roof of a house six feet behind a tree, and considerably higher than its position on the trunk had been. At the time I did not notice anything peculiar in the way in which this *Fulgorid* jumped, for there are many large species of the same family which, without being provided with long noses, can leap for a considerable distance by means of their legs only; but as I was examining my specimen (a dead one) I was struck by an indentation or crease that ran across the central region of the nose at right angles to its main axis. Then I discovered that at this point, and at this point only, it was flexible, and that if the tip of the nose and the dorsal surface of the abdomen were pressed together between the finger and thumb, and then suddenly released, the insect would not fall straight to the ground, but would be propelled for some distance through the air before doing so, just as would be the case if a piece of whalebone were treated in like manner.”^[152]

Mr. Annandale then goes on to show, or to suggest, that the *Fulgorid*—as he calls it—by pressing its snout—or lantern—against the tree-trunk, and at the same time pushing itself off from it with its legs, “would fly into the air at a tangent,” and he continues: “I have no doubt that this is substantially what occurs in the case of *Hotinus*; but in the living insect the action is far too rapid for the eye to discriminate its details, and dead specimens cannot be made to leap in this way because it is impossible to force the legs to perform their part of the action.”^[152] Such, then, is the theory, but as other members of the family jump in much the same way, to all appearance, without any such apparatus, and since the bending of the head, at such a

moment, might be correlated with the movements requisite to produce such a leap as this, it certainly wants confirmation.

Some of the finest displays of luminosity have been observed in centipedes, which although not insects, may be counted such for the purpose of this volume. Thus M. Audouin, noticing one night a light proceeding from one of his chicory-fields, “ordered his man to turn up the earth, when the scene that followed is described as truly magnificent. The soil appeared as if it had been sprinkled with molten gold, the display being intensified if the insects were trodden upon or rubbed. In the latter case streaks of light appeared, as if a bit of phosphorus had been placed upon the hands, the light being distinctly visible for twenty seconds.”^[153]

Mr. Brodhurst, again, referring to another species—*Geophilus electricus*—about an inch and a half in length, and in the daytime inconspicuous enough, says: “The light looked like moonlight, so bright was it through the trees. It was a dark night, warm and sultry. Taking a letter, I could read it. It resembled an electric light, and proceeded from two centipedes and their trails. The light illuminated the entire body of the animal, and seemed to increase its diameter three times. It flashed along both sides of the creature in sections, there being about six, from head to tail, between which the light played, moving, as it were, perpetually in two streams. The trail extended one and a half feet from each centipede over the grass and gravel walk, and it had the appearance of illuminated mucus. On securing one of the creatures for examination, I found on touching it the light was instantly extinguished.”^[153] The display is, therefore, voluntary, nor could Mr. Brodhurst ever get his centipedes to shine in captivity.

CHAPTER XXVI

Scorpions and suicide—The act proved—Intention probable—Conflicting evidence—
Scorpions and cockroaches—Concentrating backwards—Economy of poison—
Decorous feeding.

THE assertion that scorpions are occasionally luminous—if indeed it has ever been seriously made—does not appear to have received confirmation. Of fire, indeed, these creatures have a horror, but that probably relates to its property—heat—to which they are extremely sensitive. The popular belief is, that, if surrounded by fire, a scorpion will deliberately sting itself to death. Of the fact, or, at least, of the fact of the self-inflicted sting, there can be little doubt, but in regard to the motive there is room for difference of opinion. Mr. Pocock says, truly enough, that it is *à priori* improbable that the scorpion has any intention of killing itself.^[154] But what, then, is its intention in stinging itself, supposing that it deliberately does so? Nor must it be forgotten that the idea of death—of destruction—must be indissolubly associated in the scorpion's mind with the use of its sting, since it uses it with that purpose only, and that is the result which constantly attends its use. Is it, then, really so improbable that it stings itself with the same intention as that with which it stings other creatures?—or, rather, with what other possible intention can it do so, assuming the act to be a voluntary one?

Nor would it be necessary to prove the intention that the sting, thus delivered, should be fatal in its effects, and, in regard to this, Mr. Bourne has satisfied himself by experiments with some Indian scorpions that a self-inflicted wound, or even wounds inflicted by individuals of the same species on one another, have no effect. On the other hand, he found that a moderately high temperature was fatal to his scorpions, and so concludes that this has been the real cause of death in all such cases as we are here considering.^[154]

The above theory, however, hardly accords with the experience of Mr. W. G. Bidie, also of India, and that very part of it where Mr. Bourne's

experiments were made—viz. Madras. Writing to *Nature*, he says: “One morning a servant brought me a large specimen of this scorpion (the common black one of Southern India), which, having stayed out too long in its nocturnal rambles, had apparently got bewildered at daybreak and been unable to find its way home. To keep it safe the creature was at once put into a glazed entomological case. Having a few leisure minutes in the course of the forenoon, I thought I would see how my prisoner was getting on, and to have a better view of it, the case was placed in a window in the rays of the hot sun. The light and heat seemed to irritate it very much, and this recalled to my mind a story which I had read somewhere that a scorpion on being surrounded with fire had committed suicide. I hesitated about subjecting my pet to such a terrible ordeal, but taking a common botanical lens, I focussed the rays of the sun on its back” (so that Apollo may have flayed Marsyas as a mild alternative). “The moment this was done it began to run hurriedly about the case, hissing and spitting in a very fierce way. This experiment was repeated some four or five times with like results, but on trying it once again the scorpion turned up its tail and plunged the sting, quick as lightning, into its own back. The infliction of the wound was followed by a sudden escape of fluid, and a friend, standing by me, called out, ‘See! it has stung itself: it is dead.’ And sure enough in less than half a minute life was quite extinct.”

This seems plain enough. The scorpion had not died of the heat, up to the moment at which it stung itself—an act which would require some vital energy. It did sting itself, and in less than half a minute afterwards it was dead. Moreover, as the experiments with the lens were intermittent, there seems no more reason why the last one should have been fatal than the other four or five. It is, perhaps, possible to imagine that the scorpion was almost dead before, that the last heating caused it to expire, and that in the moment of doing so it stung itself by involuntary muscular action. There is nothing, however, in the narrative to suggest this, but quite the contrary.

Supposing the sting to have been a voluntary act, what could the scorpion have intended except to injure itself? Had it ever in its life used its sting with any other purpose than that of doing injury? Mr. Bidie adds: “I have written this brief note to show (1) that animals may commit suicide; (2) that the poison of certain animals may be destructive to themselves.”^[155]

Writing several years later, also to *Nature*, Dr. Allen Thomson, F.R.S., gives the following account, not, indeed, of his own experience, but that of an eye-witness in whom he feels full confidence. He says: “While residing, many years ago, during the summer months, at the baths of Sulla, in Italy, in a somewhat damp locality, my informant, together with the rest of the family, was much annoyed by the frequent intrusion of small black scorpions into the house, and their being secreted among the bedclothes, in shoes, and other articles of dress. It thus became necessary to be constantly on the watch for these troublesome creatures, and to take means for their removal and destruction. Having been informed by the natives of the place that the scorpion would destroy itself if exposed to a sudden light, my informant and her friends soon became adepts in catching the scorpions and disposing of them in the manner suggested. This consisted in confining the animal under an inverted drinking-glass or tumbler, below which a card was inserted, and then, waiting till dark, suddenly bringing the light of a candle near to the glass in which the animal was confined. No sooner was this done than the scorpion invariably showed signs of great excitement, running round and round the interior of the tumbler with reckless velocity for a number of times. This state having lasted for a minute or more, the animal suddenly became quiet, and, turning its tail on the hinder part of its body over its back, brought its recurved sting down upon the middle of the head, and, piercing it forcibly, in a few seconds became quite motionless, and, in fact, quite dead. This observation was repeated very frequently.”^[156]

Here, again, it is difficult to see how a mistake in observation can have occurred, and admitting the facts to be true, they go far beyond Mr. Bourne’s theory to account for these phenomena, which, however, has been adopted by Mr. Pocock, as the result of his own experiments. In the first place, it is not here the heat—unless by association of ideas—but the actual sight of the flame that terrifies the scorpion, and death, apparently, is inflicted as the result of that. Again, there can be no doubt as to the self-inflicted stinging, and from the manner of it, as well as its invariability, it seems to have been deliberate. Whether death was the result of it or not, we have the act, and the act, if a voluntary one, must have implied a destructive intention. It hardly seems possible, however, that the light of a candle, outside a tumbler, though held near it, can in so short a time have made the interior so hot as to kill the scorpion, whilst, on the other hand, the poison

from the creature's sting must have pierced its brain, and a few seconds afterwards it was dead.

If, then, we decide to disbelieve in the story of scorpions committing suicide when unable to escape from fire, we must explain away these two accounts, which we can do by supposing the narrators to be either dishonest or stupid. There is no other way that I can see, so if neither of these do, we ought to believe the story. However, there is plenty of evidence which points in the opposite direction, and the advantage of this is that we can take our choice.

Scorpions are interesting animals to keep in captivity, and their habits under these conditions have been carefully studied by Mr. Pocock.^[154] When supplied with sand they dig pits in it, in which they lie during the greater part of the day. The second and third pairs of legs are used for this purpose, the scorpion raising itself upon the other two pairs, as well as, to a certain extent, upon the claws and the end of the tail. In this position it kicks the sand backwards from under it, and then when the excavation is sufficiently deep, sweeps away the accumulated heap, with its tail, so that the edge of its lurking-place is on a level with the surrounding surface. It can thus, as it lies there, obtain an uninterrupted view, which the better enables it to receive with proper attention any creature of the requisite size and quality that approaches its portals. Such creatures are principally insects, spiders, centipedes, wood-lice, and the like—but here we may remember one little spider that imitates a scorpion, and may therefore approach with impunity, at least if the disparity in size be not too great, for whilst some scorpions are quite small, others attain a length of eight or nine inches, with a bulk more than in proportion to their length.

In captivity, and, no doubt, under nature too, when they happen to come across them, scorpions will eat cockroaches, but a cockroach is not altogether a defenceless creature, and sometimes a large one will give battle, and even with success. The weapons upon which, in these cases, it relies are its powerful hind legs armed, as they are, with spines which project backwards. Backwards accordingly it advances upon the scorpion, and increasing its pace suddenly, when at the requisite distance salutes the astonished enemy with a shower of kicks. So unexpected is this mode of assault that it is sometimes effective, even against so redoubtable an opponent as a scorpion, whilst a tarantula spider has been known to fly,

panic-stricken, before or rather behind a large cockroach. But such efforts, however heroic, can have only a transient success, where the conditions are so unequal. Jaws and sting must prevail against soft bodies armed only with spiny legs. “*Alla stoccata* carries it away.” Generally the poor cockroach is seized—sometimes, in the first instance, by the antennæ—as it comes inadvertently too near to the scorpion, or even trespasses upon its back. At once the tail is bent above it, and the fatal sting enters its body. Paralysis ensues, and would no doubt be quickly followed by death, even were the scorpion, thereupon, to retire. As it is, however, it is difficult to say whether the victim dies more of the sting or of being eaten.

From the latter process, at any rate, there is no recovery, as may be seen in the case of smaller cockroaches, upon whom the scorpion, from motives of economy, does not always waste its poison. It merely, when thus provident, holds the contemptible creature in its claws, whilst bringing to bear upon it its two pairs of chelæ or real jaws, which act upon the same principle as those of *Galeodes vorax*, if the reader remember. It feeds in a leisurely manner, the impatience of the cockroach not affecting it in the least. Two hours for a good-sized one—a *pièce de résistance*—is not considered too long by the scorpion.

Scorpions, it appears, use their stings in a very careful, deliberate manner. It is not a mere random thrust with them, lunged in anywhere, just as the body of an insect happens to come. On the contrary, they feel about this body, most anxiously, with their tail, till they have found a soft spot in it, and then introduce their sting in a careful manner. In fact, they sting an insect in much the way that Isaak Walton impaled a frog upon the hook—“tenderly as if they loved him”—and for the same class of reason, viz. to make a workmanlike job of it, and not break their stings against the harder parts of its body, for the point of this weapon is delicate and might get chipped against the hard shards of a beetle, or other such resisting surface.

For the same prudent reason the tail is carried aloft, over the scorpion’s back, when it walks, so that the whole organ, but especially the point of it, which is curled round again underneath, is preserved from contact with the outer world. The sting, or rather the sides of the poison vesicle just above it, are clothed with hairs, which are, no doubt, delicately tactile, and the same may be said of the tail and various other parts of the body. Touch, indeed, is

the principal sense which conveys impressions to the soul of the scorpion. Sight is defective, and hearing does not seem to exist.

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- [1.](#) *The Concise Natural History*, p. 551.
- [2.](#) In conjunction with Dr. Sandias, whose name must be understood as accompanying Grassi's—for the most part—when the latter is referred to.
- [3.](#) *Quarterly Journal of Microscopical Science*, vols. 39 and 40.
- [4.](#) A nymph is the free-moving active equivalent of the chrysalis amongst moths and butterflies.
- [5.](#) Nests built in chemical glass tubes and thus under close observation.
- [6.](#) *Chambers's Encyclopædia*.
- [7.](#) *Nature*, March 23rd, 1893.
- [8.](#) *The Cambridge Natural History*, vol. 6, p. 134.
- [9.](#) *Quarterly Journal of Microscopic Science*, vols. 39 and 40.
- [10.](#) Others call them *inquilines*.
- [11.](#) Kirby, *Marvels of Ant Life*, p. 100.
- [12.](#) If *cadaver* for corpse or carcase, why not *dies* for day, which is just as good English? Or why not all Latin, with a glossary, or—better still—a translation?
- [13.](#) Charles Janet, *Comptes Rendus*, 1897, pp. 583-5.
- [14.](#) *The American Naturalist*, December, 1901.
- [15.](#) *Ibid.*, November, 1900.
- [16.](#) *The Entomologist's Monthly Magazine* (quoted in the *American Naturalist*), August, 1896.
- [17.](#) Leland O. Howard, *The Insect Book*.
- [18.](#) Dr. Æneas Munro, *The Locust Plague and its Suppression*.
- [19.](#) E. Selous, *Bird Watching*, pp. 213-15, 271, etc.
- [20.](#) Dr. Æneas Munro, *The Locust Plague and its Suppression*.
- [21.](#) Dr. Æneas Munro, *The Locust Plague and its Suppression*.
- [22.](#) *The Agricultural Gazette of New South Wales*, March, 1900.

- [23.](#) Dr. Æneas Munro, *The Locust Plague and its Suppression*.
- [24.](#) *The Agricultural Gazette* of New South Wales, March, 1900.
- [25.](#) Leland O. Howard, *The Insect Book*. It has, however, been asserted, I know not with what truth, that Mr. Scudder was mistaken in this particular, and changed his opinion.
- [26.](#) Leland O. Howard, *The Insect Book*.
- [27.](#) *The American Naturalist*, April, 1898.
- [28.](#) *Transactions of the New Zealand Institute*, vol. 5, p. 286.
- [29.](#) *Proceedings of the Zoological Society for 1900*, pp. 837-69.
- [30.](#) *Transactions of the New Zealand Institute*, vol. 5, p. 286.
- [31.](#) *Nature*, vol. 44, p. 451.
- [32.](#) Leland O. Howard, *The Insect Book*.
- [33.](#) Plato.
- [34.](#) The above account, with the translation of the verses, is from Mr. Leland's work on Etruria. I have, however, altered some lines, in order to retain the Italian name *cavalletta* instead of the American Katydid, which jars horribly here.
- [35.](#) Vol. 57.
- [36.](#) Appleton's *Popular Scientific Monthly*, vol. v.
- [37.](#) Buckton, *Monograph of the British Cicadæ or Tettigidæ*.
- [38.](#) White's *Natural History of Selborne*, Letter liii. p. 283 (stereotyped edition).
- [39.](#) Buckton, *Monograph of the British Cicadæ or Tettigidæ*.
- [40.](#) Leland O. Howard, *The Insect Book*.
- [41.](#) *The Entomologist's Monthly Magazine*, October, 1900.
- [42.](#) Hudson, *The Naturalist in La Plata*, pp. 131, 32.
- [43.](#) Buckton, *Monograph of the British Aphides*, vol. i
- [44.](#) The Concise Knowledge Library, *Natural History*, pp. 601, 602.
- [45.](#) Darwin, *Origin of Species*, pp. 207, 208.
- [46.](#) "Letter to the Smithsonian Institute," *The American Naturalist*, September, 1874, p. 565.
- [47.](#) Belt, *The Naturalist in Nicaragua*, 1874, p. 226.

- [48.](#) February, 1898, vol. 14.
- [49.](#) Lubbock, *Ants, Bees, and Wasps*, p. 72.
- [50.](#) Kirby, *Text-book of Entomology*, p. 113.
- [51.](#) It appeared, I think, originally in the *Bombay Journal* and is reproduced in one of the leading entomological magazines.
- [52.](#) Darwin, *Origin of Species*, pp. 216, 217, 218.
- [53.](#) Rev. Farren White, *Ants and their Ways*, pp. 177-9.
- [54.](#) Lubbock, *Ants, Bees, and Wasps*, pp. 87-9.
- [55.](#) Rev. Farren White, *Ants and their Ways*, pp. 177-9.
- [56.](#) *The American Naturalist*, vol. 35, No. 419.
- [57.](#) *The American Naturalist*, July, 1901.
- [58.](#) Lubbock, *Ants, Bees, and Wasps*, pp. 78, 79.
- [59.](#) Janet, *Etudes sur les Fourmis*.
- [60.](#) Belt, *The Naturalist in Nicaragua*, pp. 72, 73, 79-81.
- [61.](#) *Journal of the Trinidad Field Club*, No. 3.
- [62.](#) As quoted in *The American Naturalist* for November, 1900.
- [63.](#) *The American Naturalist*, November, 1900.
- [64.](#) Belt, *The Naturalist in Nicaragua*, pp. 27, 28.
- [65.](#) Lubbock, *Ants, Bees, and Wasps*.
- [66.](#) The greatest being speedy extinction.
- [67.](#) Darwin, *The Descent of Man*, p. 54.
- [68.](#) *The Agricultural Gazette of New South Wales*, March, 1900.
- [69.](#) Leland O. Howard, *The Insect Book*.
- [70.](#) *The Malay Archipelago*.
- [71.](#) Darwin, *The Descent of Man*, p. 300.
- [72.](#) *The Malay Archipelago*.
- [73.](#) *Souvenirs Entomologiques*.

- [74.](#) Romanes, *Animal Intelligence*, pp. 106-107.
- [75.](#) *The American Naturalist*, February, 1902.
- [76.](#) *The Cambridge Natural History*, vol. 6.
- [77.](#) Du Chaillu, *Adventures in Equatorial Africa*, pp. 312, 313.
- [78.](#) J. G. Wood, *Homes Without Hands*, p. 452.
- [79.](#) W. F. Kirby, *Marvels of Ant life*, p. 73.
- [80.](#) Dr. Beccari's account, quoted in the *Popular Science Review*, 1875.
- [81.](#) Belt, *The Naturalist in Nicaragua*, pp. 219, 220.
- [82.](#) The ideal of what a short-billed tumbler pigeon's head should be.
- [83.](#) McCook, *The Honey-Ant of the Garden of the Gods*.
- [84.](#) *The American Naturalist*, February, 1902.
- [85.](#) Belt, *The Naturalist in Nicaragua*, pp. 28, 29.
- [86.](#) Fabre, *Insect Life*, pp. 275-283.
- [87.](#) Peckham, *The Instincts and Habits of the Solitary Wasps*.
- [88.](#) Bingley, *Animal Biography*.
- [89.](#) The real purple of the ancients was a rich red—crimson or vermilion.
- [90.](#) Romanes, *Animal Intelligence*, pp. 361, 362.
- [91.](#) Bates, *The Naturalist on the River Amazon* (second edition, 1864), p. 420.
- [92.](#) Belt, *The Naturalist in Nicaragua*, pp. 19, 20.
- [93.](#) *Proc. Zool. Society for 1891*, pp. 462, 463.
- [94.](#) *Proc. Zool. Society for 1900*, pp. 837-69.
- [95.](#) Fabre, *Insect Life*, p. 165.
- [96.](#) *Proc. Zool. Society for 1900*, pp. 837-69.
- [97.](#) There are two British species of Elephant Hawk-Moth, the large and small. The caterpillar of the former has four false eyes, that of the latter only two.
- [98.](#) Poulton, *The Colour of Animals*, pp. 258-61.
- [99.](#) Leland O. Howard, *The Insect Book*.

- [100.](#) *Proc. Zool. Society for 1900*, pp. 837-69.
- [101.](#) Leland O. Howard, *The Insect Book*.
- [102.](#) *Nature*, June 25th, 1903.
- [103.](#) Poulton, *The Colour of Animals*, pp. 258-61.
- [104.](#) Wallace, *The Malay Archipelago*, pp. 100-2.
- [105.](#) Forbes, *A Naturalist's Wanderings in the Eastern Archipelago*.
- [106.](#) Trimen, *Protective Resemblance and Mimicry in Animals*.
- [107.](#) "Contributions to an Insect Fauna of the Amazon Valley," in *Transactions of the Linnean Society*, vol. 23, p. 495.
- [108.](#) Belt, *The Naturalist in Nicaragua*, p. 109.
- [109.](#) *Ibid.*, pp. 316, 317.
- [110.](#) *Ibid.*, p. 109.
- [111.](#) "Impostors among Animals," in the *Century Magazine*, July, 1901.
- [112.](#) Bates, *The Naturalist on the River Amazon*.
- [113.](#) *Ibid.* (1864), p. 63.
- [114.](#) Wallace, *The Malay Archipelago*, p. 29.
- [115.](#) *Ibid.*, pp. 257, 258.
- [116.](#) *Ibid.*, pp. 328, 329.
- [117.](#) Bates, *The Naturalist on the River Amazon* (1864), p. 333.
- [118.](#) In 1813 Dr. Wells, first amongst the moderns, conceived, or at least formulated in writing, the idea of natural selection; but Aristotle, as is usual in such cases, had anticipated him as well as Darwin, Wallace, and one or two others, and that in a very unmistakable sentence (see footnote on first page of the "Historical Sketch" in *The Origin of Species*). It would seem, however, that no one of these conceptions was influenced by any previous one. Of sexual selection Darwin seems to have been the discoverer as well as, in the opinion of many, the demonstrator.
- [119.](#) Belt, *The Naturalist in Nicaragua*, pp. 108-12.
- [120.](#) Swallows have the highest temperature known amongst birds, viz. 111 $\frac{1}{4}$.
- [121.](#) C. Collingwood, *Rambles of a Naturalist in the China Seas*, p. 183.
- [122.](#) Scudder, *Frail Children of the Air*.
- [123.](#) See pp. [179-181](#).

- [124.](#) Scudder, *Frail Children of the Air*.
- [125.](#) My own italics.
- [126.](#) Scudder, *Frail Children of the Air*.
- [127.](#) My own italics.
- [128.](#) “Contributions to an Insect Fauna of the Amazon Valley,” in *Trans. Linn. Soc.*, vol. xxiii, p. 495.
- [129.](#) Belt, *The Naturalist in Nicaragua*, pp. 108-12.
- [130.](#) *Tropical Nature*, p. 97.
- [131.](#) *Malay Archipelago* (1898), p. 331.
- [132.](#) *Observations on Sexual Selection in Spiders*.
- [133.](#) *Occasional Papers of the Natural History Society of Wisconsin*, vol. i., Nos. 1, 2, and 3.
- [134.](#) *The Colours of Animals*.
- [135.](#) *Occasional Papers of the Natural History Society of Wisconsin*, vol. i., Nos. 1, 2, and 3.
- [136.](#) Vinson, *Aranéides des Iles de la Réunion, Maurice, et Madagascar*, pp. 268, 269.
- [137.](#) Bates, *The Naturalist on the River Amazon* (1864), p. 96.
- [138.](#) *Annals and Magazine of Natural History*, vol. x. (1872), pp. 273, 274.
- [139.](#) Captain Thomas Hutton, “Observations on the Habits of a Large Species of *Galeodes*,” *Journal of the Asiatic Society*. Reprinted in *Annals and Magazine of Natural History*, August, 1873, No. 75.
- [140.](#) The above facts are quite reliable, but having made my notes, I forgot to mention their source—one of the established entomological organs—and so cannot now refer to it.
- [141.](#) *Annals and Magazine of Natural History*, September, 1874.
- [142.](#) *The Faerie Queene*, book 2, canto vi.
- [143.](#) *Mémoires du Muséum*, vols. 18-20.
- [144.](#) *Entomologist*, vol. 14 (1881), p. 82.
- [145.](#) Quoted in *The Insect Book*, by Leland O. Howard.
- [146.](#) *Chambers’s Encyclopædia*.
- [147.](#) White, *Natural History of Selborne* (stereotyped edition), p. 84.
- [148.](#) *Ibid.*, p. 353.

- [149.](#) *Bull. Soc. Entomolog. Ital.*, 1885-7.
- [150.](#) *Cassell's Illustrated Shakespeare*, "Hamlet," Act i., Scene 5, Note 137.
- [151.](#) C. F. Holder, *Living Lights*.
- [152.](#) *Proceedings of the Zoological Society for 1900*.
- [153.](#) C. F. Holder, *Living Lights*.
- [154.](#) *Nature*, June 1st, 1893.
- [155.](#) *Ibid.*, vol. xi.
- [156.](#) *Ibid.*, vol. xx. p. 577.

Transcriber's Notes

[Title page](#) - Added missing period

Pg. [42](#) - Changed 'free-booter' > 'freebooter' in caption to match illustration index and rest of text

Pg. [106](#) - "Unto to me" > "Unto me"

Pg. [140](#) - "Tierra del Fuega" > "Tierra del Fuego"

Pg. [203](#) - 'plausibilty' > 'plausibility'

Pg. [137](#) & [142](#) - 'aphis' > 'Aphis' to match other instances

Pg. [145](#) - Corrected typo: 'Sycænidæ' > 'Lycænidæ'

Pg. [187](#) - Corrected typo: 'Bell' > 'Belt'

Pg. [217](#) - Corrected typo: "312, 33" > "312, 313" - Source checked

Pg. [222](#) - Added missing close-quote w/reference to source

Pg. [222](#) - Corrected typo: 'Amophila' > 'Ammophila'

Pg. [253](#) - Corrected typo: 'caterpilllar' > 'caterpillar'

Pg. [270](#) - Corrected typo: 'replique' > 'réplique'

Pg. [290](#) - Corrected typo: 'exisit' > 'exist'

Pg. [334](#) - Corrected typo: 'especally' > 'especially'

Except for the above, inconsistent hyphenation, as well as archaic and non-standard spelling and has been retained.

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